

Solutions to

Daron Acemoglu

Introduction to Modern Economic Growth

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Chapters 2–12

The 139 exercises that the instructor’s solutions manual of Peters and Simsek (2009) leaves unsolved, with the numerical parts computed by the Python programs in `code/`.

Each chapter opens with the list of the exercises it covers and those the manual already solves.

The exercise statements are not reproduced here: read them in Acemoglu’s *Introduction to Modern Economic Growth*.

Each solution is headed by the number of its exercise in the book; notation, the part letters (a), (b), ... follow the statements there.

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Chapter 2

The Solow Growth Model

Solutions to the fourteen exercises of Chapter 2 of Daron Acemoglu, *Introduction to Modern Economic Growth* (Princeton University Press, 2009), that are left unsolved in the instructor's solutions manual of Peters and Simsek (2009): Exercises 2.1–2.6, 2.8, 2.9, 2.10, 2.13, 2.15, 2.24, 2.25 and 2.26. The remaining thirteen exercises of the chapter (2.7, 2.11, 2.12, 2.14, 2.16–2.23 and 2.27) are solved there. Assumptions, propositions and equations cited as ASSUMPTION 2, PROPOSITION 2.5 or (2.18) refer to the book. Throughout, F is the aggregate production function, $f(k) \equiv F(k, 1, A)$ the per capita production function, $k \equiv K/L$ the capital-labor ratio, $w(k) \equiv f(k) - kf'(k)$ the wage, and $R(k) \equiv f'(k)$ the rental rate of capital. The numerical parts are computed by `code/ch2.py`.

Exercise 2.1

Fix a date t and write $\bar{L}(t) > 0$ for the (finite, inelastically supplied) labor endowment and $K(t)$ for the capital stock used in production. In a competitive equilibrium the representative firm solves (2.5),

$$\max_{K \geq 0, L \geq 0} F(K, L, A(t)) - R(t)K - w(t)L,$$

taking $(R(t), w(t))$ as given, and factor markets clear in the complementary slackness form (2.4):

$$L(t) \leq \bar{L}(t), \quad w(t) \geq 0, \quad (L(t) - \bar{L}(t))w(t) = 0.$$

Suppose, to get a contradiction, that $w(t) = 0$. Let $(K(t), L(t))$ be the equilibrium factor demands, which must be finite since they cannot exceed the endowments. Consider raising employment to $L(t) + \varepsilon$ for $\varepsilon > 0$, holding capital fixed. Profits change by

$$F(K(t), L(t) + \varepsilon, A(t)) - F(K(t), L(t), A(t)) - w(t)\varepsilon = \int_{L(t)}^{L(t)+\varepsilon} F_L(K(t), \ell, A(t)) d\ell > 0,$$

because $w(t) = 0$ and $F_L > 0$ everywhere by ASSUMPTION 1. So the firm strictly prefers more labor at *every* finite L : the maximization problem has no solution, and in particular $(K(t), L(t))$ is not profit maximizing. That contradicts firm optimization, so $w(t) > 0$.

Given $w(t) > 0$, the complementary slackness condition in (2.4) forces $L(t) - \bar{L}(t) = 0$, which is exactly the simple market clearing condition (2.3), $L(t) = \bar{L}(t)$. This is why (2.3) can be used throughout in place of (2.4).

Remark. Only $F_L > 0$ is used; neither constant returns nor the Inada conditions are needed. The identical argument with $F_K > 0$ shows $R(t) > 0$, so the capital market also clears with equality, $K(t) = \bar{K}(t)$. Note also that *some* bound on labor demand is essential to the argument: it is the finiteness of the endowment that turns “the firm always wants more labor” into a contradiction.

Exercise 2.2

Fix A and write $F(K, L)$ for $F(K, L, A)$, which by ASSUMPTION 1 is twice differentiable and homogeneous of degree 1 in (K, L) , with $F_{KK} < 0$ and $F_{LL} < 0$.

The Hessian is singular. By THEOREM 2.1 (Euler's Theorem), the partial derivatives F_K and F_L are homogeneous of degree 0. Applying Euler's Theorem to each of them,

$$KF_{KK} + LF_{KL} = 0 \quad \text{and} \quad KF_{LK} + LF_{LL} = 0.$$

Hence, for $K, L > 0$,

$$F_{KL} = -\frac{K}{L}F_{KK} > 0 \quad \text{and} \quad F_{KL} = -\frac{L}{K}F_{LL} > 0,$$

the signs following from $F_{KK} < 0$ and $F_{LL} < 0$. Multiplying the two expressions,

$$F_{KL}^2 = \left(-\frac{K}{L}F_{KK}\right)\left(-\frac{L}{K}F_{LL}\right) = F_{KK}F_{LL},$$

so that the Hessian

$$\mathbf{H} = \begin{pmatrix} F_{KK} & F_{KL} \\ F_{KL} & F_{LL} \end{pmatrix} \quad \text{has} \quad \det \mathbf{H} = F_{KK}F_{LL} - F_{KL}^2 = 0.$$

Concavity. A twice differentiable function is concave if and only if its Hessian is negative semidefinite everywhere. Here the leading principal minors are $F_{KK} < 0$ and $\det \mathbf{H} = 0$, so $\mathbf{H}$ is negative semidefinite (its eigenvalues are 0 and $F_{KK} + F_{LL} < 0$). Therefore F is concave in (K, L) .

But not strictly. Strict concavity would require $\mathbf{H}$ to be negative definite, which fails since $\det \mathbf{H} = 0$; the null direction is $(K, L)'$ itself, because $\mathbf{H} \begin{pmatrix} K \\ L \end{pmatrix} = \begin{pmatrix} KF_{KK} + LF_{KL} \\ KF_{LK} + LF_{LL} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The failure is easy to see directly: along any ray through the origin, constant returns make F linear,

$$F(\lambda K + (1 - \lambda)\lambda'K, \lambda L + (1 - \lambda)\lambda'L) = [\lambda + (1 - \lambda)\lambda']F(K, L) = \lambda F(K, L) + (1 - \lambda)F(\lambda'K, \lambda'L),$$

so the concavity inequality holds with equality for any two points on a ray. This is the sense in which the production function can be “concave but not strictly concave” even though $F_{KK} < 0$ and $F_{LL} < 0$: it curves in every direction except the direction in which the two inputs are scaled together.

Remark. The per capita function $f(k) \equiv F(k, 1)$ is strictly concave, since $f''(k) = F_{KK}(k, 1) < 0$. Strict concavity of f , not of F , is what the uniqueness arguments of the chapter use.

Exercise 2.3

Write the profit function as

$$\pi(K, L) \equiv F(K, L, A) - RK - wL.$$

Since F is homogeneous of degree 1 in (K, L) , so is π : for every $\lambda > 0$,

$$\pi(\lambda K, \lambda L) = \lambda \pi(K, L).$$

For $L > 0$ write $\kappa \equiv K/L$; then constant returns give

$$\pi(K, L) = L[f(\kappa) - R\kappa - w] \equiv L\varphi(\kappa), \quad f(\kappa) \equiv F(\kappa, 1, A).$$

Everything therefore depends on the sign of the maximized value of φ , and only three cases can arise.

Case 1: $\varphi(\kappa) > 0$ for some κ (equivalently $\pi(K_0, L_0) > 0$ for some (K_0, L_0)). Then $\pi(\lambda K_0, \lambda L_0) = \lambda\pi(K_0, L_0) \rightarrow \infty$ as $\lambda \rightarrow \infty$: profits are unbounded and no maximizer exists. This is the case in which factor prices are “too low” relative to marginal products.

Case 2: $\varphi(\kappa) < 0$ for every $\kappa \geq 0$. Then $\pi(K, L) < 0$ whenever $L > 0$. On the boundary $L = 0$ constant returns give $F(K, 0, A) = K F(1, 0, A)$, and

$$F(1, 0, A) = \lim_{L \rightarrow 0} Lf(1/L) = \lim_{k \rightarrow \infty} \frac{f(k)}{k} = 0$$

by ASSUMPTION 2, so $\pi(K, 0) = -RK \leq 0$ with equality only at $K = 0$. Since $\pi(0, 0) = 0$, the unique maximizer is $K = L = 0$ and maximal profit is zero.

Case 3: $\max_{\kappa} \varphi(\kappa) = 0$, attained at some $\kappa^* > 0$. Then $\pi(K, L) = 0$ for every (K, L) with $K/L = \kappa^*$, and $\pi(K, L) < 0$ otherwise (for $L > 0$), so the solution set is the entire ray

$$\{(K, L) : K/L = \kappa^*, L \geq 0\},$$

a continuum: the scale of the firm is completely indeterminate, only its capital-labor ratio is pinned down.

These are the only possibilities, since $\max_{\kappa} \varphi$ is either positive, negative, or zero. In equilibrium it is Case 3 that must obtain: factor supplies are fixed and finite, so Case 1 is inconsistent with market clearing and Case 2 would leave the supplies unemployed. Hence equilibrium factor prices satisfy $\varphi(\kappa^*) = 0$ with $\varphi'(\kappa^*) = f'(\kappa^*) - R = 0$, which is (2.7), and then $w = f(\kappa^*) - \kappa^* f'(\kappa^*)$, which is (2.6); profits are zero, as PROPOSITION 2.1 states. The indeterminacy of scale is why the model determines only aggregates, and why the representative firm can be taken to operate at the economy's endowment point.

Exercise 2.4

It is convenient to factor the function once and for all:

$$f(k) = k(k-1)(k-2)(k-3), \quad \text{so} \quad \frac{f(k)}{k} = (k-1)(k-2)(k-3).$$

Also

$$\begin{aligned} f'(k) &= 4k^3 - 18k^2 + 22k - 6, & f''(k) &= 12k^2 - 36k + 22, \\ w(k) &= f(k) - kf'(k) = -k^2(3k^2 - 12k + 11). \end{aligned}$$

(a) *Violations.* The underlying production function is $F(K, L) = Lf(K/L)$, so $F_K = f'(k)$, $F_{KK} = f''(k)/L$, $F_L = w(k)$ and $F_{LL} = (K^2/L^3)f''(k)$. Constant returns to scale hold by construction, and f is a polynomial, hence twice (indeed infinitely) differentiable. Everything else in ASSUMPTION 1 fails somewhere:

- *Nonnegativity of output.* $f(k) < 0$ for $k \in (0, 1) \cup (2, 3)$ — for instance $f(1/2) = -0.9375$ — so F does not map into $\mathbb{R}_+$.
- *Positive marginal products.* $f'(0) = -6 < 0$, so $F_K > 0$ fails near $k = 0$ (and again on part of $(2, 3)$). Likewise $w(k) < 0$ for $k < 2 - 1/\sqrt{3}$ and for $k > 2 + 1/\sqrt{3}$, e.g. $w(3) = -18$, so $F_L > 0$ fails.
- *Diminishing marginal products.* $f''(k) = 12k^2 - 36k + 22$ is positive outside the interval between its roots $k = (9 \pm \sqrt{15})/6$, i.e. on $(0, 0.8545) \cup (2.1455, \infty)$ — for instance $f''(0) = 22 > 0$. So $F_{KK} < 0$ and $F_{LL} < 0$ both fail there.

Of ASSUMPTION 2, only $F(0, L, A) = 0$ survives ($f(0) = 0$). Both Inada conditions for capital fail: $\lim_{k \rightarrow 0} f'(k) = -6 \neq \infty$ and $f'(k) \rightarrow +\infty$ rather than 0 as $k \rightarrow \infty$ (e.g. $f'(10) = 2414$). The Inada conditions for labor fail as well, since $F_L = w(k) \rightarrow -\infty$ as $k \rightarrow \infty$.

(b) *Three steady states.* With population growth n and depreciation δ , the fundamental law of motion (2.33) is

$$\frac{\dot{k}(t)}{k(t)} = s \frac{f(k(t))}{k(t)} - (n + \delta),$$

so an interior steady state is a solution of

$$(k - 1)(k - 2)(k - 3) = h, \quad h \equiv \frac{n + \delta}{s} > 0. \quad (*)$$

The cubic on the left has derivative $3k^2 - 12k + 11$, which vanishes at $k = 2 \mp 1/\sqrt{3}$; it therefore rises to a local maximum at $k = 2 - 1/\sqrt{3} \approx 1.42265$, falls to a local minimum at $k = 2 + 1/\sqrt{3}$, and rises thereafter. The two extreme values are

$$\left(1 - \frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{3}}\right)\left(-1 - \frac{1}{\sqrt{3}}\right) = \frac{2}{3\sqrt{3}} = \frac{2\sqrt{3}}{9} \approx 0.38490 \quad \text{and} \quad -\frac{2\sqrt{3}}{9}.$$

Hence (*) has exactly three positive roots whenever

$$0 < \frac{n + \delta}{s} < \frac{2\sqrt{3}}{9} \approx 0.38490,$$

one in $(1, 2 - 1/\sqrt{3})$, one in $(2 - 1/\sqrt{3}, 2)$, and one in $(3, \infty)$; it has exactly one (in $(3, \infty)$) when $h > 2\sqrt{3}/9$. For example, with $n = 0$ and $\delta/s = 1/4$ the three steady states are

$$k_1^* = 1.162435, \quad k_2^* = 1.730406, \quad k_3^* = 3.107160.$$

Figure 1 plots the cubic together with the horizontal line h .

(c) *Stability.* Since $\dot{k} = k s [(k - 1)(k - 2)(k - 3) - h]$, the sign of $\dot{k}$ is the sign of the vertical distance between the cubic and the line, and at a steady state

$$\left. \frac{\partial \dot{k}}{\partial k} \right|_{k=k^*} = s k^* \cdot \frac{d}{dk} \left[\frac{f(k)}{k} \right]_{k=k^*},$$

so a steady state is locally stable exactly when the cubic crosses the line *downward*. The cubic crosses upward at k_1^* (rising branch), downward at k_2^* (falling branch), and upward again at k_3^* :

$$\text{at } k_1^* : +1.1045, \quad \text{at } k_2^* : -0.7820, \quad \text{at } k_3^* : +2.6774,$$

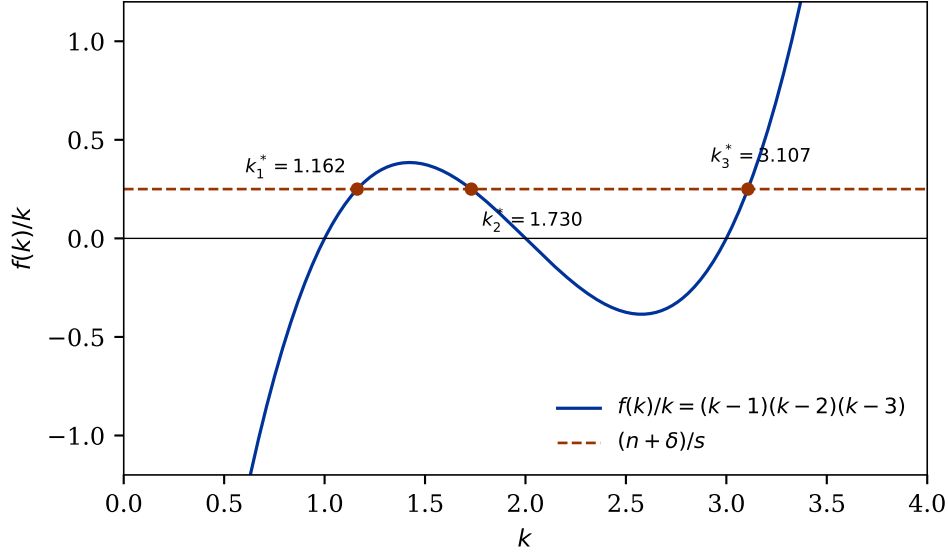


FIGURE 1. Steady states of the quartic economy: the interior steady states are the intersections of $f(k)/k = (k-1)(k-2)(k-3)$ with the horizontal line $(n+\delta)/s$, drawn here at $1/4$.

for the derivative of $f(k)/k$. Therefore k_2^* is locally asymptotically stable while k_1^* and k_3^* are unstable. In addition $k = 0$ is a rest point of the differential equation, and it is locally stable: for small $k > 0$ we have $f(k)/k < 0 < h$, so $\dot{k} < 0$ and the economy collapses to zero. The complete phase line is

$$\underbrace{0}_{\text{stable}} \longleftarrow \underbrace{k_1^*}_{\text{unstable}} \longrightarrow \underbrace{k_2^*}_{\text{stable}} \longleftarrow \underbrace{k_3^*}_{\text{unstable}} \longrightarrow \infty.$$

An economy starting with $k(0) < k_1^*$ collapses to $k = 0$; one starting in (k_1^*, k_3^*) converges to k_2^* ; one starting above k_3^* grows without bound. This is a poverty-trap configuration: history (the initial condition) decides the long-run outcome.

Finally, *no* steady state can be globally stable. Global asymptotic stability requires convergence to the same point from every initial condition; but an economy that starts at another steady state stays there forever, so the existence of more than one rest point rules global stability out immediately.

Remark. Part (c) of the exercise asks one to prove that *two* of the three steady states in (b) are locally stable. Of the three *interior* steady states exactly one — the middle one — is locally stable, as the sign pattern above shows: on the rising branches of $f(k)/k$ the economy is pushed away from the steady state. The count in the book is recovered only if the degenerate rest point $k = 0$, which the text elsewhere excludes as having “no economic content”, is included among the steady states; then two of the four rest points ($k = 0$ and k_2^*) are stable and two are unstable.

Exercise 2.5

We must show that under ASSUMPTIONS 1 and 2 the continuous-time Solow model with population growth at the rate n has a unique steady-state equilibrium, with $k^* \in (0, \infty)$ solving (2.34), $y^* = f(k^*)$ and $c^* = (1-s)f(k^*)$.

By (2.33) the equilibrium path satisfies $\dot{k}(t)/k(t) = sf(k(t))/k(t) - (n + \delta)$, so a steady state with $k > 0$ is precisely a solution of

$$\psi(k) \equiv \frac{f(k)}{k} = \frac{n + \delta}{s}. \quad (*)$$

ψ is continuous and strictly decreasing. Continuity is immediate from ASSUMPTION 1. For monotonicity, differentiate:

$$\psi'(k) = \frac{f'(k)k - f(k)}{k^2} = -\frac{w(k)}{k^2}, \quad w(k) = f(k) - kf'(k).$$

Since f is strictly concave with $f(0) = 0$ (ASSUMPTION 2), strict concavity gives $f(k) > f(0) + kf'(k) = kf'(k)$, that is $w(k) > 0$ for every $k > 0$; equivalently, $w(k) = F_L > 0$ is the wage, which is positive by ASSUMPTION 1. Hence $\psi'(k) < 0$ for all $k > 0$.

The limits. Both $f(k) \rightarrow 0$ and $k \rightarrow 0$ as $k \rightarrow 0$, so l'Hôpital's rule and ASSUMPTION 2 give

$$\lim_{k \rightarrow 0} \psi(k) = \lim_{k \rightarrow 0} f'(k) = \lim_{K \rightarrow 0} F_K = \infty, \quad \lim_{k \rightarrow \infty} \psi(k) = \lim_{k \rightarrow \infty} f'(k) = \lim_{K \rightarrow \infty} F_K = 0.$$

(The second limit also follows from $0 < f(k)/k$ and concavity.)

Existence and uniqueness. ψ is continuous on $(0, \infty)$ with limits ∞ at 0 and 0 at ∞ , so by the Intermediate Value Theorem it attains the value $(n + \delta)/s > 0$ at some $k^* \in (0, \infty)$; since ψ is strictly decreasing, that value is attained exactly once. Hence there is a unique steady-state capital-labor ratio, given by (2.34).

Finally, at k^* the level of output per capita is $y^* = f(k^*)$ by definition of f , and since a constant fraction s of income is saved, consumption per capita is $c^* = (1 - s)f(k^*)$, with the corresponding investment $sf(k^*) = (n + \delta)k^*$ exactly replacing depreciated capital and equipping the new workers.

Remark. The proof is that of PROPOSITION 2.2 with δ replaced by $n + \delta$: going from discrete to continuous time, and adding population growth, changes nothing essential. Population growth acts exactly like faster depreciation, since both erode the capital-labor ratio at a constant proportional rate.

Exercise 2.6

Let $f(k) = A\tilde{f}(k)$ with $\tilde{f}$ satisfying ASSUMPTIONS 1 and 2, and let k^* be the unique steady state of Exercise 2.5, i.e. the solution of

$$G(k^*; A, s, \delta, n) \equiv sA \frac{\tilde{f}(k^*)}{k^*} - (n + \delta) = 0.$$

Writing $\tilde{w}(k) \equiv \tilde{f}(k) - k\tilde{f}'(k) > 0$ for the wage per unit of A , the computation in Exercise 2.5 gives

$$\frac{\partial G}{\partial k^*} = sA \cdot \frac{d}{dk} \left[\frac{\tilde{f}(k)}{k} \right]_{k=k^*} = -\frac{sA \tilde{w}(k^*)}{(k^*)^2} < 0,$$

so the Implicit Function Theorem applies at every parameter configuration and yields $\partial k^*/\partial x = -(\partial G/\partial x)/(\partial G/\partial k^*)$ for each parameter x . Since $\partial G/\partial A = s\tilde{f}(k^*)/k^* > 0$, $\partial G/\partial s = A\tilde{f}(k^*)/k^* > 0$, $\partial G/\partial \delta = -1$ and $\partial G/\partial n = -1$,

$$\frac{\partial k^*}{\partial A} = \frac{k^* \tilde{f}(k^*)}{A \tilde{w}(k^*)} > 0, \quad \frac{\partial k^*}{\partial s} = \frac{k^* \tilde{f}(k^*)}{s \tilde{w}(k^*)} > 0, \quad \frac{\partial k^*}{\partial \delta} = \frac{\partial k^*}{\partial n} = -\frac{(k^*)^2}{sA \tilde{w}(k^*)} < 0.$$

This proves the four statements about k^* .

For output, $y^* = A\tilde{f}(k^*)$, so for $x \in \{s, \delta, n\}$ the chain rule gives

$$\frac{\partial y^*}{\partial x} = A\tilde{f}'(k^*)\frac{\partial k^*}{\partial x},$$

which has the same sign as $\partial k^*/\partial x$ because $\tilde{f}' > 0$: $\partial y^*/\partial s > 0$ and $\partial y^*/\partial \delta, \partial y^*/\partial n < 0$. For A there is in addition a direct effect, and both effects push the same way:

$$\frac{\partial y^*}{\partial A} = \tilde{f}(k^*) + A\tilde{f}'(k^*)\frac{\partial k^*}{\partial A} > 0.$$

The economics is transparent. A higher saving rate or a better technology raises investment at every capital-labor ratio and so shifts the $sf(k)/k$ schedule up, moving its intersection with the horizontal line $n + \delta$ to the right. A higher depreciation rate or a higher rate of population growth raises the amount of investment needed merely to hold k constant — capital must be replaced, and new workers must be equipped — and so moves the intersection to the left. The only new result relative to PROPOSITION 2.3 is the effect of n , and it enters exactly as δ does: countries with faster population growth are, other things equal, poorer in per capita terms.

Exercise 2.8

Throughout, F is twice differentiable and exhibits constant returns to scale, $f(k) = F(k, 1, A)$ is concave (so $f'' \leq 0$) with $f(0) = 0$, and the Inada conditions of ASSUMPTION 2 hold. What is *not* assumed is $F_{KK} < 0$, $F_{LL} < 0$, or even $F_K > 0$ and $F_L > 0$. The first thing to notice is that the missing sign restrictions on the first derivatives come for free.

Step 1: the marginal product of capital is nonnegative. Concavity makes f' nonincreasing. If $f'(k_0) < 0$ for some k_0 , then $f'(k) \leq f'(k_0) < 0$ for every $k \geq k_0$, contradicting the Inada condition $\lim_{k \rightarrow \infty} f'(k) = 0$. Hence $f'(k) \geq 0$, that is $F_K \geq 0$, for all $k > 0$. (One cannot get more: a technology that is affine from some k_0 on has $f' \equiv 0$ beyond k_0 and still satisfies every hypothesis. As the rest of the argument shows, $f' \geq 0$ is all that is needed.)

Step 2: the wage is strictly positive. Write $w(k) = f(k) - kf'(k)$. Since $f(0) = 0$,

$$w(k) = \int_0^k [f'(x) - f'(k)] dx,$$

and the integrand is nonnegative because f' is nonincreasing. It is moreover strictly positive on a set of positive measure: $f'(x) \rightarrow \infty$ as $x \rightarrow 0$ while $f'(k) < \infty$, so $f'(x) > f'(k)$ for all small x . Hence $w(k) > 0$ for every $k > 0$; that is, $F_L > 0$, and the “not necessarily strictly” concave technology still pays a positive wage.

PROPOSITION 2.2: existence and uniqueness of the steady state. Exactly as in Exercise 2.5, $\psi(k) \equiv f(k)/k$ satisfies $\psi'(k) = -w(k)/k^2 < 0$ by Step 2, so ψ is *strictly* decreasing even though f is only weakly concave. Its limits are ∞ at 0 and 0 at ∞ by ASSUMPTION 2 and l'Hôpital's rule. The Intermediate Value Theorem therefore gives a unique $k^* \in (0, \infty)$ with $f(k^*)/k^* = \delta/s$, and then $y^* = f(k^*)$ and $c^* = (1-s)f(k^*)$. Nothing in the argument used strict concavity; it used only that $w > 0$, which Step 2 delivers.

PROPOSITION 2.5: global stability. Let $g(k) \equiv sf(k) + (1-\delta)k$, so that the discrete time law of motion is $k(t+1) = g(k(t))$ and $k^* = g(k^*)$. By Step 1, $g'(k) = sf'(k) + 1 - \delta > 0$. By Step 2,

$\delta = sf(k^*)/k^* > sf'(k^*)$, whence

$$g'(k^*) = sf'(k^*) + 1 - \delta \in (0, 1),$$

and local stability follows from COROLLARY 2.1. For global stability, take $k(t) \in (0, k^*)$. Since g is strictly increasing, $k(t+1) = g(k(t)) < g(k^*) = k^*$; and since ψ is strictly decreasing,

$$\frac{k(t+1) - k(t)}{k(t)} = s \frac{f(k(t))}{k(t)} - \delta > s \frac{f(k^*)}{k^*} - \delta = 0.$$

So $k(t+1) \in (k(t), k^*)$: the sequence is increasing and bounded above by k^* , hence convergent, and its limit is a fixed point of g in $(0, k^*]$, which can only be k^* by uniqueness. The argument starting from $k(0) > k^*$ is identical with the inequalities reversed. Monotone global convergence therefore survives the weakening of ASSUMPTION 1.

PROPOSITION 2.6 *must be weakened to weak monotonicity*. The proposition asserts that $\{w(t)\}$ is an increasing and $\{R(t)\}$ a decreasing sequence when $k(0) < k^*$. With $f'' \leq 0$ instead of $f'' < 0$ we get

$$R'(k) = f''(k) \leq 0 \quad \text{and} \quad w'(k) = -kf''(k) \geq 0,$$

so the correct statement is that $\{w(t)\}$ is *nondecreasing* and $\{R(t)\}$ is *nonincreasing* (with the inequalities reversed when $k(0) > k^*$). Both can be locally constant: on any interval where f is affine — the production function is then locally an AK technology — the capital-labor ratio keeps rising while factor prices do not move at all. Strict monotonicity is restored as soon as $f'' < 0$ on the relevant range, which is what ASSUMPTION 1 guarantees.

Exercise 2.9

Let ASSUMPTIONS 1 and 2 hold and suppose $k(0) < k^*$. By PROPOSITION 2.5 the capital-labor ratio increases monotonically to k^* : $k(t+1) > k(t)$ for every t , with $k(t) \rightarrow k^*$.

In equilibrium factor prices are the marginal products, and with constant returns they depend on the capital-labor ratio alone:

$$\begin{aligned} R(t) &= F_K(K(t), L(t), A) = f'(k(t)), \\ w(t) &= F_L(K(t), L(t), A) = f(k(t)) - k(t)f'(k(t)) = w(k(t)). \end{aligned}$$

Differentiating the two expressions in k ,

$$\frac{dR}{dk} = f''(k) < 0, \quad \frac{dw}{dk} = f'(k) - f'(k) - kf''(k) = -kf''(k) > 0,$$

both strict by ASSUMPTION 1, which has $f''(k) = F_{KK}(k, 1, A) < 0$.

So $R(\cdot)$ is a strictly decreasing and $w(\cdot)$ a strictly increasing function of k . Composing with the strictly increasing sequence $\{k(t)\}$ gives

$$w(t+1) = w(k(t+1)) > w(k(t)) = w(t), \quad R(t+1) = f'(k(t+1)) < f'(k(t)) = R(t),$$

that is, $\{w(t)\}_{t=0}^\infty$ is an increasing and $\{R(t)\}_{t=0}^\infty$ a decreasing sequence, and they converge to $w(k^*)$ and $f'(k^*)$. If instead $k(0) > k^*$, PROPOSITION 2.5 gives a strictly decreasing $\{k(t)\}$ and the two conclusions are reversed.

The economics is the usual scarcity argument: capital deepening makes capital abundant relative to labor, so the marginal product of capital — and with it the rental rate and the interest rate $r = R - \delta$ — falls, while the marginal product of labor, and hence the wage, rises. Along the transition the entire increase in output per worker that is not saved accrues to labor.

Exercise 2.10

Part 1. The linear equation $\dot{x}(t) = ax(t)$ with $x(0)$ given has the unique solution $x(t) = x(0) \exp(at)$. If $a < 0$ then $\exp(at) \rightarrow 0$, so $x(t) \rightarrow 0$ for every initial condition: the steady state $x^* = 0$ is globally asymptotically stable. (If $a > 0$ the solution diverges unless $x(0) = 0$, and if $a = 0$ every point is a steady state.)

Part 2. Let g be differentiable in a neighborhood of x^* , with $g(x^*) = 0$ and $g'(x^*) < 0$. This is the scalar case ($n = 1$) of THEOREM 2.5: the Jacobian is the number $g'(x^*)$, its unique eigenvalue is $g'(x^*) < 0$, which has negative real part, so x^* is locally asymptotically stable. The direct argument is just as short. Since g is continuous with $g(x^*) = 0$ and $g'(x^*) < 0$, there is $\varepsilon > 0$ with

$$g(x) < 0 \text{ for } x \in (x^*, x^* + \varepsilon) \quad \text{and} \quad g(x) > 0 \text{ for } x \in (x^* - \varepsilon, x^*),$$

so on that neighborhood $\dot{x}$ always points toward x^* ; Part 3 applied to the interval $(x^* - \varepsilon, x^* + \varepsilon)$ then gives convergence.

Part 3. Let g be continuously differentiable with $g(x^*) = 0$, $g(x) < 0$ for $x > x^*$ and $g(x) > 0$ for $x < x^*$. Take any $x(0) < x^*$ (the case $x(0) > x^*$ is symmetric, and $x(0) = x^*$ is trivial).

Because g is continuously differentiable it is locally Lipschitz, so by the Picard–Lindelöf theorem the solution through any point is unique. The constant function $x \equiv x^*$ solves the equation, so the trajectory starting below x^* can never reach x^* in finite time, and *a fortiori* never crosses it: $x(t) < x^*$ for all t . Consequently $g(x(t)) > 0$ and $\dot{x}(t) > 0$ for all t , so $x(\cdot)$ is strictly increasing and bounded above by x^* . A monotone bounded function converges; let $L \equiv \lim_{t \rightarrow \infty} x(t) \leq x^*$.

Suppose $L < x^*$. By continuity $\dot{x}(t) = g(x(t)) \rightarrow g(L) > 0$, so there are T and $\eta > 0$ with $\dot{x}(t) \geq \eta$ for all $t \geq T$, whence $x(t) \geq x(T) + \eta(t - T) \rightarrow \infty$, contradicting $x(t) \leq x^*$. Therefore $L = x^*$, and $x(t) \rightarrow x^*$ from every initial condition: x^* is globally asymptotically stable.

Remark. Part 3 is what makes the continuous-time Solow model so much better behaved than its discrete-time counterpart. In discrete time a one-dimensional map can jump over its fixed point, so a sign condition on g alone does not preclude two-cycles; EXERCISE 2.21 constructs exactly such a cycle. In continuous time trajectories are continuous and cannot cross a rest point, which is why the analogue of Part 3 fails in discrete time.

Exercise 2.13

The law of motion. Labor income per worker is the wage $w(k) = f(k) - kf'(k)$, of which a fraction s is saved; capital income $f'(k)k$ is entirely consumed. Aggregate saving is therefore $sw(k(t))L(t)$, and with population growth at the rate n and depreciation δ the capital-labor ratio obeys

$$\dot{k}(t) = sw(k(t)) - (n + \delta)k(t), \quad \text{so} \quad \frac{\dot{k}(t)}{k(t)} = s\phi(k(t)) - (n + \delta), \quad \phi(k) \equiv \frac{w(k)}{k}.$$

Interior steady states are the solutions of

$$\phi(k) = \frac{w(k)}{k} = \frac{n + \delta}{s}. \quad (*)$$

The contrast with the baseline model is exactly the replacement of $f(k)/k$ by $w(k)/k$. In the baseline model $f(k)/k$ is strictly decreasing, which is what delivered uniqueness in Exercise 2.5. Nothing of the sort holds for $w(k)/k$.

When is ϕ increasing? Differentiating,

$$\phi'(k) = \frac{k w'(k) - w(k)}{k^2} = \frac{-k^2 f''(k) - w(k)}{k^2},$$

using $w'(k) = -k f''(k)$. Now recall that for a constant returns technology the elasticity of substitution (2.37) can be written as

$$\sigma(k) = \frac{f'(k) w(k)}{-k f(k) f''(k)},$$

so that $-k^2 f''(k) = \alpha_K(k) w(k)/\sigma(k)$ with $\alpha_K(k) \equiv k f'(k)/f(k)$ the share of capital in national income. Substituting,

$$\phi'(k) = \frac{w(k)}{k^2} \left[\frac{\alpha_K(k)}{\sigma(k)} - 1 \right], \quad \text{hence} \quad \phi'(k) > 0 \iff \sigma(k) < \alpha_K(k).$$

So $w(k)/k$ rises precisely where capital and labor are hard to substitute relative to the capital share. Intuitively, when σ is small an increase in capital raises the wage more than proportionately, because labor becomes the scarce complementary factor; saving out of wages then feeds back into more capital, and the economy can have several rest points.

An explicit example. Let

$$f(k) = A[\gamma k^\rho + (1 - \gamma)]^{1/\rho} + Bk^\alpha, \quad \rho = \frac{\sigma - 1}{\sigma} < 0, \quad \alpha \in (0, 1), \quad B > 0 :$$

a CES technology with a low elasticity of substitution plus a small Cobb–Douglas term. Both terms are increasing, strictly concave and homogeneous of degree one in (K, L) when written as $Lf(K/L)$, so ASSUMPTION 1 holds. The CES term alone would violate ASSUMPTION 2, since $f'_{\text{CES}}(0) = A\gamma^{1/\rho} < \infty$ and $w_{\text{CES}}(\infty) = A(1 - \gamma)^{1/\rho} < \infty$; the Cobb–Douglas term repairs both, because Bk^α contributes $B\alpha k^{\alpha-1} \rightarrow \infty$ to $f'(k)$ as $k \rightarrow 0$ and $(1 - \alpha)Bk^\alpha \rightarrow \infty$ to $w(k)$ as $k \rightarrow \infty$. So f satisfies ASSUMPTIONS 1 and 2 in full, and

$$w(k) = A(1 - \gamma)[\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho} + (1 - \alpha)Bk^\alpha.$$

Take $A = 1$, $\gamma = 1/2$, $\sigma = 1/4$ (so $\rho = -3$), $B = 0.02$ and $\alpha = 1/2$. Then $\phi(k) = w(k)/k$ falls from $+\infty$ to a local minimum $\phi = 0.03005$ at $k = 0.1509$, rises to a local maximum $\phi = 0.60360$ at $k = 1.4340$, and falls back to 0; the turning points are exactly where $\sigma(k)$ crosses $\alpha_K(k)$:

k	0.05	0.30	1.00	3.00
$\sigma(k)$	1.8871	0.3581	0.2537	0.3264
$\alpha_K(k)$	0.9667	0.9603	0.5000	0.0483
$\phi'(k)$	-0.438	+0.286	+0.495	-0.115

Consequently (*) has *three* solutions for every $(n + \delta)/s$ strictly between 0.03005 and 0.60360. With $(n + \delta)/s = 0.045$ they are

$$k_1^* = 0.04972, \quad k_2^* = 0.27684, \quad k_3^* = 29.19752,$$

and since $\dot{k}$ has the sign of $\phi(k) - (n + \delta)/s$, the outer two are locally asymptotically stable ($\phi'(k_1^*) = -0.442$, $\phi'(k_3^*) = -0.002$) while the middle one is unstable ($\phi'(k_2^*) = +0.240$). Figure 2 plots ϕ and the horizontal line. The economy converges to the poor or to the rich steady state according to whether it starts below or above k_2^* : with saving out of wages alone, the Solow model produces a poverty trap.

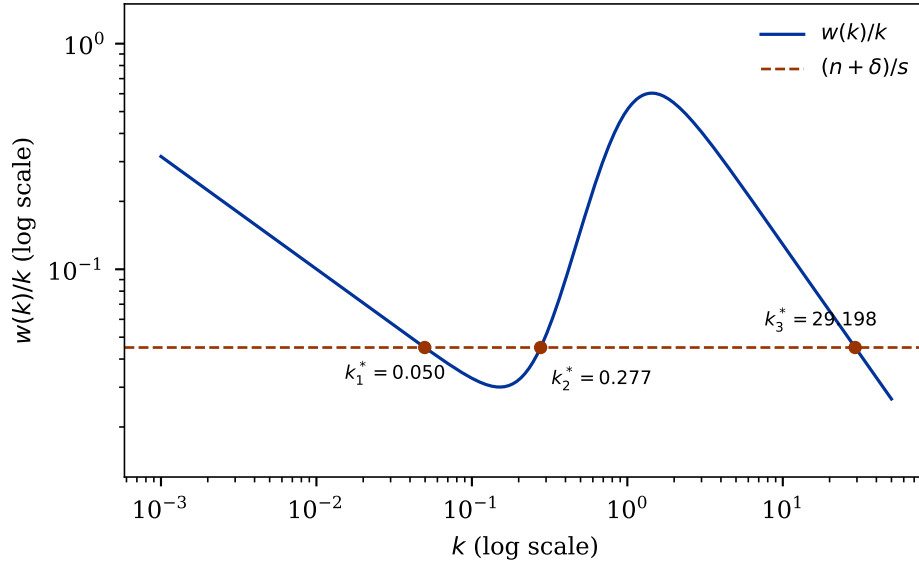


FIGURE 2. Saving out of labor income only: steady states are the intersections of $w(k)/k$ with the horizontal line $(n + \delta)/s$, drawn here at 0.045, for the technology of the text.

Remark. The comparison with EXERCISE 2.12 is instructive. When only capital income is saved, the law of motion is $\dot{k}/k = s_K f'(k) - (n + \delta)$, and f' is strictly decreasing by ASSUMPTION 1: the steady state is unique and globally stable, whatever the elasticity of substitution. It is the wage — the factor price that *rises* with capital — that can generate the self-reinforcing accumulation behind multiplicity.

Exercise 2.15

With constant returns, output per worker and the wage are both functions of the capital-labor ratio alone:

$$y = f(k), \quad w = F_L = f(k) - k f'(k),$$

the second equality being the competitive wage. Along the family of economies indexed by k ,

$$\frac{dy}{dk} = f'(k), \quad \frac{dw}{dk} = f'(k) - f'(k) - k f''(k) = -k f''(k) > 0,$$

so w is a strictly increasing function of k and can be inverted; by the chain rule,

$$\frac{\partial y}{\partial w} = \frac{dy/dk}{dw/dk} = \frac{f'(k)}{-k f''(k)}.$$

Therefore

$$\varepsilon_{y,w} = \frac{\partial y}{\partial w} \cdot \frac{w}{y} = \frac{f'(k)}{-kf''(k)} \cdot \frac{f(k) - kf'(k)}{f(k)} = \frac{f'(k)[f(k) - kf'(k)]}{-kf(k)f''(k)}. \quad (*)$$

It remains to check that the right-hand side of (*) is the elasticity of substitution (2.37). With constant returns, $F_K = f'(k)$ and $F_L = f(k) - kf'(k)$, so

$$\frac{\partial \log(F_K/F_L)}{\partial \log k} = k \frac{d}{dk} \left[\log f'(k) - \log(f(k) - kf'(k)) \right] = k \left[\frac{f''(k)}{f'(k)} + \frac{kf''(k)}{f(k) - kf'(k)} \right].$$

Putting the bracket over a common denominator,

$$\frac{\partial \log(F_K/F_L)}{\partial \log k} = kf''(k) \frac{[f(k) - kf'(k)] + kf'(k)}{f'(k)[f(k) - kf'(k)]} = \frac{kf(k)f''(k)}{f'(k)[f(k) - kf'(k)]},$$

and hence, by the definition (2.37),

$$\sigma = - \left[\frac{\partial \log(F_K/F_L)}{\partial \log(K/L)} \right]^{-1} = \frac{f'(k)[f(k) - kf'(k)]}{-kf(k)f''(k)},$$

which is exactly (*). Therefore $\varepsilon_{y,w} = \sigma$.

The result is the basis of the empirical strategy of Arrow et al. (1961) taken up in EXERCISE 2.16: regressing the logarithm of output per capita on the logarithm of the wage across economies (or across time) estimates the elasticity of substitution directly, and a slope of 1 — which is what those authors found to be a good approximation — says that the technology is Cobb–Douglas. It also gives a clean reading of σ : it measures how much richer, in output per worker, an economy must be in order to sustain a given proportional increase in its wage.

Exercise 2.24

Normalize the technology terms to $A_H = A_K = A_L = 1$ (they only rescale the arguments and the level, and none of the limits below depends on them) and write the per capita form of (2.38) as

$$f(k) = [\gamma k^\rho + (1 - \gamma)]^{1/\rho}, \quad \rho \equiv \frac{\sigma - 1}{\sigma} \in (-\infty, 1), \quad \gamma \in (0, 1),$$

so that $\sigma > 1 \iff \rho \in (0, 1)$, $\sigma < 1 \iff \rho < 0$ and $\sigma = 1$ is the Cobb–Douglas limit $f(k) = k^\gamma$. Differentiating,

$$f'(k) = \gamma k^{\rho-1} [\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho}, \quad w(k) = f(k) - kf'(k) = (1 - \gamma) [\gamma k^\rho + (1 - \gamma)]^{(1-\rho)/\rho}.$$

Recall that ASSUMPTION 2 requires four limits — $F_K \rightarrow \infty$ as $K \rightarrow 0$, $F_K \rightarrow 0$ as $K \rightarrow \infty$, $F_L \rightarrow \infty$ as $L \rightarrow 0$, $F_L \rightarrow 0$ as $L \rightarrow \infty$ — together with $F(0, L, A) = 0$. In per capita terms $F_K = f'(k)$ and $F_L = w(k)$, and $L \rightarrow 0$ with K fixed means $k \rightarrow \infty$, while $L \rightarrow \infty$ means $k \rightarrow 0$.

Case $\sigma > 1$ (i.e. $0 < \rho < 1$). As $k \rightarrow 0$ the bracket tends to $1 - \gamma > 0$, so

$$f(0) = (1 - \gamma)^{1/\rho} > 0,$$

and capital is not essential: $F(0, L, A) > 0$, the last requirement of ASSUMPTION 2, fails. As $k \rightarrow \infty$ the bracket behaves like γk^ρ , so

$$f'(k) \longrightarrow \gamma k^{\rho-1} (\gamma k^\rho)^{(1-\rho)/\rho} = \gamma^{1/\rho} > 0,$$

so $\lim_{K \rightarrow \infty} F_K = 0$ fails too: the marginal product of capital is bounded below by a positive number, and the technology is asymptotically linear in capital — an AK technology in the limit, which is exactly why EXERCISE 2.23(a) finds sustained growth in this case. Finally, as $k \rightarrow 0$ (that is, $L \rightarrow \infty$ for given K), the bracket tends to $1 - \gamma$ and

$$w(k) \rightarrow (1 - \gamma)(1 - \gamma)^{(1-\rho)/\rho} = (1 - \gamma)^{1/\rho} > 0,$$

so $\lim_{L \rightarrow \infty} F_L = 0$ fails as well: labor remains productive at the margin however abundant it is, because output does not require capital. (The two remaining limits, $f'(0) = \infty$ and $w(\infty) = \infty$, do hold, the latter because $w(k) \sim (1 - \gamma)\gamma^{(1-\rho)/\rho}k^{1-\rho}$ with $1 - \rho > 0$.)

Case $\sigma < 1$ (i.e. $\rho < 0$). Now $k^\rho \rightarrow \infty$ as $k \rightarrow 0$, so the bracket diverges and, since $1/\rho < 0$, $f(0) = 0$: capital is essential, as ASSUMPTION 2 wants. But

$$f'(k) \rightarrow \gamma k^{\rho-1}(\gamma k^\rho)^{(1-\rho)/\rho} = \gamma^{1/\rho} < \infty \quad \text{as } k \rightarrow 0,$$

so the Inada condition $\lim_{K \rightarrow 0} F_K = \infty$ fails. At the other end, as $k \rightarrow \infty$ the bracket tends to $1 - \gamma$ and

$$w(k) \rightarrow (1 - \gamma)^{1/\rho} < \infty,$$

so $\lim_{L \rightarrow 0} F_L = \infty$ fails as well. (The two remaining limits, $f'(\infty) = 0$ and $w(0) = 0$, do hold.)

Case $\sigma = 1$. Then $f(k) = k^\gamma$ with $\gamma \in (0, 1)$: $f(0) = 0$, $f'(k) = \gamma k^{\gamma-1} \rightarrow \infty$ as $k \rightarrow 0$ and $\rightarrow 0$ as $k \rightarrow \infty$, and $w(k) = (1 - \gamma)k^\gamma \rightarrow 0$ as $k \rightarrow 0$ and $\rightarrow \infty$ as $k \rightarrow \infty$. All the requirements of ASSUMPTION 2 hold.

Hence ASSUMPTION 2 holds for the CES technology if and only if $\sigma = 1$. The failure is economically meaningful rather than technical: with $\sigma > 1$ the two factors are good enough substitutes that capital alone can produce, and accumulating it never drives its marginal product to zero; with $\sigma < 1$ they are such poor substitutes that the first units of capital are not infinitely productive, and an economy with a fixed labor force cannot make the wage arbitrarily large.

Remark. ASSUMPTION 1, by contrast, holds for every finite $\sigma > 0$. Logarithmic differentiation of f' gives

$$\frac{f''(k)}{f'(k)} = \frac{\rho - 1}{k} + \frac{(1 - \rho)\gamma k^{\rho-1}}{\gamma k^\rho + (1 - \gamma)} = -\frac{(1 - \rho)(1 - \gamma)}{k[\gamma k^\rho + (1 - \gamma)]} < 0 \quad \text{for all } \rho < 1,$$

so $f'' < 0$, and $F_{KK}, F_{LL} < 0$ follow. The text on page 55 states instead that “the CES production function with $\sigma > 1$ violates Assumption 1”; what the computation above shows, and what this exercise asks one to prove, is that what fails is ASSUMPTION 2.

Exercise 2.25

With Harrod-neutral technological progress at the rate g and population growth at the rate n , the effective capital-labor ratio $k(t) = K(t)/(A(t)L(t))$ obeys (2.51), and the BGP value k^* solves (2.52),

$$H(k^*; s, \delta, n, g) \equiv s \frac{f(k^*)}{k^*} - (\delta + g + n) = 0.$$

The level of technology does not enter. $A(0)$ appears nowhere in this equation — it has been normalized away by the definition of k — so

$$\frac{\partial k^*(A(0), s, \delta, n, g)}{\partial A(0)} = 0.$$

The other comparative statics. Exactly as in Exercise 2.5,

$$\frac{\partial H}{\partial k^*} = s \cdot \frac{d}{dk} \left[\frac{f(k)}{k} \right]_{k=k^*} = -\frac{s w(k^*)}{(k^*)^2} < 0,$$

where $w(k) = f(k) - k f'(k) > 0$. The Implicit Function Theorem applies and gives, with $\partial H / \partial s = f(k^*) / k^* > 0$ and $\partial H / \partial \delta = \partial H / \partial n = \partial H / \partial g = -1$,

$$\frac{\partial k^*}{\partial s} = \frac{k^* f(k^*)}{s w(k^*)} > 0, \quad \frac{\partial k^*}{\partial n} = \frac{\partial k^*}{\partial \delta} = \frac{\partial k^*}{\partial g} = -\frac{(k^*)^2}{s w(k^*)} < 0.$$

Output per capita. Along the BGP, income per capita is $y^*(t) = A(t)f(k^*) = A(0)e^{gt}f(k^*)$ by (2.50). Hence, for each t ,

$$\frac{\partial y^*}{\partial A(0)} = e^{gt} f(k^*) > 0,$$

which is the one comparative static that is new relative to PROPOSITION 2.8: a higher initial technology level leaves the effective capital-labor ratio untouched but raises income per capita proportionately, at every date. For the remaining parameters the effect runs through k^* alone,

$$\frac{\partial y^*}{\partial x} = A(0)e^{gt} f'(k^*) \frac{\partial k^*}{\partial x}, \quad x \in \{s, \delta, n\},$$

which is positive for s and negative for δ and n , since $f' > 0$.

Remark. The proposition lists no comparative static with respect to g , and for good reason: a higher g lowers k^* and hence the level of income per capita at any fixed date, but it raises its growth rate, so $\partial y^*(t) / \partial g$ changes sign as t grows. This is the first appearance of a theme that recurs throughout the book — level effects and growth effects need not point the same way.

Exercise 2.26

Let ASSUMPTIONS 1 and 2 hold and consider the law of motion (2.51) of the effective capital-labor ratio,

$$\dot{k}(t) = G(k(t)), \quad G(k) \equiv s f(k) - (\delta + g + n)k.$$

G is continuously differentiable on $(0, \infty)$ by ASSUMPTION 1, and by PROPOSITION 2.11 it has a unique zero $k^* \in (0, \infty)$, characterized by $f(k^*) / k^* = (\delta + g + n) / s$.

Write $G(k) = k[s\psi(k) - (\delta + g + n)]$ with $\psi(k) \equiv f(k) / k$. As established in Exercise 2.5, ψ is strictly decreasing, with $\psi(k) \rightarrow \infty$ as $k \rightarrow 0$ and $\psi(k) \rightarrow 0$ as $k \rightarrow \infty$. Therefore

$$k < k^* \implies \psi(k) > \psi(k^*) = \frac{\delta + g + n}{s} \implies G(k) > 0,$$

$$k > k^* \implies \psi(k) < \psi(k^*) \implies G(k) < 0.$$

That is exactly the hypothesis of part 3 of COROLLARY 2.2 (proved in Exercise 2.10), which therefore gives global asymptotic stability: starting from any $k(0) > 0$, $k(t) \rightarrow k^*$. Moreover the convergence is monotone, since $\dot{k}$ keeps a constant sign along the path: an economy that starts

below its BGP accumulates effective capital throughout the transition, and one that starts above decumulates it.

Local stability can also be read off the derivative at the rest point. Using $sf(k^*) = (\delta + g + n)k^*$,

$$G'(k^*) = sf'(k^*) - (\delta + g + n) = sf'(k^*) - \frac{sf(k^*)}{k^*} = -\frac{sw(k^*)}{k^*} < 0,$$

because the wage $w(k^*) = f(k^*) - k^*f'(k^*)$ is strictly positive. So the linearization has a strictly negative eigenvalue, confirming local asymptotic stability by part 2 of COROLLARY 2.2, and giving in addition the speed of convergence: near the BGP the gap $k(t) - k^*$ closes at the exponential rate $sw(k^*)/k^* = (\delta + g + n)\alpha_L(k^*)$, where $\alpha_L = w/f$ is the share of labor in national income.

Since $k(t) \rightarrow k^*$ and $y(t) = A(t)f(k(t))$, output per capita converges to the balanced growth path $A(0)e^{gt}f(k^*)$ and grows asymptotically at the rate g : with Harrod-neutral progress, transitional dynamics and long-run growth separate cleanly, and the model behaves exactly as it did without technological change once variables are measured per efficiency unit of labor.

Chapter 3

The Solow Model and the Data

The seven exercises of Chapter 3 left unsolved in the instructor's manual: Exercises 3.3–3.8 and 3.11. The manual solves 3.1, 3.2, 3.9 and 3.10. Notation follows the book: $k \equiv K/(AL)$ is the effective capital-labor ratio, $\hat{y} \equiv Y/(AL)$, $y \equiv Y/L$, $y^*(t) \equiv A(t)f(k^*)$ is the steady-state (BGP) level of output per capita, and $\varepsilon_k(k) \equiv f'(k)k/f(k)$ is the elasticity of the per capita production function (3.9). The numerical parts are computed by `code/ch3.py`.

Exercise 3.3

Write the technology in labor-augmenting form, $Y = K^\alpha(AL)^{1-\alpha}$, so that in terms of the effective capital-labor ratio $k = K/(AL)$,

$$\hat{y} = k^\alpha, \quad y = Ak^\alpha, \quad \frac{\dot{k}}{k} = sk^{\alpha-1} - (\delta + g + n), \quad k^* = \left(\frac{s}{\delta + g + n} \right)^{\frac{1}{1-\alpha}}.$$

The exact equation. Differentiating $y = Ak^\alpha$ logarithmically and substituting the law of motion, with no approximation at all,

$$\frac{\dot{y}}{y} = g + \alpha \frac{\dot{k}}{k} = g + \alpha [sk^{\alpha-1} - (\delta + g + n)] = g + \alpha(\delta + g + n) \left[\left(\frac{k}{k^*} \right)^{\alpha-1} - 1 \right],$$

the last step using $s(k^*)^{\alpha-1} = \delta + g + n$. Since $y/y^* = (k/k^*)^\alpha$, we have $\log k - \log k^* = \alpha^{-1}(\log y - \log y^*)$, and therefore

$$\boxed{\frac{\dot{y}(t)}{y(t)} = g + \alpha(\delta + g + n) \left[\exp\left(-\frac{1-\alpha}{\alpha} (\log y(t) - \log y^*(t)) \right) - 1 \right]} \quad (*)$$

which is the exact analogue of (3.10). Expanding the exponential to first order around $\log y = \log y^*$ returns $\dot{y}/y \approx g - (1-\alpha)(\delta + g + n)(\log y - \log y^*)$, which is (3.11): the book's equation is the linearization of (*).

The rate of convergence is not a constant. Write $z \equiv \log y(t) - \log y^*(t)$ for the gap and define the convergence coefficient implicitly by $\dot{y}/y = g - \beta(z)z$, so that from (*)

$$\beta(z) = \alpha(\delta + g + n) \frac{1 - e^{-cz}}{z}, \quad c \equiv \frac{1-\alpha}{\alpha} > 0.$$

The function $\varphi(u) \equiv (1 - e^{-u})/u$ has $\varphi'(u) = e^{-u}[u + 1 - e^u]/u^2 < 0$ for $u \neq 0$, because $e^u > 1 + u$; and $\varphi(u) \rightarrow 1$ as $u \rightarrow 0$. Hence

$$\beta \text{ is strictly decreasing in } z, \quad \beta(0) = \alpha(\delta + g + n)c = (1-\alpha)(\delta + g + n),$$

$$\beta(z) \rightarrow \infty \text{ as } z \rightarrow -\infty, \quad \beta(z) \rightarrow 0 \text{ as } z \rightarrow +\infty.$$

So for a country *below* its steady state ($z < 0$, the empirically relevant case) the true speed of convergence *exceeds* the constant $(1-\alpha)(\delta + g + n)$ of the linearized equation, and it does so by more the poorer the country is; for a country above its steady state convergence is slower than the

linear approximation says. With the parameter values of EXAMPLE 3.1 ($\alpha = 1/3$, $\delta + g + n = 0.08$, so $(1 - \alpha)(\delta + g + n) = 0.0533$):

y/y^*	0.10	0.25	0.50	0.80	1.25	2.00	10.0
β	1.1465	0.2885	0.1154	0.0672	0.0430	0.0289	0.0115
$\beta/\beta(0)$	21.5	5.41	2.16	1.26	0.81	0.54	0.21

A country at a tenth of its steady-state income converges twenty times faster than the log-linear coefficient suggests. Figure 3 plots β .

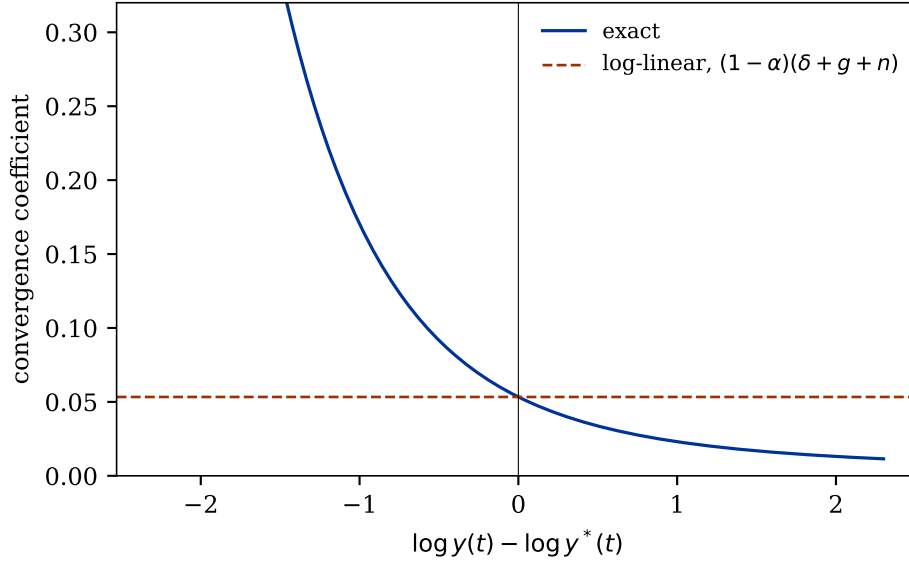


FIGURE 3. The exact convergence coefficient β as a function of the gap $\log y - \log y^*$, against the constant coefficient of the log-linearized equation (3.11).

Remark. This is a first hint of why growth regressions of the form (3.12)–(3.14), which impose a *constant* coefficient b_1 on $\log y_{i,t-1}$, are hard to interpret. Even within the Solow model with a common technology, the correct coefficient varies systematically with how far a country is from its own steady state, so the estimated b_1 is some average of very different speeds — and the average depends on the composition of the sample.

Exercise 3.4

Log-linear answer. By (3.11), each country's gap $z_i(t) \equiv \log y_i(t) - \log y^*(t)$ obeys $\dot{z}_i = -\beta z_i$ with $\beta = (1 - \alpha)(\delta + g + n)$. Since the two countries share $y^*(t)$, the *relative* gap $D(t) \equiv \log y_1(t) - \log y_2(t) = z_1(t) - z_2(t)$ obeys the same linear equation,

$$\dot{D} = -\beta D \quad \implies \quad D(t) = D(0)e^{-\beta t}.$$

With $\alpha = 1/3$, $g = 0.02$, $n = 0.01$ and $\delta = 0.05$ we get $\beta = \frac{2}{3} \times 0.08 = 0.0533$ per year. Starting from $D(0) = \log 2 = 0.6931$ and requiring $D(T) = \log 1.1 = 0.0953$,

$$T = \frac{1}{\beta} \log \frac{\log 2}{\log 1.1} = \frac{\log(7.273)}{0.0533} = \frac{1.9843}{0.0533} \approx 37.2 \text{ years.}$$

The half-life of the gap is $\log 2/\beta = 13.0$ years, which is the “little more than 10 years” quoted in EXAMPLE 3.1.

Exact answer. The Cobb–Douglas model can be solved in closed form, so the approximation can be checked. With $x \equiv k^{1-\alpha}$ the law of motion is linear, $\dot{x} = (1 - \alpha)[s - (\delta + g + n)x]$, so

$$\frac{x(t)}{x^*} = 1 + \left(\frac{x(0)}{x^*} - 1\right)e^{-\beta t}, \quad \frac{y(t)}{y^*(t)} = \left(\frac{x(t)}{x^*}\right)^{\frac{\alpha}{1-\alpha}},$$

with the *same* $\beta = (1 - \alpha)(\delta + g + n)$ now appearing exactly. The log gap between the two countries is no longer exactly exponential, because the map from x to y is concave, so the answer depends on where the pair sits relative to the common steady state:

country 1 starts at $y_1(0)/y^*$	0.8	1.0	2.0
years until $y_1/y_2 = 1.1$	21.4	27.4	49.9

The log-linear figure of 37 years is of the right order but not accurate: when both countries are below the steady state the true convergence is faster (Exercise 3.3), and when both are above it is slower.

Either way the message is the one the book draws: the basic Solow model with a capital share of one third predicts that a twofold income gap between otherwise identical countries is largely gone within a few decades. That is far faster than anything in the data — the gaps documented in Chapter 1 have persisted for a century and more — which is one of the motivations for the augmented model of Section 3.3 (where a larger effective capital share slows convergence) and, ultimately, for models in which y^* itself differs permanently across countries.

Exercise 3.5

Index economies by i and let each be a Solow economy with its own saving rate s_i , depreciation rate δ_i , population growth rate n_i and technology level $A_i(t)$, with a common rate of technological progress g (so that a stable world income distribution is possible). Each economy in isolation obeys (3.7), so its steady-state effective capital-labor ratio k_i^* solves

$$s_i \frac{f(k_i^*)}{k_i^*} = \delta_i + g + n_i,$$

and its steady-state output per capita is $y_i^*(t) = A_i(t)f(k_i^*)$.

Repeating the derivation of (3.10) economy by economy — nothing in it used cross-country homogeneity — gives the country-specific convergence equation

$$\frac{\dot{y}_i(t)}{y_i(t)} \approx g - (1 - \varepsilon_k(k_i^*))(\delta_i + g + n_i)(\log y_i(t) - \log y_i^*(t)).$$

Writing $\beta_i \equiv (1 - \varepsilon_k(k_i^*))(\delta_i + g + n_i)$ and discretizing between $t - 1$ and t ,

$$g_{i,t,t-1} \approx \underbrace{[g + \beta_i \log y_i^*(t - 1)]}_{b_i^0} - \beta_i \log y_{i,t-1}. \quad (*)$$

This is (3.13), $g_{i,t,t-1} = b_i^0 + b_1 \log y_{i,t-1} + \varepsilon_{i,t}$, with $b_1 = -\beta_i$: the constant term is country specific, precisely because the steady state y_i^* is. Two observations complete the derivation.

The slope is common when the elasticity is. With a Cobb–Douglas technology $\varepsilon_k \equiv \alpha$ and $\beta_i = (1 - \alpha)(\delta_i + g + n_i)$, so the slope varies across countries only through $\delta_i + n_i$. Since $\delta + n$ moves within a narrow range across countries while y_i^* varies enormously, treating b_1 as common and b_i^0 as country specific — exactly what (3.13) does — is the natural first approximation. (That b_1 is in principle country specific is the caveat the book records after (3.13).)

The intercept is a function of observable country characteristics. With Cobb–Douglas technology $y_i^*(t) = A_i(t)[s_i/(\delta_i + g + n_i)]^{\alpha/(1-\alpha)}$, so

$$b_i^0 = g + (1 - \alpha)(\delta_i + g + n_i) \left[\log A_i(t-1) + \frac{\alpha}{1-\alpha} (\log s_i - \log(\delta_i + g + n_i)) \right].$$

Modeling b_i^0 as a linear function of $\log s_i$, $\log(\delta_i + g + n_i)$ and proxies for $\log A_i$ — schooling, institutions, openness, and the rest — delivers the empirical specification (3.14), $g_{i,t-1} = X_{i,t}^T \beta + b_1 \log y_{i,t-1} + \varepsilon_{i,t}$. Conditional convergence is then the statement that $b_1 < 0$ once these characteristics are held fixed, whereas unconditional convergence — the regression (3.12) with a common intercept — need not show $b_1 < 0$ at all: omitting b_i^0 puts it in the error term, and since economies with low y_i^* also tend to have low $y_{i,t-1}$, the resulting $\text{Cov}(b_i^0, \log y_{i,t-1}) > 0$ biases the estimate of b_1 upward.

Exercise 3.6

The system to be studied is

$$\dot{k} = G_k(k, h) \equiv s_k f(k, h) - (\delta_k + g + n)k, \quad \dot{h} = G_h(k, h) \equiv s_h f(k, h) - (\delta_h + g + n)h,$$

on $\mathbb{R}_{++}^2$, where $f(k, h) = F(k, h, 1)$ is strictly increasing and strictly concave in (k, h) by ASSUMPTION 1' and satisfies the Inada conditions of ASSUMPTION 2'. By PROPOSITION 3.1 there is exactly one rest point $(k^*, h^*) \in \mathbb{R}_{++}^2$. We must show that every solution starting in $\mathbb{R}_{++}^2$ converges to it.

Step 1: the system is cooperative, hence the flow is monotone. The off-diagonal entries of the Jacobian are

$$\frac{\partial G_k}{\partial h} = s_k f_h(k, h) > 0, \quad \frac{\partial G_h}{\partial k} = s_h f_k(k, h) > 0,$$

so the vector field is *cooperative*: each variable helps the other accumulate — more human capital raises output and therefore physical investment, and conversely. By Kamke's comparison theorem the flow preserves the componentwise order: if $(k_1(0), h_1(0)) \leq (k_2(0), h_2(0))$ then $(k_1(t), h_1(t)) \leq (k_2(t), h_2(t))$ for all $t \geq 0$. Two consequences will be used repeatedly. First, a solution that starts above (below) (k^*, h^*) stays above (below) it, since (k^*, h^*) is itself a solution. Second, if the vector field is nonpositive in both coordinates at the initial point, then the solution is nonincreasing in both coordinates for all t : the solution started at time $\tau > 0$ lies below the one started at time 0, and both solve the same autonomous equation.

Step 2: a supersolution and a subsolution. Recall from the proof of PROPOSITION 3.1 that the loci $\{G_k = 0\}$ and $\{G_h = 0\}$ are increasing curves in (k, h) space that cross exactly once, at (k^*, h^*) , with $\{G_k = 0\}$ steeper there; moreover $\{G_k = 0\}$ lies below $\{G_h = 0\}$ for small (k, h) and above it for large (k, h) . Since G_k is increasing in h and G_h is increasing in k ,

$$G_k < 0 \text{ to the right of } \{G_k = 0\}, \quad G_h < 0 \text{ above } \{G_h = 0\}.$$

For $k > k^*$ the region between the two curves is therefore a set on which $G_k < 0$ and $G_h < 0$; it is nonempty and, because both curves are increasing and unbounded (by ASSUMPTION 2', $f(k, h)/k \rightarrow 0$ as $k \rightarrow \infty$ forces $h \rightarrow \infty$ along $\{G_k = 0\}$), it contains points $(\bar{k}, \bar{h})$ with $\bar{k} \geq k(0)$ and $\bar{h} \geq h(0)$, as large as we please. Symmetrically, for $k < k^*$ the region between the two curves has $G_k > 0$ and $G_h > 0$ and contains points $(\underline{k}, \underline{h})$ with $\underline{k} \leq k(0)$, $\underline{h} \leq h(0)$, as small as we please.

Step 3: the sandwich. Consider the solution started at $(\bar{k}, \bar{h})$. By Step 1 it is nonincreasing in both coordinates, and it is bounded below by (k^*, h^*) — it started above the rest point, so it stays above. A monotone bounded trajectory converges, and its limit is a rest point of the system; by uniqueness, the limit is (k^*, h^*) . Exactly the same argument applied at $(\underline{k}, \underline{h})$ gives a trajectory that is nondecreasing, bounded above by (k^*, h^*) , and convergent to it. Finally, by monotonicity of the flow, the solution started at $(k(0), h(0))$ is trapped between these two for all $t \geq 0$:

$$(\underline{k}(t), \underline{h}(t)) \leq (k(t), h(t)) \leq (\bar{k}(t), \bar{h}(t)),$$

and both bounds converge to (k^*, h^*) . Hence $(k(t), h(t)) \rightarrow (k^*, h^*)$ from any initial condition in $\mathbb{R}_{++}^2$, which is the assertion of PROPOSITION 3.2.

Remark. The phase diagram in Figure 3.1 of the book is exactly this argument in pictures: the two loci cut the positive quadrant into four regions, in two of which both variables move toward the steady state, while in the other two the trajectory must cross into one of those. What the monotone-flow argument adds is the step the picture cannot supply — that no trajectory can cycle around the rest point instead of converging to it. Cooperativity is what rules that out; for planar cooperative systems there are no closed orbits.

Exercise 3.7

The steady state solves (3.17)–(3.18),

$$\Phi_1(k, h) \equiv s_k f(k, h) - (\delta_k + g + n)k = 0, \quad \Phi_2(k, h) \equiv s_h f(k, h) - (\delta_h + g + n)h = 0.$$

Write J for the Jacobian of (Φ_1, Φ_2) in (k, h) , evaluated at (k^*, h^*) :

$$J = \begin{pmatrix} s_k f_k - (\delta_k + g + n) & s_k f_h \\ s_h f_k & s_h f_h - (\delta_h + g + n) \end{pmatrix} = \begin{pmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{pmatrix}.$$

Signs of the entries and of $\det J$. The off-diagonal entries are positive. For the diagonal ones, substitute $\delta_k + g + n = s_k f(k^*, h^*)/k^*$ from (3.17):

$$J_{11} = s_k \left[f_k(k^*, h^*) - \frac{f(k^*, h^*)}{k^*} \right] < 0,$$

because strict concavity of f in k together with $f(0, h^*) \geq 0$ gives $f > k f_k$; similarly $J_{22} < 0$. For the determinant, substituting both steady-state conditions,

$$\det J = (\delta_k + g + n)(\delta_h + g + n) - s_k f_k(\delta_h + g + n) - s_h f_h(\delta_k + g + n) = \frac{s_k s_h f}{k^* h^*} [f - k^* f_k - h^* f_h] > 0,$$

the bracket being positive by strict concavity of f (this is the same inequality that made the two loci cross only once in PROPOSITION 3.1).

The comparative statics. By the Implicit Function Theorem, $\partial(k^*, h^*)^T / \partial\theta = -J^{-1} \partial(\Phi_1, \Phi_2)^T / \partial\theta$ with

$$J^{-1} = \frac{1}{\det J} \begin{pmatrix} J_{22} & -J_{12} \\ -J_{21} & J_{11} \end{pmatrix}.$$

Since $\partial\Phi/\partial s_k = (f, 0)^T$, $\partial\Phi/\partial s_h = (0, f)^T$ and $\partial\Phi/\partial n = (-k^*, -h^*)^T$,

$$\begin{aligned} \frac{\partial k^*}{\partial s_k} &= \frac{-J_{22}f}{\det J} > 0, & \frac{\partial h^*}{\partial s_k} &= \frac{J_{21}f}{\det J} = \frac{s_h f_k f}{\det J} > 0, \\ \frac{\partial k^*}{\partial s_h} &= \frac{J_{12}f}{\det J} = \frac{s_k f_h f}{\det J} > 0, & \frac{\partial h^*}{\partial s_h} &= \frac{-J_{11}f}{\det J} > 0, \\ \frac{\partial k^*}{\partial n} &= \frac{J_{22}k^* - J_{12}h^*}{\det J} < 0, & \frac{\partial h^*}{\partial n} &= \frac{J_{11}h^* - J_{21}k^*}{\det J} < 0, \end{aligned}$$

the last two because $J_{11}, J_{22} < 0$ while $J_{12}, J_{21} > 0$. Since f is increasing in both arguments, output per effective unit of labor $\hat{y}^* = f(k^*, h^*)$ inherits the signs: it rises with either saving rate and falls with population growth.

Interpretation. The novelty relative to the basic model is the *cross* effect: a higher saving rate in physical capital raises the steady-state stock of *human* capital as well, and vice versa. The mechanism is the complementarity of the two factors in f — more physical capital raises output, and since a constant fraction s_h of output is invested in schooling, it raises human capital investment too. This is the “multiplier” that makes the augmented model fit cross-country income differences so much better than the basic one: the elasticity of steady-state income with respect to s_k is $\alpha/(1 - \alpha - \beta)$ rather than $\alpha/(1 - \alpha)$, as (3.22)–(3.23) show explicitly for the Cobb–Douglas case.

Exercise 3.8

The accounting equation. With $Y = F(K, H, AL)$, differentiate with respect to time and divide by Y :

$$\frac{\dot{Y}}{Y} = \frac{F_K K}{Y} \frac{\dot{K}}{K} + \frac{F_H H}{Y} \frac{\dot{H}}{H} + \frac{F_{AL} AL}{Y} \left(\frac{\dot{A}}{A} + \frac{\dot{L}}{L} \right),$$

where F_{AL} is the derivative with respect to the third argument. With competitive factor markets the elasticities are the observed income shares: $R = F_K$ gives $\alpha_K = F_K K/Y$; the rental rate of human capital $R_H = F_H$ gives $\alpha_H = F_H H/Y$; and the wage per worker is $w = AF_{AL}$, so $\alpha_L = wL/Y = F_{AL} AL/Y$. Constant returns in the three arguments make the three shares sum to one. Writing g_X for the growth rate of X ,

$$\boxed{g_Y = \alpha_K g_K + \alpha_H g_H + \alpha_L (g_A + g_L)}$$

so that the estimate of TFP growth — the part of output growth not accounted for by measured inputs — is

$$\hat{x} = g_Y - \alpha_K g_K - \alpha_H g_H - \alpha_L g_L, \quad \text{and} \quad g_A = \frac{\hat{x}}{\alpha_L}.$$

Two differences from the fundamental growth accounting equation (3.3) of the basic model deserve emphasis. First, the accumulation of human capital is now an *input*, so growth in schooling and skills is no longer part of the residual. Second, because technology is labor augmenting, the residual $\hat{x}$ measures $\alpha_L g_A$, not g_A : with a labor share of about 2/3, the implied rate of technological progress is one and a half times the measured residual.

Using data. One needs (i) output growth g_Y , from national accounts; (ii) capital stock growth g_K , from a perpetual inventory calculation on investment series; (iii) growth in efficiency units of labor g_H , constructed as in Section 3.5 from schooling data and Mincerian returns, and g_L from employment; and (iv) the three factor shares. The last item is the binding constraint, and it is where the exercise bites: national accounts report a split of value added into capital income and labor income, but labor income here is the sum of two conceptually different payments, $R_H H$ and wL . Some identifying assumption is unavoidable. The two used in practice are

- impose a functional form and calibrate: with $Y = K^\alpha H^\beta (AL)^{1-\alpha-\beta}$ as in (3.21), the shares are the constants α , β and $1 - \alpha - \beta$, and β is pinned down from microeconomic (Mincerian) estimates of the return to schooling; or
- avoid the split altogether by folding human capital into effective labor, $Y = F(K, AH)$, as in (3.29): then only two shares are needed, α_K and $1 - \alpha_K$, and the accounting equation collapses to $g_Y = \alpha_K g_K + (1 - \alpha_K)(g_A + g_H)$, at the price of imposing that raw labor and human capital are perfect substitutes in efficiency units.

Either way the residual absorbs everything the measurement misses — quality of schooling, capacity utilization, reallocation across sectors and firms, and pure measurement error — so it remains, in Abramovitz’s phrase, a measure of our ignorance; adding human capital narrows that ignorance rather than eliminating it.

Exercise 3.11

(a) *The accounting.* Work in per worker terms, $y_j \equiv Y_j/L_j$ and $\kappa_j \equiv K_j/L_j$. Constant returns make output per worker a function of the capital-labor ratio and technology alone, $y = \phi(\kappa; A)$, and competitive markets make the elasticity of that function equal to the *observed* capital share:

$$\frac{\partial \log y}{\partial \log \kappa} = \frac{F_K K}{Y} = \alpha^K.$$

This is all that is needed: no functional form, no assumption about how A enters. Order the countries by their capital-labor ratio, $\kappa_{(1)} \leq \kappa_{(2)} \leq \dots \leq \kappa_{(J)}$. If two adjacent countries had the same technology, the whole of their income difference would have to come from capital deepening,

$$\Delta \log y_{\text{capital}} = \bar{\alpha}^K \Delta \log \kappa, \quad \bar{\alpha}^K = \frac{1}{2}(\alpha_{(i)}^K + \alpha_{(i+1)}^K),$$

a Divisia (Törnqvist) index that uses the average of the two observed shares over the interval. The residual is the estimated technology gap,

$$\log \frac{A_{(i+1)}}{A_{(i)}} = (\log y_{(i+1)} - \log y_{(i)}) - \bar{\alpha}^K (\log \kappa_{(i+1)} - \log \kappa_{(i)}),$$

and cumulating these differences along the ordered cross section gives every country’s productivity relative to a numeraire country (the United States, say). Equivalently, and in one step for any pair,

$$\log \frac{A_j}{A_{\text{US}}} = \log \frac{y_j}{y_{\text{US}}} - \bar{\alpha}^K \log \frac{\kappa_j}{\kappa_{\text{US}}}.$$

The contribution of technology to cross-country income differences is then measured by the dispersion of $\log A_j$ relative to the dispersion of $\log y_j$, and the “predicted” income of country j

under a common technology is $\log \hat{y}_j = \log y_{US} + \bar{\alpha}^K \log(\kappa_j/\kappa_{US})$. (With a common Cobb–Douglas technology $\alpha_j^K \equiv \alpha$ and this is exactly the calibration of Section 3.5.) Only if the shares vary across countries does the chaining matter, and then the index is exact only up to the usual path dependence, which is second order when the share varies slowly.

(b) *What do large A differences mean?* Not necessarily that countries have access to different production possibility sets. The residual A_j is defined as whatever makes the accounting identity hold, so it absorbs every reason why a country produces less from the *measured* inputs than the numeraire country does:

- unmeasured or mismeasured inputs — human capital (part (c)), land and natural resources, capital quality and capacity utilization, hours per worker;
- allocative efficiency — misallocation of capital and labor across firms and sectors, distortions, credit constraints, monopoly power, informality;
- organization and institutions — the efficiency with which given inputs are combined;
- genuinely different technical knowledge or its adoption.

Only the last is technology in the engineering sense. Since a country can, in principle, adopt known technologies, the finding of very large A gaps is better read as a statement that *something other than the accumulation of measured factors* accounts for most of cross-country income differences — which is precisely what motivates Chapter 4’s search for fundamental causes.

(c) *What data are needed with $F(K, H, A)$?* Efficiency units of labor must be measured: the schooling distribution of the workforce (attainment by level), an estimate of how schooling maps into productivity (Mincerian returns, ideally allowing the return to vary by level and country), experience profiles, and — if one wants α_H separately from α_L — the split of labor income into payments to raw labor and to skills. Data on schooling quality (test scores, expenditure per pupil, teacher quality) are needed to avoid the bias in part (d), as are hours worked if labor input is to be measured consistently.

(d) *Quality of schooling shows up as productivity.* Suppose the truth is $Y_j = K_j^\alpha (AH_j)^{1-\alpha}$ with a common A , and that the true stock of efficiency units is $H_j = Q_j \hat{H}_j$, where $\hat{H}_j$ is the measure constructed from years of schooling as in Section 3.5 and $Q_j > 0$ captures the quality of a year of schooling. The calibration of Section 3.5 computes productivity as the residual from the measured stock $\hat{H}_j$:

$$\hat{A}_j \equiv \left[\frac{Y_j}{K_j^\alpha \hat{H}_j^{1-\alpha}} \right]^{\frac{1}{1-\alpha}} = \left[\frac{K_j^\alpha (AQ_j \hat{H}_j)^{1-\alpha}}{K_j^\alpha \hat{H}_j^{1-\alpha}} \right]^{\frac{1}{1-\alpha}} = AQ_j.$$

So the estimated productivity of country j relative to the United States is

$$\frac{\hat{A}_j}{\hat{A}_{US}} = \frac{Q_j}{Q_{US}},$$

even though all countries share the same technology: *every* estimated productivity difference is schooling quality in disguise. The bias goes the empirically awkward way, too. Poor countries are generally thought to have lower Q , so $\hat{A}_j$ is understated for them, and the calibration attributes to

“technology” what is really an unmeasured input — which is exactly the criticism of the Hall–Jones and Klenow–Rodríguez calibrations that the quality-adjustment literature (Caselli, 2005) takes up.

Chapter 4

Fundamental Determinants of Differences in Economic Performance

The two exercises of Chapter 4 left unsolved in the instructor's manual: Exercises 4.1 and 4.2, both on the Simon–Kremer model of Section 4.2. The manual solves 4.3. Throughout, world output is $Y(t) = L(t)^\alpha (A(t)Z)^{1-\alpha}$ with Z normalized to 1, so that $Y = L^\alpha A^{1-\alpha}$, and population responds to output through (4.2), $L(t) = \phi Y(t)$.

Exercise 4.1

Population as a function of the stock of ideas. Substituting the production function into (4.2),

$$L = \phi Y = \phi L^\alpha A^{1-\alpha} \quad \implies \quad L^{1-\alpha} = \phi A^{1-\alpha} \quad \implies \quad L(t) = \phi^{\frac{1}{1-\alpha}} A(t). \quad (*)$$

The Malthusian channel thus ties population to the stock of knowledge one for one: any improvement in technology is absorbed by a proportional increase in the number of people it has to feed.

Equation (4.3). Each individual discovers ideas at the rate λ , so by (4.1) $\dot{A} = \lambda L$. Substituting (*),

$$\dot{A}(t) = \lambda \phi^{\frac{1}{1-\alpha}} A(t),$$

which is (4.3): the *level* of the flow of new ideas is proportional to the existing stock, because the population that generates them is.

Equation (4.4). This is a linear differential equation with constant coefficient; separating variables, $d \log A = \lambda \phi^{1/(1-\alpha)} dt$, so with the initial condition $A(0) > 0$,

$$A(t) = \exp(\lambda \phi^{1/(1-\alpha)} t) A(0),$$

which is (4.4). Technology grows exponentially at the constant rate $g \equiv \lambda \phi^{1/(1-\alpha)}$, and so do population, by (*), and world output, since

$$Y = L^\alpha A^{1-\alpha} = (\phi^{\frac{1}{1-\alpha}} A)^\alpha A^{1-\alpha} = \phi^{\frac{\alpha}{1-\alpha}} A.$$

Remark. Income per capita, however, does not grow at all: $y = Y/L = 1/\phi$ at every date, directly from (4.2). The model produces a growing *world* — more people, more ideas, more output — with living standards pinned down by the Malthusian relation between income and population. That is the sense in which economies of scale in population are a theory of world growth rather than of prosperity, and it is the observation that Exercise 4.3(a) formalizes.

Exercise 4.2

Derivation. Simon and Kremer replace (4.1) by the assumption that population determines the *proportional* rate of improvement,

$$\frac{\dot{A}(t)}{A(t)} = \lambda L(t).$$

The static relations are unchanged, so (*) of Exercise 4.1 still gives $L(t) = \phi^{1/(1-\alpha)} A(t)$ and therefore

$$\dot{A}(t) = \lambda \phi^{\frac{1}{1-\alpha}} A(t)^2.$$

This is separable: $dA/A^2 = \lambda \phi^{1/(1-\alpha)} dt$, and integrating from 0 to t ,

$$\frac{1}{A(0)} - \frac{1}{A(t)} = \lambda \phi^{\frac{1}{1-\alpha}} t \quad \implies \quad A(t) = \frac{1}{A(0)^{-1} - \lambda \phi^{1/(1-\alpha)} t},$$

which is (4.5).

How the behavior differs. Under (4.4) the growth rate of technology is the constant $\lambda \phi^{1/(1-\alpha)}$. Under (4.5) it is

$$\frac{\dot{A}(t)}{A(t)} = \lambda \phi^{\frac{1}{1-\alpha}} A(t) = \frac{\lambda \phi^{1/(1-\alpha)}}{A(0)^{-1} - \lambda \phi^{1/(1-\alpha)} t},$$

which is increasing in t : growth accelerates. Indeed the acceleration is violent. The denominator of (4.5) vanishes at

$$T^* = \frac{1}{\lambda \phi^{1/(1-\alpha)} A(0)} < \infty,$$

so technology, population and output all become infinite in *finite* time; the solution does not merely grow fast, it blows up. Starting from a small $A(0)$, T^* is large and the early path is nearly flat, which is what gives the model its appeal as a story of millennia of stagnation followed by a takeoff.

Why. The difference is in how an idea is assumed to interact with the stock of ideas already available. Under (4.1) a discovery adds a fixed increment to the pool: a person is worth λ units of A per unit of time regardless of how much is already known. Under the Simon–Kremer assumption a discovery raises the stock *proportionally*: each person contributes λ percentage points of growth per unit of time, so a discovery is worth more the more is already known — the “standing on the shoulders of giants” externality. Since the Malthusian channel makes population itself proportional to A , the second assumption makes the growth rate proportional to A , and a differential equation whose growth rate grows linearly in the state explodes in finite time.

Which is more plausible? Neither is comfortable, and they fail in different places.

Taken as a description of the very long run, the Simon–Kremer version is the better one. It is essentially the only simple mechanism that delivers the qualitative shape of the historical record — hundreds of millennia of near-stagnation followed by an acceleration — and Kremer’s evidence that the growth rate of world population has risen with the *level* of world population over a million years is direct support for the proportional specification.

Taken literally as a model, however, it cannot be right. Finite-time blow-up is not a prediction about any observable world, so the specification must fail at some scale; and the mechanism is a pure scale effect, so it predicts that larger populations grow faster — a prediction rejected when the units are countries (the European population was below that of Asia for two millennia, yet modern growth began in Europe), and rescued only by insisting that the relevant unit is the world as a whole.

The assumption in (4.1) has the more defensible microfoundation — independent discoveries arriving at a constant rate per person, which is exactly the probabilistic story the section tells — but it can only generate steady exponential growth, never a takeoff.

The binding weakness, though, is shared and lies elsewhere: it is (4.2). Both models force income per capita to equal $1/\phi$ at every date, so neither can speak to living standards at all, and the Malthusian link between income and population on which they rest broke down precisely when modern growth began, with the demographic transition reversing the sign of the relation between income and fertility. A mechanism that is silent about income per capita and about *where* growth started is, as the chapter concludes, at best a theory of world growth — not a fundamental cause of cross-country differences.

Chapter 5

Foundations of Neoclassical Growth

The six exercises of Chapter 5 left unsolved in the instructor's manual: Exercises 5.3–5.8. The manual solves 5.1, 5.2 and 5.9–5.14. Notation follows the book: $\mathcal{H}$ is the set of households, $p = (p_1, \dots, p_N)$ the price vector, w^h household h 's income, $v^h(p, w^h)$ its indirect utility, ω^h its endowment and θ_f^h its share in firm f .

Exercise 5.3

Each household $h \in \mathcal{H}$ has indirect utility of the Gorman form (5.3),

$$v^h(p, w^h) = a^h(p) + b(p)w^h,$$

with a common $b(\cdot)$, and a positive demand for every commodity. We must show that aggregate demand is the demand of a single household with indirect utility $v(p, w) = a(p) + b(p)w$, where $a(p) \equiv \int_{h \in \mathcal{H}} a^h(p) dh$ and $w \equiv \int_{h \in \mathcal{H}} w^h dh$.

Step 1: what homogeneity forces. Any indirect utility function is homogeneous of degree 0 in (p, w) , so for every $\lambda > 0$ and every w^h ,

$$a^h(\lambda p) + b(\lambda p)\lambda w^h = a^h(p) + b(p)w^h.$$

Both sides are affine in w^h , and the household's income can be varied freely, so the constant terms and the slopes must agree separately:

$$a^h(\lambda p) = a^h(p) \quad \text{and} \quad b(\lambda p) = \lambda^{-1}b(p),$$

that is, a^h is homogeneous of degree 0 and b of degree -1 . Since v^h is strictly increasing in income, $b(p) > 0$.

Step 2: individual demands. Assume a^h and b differentiable. By Roy's Identity, household h 's demand for commodity j is

$$x_j^h(p, w^h) = -\frac{\partial v^h / \partial p_j}{\partial v^h / \partial w^h} = -\frac{1}{b(p)} \frac{\partial a^h(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w^h. \quad (*)$$

(The hypothesis that each household consumes a positive amount of every commodity is what makes the demands interior, so that Roy's Identity applies and no household sits at a corner where the envelope argument would fail.) Note the structure: the Engel curve of household h for commodity j is a straight line in w^h whose *slope*, $-b^{-1} \partial b / \partial p_j$, does not depend on h . All the household-specific information sits in the intercept.

Step 3: aggregation. Integrating $(*)$ over $\mathcal{H}$ and exchanging differentiation and integration,

$$\begin{aligned} x_j(p) &\equiv \int_{h \in \mathcal{H}} x_j^h(p, w^h) dh = -\frac{1}{b(p)} \frac{\partial}{\partial p_j} \int_{h \in \mathcal{H}} a^h(p) dh - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} \int_{h \in \mathcal{H}} w^h dh \\ &= -\frac{1}{b(p)} \frac{\partial a(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w. \end{aligned}$$

Now consider the hypothetical household with indirect utility $v(p, w) = a(p) + b(p)w$. It is a legitimate indirect utility function: a is homogeneous of degree 0 and b of degree -1 by Step 1, so v is homogeneous of degree 0; it is increasing in w and decreasing in each p_j wherever the individual ones are. Applying Roy's Identity to it gives demands

$$-\frac{1}{b(p)} \frac{\partial a(p)}{\partial p_j} - \frac{1}{b(p)} \frac{\partial b(p)}{\partial p_j} w,$$

which is exactly $x_j(p)$. Hence aggregate demand is generated by the maximization problem of the single household with indirect utility $v(p, w)$: the economy admits a representative household, and THEOREM 5.2 is proved.

Step 4: the representative household is strong. Aggregate demand depends on the distribution of income only through the aggregate $w = \int w^h dh$, so any redistribution of income across households leaves it unchanged. The converse also holds, which is why the Gorman form is not merely sufficient but necessary: if two households h and h' had Engel curves of different slopes for commodity j , that is $b_h^{-1} \partial b_h / \partial p_j \neq b_{h'}^{-1} \partial b_{h'} / \partial p_j$, then transferring a unit of income from h' to h would change aggregate demand for j , and no single household's demand function could reproduce aggregate demand at all income distributions.

Remark. Nothing in the argument used $N < \infty$ beyond writing the derivatives coordinate by coordinate, and nothing used $\mathcal{H}$ finite: the integral is a Lebesgue integral and covers a continuum of households. What the theorem does *not* deliver is a *normative* representative household — the utility $v(p, w)$ reproduces aggregate demand but need not rank allocations the way any Pareto criterion would. That requires the additional structure of THEOREM 5.3, in particular $a^h(p) = a^h$ independent of p for the converse direction.

Exercise 5.4

Let the commodity space be indexed by $j \in J \equiv [0, 1]$ with the Lebesgue measure (nothing depends on the particular index set; $J = \mathbb{R}_+$ for a continuous-time horizon reads the same way). A consumption bundle is a measurable function $x : J \rightarrow \mathbb{R}$, and we take the commodity space to be $L^\infty(J)$ with prices in its predual $L^1(J)$, so that the value of a bundle is

$$p \cdot x = \int_J p(j)x(j) dj,$$

which is finite for $p \in L^1$, $x \in L^\infty$. (Any dual pair with a bilinear form of this kind will do; what matters is that values are given by integrals and that the dual space is rich enough to separate points.) Household h 's preferences have the Gorman representation $v^h(p, w^h) = a^h(p) + b(p)w^h$, each household demands a positive amount of (almost) every commodity, and the aggregate production possibilities set $Y \subset L^\infty(J)$ is convex. The claim is that THEOREM 5.3 survives verbatim:

- (1) any feasible allocation that maximizes $v(p, w) = \int_{\mathcal{H}} a^h(p) dh + b(p)w$, with $w = \int_{\mathcal{H}} w^h dh$, is Pareto optimal;
- (2) if in addition $a^h(p) = a^h$ for all p and all h , then every Pareto optimal allocation maximizes $v(p, w)$.

What carries over unchanged. The proof of THEOREM 5.3 is an argument about the *objective*

$$\max_{h \in \mathcal{H}} \sum \alpha^h v^h(p, w^h) = \max \left[\int_{\mathcal{H}} \alpha^h a^h(p) dh + b(p) \int_{\mathcal{H}} \alpha^h w^h dh \right]$$

subject to market clearing and to net supplies being profit maximizing at p , exactly as in (5.7). Every step of it is linear in the commodity index: sums $\sum_{j=1}^N p_j y_j$ become integrals $\int_J p(j) y(j) dj$, and Roy's Identity becomes the same formula with the functional (Gâteaux) derivatives $\delta a^h(p)/\delta p(j)$ and $\delta b(p)/\delta p(j)$ in place of partial derivatives. The linearity of the Gorman form in w^h — which is what makes the Pareto weights enter only through the weighted aggregate income and lets one substitute income transfers for commodity reallocations — is a statement about incomes, not about commodities, so it is untouched. With $b(p) > 0$, the objective is still increasing in aggregate income, and part 1 follows as before; with $a^h(p) = a^h$ constant, the argument by contradiction that produced (5.9)–(5.11) goes through with integrals.

What genuinely needs an extra assumption. Two things that are automatic in $\mathbb{R}^N$ are not automatic here.

- *Existence of supporting prices.* In $\mathbb{R}^N$ a convex set can always be separated from a disjoint convex set. In an infinite-dimensional space the separating hyperplane theorem requires a topological condition — for instance that Y (or the set of preferred allocations) have nonempty interior in the sup-norm topology, or that preferences satisfy Mas-Colell's properness condition. Without it a Pareto optimum need not be supported by any price system in the chosen dual space, and the phrase “at price vector p ” in the theorem would be empty. This is the same hypothesis the book adds in THEOREMS 5.6 and 5.7 when it proves the Second Welfare Theorem with infinitely many commodities; here it is needed for exactly the same reason.
- *Integrability.* The aggregates $a(p) = \int_{\mathcal{H}} a^h(p) dh$ and $w = \int_{\mathcal{H}} w^h dh$ must be finite, and the interchange of $\delta/\delta p(j)$ with $\int_{\mathcal{H}} dh$ must be legitimate — guaranteed, for instance, by a uniform integrable bound on the family $\{\delta a^h(p)/\delta p(j)\}$. With a continuum of commodities one must also require $p \cdot x^h < \infty$ for each household, which Exercise 5.5 delivers from local nonsatiation.

Conclusion. With these two provisos — a price system exists in the dual, and the household aggregates are well defined — THEOREM 5.3 holds word for word with $\sum_{j=1}^N$ replaced by $\int_J dj$. The economics of the result is therefore robust to the infinite-dimensional setting that growth models live in: it is the *linearity of Gorman preferences in income*, not the finiteness of the commodity space, that makes the representative household normative.

Exercise 5.5

“Household h chooses x^h at prices p ” means that x^h solves

$$\max_{x \in X^h} U^h(x) \quad \text{subject to} \quad p \cdot x \leq w^h, \quad w^h \equiv p \cdot \left(\omega^h + \sum_{f \in \mathcal{F}} \theta_f^h y^f \right),$$

and local nonsatiation is DEFINITION 5.3: at every $x \in X^h$, U^h is strictly increasing in at least one of its arguments, and $U^h(x) < \infty$.

Step 1: the income of a locally nonsatiated household that makes a choice is finite. Suppose $w^h = \infty$. Then the budget constraint does not bind anywhere and the problem reduces to maximizing U^h over X^h . But local nonsatiation says that at *every* point of X^h there is a coordinate j' in which U^h is strictly increasing, so for any candidate x the bundle $x(\varepsilon)$ that adds $\varepsilon > 0$ to coordinate j' — which is in X^h for small ε , this being exactly the construction used in the proof of THEOREM 5.5 — satisfies $U^h(x(\varepsilon)) > U^h(x)$ and is still affordable. No maximum exists, so no bundle is chosen, contradicting the hypothesis. Hence $w^h < \infty$.

Step 2: the chosen bundle is affordable, hence of finite value. Since x^h is in the budget set,

$$p \cdot x^h \leq w^h < \infty.$$

Step 3: in fact the budget holds with equality. Suppose $p \cdot x^h < w^h$. By local nonsatiation there is a coordinate j' in which U^h is strictly increasing at x^h ; for ε small enough the bundle $x^h(\varepsilon)$ obtained by adding ε to that coordinate still costs less than w^h and yields strictly higher utility, contradicting optimality. Therefore

$$p \cdot x^h = w^h < \infty.$$

Remark. In a finite-dimensional economy this is a triviality — a finite sum of finite terms. Its content is in the infinite-dimensional case relevant to growth theory, where $p \cdot x = \sum_{t=0}^{\infty} p_t x_t$ is an infinite series that may diverge. The result says that at any chosen bundle the series converges, and this is precisely what the proof of the First Welfare Theorem needs: (5.17)–(5.18) are summed over households, and the sums are meaningless unless each term is finite. It is also the reason DEFINITION 5.3 builds $U^h(x^h) < \infty$ into local nonsatiation: an infinite-utility plan would allow a household to be indifferent to budget constraints, and both the definition of a competitive equilibrium and the welfare theorems would lose their force.

Exercise 5.6

(a) *Demands and indirect utility.* Write $\hat{x}_j^h \equiv x_j^h - \xi_j^h$ and $\hat{w}^h \equiv w^h - \sum_j p_j \xi_j^h > 0$ (positive by the assumption $\sum_j p_j \bar{\xi} < w^h$). The problem

$$\max \left[\sum_{j=1}^N (\hat{x}_j^h)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad \text{s.t.} \quad \sum_{j=1}^N p_j \hat{x}_j^h \leq \hat{w}^h$$

is the textbook CES problem. The first-order conditions give $(\hat{x}_j^h)^{-1/\sigma} = \mu p_j$ for a multiplier $\mu > 0$, hence $\hat{x}_j^h \propto p_j^{-\sigma}$, and the budget constraint fixes the factor:

$$\hat{x}_j^h = \frac{\hat{w}^h p_j^{-\sigma}}{P}, \quad P \equiv \sum_{k=1}^N p_k^{1-\sigma}, \quad \text{that is} \quad x_j^h = \xi_j^h + \frac{(w^h - \sum_k p_k \xi_k^h) p_j^{-\sigma}}{\sum_k p_k^{1-\sigma}}.$$

Substituting back,

$$\sum_j (\hat{x}_j^h)^{\frac{\sigma-1}{\sigma}} = \left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} \sum_j p_j^{1-\sigma} = \left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} P,$$

so that

$$v^h(p, w^h) = \left[\left(\frac{\hat{w}^h}{P} \right)^{\frac{\sigma-1}{\sigma}} P \right]^{\frac{\sigma}{\sigma-1}} = \hat{w}^h P^{\frac{1}{\sigma-1}} = \frac{-\sum_j p_j \xi_j^h + w^h}{[\sum_j p_j^{1-\sigma}]^{\frac{1}{1-\sigma}}},$$

which is (5.5). It is of the Gorman form (5.3) with

$$b(p) = P^{\frac{1}{\sigma-1}} > 0 \quad \text{and} \quad a^h(p) = -\left(\sum_j p_j \xi_j^h\right) P^{\frac{1}{\sigma-1}},$$

and the household-specific term enters only through a^h , exactly as THEOREM 5.2 requires. It is homogeneous of degree 0: $\hat{w}^h$ is homogeneous of degree 1 in (p, w) and $P^{1/(\sigma-1)}$ of degree -1 .

(b) *The representative household.* The utility function (5.6) is the same CES form with $\xi_j \equiv \int_{\mathcal{H}} \xi_j^h dh$, so the computation in (a) applies verbatim and gives

$$v(p, w) = \frac{-\sum_j p_j \xi_j + w}{[\sum_j p_j^{1-\sigma}]^{\frac{1}{1-\sigma}}} = \int_{\mathcal{H}} a^h(p) dh + b(p)w,$$

using $\int_{\mathcal{H}} a^h(p) dh = -(\sum_j p_j \xi_j) P^{1/(\sigma-1)}$ and $w = \int_{\mathcal{H}} w^h dh$. This is precisely the indirect utility of the representative household constructed in THEOREM 5.2, so (5.6) is the utility function that rationalizes aggregate demand.

(c) *Dropping the outer exponent.* Let now $\tilde{U}^h = \sum_j (x_j^h - \xi_j^h)^{(\sigma-1)/\sigma}$. Take $\sigma > 1$, so that $(\sigma-1)/\sigma > 0$ and $\tilde{U}^h = (U^h)^{(\sigma-1)/\sigma}$ is a strictly increasing transformation of the utility in part (a). Demands are therefore unchanged, and the indirect utility function is the same transformation of (5.5):

$$\tilde{v}^h(p, w^h) = [\hat{w}^h P^{\frac{1}{\sigma-1}}]^{\frac{\sigma-1}{\sigma}} = (w^h - \sum_j p_j \xi_j^h)^{\frac{\sigma-1}{\sigma}} P^{\frac{1}{\sigma}}.$$

It is still homogeneous of degree 0 — the first factor has degree $(\sigma-1)/\sigma$ and the second $(1-\sigma)/\sigma$ — but it is *not* of the Gorman form: it is a strictly concave, not affine, function of w^h , so the Engel curves are no longer straight lines. The monotone transformation $z \mapsto z^{\sigma/(\sigma-1)}$ restores (5.5) and with it the Gorman form.

Is that enough? For the positive question — does the economy admit a (strong) representative household? — yes. Demand is an ordinal object: it is invariant to strictly increasing transformations of utility, so what matters is whether *some* representation of preferences takes the Gorman form, not whether the one we happened to write down does. This is the point the book makes right after THEOREM 5.2, and part (c) is the example that makes it concrete: aggregate demand here is exactly as in part (b).

Two caveats keep the answer from being an unqualified yes. First, under uncertainty utility is cardinal: the transformation $z \mapsto z^{\sigma/(\sigma-1)}$ changes the household's attitude to risk, so two representations that are equivalent for a deterministic exchange economy are *not* equivalent once expected utility is taken over lotteries — the aggregation result survives only for the representation that is also the Bernoulli utility index. Second, for a *normative* representative household the transformation matters as well: THEOREM 5.3(2) requires the Gorman representation to have $a^h(p) = a^h$ independent of prices, which is a restriction on a particular cardinalization, and Pareto weights attached to $\tilde{v}^h$ and to v^h define different social objectives. Aggregation of demand is ordinal; welfare analysis is not.

Exercise 5.7

Survival. Death arrives as the first event of a Poisson process with intensity ν , so the random date of death T is exponentially distributed:

$$\Pr\{T > t\} = e^{-\nu t}, \quad \mathbb{E}[T] = \frac{1}{\nu} < \infty.$$

The individual's expected lifetime is $1/\nu$, which is how one calibrates ν .

Expected utility. Normalize the flow utility after death to $u(0) = 0$, and let $[c(t)]_{t \geq 0}$ be the consumption plan conditional on being alive. Discounting the future at the pure rate ρ , expected lifetime utility at date 0 is

$$U_0 = \mathbb{E} \left[\int_0^T e^{-\rho t} u(c(t)) dt \right] = \mathbb{E} \left[\int_0^\infty e^{-\rho t} u(c(t)) \mathbf{1}\{T > t\} dt \right].$$

The integrand is nonnegative and measurable, so Fubini's theorem allows the expectation to be taken inside:

$$U_0 = \int_0^\infty e^{-\rho t} u(c(t)) \Pr\{T > t\} dt = \int_0^\infty e^{-\rho t} e^{-\nu t} u(c(t)) dt = \int_0^\infty e^{-(\rho+\nu)t} u(c(t)) dt.$$

This is the objective of an infinitely lived individual with discount rate

$\rho_{\text{eff}} = \rho + \nu$

Since the constraint set — the flow budget constraint and the no-Ponzi condition — is unaffected by the *form* of the objective, the individual's problem is identical to that of an infinitely lived agent who discounts at $\rho + \nu$. Her behavior, in particular her Euler equation $\dot{c}/c = (r(t) - \rho - \nu)/\theta$ for the CRRA case, is that of the infinite-horizon model with the larger discount rate.

Consistency with (5.12). In discrete time the book obtains $\beta = \hat{\beta}(1 - \nu)$; writing $\hat{\beta} = e^{-\rho\Delta}$ and $1 - \nu \approx e^{-\nu\Delta}$ for a period of length Δ gives $\beta \approx e^{-(\rho+\nu)\Delta}$, which is the same statement: the probability of not surviving acts exactly like impatience.

Remark. The result says that mortality raises the *effective discount rate*, not that it makes people poorer. Whether it also changes consumption growth depends on the asset market. With actuarially fair annuities — the Blanchard–Yaari setting of Chapter 9 — a surviving individual earns $r + \nu$ rather than r on her savings, because the assets of those who die are shared among the survivors, and the two effects cancel in the individual Euler equation: $\dot{c}/c = [(r + \nu) - (\rho + \nu)]/\theta = (r - \rho)/\theta$. What mortality then changes is the *aggregate*, through the arrival of newborns who hold no assets, which is exactly why the perpetual youth model breaks Ricardian equivalence while the individual problem looks infinite horizon.

Exercise 5.8

(a) *Different utility functions across generations.* Yes — the dynastic recursion still collapses into an infinite-horizon maximization, but the problem loses stationarity. Let generation t value its own consumption through u_t and the continuation value of its descendant with factor β :

$$V_t(y_t) = \max_{c_t + b_t \leq y_t} \{u_t(c_t) + \beta V_{t+1}(b_t + w)\}.$$

Iterating forward,

$$V_0 = \max \sum_{t=0}^{\infty} \beta^t u_t(c_t),$$

provided the sum converges (e.g. if the u_t are uniformly bounded). So a dynasty of one-period-lived altruistic individuals again behaves like a single infinitely lived agent.

Two things should be said about what does and does not survive. The objective remains *time-consistent*: viewed from date t , the dynasty maximizes $\sum_{s \geq t} \beta^{s-t} u_s(c_s)$, which ranks continuation plans exactly as $\sum_{s \geq 0} \beta^s u_s(c_s)$ does, because the two differ by the positive factor β^t and by terms that are already sunk. Geometric discounting, not the identity of the period utility functions, is what delivers consistency (compare EXERCISE 5.1). What is lost is *stationarity*: the maximization problem is no longer autonomous, so the stationary dynamic programming of Chapter 6 does not apply directly and one needs its nonstationary version (Section 6.7), with value functions indexed by time. Balanced growth also imposes a restriction: if $u_t(c) = \gamma^t u(c)$ for a constant $\gamma > 0$ — taste change at a constant rate — the objective is $\sum (\beta\gamma)^t u(c_t)$ and stationarity is restored with the modified discount factor $\beta\gamma$, which must satisfy $\beta\gamma < 1$. For general sequences $\{u_t\}$ neither a steady state nor a balanced growth path need exist.

(b) *Caring about grandchildren directly.* Now

$$V_t = u(c_t) + \beta V_{t+1} + \delta V_{t+2}, \quad \beta, \delta > 0.$$

Expanding, the weight a_s that generation t attaches to $u(c_{t+s})$ obeys the linear recursion

$$a_0 = 1, \quad a_1 = \beta, \quad a_s = \beta a_{s-1} + \delta a_{s-2} \quad (s \geq 2),$$

whose solution is a combination of the powers of the roots of $x^2 = \beta x + \delta$, the dominant one being $\varphi = (\beta + \sqrt{\beta^2 + 4\delta})/2$. Three consequences follow.

First, the objective is still an infinite discounted sum, $V_t = \sum_{s \geq 0} a_s u(c_{t+s})$, and it converges if and only if $\varphi < 1$, which is equivalent to

$$\beta + \delta < 1.$$

So the extra altruism must be paid for out of the total weight on the future.

Second — and this is the substantive difference — the discounting is *not* geometric. The ratios of successive weights are

$$\frac{a_1}{a_0} = \beta, \quad \frac{a_2}{a_1} = \beta + \frac{\delta}{\beta} > \beta,$$

and they keep changing (converging to φ only in the limit). A sequence of weights that falls at a varying rate is exactly the structure of the hyperbolic discounting of EXERCISE 5.1(b), and it has the same consequence: the plan that generation t would like its descendants to follow is not the plan that generation $t+1$ chooses, because generation t discounts dates $t+1$ and $t+2$ in the ratio β to $\beta^2 + \delta$, while generation $t+1$ discounts the same two dates in the ratio 1 to β . Preferences are *time-inconsistent*, and the dynasty can no longer be represented as a single maximizing agent.

Third, the modeling consequence: the right solution concept is not a maximization problem but an intergenerational *game*, in which each generation takes the decision rules of its descendants as given; the natural equilibrium notion is a Markov-perfect (or “consistent-planning”) equilibrium,

and saving behavior in it generally differs from the commitment solution — the dynasty saves less than generation t would wish, since it cannot bind its descendant to treat the grandchild as generously as it would like. The economics is the same as in the individual hyperbolic-discounting model: an extra, non-geometric concern for a more distant date creates a conflict between successive decision makers.

Chapter 6

Infinite-Horizon Optimization and Dynamic Programming

The fourteen exercises of Chapter 6 left unsolved in the instructor's manual: Exercises 6.1, 6.4, 6.5, 6.6, 6.10, 6.11, 6.13, 6.14, 6.15, 6.16, 6.17, 6.19, 6.20 and 6.21. The manual solves 6.2, 6.3, 6.7, 6.8, 6.9, 6.12 and 6.18. Notation follows the book: PROBLEM 6.2 is the sequence problem, PROBLEM 6.3 the Bellman equation, T the operator (6.18), $\Phi(x(0))$ the set of feasible plans, and π the policy function. Assumptions numbered 6.1–6.5 are the dynamic programming assumptions; Assumptions 1, 2 and 3' are those on technology and preferences. The numerical part is computed by `code/ch6.py`.

Exercise 6.1

Following EXAMPLE 6.1, the state is $x = k$, the control is next period's capital $y = k'$, and

$$U(k, k') = u(g(k) - k'), \quad g(k) \equiv f(k) + (1 - \delta)k, \quad G(k) = [0, g(k)],$$

the constraint $k' \leq g(k)$ being $c \geq 0$. Assumptions 1 and 2 of Chapter 2 restrict f only, so throughout we also use the standard assumptions on the utility function collected in Assumption 3': u is continuous, strictly increasing, strictly concave, and (for ASSUMPTION 6.5) continuously differentiable.

A compact state space. First note that the state can be confined to a compact interval. By Assumption 2, $\lim_{k \rightarrow \infty} f(k)/k = 0$, so there is a unique $\bar{k} > 0$ with $f(\bar{k}) = \delta\bar{k}$: at $\bar{k}$ the whole of output is needed to replace depreciated capital, and no feasible plan starting below $\bar{k}$ can ever exceed it. Hence, with $\vec{k} \equiv g(\max\{k(0), \bar{k}\})$, every feasible plan from $k(0)$ remains in $X \equiv [0, \vec{k}]$, a compact set.

ASSUMPTION 6.1. $G(k) \ni 0$ is nonempty for every $k \in X$. Since X is compact, g is continuous and u is continuous, U is bounded on X_G (a continuous function on a compact set), say $|U| \leq \bar{U} < \infty$. Then for any feasible plan $\sum_{t=0}^n \beta^t U(k(t), k(t+1))$ is a Cauchy sequence, since the tail is bounded by $\bar{U}\beta^{n+1}/(1 - \beta)$; the limit exists and is finite.

ASSUMPTION 6.2. X is a compact subset of $\mathbb{R}$. $G(k) = [0, g(k)]$ is nonempty- and compact-valued, and it is continuous: both endpoints are continuous functions of k , which gives upper and lower hemicontinuity. U is continuous on $X_G = \{(k, k') : 0 \leq k' \leq g(k)\}$ as the composition of the continuous functions u and $(k, k') \mapsto g(k) - k'$.

ASSUMPTION 6.3. Take $(k, k'), (\tilde{k}, \tilde{k}') \in X_G$ and $\alpha \in (0, 1)$. Since f is concave (Assumption 1), so is g , and since u is increasing and concave,

$$\begin{aligned} U(\alpha k + (1 - \alpha)\tilde{k}, \alpha k' + (1 - \alpha)\tilde{k}') &= u(g(\alpha k + (1 - \alpha)\tilde{k}) - \alpha k' - (1 - \alpha)\tilde{k}') \\ &\geq u(\alpha[g(k) - k'] + (1 - \alpha)[g(\tilde{k}) - \tilde{k}']) \\ &\geq \alpha U(k, k') + (1 - \alpha)U(\tilde{k}, \tilde{k}'), \end{aligned}$$

the first inequality by concavity of g and monotonicity of u , the second by concavity of u . If $k \neq \tilde{k}$ the first inequality is strict, because f is strictly concave (Assumption 1 has $f'' < 0$) and u is strictly increasing — which is exactly the strict inequality required by ASSUMPTION 6.3. Convexity of G is the same computation: if $k' \leq g(k)$ and $\tilde{k}' \leq g(\tilde{k})$ then $\alpha k' + (1 - \alpha)\tilde{k}' \leq \alpha g(k) + (1 - \alpha)g(\tilde{k}) \leq g(\alpha k + (1 - \alpha)\tilde{k})$.

ASSUMPTION 6.4. $U(\cdot, k')$ is strictly increasing because $\partial U / \partial k = u'(c)g'(k) = u'(c)[f'(k) + 1 - \delta] > 0$, using $u' > 0$ and $f' > 0$ (Assumption 1). And g is increasing, so $k \leq \tilde{k}$ implies $G(k) = [0, g(k)] \subset [0, g(\tilde{k})] = G(\tilde{k})$: a larger capital stock relaxes the constraint set.

ASSUMPTION 6.5. On the interior of X_G — where $c = g(k) - k' > 0$ — U is continuously differentiable, since u is continuously differentiable and f is twice differentiable by Assumption 1.

All five assumptions therefore hold, and THEOREMS 6.1–6.6 apply to the optimal growth model, which is what licenses the recursive treatment in Sections 6.6 and 6.8.

Remark. Only two properties of the technology did real work: concavity of f (for ASSUMPTION 6.3) and the Inada condition $f(k)/k \rightarrow 0$ (for compactness of X). The second is indispensable in this form of the argument: an AK technology, which violates Assumption 2, has an unbounded state space, and the contraction-based theory of this chapter does not apply directly — which is why the book turns to continuous-time methods for endogenous growth models.

Exercise 6.4

Recall the setting: $\dot{x}(t) = f(x(t))$ with $x(0) = c$, where f is Lipschitz with constant M , and for $g \in C[0, s]$,

$$Tg(z) = c + \int_0^z f(g(x)) dx, \quad z \in [0, s].$$

(a) T maps $C[0, s]$ into itself, and its fixed points are the solutions. If g is continuous then $f \circ g$ is continuous, so the integral is well defined and Tg is continuously differentiable, hence continuous: $T : C[0, s] \rightarrow C[0, s]$. If $Tg = g$ then $g(z) = c + \int_0^z f(g(x)) dx$, so $g(0) = c$ and, by the Fundamental Theorem of Calculus, g is differentiable with $\dot{g}(z) = f(g(z))$: g solves (6.7). Conversely, integrating (6.7) shows every continuous solution is a fixed point. For the contraction property, Lipschitz continuity gives, for every $z \in [0, s]$,

$$|Tg(z) - T\tilde{g}(z)| \leq \int_0^z |f(g(x)) - f(\tilde{g}(x))| dx \leq M \int_0^z |g(x) - \tilde{g}(x)| dx \leq Mz\|g - \tilde{g}\|, \quad (*)$$

so $\|Tg - T\tilde{g}\| \leq Ms\|g - \tilde{g}\|$. For $s < 1/M$ this is a contraction, and since $C[0, s]$ with the sup norm is complete, THEOREM 6.7 gives a unique fixed point: (6.7) has exactly one continuous solution on $[0, s]$ when $s < 1/M$.

(b) *The second iterate.* The useful form of (*) is the *pointwise* bound $|Tg(z) - T\tilde{g}(z)| \leq Mz\|g - \tilde{g}\|$. Applying T again and using it inside the integral,

$$|T^2g(z) - T^2\tilde{g}(z)| \leq M \int_0^z |Tg(x) - T\tilde{g}(x)| dx \leq M \int_0^z Mx\|g - \tilde{g}\| dx = \frac{M^2 z^2}{2} \|g - \tilde{g}\|,$$

and taking the supremum over $z \in [0, s]$ gives $\|T^2g - T^2\tilde{g}\| \leq M^2 \frac{s^2}{2} \|g - \tilde{g}\|$.

(c) *Induction.* Suppose $|T^n g(z) - T^n \tilde{g}(z)| \leq M^n \frac{z^n}{n!} \|g - \tilde{g}\|$ for all $z \in [0, s]$, which is (*) for $n = 1$. Then

$$|T^{n+1}g(z) - T^{n+1}\tilde{g}(z)| \leq M \int_0^z \frac{M^n x^n}{n!} \|g - \tilde{g}\| dx = \frac{M^{n+1} z^{n+1}}{(n+1)!} \|g - \tilde{g}\|,$$

completing the induction; the sup over $[0, s]$ gives the stated bound.

(d) *Any interval.* Fix $s \in \mathbb{R}_+$ arbitrary. Since $M^n s^n / n! \rightarrow 0$ as $n \rightarrow \infty$ — the terms of the convergent series for e^{Ms} — there is n with $M^n s^n / n! < 1$, so T^n is a contraction on the complete metric space $C[0, s]$. By EXERCISE 6.2, if some iterate T^n is a contraction then T itself has a unique fixed point. Hence (6.7) has a unique continuous solution on $[0, s]$, for *every* s , not merely for $s < 1/M$.

Remark. This is the Picard–Lindelöf theorem, proved by contraction. The restriction $s < 1/M$ in EXAMPLE 6.3 is an artifact of using T rather than its iterates: the factorial in part (c) beats the geometric growth of $M^n s^n$, so the solution never breaks down at a finite horizon. The same device — show that some iterate is a contraction even when the map itself is not — is what makes the contraction method usable in dynamic programming problems where the one-step operator is not a contraction, for instance with $\beta = 1$ but a bounded number of periods of “effective” discounting.

Exercise 6.5

The hypotheses are that $\varphi : \mathbb{R} \times [a, b] \rightarrow \mathbb{R}$ is continuously differentiable with $0 < m \leq \partial\varphi(y, x)/\partial y \leq M < \infty$ for all (y, x) .

(a) *T is a contraction.* For $y, \tilde{y}$ and any $x \in [a, b]$, the Mean Value Theorem gives a ξ between $y(x)$ and $\tilde{y}(x)$ with $\varphi(y(x), x) - \varphi(\tilde{y}(x), x) = \varphi_y(\xi, x)[y(x) - \tilde{y}(x)]$. Therefore

$$Ty(x) - T\tilde{y}(x) = [y(x) - \tilde{y}(x)] \left[1 - \frac{\varphi_y(\xi, x)}{M} \right],$$

and since $m \leq \varphi_y \leq M$ we have $0 \leq 1 - \varphi_y/M \leq 1 - m/M$. Hence

$$\|Ty - T\tilde{y}\| \leq \left(1 - \frac{m}{M}\right) \|y - \tilde{y}\|, \quad \beta \equiv 1 - \frac{m}{M} \in [0, 1),$$

so T is a contraction with modulus β . That T maps continuous functions to continuous functions is immediate from continuity of φ .

(b) *The theorem.* $C([a, b])$ with the sup norm is a complete metric space, so by THEOREM 6.7 T has a unique fixed point y^* . By construction

$$Ty^* = y^* \iff \frac{\varphi(y^*(x), x)}{M} = 0 \iff \varphi(y^*(x), x) = 0 \quad \text{for all } x \in [a, b],$$

which is the existence and uniqueness assertion of the Implicit Function Theorem. Differentiability follows once existence is in hand: since φ is continuously differentiable and $\varphi_y \geq m > 0$, the identity

$\varphi(y^*(x), x) = 0$ can be differentiated to give

$$\frac{dy^*(x)}{dx} = -\frac{\varphi_x(y^*(x), x)}{\varphi_y(y^*(x), x)},$$

which is continuous, so $y^* \in C^1([a, b])$.

Remark. The exercise suggests working directly in $C^1([a, b])$. That is the natural guess but the wrong space: with the C^1 norm $\|y\| + \|\dot{y}\|$, T need not be a contraction, because differentiating Ty brings in φ_x and the derivative $\dot{y}$ with a coefficient that is not bounded by β . The clean route is the one above — contract on $C([a, b])$, where the sup-norm estimate is exactly the one-dimensional bound $|1 - \varphi_y/M| \leq 1 - m/M$, and recover continuous differentiability afterwards from the implicit relation. The uniform bound $\varphi_y \geq m > 0$ is what makes the modulus strictly less than one, and it is the same hypothesis that makes the implicit function well defined globally on $[a, b]$ rather than only locally.

Exercise 6.6

Recall

$$TV(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y)\}, \quad V \in C(X).$$

Under ASSUMPTION 6.2, X is compact, G is nonempty- and compact-valued and continuous, and U is continuous; so the maximum is attained (Weierstrass) and $TV \in C(X)$ by Berge's Maximum Theorem. Since X is compact, every $V \in C(X)$ is bounded, and $C(X)$ with the sup norm is a complete metric space. We verify Blackwell's two sufficient conditions (THEOREM 6.9).

Monotonicity. Let $V, W \in C(X)$ with $V(y) \leq W(y)$ for all y . For every x and every $y \in G(x)$,

$$U(x, y) + \beta V(y) \leq U(x, y) + \beta W(y) \leq TW(x),$$

the last step because $TW(x)$ is the maximum over $y \in G(x)$. Taking the maximum of the left-hand side over $y \in G(x)$ gives $TV(x) \leq TW(x)$ for all x , that is $TV \leq TW$.

Discounting. For a constant $c \geq 0$,

$$T(V + c)(x) = \max_{y \in G(x)} \{U(x, y) + \beta V(y) + \beta c\} = \left[\max_{y \in G(x)} \{U(x, y) + \beta V(y)\} \right] + \beta c = TV(x) + \beta c,$$

because the additive constant βc does not affect which y attains the maximum. So $T(V + c) \leq TV + \beta c$, with $\beta \in [0, 1)$ the discount factor.

Both of Blackwell's conditions hold, so by THEOREM 6.9 T is a contraction on $C(X)$ with modulus β , and THEOREM 6.7 then delivers the unique fixed point $V \in C(X)$ that solves PROBLEM 6.3.

Remark. The proof shows that discounting does all the work: the modulus of the contraction *is* the discount factor. This is why $\beta < 1$ is not a technical convenience but the substantive hypothesis of the whole theory — with $\beta = 1$ the operator satisfies monotonicity but not discounting, the fixed point need not be unique or even exist, and problems like EXERCISE 6.15 have no solution at all.

Exercise 6.10

It is enough to exhibit one. Take the log/Cobb–Douglas optimal growth model of EXAMPLE 6.4 with full depreciation:

$$\max \sum_{t=0}^{\infty} \beta^t \log c(t) \quad \text{s.t.} \quad k(t+1) = k(t)^\alpha - c(t), \quad k(0) > 0.$$

The Euler equation (6.50) here is

$$\frac{1}{c(t)} = \beta \alpha k(t+1)^{\alpha-1} \frac{1}{c(t+1)}. \quad (*)$$

The Euler equations alone leave one degree of freedom. Let $u(t) \equiv c(t)/k(t)^\alpha$ be the consumption share, so that $k(t+1) = (1 - u(t))k(t)^\alpha$. Substituting into (*) and simplifying, the Euler equation is equivalent to the first-order difference equation

$$u(t+1) = \frac{\alpha\beta u(t)}{1 - u(t)},$$

with fixed points $u^* = 1 - \alpha\beta$ — the familiar optimal policy, $c = (1 - \alpha\beta)k^\alpha$ — and $u = 0$. The derivative of the map at u^* is $\alpha\beta/(1 - u^*)^2 = 1/(\alpha\beta) > 1$, so u^* is *unstable* going forward, while $u = 0$ is stable (derivative $\alpha\beta < 1$). Consequently, for any $u(0) \neq u^*$ the resulting path satisfies *every* Euler equation, and:

- if $u(0) > u^*$ the consumption share rises until it exceeds 1 and capital would have to be negative: the plan is infeasible after finitely many periods;
- if $u(0) < u^*$ the path is feasible forever, but $u(t) \rightarrow 0$ geometrically — the economy saves an ever larger fraction of output and consumption vanishes relative to output.

The second family is the answer: these plans are feasible, satisfy all the Euler equations, and are strictly suboptimal.

They fail transversality. The transversality condition (6.51) here is

$$\lim_{t \rightarrow \infty} \beta^t f'(k(t)) u'(c(t)) k(t) = \lim_{t \rightarrow \infty} \beta^t \frac{\alpha k(t)^\alpha}{c(t)} = \lim_{t \rightarrow \infty} \frac{\alpha\beta^t}{u(t)} = 0.$$

On the optimal path $u(t) \equiv u^*$ and the expression is $\alpha\beta^t/u^* \rightarrow 0$. On an over-saving path $u(t+1) \approx \alpha\beta u(t)$ once u is small, so $u(t) \sim C(\alpha\beta)^t$ and

$$\frac{\alpha\beta^t}{u(t)} \sim \frac{\alpha}{C} \left(\frac{1}{\alpha}\right)^t \rightarrow \infty,$$

so transversality fails as badly as possible.

And they are strictly worse. With $\alpha = 1/3$, $\beta = 0.9$ and $k(0) = 1$, so that $u^* = 0.7$:

	optimal, $u(0) = 0.70$	$u(0) = 0.60$	$u(0) = 0.50$
value $\sum \beta^t \log c(t)$	-8.7266	-98.1473	-105.2498
loss relative to the optimum	—	89.42	96.52
$\beta^t \alpha / u(t)$ at $t = 100$	4.3×10^{-7}	4.1×10^{46}	9.8×10^{46}

Figure 4 plots the consumption share and the transversality expression along one such path.

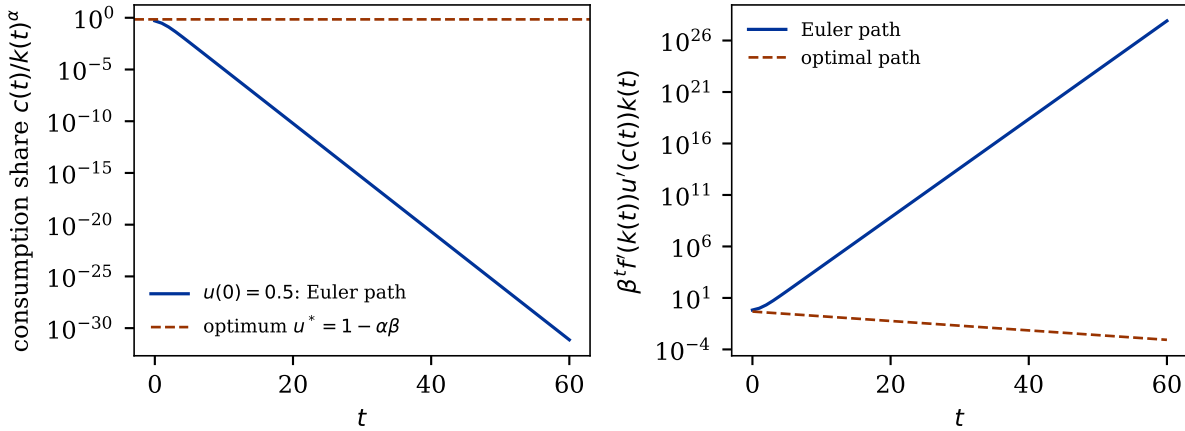


FIGURE 4. An Euler path that is not optimal. Left: the consumption share $u(t) = c(t)/k(t)^\alpha$ drifts away from the optimal $u^* = 1 - \alpha\beta$ and vanishes. Right: the transversality expression $\beta^t f'(k(t)) u'(c(t)) k(t)$, which tends to zero on the optimal path and diverges on the Euler path.

The economics is transparent: the household keeps accumulating capital it never consumes. Each Euler equation is a statement about a *local, one-period* deviation — shifting a little consumption between t and $t+1$ cannot improve matters — and every such deviation is indeed unprofitable along these paths. What they miss is the *global* deviation of consuming a fraction of the terminal capital stock, which is what the transversality condition prices: when $\lim \beta^t D_x U(x^*(t), x^*(t+1)) \cdot x^*(t+1) > 0$, the plan leaves asymptotically valuable capital unconsumed, and shrinking the whole capital path by a factor $1 - \varepsilon$ raises utility. That is exactly the perturbation used in the proof of THEOREM 6.10.

Remark. The instability of the Euler map is not a defect of the example but the general situation: the Euler equations form a second-order difference equation in the state, whose steady state is a saddle, so the initial condition $k(0)$ pins down only one of the two degrees of freedom and transversality selects the saddle path. It also means that one cannot compute the optimum by iterating the Euler equation forward — even starting exactly at u^* , rounding error grows by a factor $1/(\alpha\beta)$ per period. The accompanying program therefore imposes the optimal path exactly rather than iterating it.

Exercise 6.11

Shifting consumption up is always affordable. The only constraint is the flow budget constraint

$$a(t+1) = (1 + r(t))a(t) + w(t) - c(t), \quad a(0) \text{ given.}$$

Given a path $\{c(t)\}$ with assets $\{a(t)\}$, define $c'(t) \equiv c(t) + \gamma$ and let $\{a'(t)\}$ be generated by the same flow constraint from the same $a(0)$. The difference $d(t) \equiv a'(t) - a(t)$ satisfies $d(0) = 0$ and $d(t+1) = (1 + r(t))d(t) - \gamma$, so

$$d(t) = -\gamma \sum_{s=0}^{t-1} \prod_{j=s+1}^{t-1} (1 + r(j)) < 0.$$

Every equation of the flow constraint holds by construction, so $\{c'(t)\}$ is feasible: the extra consumption is simply borrowed, and the debt is rolled over forever. With $r(t) \geq 0$ the debt grows at least linearly, and $a'(t) \rightarrow -\infty$.

Hence no optimum exists. Since u is strictly increasing,

$$\sum_{t=0}^{\infty} \beta^t u(c(t) + \gamma) > \sum_{t=0}^{\infty} \beta^t u(c(t)) \quad \text{for every } \gamma > 0,$$

so no feasible path can be optimal: whatever is proposed, adding γ to consumption at every date is also feasible and strictly better. If in addition u is unbounded above, letting $\gamma \rightarrow \infty$ makes the supremum $+\infty$; if u is bounded above by $\bar{u}$, the supremum is $\bar{u}/(1 - \beta)$ and is still not attained. Either way the maximization problem is not well posed — ASSUMPTION 6.1 fails, and with it the entire dynamic programming apparatus.

Why market economies exclude these paths. The path finances consumption by borrowing and never repaying: the present value of the consumer's debt does not go to zero, so in present-value terms he is consuming resources he never earns. Nobody would take the other side of that trade. Formally, the lender's own optimization imposes that the value of the loan be recovered, so equilibrium requires a *no-Ponzi-game* condition

$$\lim_{t \rightarrow \infty} \frac{a(t)}{\prod_{j=0}^{t-1} (1 + r(j))} \geq 0,$$

or, equivalently and more primitively, the *natural debt limit* (6.44): the consumer can borrow at most the present value of his future labor income, since that is all he can ever repay with nonnegative consumption. Either restriction restores a compact feasible set and a well-posed problem, and in general equilibrium the restriction is not an extra assumption at all — aggregate assets must equal the aggregate capital stock, which is finite, so debts must be backed by someone's assets.

Remark. This is the discrete-time counterpart of the transversality condition of EXERCISE 6.10, seen from the household's side. There the danger was accumulating capital forever and never consuming it; here it is borrowing forever and never repaying. Both are ruled out by the same kind of terminal condition, and both show that the Euler equations alone do not characterize optimal behavior in infinite horizon.

Exercise 6.13

Write $q(t) \equiv \prod_{j=0}^{t-1} (1 + r(j))^{-1}$ for the price at date 0 of a unit of the good at date t (with $q(0) = 1$), and let

$$W(t) \equiv \sum_{s=0}^{\infty} \left(\prod_{j=0}^s \frac{1}{1 + r(t+j)} \right) w(t+s)$$

be the present value at t of current and future labor income, so that $\bar{W} = \max_t W(t)$.

(a) *The natural debt limit.* Iterating the flow budget constraint forward from t and dividing by the accumulated interest factors,

$$\sum_{s=0}^T \left(\prod_{j=0}^s \frac{1}{1 + r(t+j)} \right) c(t+s) = a(t) + \sum_{s=0}^T \left(\prod_{j=0}^s \frac{1}{1 + r(t+j)} \right) w(t+s) - \left(\prod_{j=0}^T \frac{1}{1 + r(t+j)} \right) a(t+T+1).$$

The debt is repayable only if the household can make the left-hand side nonnegative for every T while letting the last term vanish. Since consumption cannot be negative, the most the household can ever dedicate to repayment is its entire future labor income; taking $T \rightarrow \infty$ and imposing $c \geq 0$ gives

$$a(t) \geq -\sum_{s=0}^{\infty} \left(\prod_{j=0}^s \frac{1}{1+r(t+j)} \right) w(t+s) = -W(t),$$

which is (6.44). It binds exactly when the household plans zero consumption forever, which is why it is called *natural*: it is implied by feasibility alone, not by any extra market restriction.

(b) *Bounds on assets.* The lower bound is immediate: $a(t) \geq -W(t) \geq -\bar{W}$. For the upper bound, work with total wealth $\Omega(t) \equiv a(t) + W(t) \geq 0$, which evolves as $\Omega(t+1) = (1+r(t))(\Omega(t) - c(t))$. So Ω is nonincreasing as soon as the household consumes at least the interest on its wealth,

$$c(t) \geq \frac{r(t)}{1+r(t)} \Omega(t),$$

and a sufficient condition for that at the optimum is $\beta(1+r(t)) \leq 1$ for all t : with the Euler equation (6.45), $\beta(1+r(t+1)) \leq 1$ makes consumption nonincreasing, which (given the intertemporal budget constraint) requires consuming at least the annuity value of wealth. Under that condition $\Omega(t) \leq \Omega(0) = a(0) + W(0) \leq a(0) + \bar{W}$, and since $W(t) \geq 0$,

$$a(t) = \Omega(t) - W(t) \leq a(0) + \bar{W}.$$

So the conditions are: $\bar{W} < \infty$ (finite lifetime resources) and $\beta(1+r(t)) \leq 1$ for all t (the household is at least as impatient as the market). Both are needed — EXERCISE 6.12(b) shows that with $r > \beta^{-1} - 1$ assets grow without bound and no such $\bar{a}$ exists.

(c) *The level of initial consumption.* The Euler equation (6.45) determines the whole path from its first element: define recursively $c(t) = h_t(c(0))$ by $u'(c(t)) = \beta(1+r(t+1))u'(c(t+1))$, so each h_t is strictly increasing in $c(0)$ because u' is strictly decreasing. The transversality condition — equivalently the no-Ponzi condition holding with equality, $\lim_{t \rightarrow \infty} q(t)a(t) = 0$ — turns the flow constraints into the single intertemporal budget constraint

$$\sum_{t=0}^{\infty} q(t) h_t(c(0)) = a(0) + \sum_{t=0}^{\infty} q(t) w(t) \equiv \Omega(0), \quad (*)$$

which determines $c(0)$ implicitly. The left-hand side is continuous and strictly increasing in $c(0)$ (from 0 to ∞), so the solution exists and is unique. Differentiating (*),

$$\frac{\partial c(0)}{\partial a(0)} = \left[\sum_{t=0}^{\infty} q(t) h'_t(c(0)) \right]^{-1} > 0 :$$

an increase in initial assets raises initial consumption. With CRRA utility $h_t(c(0)) = c(0) \prod_{j=1}^t [\beta(1+r(j))]^{1/\theta}$, so $c(0) = \Omega(0) / \sum_t q(t) \prod_{j \leq t} [\beta(1+r(j))]^{1/\theta}$ and the derivative is the marginal propensity to consume out of wealth, a number in $(0, 1]$.

(d) *Retiming earnings.* Let $\{\tilde{w}(t)\}$ satisfy $\tilde{w}(t) > w(t)$ for $t < T$, $\tilde{w}(t) \leq w(t)$ for $t \geq T$, and $\sum_t (1+r)^{-t} w(t) = \sum_t (1+r)^{-t} \tilde{w}(t)$. Then $\Omega(0)$ is unchanged, so by (*) the whole consumption path $\{c(t)\}$ — not merely $c(0)$ — is unchanged. What changes is the asset path: with income

arriving earlier, the household saves the difference, so

$$\tilde{a}(t) - a(t) = \sum_{s < t} (1+r)^{t-s} [\tilde{w}(s) - w(s)] > 0 \quad \text{for } 0 < t \leq T,$$

and the extra assets are run down thereafter, reaching zero in present value again.

The intuition is the permanent income hypothesis in its purest form. With perfect capital markets, a complete set of borrowing and lending opportunities at the market interest rate, and no binding constraint, the household cares only about the *present value* of its earnings; the timing affects only the portfolio decision, not consumption. Two qualifications are worth noting, because they are where the result breaks in applications: if the natural debt limit (6.44) or a tighter borrowing constraint binds, a shift of income toward the present relaxes it and does raise consumption; and if the interest rate at which the household can borrow exceeds that at which it can lend, the equivalence fails as well. Here neither applies — the change makes the household unambiguously less constrained early on.

Exercise 6.14

Conditions. The natural hypotheses are the stationary ones imposed *uniformly in t*:

- (1) $X \subset \mathbb{R}^K$ is compact, and for every t , $G(t, \cdot)$ is nonempty-valued, compact-valued and continuous, with the graph $X_G(t) = \{(x, y) : y \in G(t, x)\}$ closed;
- (2) $U(t, \cdot, \cdot)$ is continuous on $X_G(t)$ for each t , and *uniformly bounded*, in the sense that $\bar{U} \equiv \sup_t \sup_{(x,y) \in X_G(t)} |U(t, x, y)|$ is finite;
- (3) $\beta \in [0, 1)$.

Let $C(\mathbb{Z}_+ \times X)$ be the space of functions $V(t, x)$, continuous in x for each t and bounded, with the sup norm $\|V\| = \sup_{t,x} |V(t, x)|$. This is a complete metric space.

T is a well-defined self-map. For each (t, x) the maximand is continuous in y on the compact set $G(t, x)$, so the maximum is attained (Weierstrass); by Berge's Maximum Theorem $TV(t, \cdot)$ is continuous; and $\|TV\| \leq \bar{U} + \beta\|V\| < \infty$, so $TV \in C(\mathbb{Z}_+ \times X)$. Uniform boundedness of U is exactly what is needed here — without it the sup norm over all t is infinite and the space is not the right one.

T is a contraction. Blackwell's conditions hold verbatim as in Exercise 6.6, because t is carried along as a parameter: if $V \leq W$ pointwise then $TV \leq TW$; and $T(V + c)(t, x) = TV(t, x) + \beta c$ for any constant c . By THEOREM 6.9, T is a contraction of modulus β , so by THEOREM 6.7 it has a unique fixed point $V \in C(\mathbb{Z}_+ \times X)$, the value function of the nonstationary problem.

The equivalents of THEOREMS 6.2–6.6. Write $\Phi(t, x)$ for the set of feasible plans from state x at date t , and $V^*(t, x)$ for the supremum of $\sum_{s \geq t} \beta^{s-t} U(s, x(s), x(s+1))$.

- *Principle of optimality (6.2).* $V^*(t, x^*(t)) = U(t, x^*(t), x^*(t+1)) + \beta V^*(t+1, x^*(t+1))$ for an optimal plan, and any feasible plan satisfying this relation at every t attains $V^*(t, x)$. The proof is the stationary one with the date carried along: truncate the plan at n , use uniform boundedness to control the tail by $\bar{U}\beta^{n+1}/(1-\beta)$, and pass to the limit.

- *Existence (6.3)*. The fixed point V exists and is unique by the contraction argument above, and $V = V^*$; the maximum in the Bellman equation is attained, which defines a (possibly set-valued) optimal policy $\pi(t, x)$. Equivalently one can argue directly on the sequence problem: $\Phi(t, x)$ is compact in the product topology (Tychonoff), the objective is continuous, and Weierstrass applies.
- *Concavity (6.4)*. If in addition $U(t, \cdot, \cdot)$ is concave and $G(t, \cdot)$ has convex graph for each t , then $V(t, \cdot)$ is concave for each t . Proof: the set of V that are concave in x for every t is a closed subset of $C(\mathbb{Z}_+ \times X)$, and T maps it into the (smaller) set of functions strictly concave in x when U is strictly concave; apply THEOREM 6.8(2).
- *Monotonicity (6.5)*. If $U(t, \cdot, y)$ is strictly increasing and $G(t, \cdot)$ is monotone for each t , the same closed-subset argument gives $V(t, \cdot)$ strictly increasing.
- *Differentiability (6.6)*. If $U(t, \cdot, \cdot)$ is continuously differentiable and the optimal policy is interior, $x \in \text{Int } X$ and $\pi(t, x) \in \text{Int } G(t, x)$, then $V(t, \cdot)$ is differentiable at x with $D_x V(t, x) = D_x U(t, x, \pi(t, x))$. The Benveniste–Scheinkman argument is unchanged: $W(z) \equiv U(t, z, \pi(t, x)) + \beta V(t+1, \pi(t, x))$ is differentiable, lies below $V(t, \cdot)$, and touches it at x .

Remark. The one substantive difference from the stationary theory is uniformity. Continuity and boundedness must hold uniformly in t , since otherwise $C(\mathbb{Z}_+ \times X)$ with the sup norm is not the relevant space and the contraction modulus is not β . Economically this rules out problems in which the environment deteriorates or improves without bound — for instance a wage sequence that diverges, which is precisely why EXAMPLE 6.5 needs $\bar{W} < \infty$. The book’s own treatment in Section 6.7 proves the analogous results by working directly with the sequence problem; the contraction route above is shorter but buys that brevity with the uniform bounds.

Exercise 6.15

Note two features: there is no discounting ($\beta = 1$) and no production, so the problem is to divide a fixed cake $k(0)$ among infinitely many dates. Feasibility requires $c(t) \geq 0$ and $\sum_{t=0}^{\infty} c(t) \leq k(0)$. Normalize $u(0) = 0$; this is without loss, since if $u(0) > 0$ then every plan that consumes nothing in infinitely many periods already has value $+\infty$ and the problem is ill posed for a still cruder reason, while if $u(0) < 0$ the same is true with $-\infty$.

The supremum is not attained. Consider the plans

$$c_n(t) = \begin{cases} k(0)/n, & t = 0, \dots, n-1, \\ 0, & t \geq n, \end{cases}$$

which are feasible and give value $V_n = n u(k(0)/n)$. Because u is strictly concave with $u(0) = 0$, the ratio $u(c)/c$ is strictly decreasing in c , so

$$V_n = k(0) \cdot \frac{u(k(0)/n)}{k(0)/n}$$

is strictly increasing in n , with

$$\lim_{n \rightarrow \infty} V_n = k(0) \cdot \lim_{c \downarrow 0} \frac{u(c)}{c} = k(0) u'(0^+) \in (0, \infty].$$

On the other hand every feasible plan satisfies, again by strict concavity and $u(0) = 0$,

$$\sum_t u(c(t)) < u'(0^+) \sum_t c(t) \leq u'(0^+)k(0)$$

whenever $u'(0^+) < \infty$ and some $c(t) > 0$. So the supremum equals $u'(0^+)k(0)$ and is approached by the sequence $\{c_n\}$ but attained by no feasible plan: any candidate can be improved by splitting its consumption into more, smaller pieces. If $u'(0^+) = \infty$ the same computation shows the supremum is $+\infty$, and again no plan attains it. In both cases there is no solution.

Why. Without discounting there is no cost to postponing consumption, while strict concavity means that the *average* utility per unit consumed, $u(c)/c$, rises as the portions shrink. Spreading the cake ever more thinly over ever more dates therefore always raises total utility, and the “optimum” would be to consume infinitesimal amounts forever — which is not a feasible plan. In the language of the chapter, ASSUMPTION 6.1 fails: with $\beta = 1$ the limit $\lim_n \sum_{t=0}^n U(x(t), x(t+1))$ need not behave well, the objective is not continuous in the product topology (the tail of a plan matters in the limit), and the Bellman operator satisfies Blackwell’s monotonicity but not the discounting condition of Exercise 6.6, so it is not a contraction. Boundedness of u does not rescue matters — it makes the supremum finite when $u'(0^+) < \infty$, but attainment is what fails.

Remark. The example isolates discounting as the substantive assumption. Economists sometimes treat $\beta < 1$ as a mere technical device; here it is what makes the problem have a solution at all. When one does want to study undiscounted problems — the golden rule, or the Ramsey problem with $\beta = 1$ — the optimality criterion itself has to be weakened, to overtaking or catching-up criteria, as the footnote to ASSUMPTION 6.1 notes.

Exercise 6.16

(a) *Recursive formulation.* With k the state and next period’s capital k' the control,

$$V(k) = \max_{k' \in G(k)} \{u(f(k) - k') + \beta V(k')\}, \quad G(k) = [0, f(k)].$$

As in Exercise 6.1, the state can be confined to the compact interval $[0, \bar{k}]$, where $\bar{k}$ solves $f(\bar{k}) = \bar{k}$ (full depreciation makes the maximum sustainable stock the fixed point of f) and $\bar{k} = f(\max\{k(0), \bar{k}\})$.

(b) *Existence, uniqueness and shape.* The problem satisfies ASSUMPTIONS 6.1–6.4 by the argument of Exercise 6.1, with $g(k) = f(k)$. Hence:

- THEOREM 6.3 gives a unique continuous bounded V solving the Bellman equation, and an optimal plan exists;
- THEOREM 6.4 gives strict concavity of V , since $U(k, k') = u(f(k) - k')$ satisfies ASSUMPTION 6.3;
- COROLLARY 6.1 then gives a *unique* optimal plan, generated by a continuous policy function $k' = \pi(k)$, equivalently $c = \pi_c(k) = f(k) - \pi(k)$;
- THEOREM 6.5 gives that V is strictly increasing.

That π is nondecreasing follows from the revealed-preference argument used for PROPOSITION 6.1: if $k' > k$ but $\pi(k) > \pi(k')$, adding the two optimality inequalities gives

$$u(z - \pi(k')) - u(z - \pi(k)) \leq u(z' - \pi(k')) - u(z' - \pi(k)), \quad z \equiv f(k) < z' \equiv f(k'),$$

which contradicts strict concavity of u , since the two consumption differences are equal in size but taken at strictly higher levels on the right. Monotonicity of the consumption function follows from the Euler equation as in Exercise 6.17.

(c) *Differentiability.* THEOREM 6.6 gives differentiability of V at any k such that u and f are continuously differentiable near the relevant point (ASSUMPTION 6.5) and the optimal policy is interior, $\pi(k) \in \text{Int } G(k)$, that is $0 < \pi(k) < f(k)$ — equivalently both consumption and investment are strictly positive. Interiority is guaranteed if u and f satisfy Inada conditions ($u'(0) = \infty$ rules out $c = 0$; $f'(0) = \infty$ rules out $k' = 0$). Then

$$V'(k) = \frac{\partial U(k, \pi(k))}{\partial k} = f'(k) u'(f(k) - \pi(k)).$$

Without differentiability of u (or at corner solutions) V can have kinks, though it always has left and right derivatives by concavity.

(d) *The Euler equation.* The first-order condition of the Bellman problem is $u'(c) = \beta V'(k')$, and the envelope condition just derived is $V'(k) = f'(k) u'(c)$. Combining, with $c(t) = f(k(t)) - k(t+1)$,

$$\boxed{u'(c(t)) = \beta f'(k(t+1)) u'(c(t+1))}$$

the familiar condition that the marginal utility given up today equals the discounted marginal utility gained tomorrow, multiplied by the marginal product of the capital carried forward.

(e) *Is it enough?* No. Substituting $c(t) = f(k(t)) - k(t+1)$, the Euler equation is a *second order* difference equation in $\{k(t)\}$,

$$u'(f(k(t)) - k(t+1)) = \beta f'(k(t+1)) u'(f(k(t+1)) - k(t+2)),$$

so the initial condition $k(0)$ leaves a one-parameter family of solutions, as EXERCISE 6.10 shows explicitly. The missing condition is the transversality condition

$$\lim_{t \rightarrow \infty} \beta^t f'(k(t)) u'(c(t)) k(t) = 0,$$

which by THEOREM 6.10 is necessary and sufficient, together with the Euler equations, for optimality of an interior plan. Intuitively it says that the present value of the capital stock left over at the end of time must be zero: capital is worth holding only for the consumption it eventually delivers, so a plan that keeps accumulating capital of positive present value forever is wasting resources. It is the infinite-horizon analogue of the terminal condition $k(T+1) = 0$ in a finite-horizon problem — one would never end a finite life with capital left in the ground.

Exercise 6.17

The consumption function is nondecreasing. By PROPOSITION 6.1 the policy function $s(k)$ — the capital carried into the next period — is nondecreasing, and by PROPOSITION 6.2 the value function V is differentiable; it is strictly concave by THEOREM 6.4, so V' is strictly decreasing. The

first-order condition of the recursive problem (6.48) is

$$u'(c(k)) = \beta V'(s(k)).$$

Take $k' > k$. Then $s(k') \geq s(k)$, so $V'(s(k')) \leq V'(s(k))$, and therefore

$$u'(c(k')) = \beta V'(s(k')) \leq \beta V'(s(k)) = u'(c(k)).$$

Since u is strictly concave, u' is strictly decreasing, and the inequality inverts: $c(k') \geq c(k)$. Hence $c(\cdot)$ is nondecreasing, as claimed.

Monotone convergence of consumption. Suppose $k(0) < k^*$. By PROPOSITION 6.3 the capital stock increases monotonically to the steady state, $k(t) \uparrow k^*$. Composing with the nondecreasing and continuous function $c(\cdot)$,

$$c(t) = c(k(t)) \text{ is nondecreasing, and } c(t) \rightarrow c(k^*)$$

by continuity of $c(\cdot)$ (COROLLARY 6.1). The limit is the steady-state consumption level: at k^* the resource constraint with $k(t+1) = k(t) = k^*$ gives

$$c^* = f(k^*) + (1 - \delta)k^* - k^* = f(k^*) - \delta k^*.$$

Thus $c(t) \uparrow c^*$. If instead $k(0) > k^*$, PROPOSITION 6.3 gives $k(t) \downarrow k^*$ and the same argument gives $c(t) \downarrow c^*$.

Remark. The result is worth contrasting with the Solow model. There the saving rate is constant, so consumption is a fixed fraction of output and inherits its monotonicity mechanically. Here the saving rate is chosen, and could in principle move in either direction; what pins the sign down is the combination of two facts — the economy accumulates capital monotonically, and the marginal value of capital falls as capital rises. Note also what is *not* claimed: the saving *rate* need not be monotone along the transition, and generally is not.

Exercise 6.19

Suppose, to get a contradiction, that there are arbitrarily small $\varepsilon > 0$ with $s(\varepsilon) \leq \varepsilon$. Along the optimal path starting from $k = \varepsilon$, the Euler equation (6.50) holds:

$$u'(c(\varepsilon)) = \beta \left[f'(s(\varepsilon)) + 1 - \delta \right] u'(c(s(\varepsilon))).$$

By PROPOSITION 6.4 (Exercise 6.17) the consumption function is nondecreasing, so $s(\varepsilon) \leq \varepsilon$ implies $c(s(\varepsilon)) \leq c(\varepsilon)$ and hence $u'(c(s(\varepsilon))) \geq u'(c(\varepsilon)) > 0$, because u' is decreasing. Dividing the Euler equation by $u'(c(\varepsilon))$,

$$1 = \beta \left[f'(s(\varepsilon)) + 1 - \delta \right] \frac{u'(c(s(\varepsilon)))}{u'(c(\varepsilon))} \geq \beta \left[f'(s(\varepsilon)) + 1 - \delta \right],$$

that is

$$f'(s(\varepsilon)) \leq \frac{1}{\beta} - 1 + \delta < \infty.$$

But $0 < s(\varepsilon) \leq \varepsilon$, so letting $\varepsilon \rightarrow 0$ forces $s(\varepsilon) \rightarrow 0$, and Assumption 2 (the Inada condition $\lim_{k \rightarrow 0} f'(k) = \infty$) gives $f'(s(\varepsilon)) \rightarrow \infty$ — contradicting the bound just derived. Hence for all sufficiently small ε we must have $s(\varepsilon) > \varepsilon$.

Intuition. Near zero the marginal product of capital is arbitrarily large, so an extra unit of saving returns arbitrarily much tomorrow. A household with a finite rate of time preference — the right-hand side $\beta^{-1} - 1 + \delta$ is a fixed, finite number, the effective required return — will always take that trade: at very low capital the return to accumulation exceeds the required return by an unbounded margin. Thus the economy cannot optimally run its capital stock down toward zero, and $k = 0$, though a rest point of the resource constraint, is never approached from a positive initial stock. This is precisely what the proof of PROPOSITION 6.3 needs to conclude that the monotone sequence $\{k(t)\}$ converges to the *positive* steady state k^* rather than to zero.

Exercise 6.20

The claims of PROPOSITION 6.1 are: there exist a unique value function V and consumption function $c(k)$, with $s(k) = f(k) + (1 - \delta)k - c(k)$; V is strictly increasing and strictly concave; and $s(k)$ is nondecreasing. Assume u is continuous, strictly increasing and strictly concave, but *not* necessarily differentiable, and let f satisfy Assumptions 1 and 2.

Step 1: the dynamic programming assumptions still hold. Exercise 6.1 verified ASSUMPTIONS 6.1–6.4 using only continuity, monotonicity and strict concavity of u together with concavity and monotonicity of f ; differentiability entered only in ASSUMPTION 6.5. So ASSUMPTIONS 6.1–6.4 hold as before, while ASSUMPTION 6.5 is dropped.

Step 2: existence, uniqueness, concavity, monotonicity. THEOREMS 6.3, 6.4 and 6.5 and COROLLARY 6.1 require only ASSUMPTIONS 6.1–6.4. They give, in order: a unique continuous bounded V solving the Bellman equation (6.48), with an optimal plan existing; V strictly concave; V strictly increasing; and a unique optimal plan generated by a continuous policy function. None of these arguments differentiates anything: the contraction argument uses Blackwell's conditions (Exercise 6.6), and strict concavity is obtained from THEOREM 6.8(2), by showing that T maps the closed set of weakly concave functions into the set of strictly concave ones.

Step 3: $s(k)$ is nondecreasing. This is the only part of PROPOSITION 6.1 whose proof might seem to need calculus, and it does not: the book's argument is a revealed-preference argument. Suppose $k' > k$ but $s(k) > s(k')$. Both $s(k)$ and $s(k')$ are feasible at both states, since $f(k) + (1 - \delta)k < f(k') + (1 - \delta)k'$ and $s(k') < s(k)$. Optimality at k and at k' gives

$$\begin{aligned} u(z - s(k)) + \beta V(s(k)) &\geq u(z - s(k')) + \beta V(s(k')), \\ u(z' - s(k')) + \beta V(s(k')) &\geq u(z' - s(k)) + \beta V(s(k)), \end{aligned}$$

with $z \equiv f(k) + (1 - \delta)k < z' \equiv f(k') + (1 - \delta)k'$. Adding and rearranging eliminates V :

$$u(z - s(k')) - u(z - s(k)) \leq u(z' - s(k')) - u(z' - s(k)).$$

But the two consumption increments are equal, $(z - s(k')) - (z - s(k)) = (z' - s(k')) - (z' - s(k)) = s(k) - s(k') > 0$, while the right-hand pair is evaluated at strictly higher consumption levels ($z' > z$). Strict concavity of u makes equal increments worth strictly less at higher levels, so the right-hand side is *strictly smaller* than the left — a contradiction. Hence $s(k)$ is nondecreasing.

This completes the proof of PROPOSITION 6.1 using only continuity, monotonicity and strict concavity.

Remark. What is genuinely lost without differentiability is PROPOSITION 6.2 — that V is differentiable — and with it the Euler equation (6.50) in its classical form. Concavity still gives V left and right derivatives everywhere, and the first-order condition becomes the inclusion $u'_-(c) \geq \beta V'_+(s)$ and $u'_+(c) \leq \beta V'_-(s)$, or in the language of convex analysis, $0 \in \partial[u(z - s) + \beta V(s)]$ with ∂ the superdifferential. All the qualitative conclusions of Sections 6.8 — uniqueness of the steady state, monotone convergence — survive, because they rest on the monotonicity of $s(\cdot)$ established above rather than on calculus.

Exercise 6.21

The two problems generate the same difference equations. In the competitive equilibrium, firms hire capital and labor competitively, so

$$R(t) = f'(k(t)), \quad 1 + r(t+1) = f'(k(t)) + 1 - \delta, \quad w(t) = f(k(t)) - k(t)f'(k(t)),$$

and asset market clearing requires $a(t) = k(t)$. Substituting the factor prices into the household's flow budget constraint (6.54),

$$a(t+1) = (1 + r(t))a(t) + w(t) - c(t) = k(t) + [f'(k(t)) - \delta]k(t) + f(k(t)) - k(t)f'(k(t)) - c(t),$$

that is

$$k(t+1) = f(k(t)) + (1 - \delta)k(t) - c(t),$$

which is exactly the resource constraint (6.47) of the optimal growth problem. Substituting the same factor prices into the household's Euler equation (6.55) gives (6.56),

$$u'(c(t)) = \beta [f'(k(t+1)) + 1 - \delta] u'(c(t+1)),$$

which is identical to the planner's Euler equation (6.50). The two problems also share the initial condition $k(0)$, and the household's no-Ponzi condition with $a(t) = k(t)$, $\lim_{t \rightarrow \infty} \beta^t u'(c(t))k(t+1) = 0$, is the planner's transversality condition (6.51).

Uniqueness closes the argument. By THEOREM 6.10, for this problem the Euler equations together with the transversality condition are necessary *and* sufficient for optimality, and by COROLLARY 6.1 — the objective being strictly concave and the constraint set convex — the optimal plan is unique. So there is exactly one sequence $\{k(t), c(t)\}$ satisfying (i) the resource constraint, (ii) the Euler equation, (iii) $k(0)$ given, and (iv) transversality. The competitive equilibrium path satisfies all four, and so does the optimal growth path; therefore they coincide.

Both directions. The argument above takes an equilibrium and shows it solves the planner's problem. The converse is equally immediate: given the optimal path $\{k(t), c(t)\}$, define prices by the marginal product conditions above and set $a(t) = k(t)$. Then firms maximize profits by construction (they take prices as given and earn zero profits, by Euler's Theorem and constant returns), the household's budget constraint holds at every date by the computation above, its Euler equation is the planner's, and its transversality condition holds. Hence the optimal allocation, supported by these prices, is a competitive equilibrium.

Remark. This is the First and Second Welfare Theorems of Chapter 5 in action, made concrete: the equivalence holds because there are no externalities, no distortions and no missing markets, and because the natural debt limit rules out the Ponzi schemes of EXERCISE 6.11. The practical payoff is large — it is why one

may compute equilibria of the neoclassical growth model by solving a single planner's dynamic programming problem, a device used throughout the rest of the book. It is also worth noting where it stops: as soon as Chapter 11 onward introduces externalities or monopolistic competition, the planner's and the equilibrium paths separate, and this shortcut is no longer available.

Chapter 7

An Introduction to the Theory of Optimal Control

The sixteen exercises of Chapter 7 left unsolved in the instructor's manual: Exercises 7.3, 7.4, 7.6–7.9, 7.11–7.16, 7.20, 7.22, 7.27 and 7.29. The manual solves 7.1, 7.2, 7.5, 7.10, 7.17–7.19, 7.21, 7.23–7.26 and 7.28. Notation follows the book: x is the state, y the control, λ the costate, $H(t, x, y, \lambda) = f + \lambda g$ the Hamiltonian (7.16), $M(t, x, \lambda) = \max_y H$ the maximized Hamiltonian (7.20), and $\hat{H} = e^{\rho t} H$ the current-value Hamiltonian with costate $\mu = e^{\rho t} \lambda$. The numerical part is computed by `code/ch7.py`.

Exercise 7.3

Put the problem in control form by treating the derivative as the control: set

$$y(t) \equiv \dot{x}(t), \quad f(t, x, y) \equiv F(t, x, y), \quad g(t, x, y) \equiv y,$$

so the state equation $\dot{x} = g(t, x, y) = y$ holds by definition and the problem is exactly the one covered by THEOREM 7.2. The Hamiltonian (7.16) is

$$H(t, x, y, \lambda) = F(t, x, y) + \lambda(t)y.$$

THEOREM 7.2 gives a continuously differentiable costate λ with, for all $t \in [0, t_1]$,

$$H_y = 0 : \quad \frac{\partial F(t, x, \dot{x})}{\partial \dot{x}} + \lambda(t) = 0, \tag{*}$$

$$\dot{\lambda} = -H_x : \quad \dot{\lambda}(t) = -\frac{\partial F(t, x, \dot{x})}{\partial x}. \tag{**}$$

From (*), $\lambda(t) = -F_{\dot{x}}(t, x(t), \dot{x}(t))$, which is differentiable in t along the (continuously differentiable) solution. Differentiating and substituting into (**),

$$-\frac{d}{dt} \left[\frac{\partial F(t, x(t), \dot{x}(t))}{\partial \dot{x}} \right] = -\frac{\partial F(t, x(t), \dot{x}(t))}{\partial x},$$

that is,

$$\boxed{\frac{\partial F(t, x(t), \dot{x}(t))}{\partial x} - \frac{d}{dt} \left[\frac{\partial F(t, x(t), \dot{x}(t))}{\partial \dot{x}} \right] = 0}$$

which is the Euler–Lagrange equation. The costate has a clean reading here: $\lambda(t) = -F_{\dot{x}}$ is (minus) the marginal value of the rate of change of the state, and the Euler–Lagrange equation is the statement that its rate of change must equal the marginal value of the state itself.

Remark. The book writes the last term as $\partial^2 F / \partial \dot{x} \partial t$. That notation is correct only if the symbol is read as the *total* derivative along the solution path,

$$\frac{d}{dt} F_{\dot{x}} = F_{\dot{x}t} + F_{\dot{x}x} \dot{x}(t) + F_{\dot{x}\dot{x}} \ddot{x}(t),$$

and not as the partial derivative of $F_{\dot{x}}$ with respect to its explicit time argument, which would drop the last two terms. It is the total derivative that the variational argument produces, as (*) and (**) show.

Exercise 7.4

Here $F(t, x, \dot{x}) = -\sqrt{1 + \dot{x}^2}$, which does not depend on x or on t explicitly, so

$$F_x = 0, \quad F_{\dot{x}} = -\frac{\dot{x}}{\sqrt{1 + \dot{x}^2}}.$$

The Euler–Lagrange equation of Exercise 7.3, $F_x - \frac{d}{dt}F_{\dot{x}} = 0$, therefore reduces to

$$\frac{d}{dt} \left[\frac{x'(t)}{\sqrt{1 + (x'(t))^2}} \right] = 0.$$

(The same conclusion follows from the general fact that when F does not depend on x , the costate is constant: $\dot{\lambda} = -F_x = 0$ and $\lambda = -F_{\dot{x}}.$)

Carrying out the differentiation,

$$\frac{d}{dt} \left[x'(1 + (x')^2)^{-1/2} \right] = x''(1 + (x')^2)^{-1/2} - x' \cdot x'x''(1 + (x')^2)^{-3/2} = \frac{x''(t)}{(1 + (x'(t))^2)^{3/2}}.$$

Since $(1 + (x')^2)^{3/2} > 0$ always, the Euler–Lagrange equation holds if and only if

$$x''(t) = 0 \quad \text{for all } t,$$

so $x(t) = a + bt$: the extremal is a straight line, and the constants are pinned down by the endpoints, $b = (u_1 - u_0)/(z_1 - z_0)$ and $a = u_0$ after the normalization $(z_0, u_0) = (0, 0)$.

Finally, the extremal really is the minimizer of length rather than merely a stationary point: F does not depend on x at all, so the maximized Hamiltonian $M(t, x, \lambda)$ is constant — in particular concave — in x , and THEOREM 7.6 applies, giving a global maximum of $-\int \sqrt{1 + (x')^2}$, that is a global minimum of arc length. Equivalently, $p \mapsto -\sqrt{1 + p^2}$ is strictly concave, so the stationary path is the unique minimizer.

Remark. The expression printed in the book, $d[x'(t)(1 + (x'(t))^2)]/dt = 0$, has the bracket raised to the power +1; the derivative of $-\sqrt{1 + \dot{x}^2}$ gives the power $-1/2$, as above. The conclusion $x'' = 0$ is unaffected — indeed the misprinted expression also implies $x'' = 0$, since $\frac{d}{dt}[x'(1 + (x')^2)] = x''(1 + 3(x')^2)$ and $1 + 3(x')^2 > 0$ — but only the form with the power $-1/2$ is the Euler–Lagrange equation of this problem.

Exercise 7.6

Setting. The problem is $\max \int_0^1 e^{-\rho t} u(c) dt$ subject to $\dot{a} = ra + w - c$, $a(t) \geq 0$, $a(0) > 0$ given, and — as the book argues — $a(1) = 0$. THEOREM 7.2 gives the costate equation and the first-order condition

$$e^{-\rho t} u'(c(t)) = \lambda(t), \quad \dot{\lambda}(t) = -r\lambda(t) \implies \lambda(t) = \lambda(0)e^{-rt},$$

so that consumption satisfies

$$u'(c(t)) = u'(c(0))e^{(\rho-r)t}. \quad (*)$$

Integrating the flow constraint with $a(1) = 0$ gives the intertemporal budget constraint

$$\int_0^1 e^{-rt} c(t) dt = a(0) + \int_0^1 e^{-rt} w dt = a(0) + \frac{w(1 - e^{-r})}{r} \equiv \Omega. \quad (**)$$

(a) *Initial consumption.* Equation (*) writes every $c(t)$ as an increasing function of $c(0)$: $c(t) = h_t(c(0))$ with $h_t \equiv (u')^{-1}[u'(c(0))e^{(\rho-r)t}]$, increasing because u' is strictly decreasing. Substituting into (**),

$$\int_0^1 e^{-rt} h_t(c(0)) dt = \Omega,$$

whose left-hand side is continuous and strictly increasing in $c(0)$, from 0 to ∞ : $c(0)$ is uniquely determined. With the CRRA specification $u(c) = c^{1-\theta}/(1-\theta)$ this can be solved in closed form. Then (*) gives $c(t) = c(0)e^{(r-\rho)t/\theta}$, and (**) gives

$$c(0) \int_0^1 e^{-\zeta t} dt = \Omega, \quad \zeta \equiv r - \frac{r-\rho}{\theta},$$

so that

$$c(0) = \frac{\zeta}{1-e^{-\zeta}} \left[a(0) + \frac{w(1-e^{-r})}{r} \right]$$

(with $c(0) = \Omega$ in the knife-edge case $\zeta = 0$). For log utility ($\theta = 1$), $\zeta = \rho$ and $c(0) = \rho\Omega/(1-e^{-\rho})$.

(b) *A rise in initial assets.* Differentiating,

$$\frac{\partial c(0)}{\partial a(0)} = \frac{\zeta}{1-e^{-\zeta}} > 0,$$

and since $c(t) = c(0)e^{(r-\rho)t/\theta}$ with the growth rate independent of wealth, the *entire* consumption path shifts up by the same proportional factor: preferences are homothetic, so wealth effects are neutral across dates. The slope of the path, $(r-\rho)/\theta$, is unchanged — consumption grows if $r > \rho$ and falls if $r < \rho$, whatever the level of assets. (For general u the same argument gives $\partial c(0)/\partial a(0) = [\int_0^1 e^{-rt} h'_t dt]^{-1} > 0$, though the path need no longer scale proportionally unless u is CRRA.)

(c) *A time-varying earnings profile.* Replacing w by $[w(t)]_{t=0}^1$ changes *only* the constant Ω , which becomes $a(0) + \int_0^1 e^{-rt} w(t) dt$: the Euler equation (*) is untouched, because labor income does not enter the first-order conditions. Hence

- if the present value of earnings is unchanged, the consumption path is exactly unchanged — only the asset path moves, the household saving when income arrives early and borrowing when it arrives late;
- if the present value rises, the whole path shifts up proportionally, as in part (b).

This is the permanent income logic: with a perfect capital market the timing of income is irrelevant, only its present value matters.

The qualification matters here, though, because EXAMPLE 7.1 imposes $a(t) \geq 0$: the household cannot borrow at all. If the earnings profile is back-loaded enough that the unconstrained solution would require $a(t) < 0$ on some interval, the constraint binds there. On such an interval the household simply consumes its income, $c(t) = w(t)$ and $a(t) = 0$, and the Euler equation holds with an inequality ($e^{-\rho t} u'(c(t)) \geq \lambda(t)$, the multiplier on $a \geq 0$ being positive). In that case the timing of income *does* affect consumption, and a shift of income toward the present raises welfare even holding its present value fixed.

Exercise 7.7

Let x_1, x_2 be in the (convex) domain, $\alpha \in (0, 1)$, and let y_1, y_2 attain the maxima defining $M(x_1)$ and $M(x_2)$. Then $(\alpha x_1 + (1 - \alpha)x_2, \alpha y_1 + (1 - \alpha)y_2)$ is a feasible pair, so

$$M(\alpha x_1 + (1 - \alpha)x_2) \geq \varphi(\alpha x_1 + (1 - \alpha)x_2, \alpha y_1 + (1 - \alpha)y_2) \geq \alpha \varphi(x_1, y_1) + (1 - \alpha)\varphi(x_2, y_2),$$

and the right-hand side is $\alpha M(x_1) + (1 - \alpha)M(x_2)$: the first inequality holds because M is a maximum and the second by joint concavity of φ . Hence M is concave.

If φ is *strictly* jointly concave and $x_1 \neq x_2$, the second inequality is strict — strict concavity applies because the two points (x_1, y_1) and (x_2, y_2) differ (their first coordinates do) — and therefore M is strictly concave.

If the maximum is not attained, the same argument runs with ε -maximizers: choose y_i with $\varphi(x_i, y_i) \geq M(x_i) - \varepsilon$, conclude $M(\alpha x_1 + (1 - \alpha)x_2) \geq \alpha M(x_1) + (1 - \alpha)M(x_2) - \varepsilon$, and let $\varepsilon \rightarrow 0$.

Remark. This little lemma is what makes Arrow's sufficiency conditions usable. THEOREMS 7.6 and 7.8 require concavity of the *maximized* Hamiltonian $M(t, x, \lambda)$ in x , which is not an assumption one can check directly; the lemma reduces it to joint concavity of $H(t, x, y, \lambda)$ in (x, y) , which is usually immediate from concavity of f and linearity or concavity of g — provided $\lambda(t) \geq 0$, since $H = f + \lambda g$ inherits concavity from g only when the multiplier is nonnegative.

Exercise 7.8

Claim. Consider $\max \int_0^{t_1} f(t, x, y) dt$ subject to $\dot{x} = g(t, x, y)$, $x(0) = x_0$ and $x(t_1) = x_1$, with f, g continuously differentiable. Suppose the admissible pair $(\hat{x}, \hat{y})$ satisfies the necessary conditions of THEOREM 7.2: $H_y(t, \hat{x}, \hat{y}, \lambda) = 0$, $\dot{\lambda} = -H_x(t, \hat{x}, \hat{y}, \lambda)$ and $\hat{\dot{x}} = g$. If X is convex and $M(t, x, \lambda) \equiv \max_y H(t, x, y, \lambda)$ is concave in x for every $t \in [0, t_1]$, then $(\hat{x}, \hat{y})$ attains the global maximum; if M is strictly concave in x , it is the unique solution.

Proof. Let (x, y) be any admissible pair, so that $x(0) = x_0$ and — this is the only change from THEOREM 7.6 — $x(t_1) = x_1 = \hat{x}(t_1)$. By definition of M and of the Hamiltonian,

$$f(t, x, y) = H(t, x, y, \lambda) - \lambda g(t, x, y) \leq M(t, x, \lambda) - \lambda(t)\dot{x}(t),$$

with equality along $(\hat{x}, \hat{y})$ because $\hat{y}$ maximizes H (which is what $H_y = 0$ plus concavity gives). Integrating and subtracting,

$$W(x, y) - W(\hat{x}, \hat{y}) \leq \int_0^{t_1} [M(t, x, \lambda) - M(t, \hat{x}, \lambda)] dt - \int_0^{t_1} \lambda(t) [\dot{x}(t) - \dot{\hat{x}}(t)] dt.$$

Concavity of M in x gives $M(t, x, \lambda) - M(t, \hat{x}, \lambda) \leq M_x(t, \hat{x}, \lambda) [x(t) - \hat{x}(t)]$, and by the envelope theorem (EXERCISE 7.2) $M_x(t, \hat{x}, \lambda) = H_x(t, \hat{x}, \hat{y}, \lambda) = -\dot{\lambda}(t)$. Hence

$$W(x, y) - W(\hat{x}, \hat{y}) \leq - \int_0^{t_1} \dot{\lambda} [x - \hat{x}] dt - \int_0^{t_1} \lambda [\dot{x} - \dot{\hat{x}}] dt = - \left[\lambda(t) (x(t) - \hat{x}(t)) \right]_0^{t_1},$$

integrating by parts. At $t = 0$ both paths start at x_0 , so the lower limit vanishes. At $t = t_1$ the *terminal constraint* gives $x(t_1) = \hat{x}(t_1) = x_1$, so the upper limit vanishes as well. Therefore

$$W(x, y) \leq W(\hat{x}, \hat{y}),$$

which is the assertion. If M is strictly concave in x , the concavity inequality is strict whenever $x(t) \neq \hat{x}(t)$ on a set of positive measure, so any other admissible pair is strictly worse and the solution is unique.

Remark. The comparison with THEOREM 7.6 is instructive: there the boundary term is killed by the transversality condition $\lambda(t_1) = 0$, which says that leftover state is worthless; here it is killed by the constraint $x(t_1) = x_1$, which says that every admissible path ends at the same place, so the value of leftover state is irrelevant. Exactly one of the two is needed, which is why THEOREMS 7.2 and 7.1 differ only in this boundary condition.

Exercise 7.9

The problem is that of THEOREM 7.2 with the terminal condition weakened to $x(t_1) \geq x_1$. Let $(\hat{x}, \hat{y})$ be an interior continuous solution and consider, as in the proofs of THEOREMS 7.1 and 7.2, the variations $y(t, \varepsilon) = \hat{y}(t) + \varepsilon \eta(t)$ with the induced state path $x(t, \varepsilon)$, $x(0, \varepsilon) = x_0$. Writing $W(\varepsilon)$ for the value of the objective and introducing a continuously differentiable costate λ , the same integration by parts as in the proof of THEOREM 7.1 gives

$$\left. \frac{dW}{d\varepsilon} \right|_{\varepsilon=0} = \int_0^{t_1} \left[H_x(t, \hat{x}, \hat{y}, \lambda) + \dot{\lambda}(t) \right] x_\varepsilon(t, 0) dt + \int_0^{t_1} H_y(t, \hat{x}, \hat{y}, \lambda) \eta(t) dt - \lambda(t_1) x_\varepsilon(t_1, 0). \quad (*)$$

The interior conditions. Restrict attention first to variations with $x_\varepsilon(t_1, 0) = 0$ — feasible whatever the terminal constraint, since they do not move the endpoint. Choosing λ to solve the linear terminal-value problem $\dot{\lambda} = -H_x(t, \hat{x}, \hat{y}, \lambda)$ kills the first integral, and optimality then requires $\int H_y \eta dt = 0$ for all such η , whence $H_y = 0$ for all t , that is (7.11); and $\dot{\lambda} = -H_x$ is (7.12). These are exactly the conditions of THEOREM 7.2.

The terminal condition. With (7.11) and (7.12) in hand, $(*)$ reduces to

$$\left. \frac{dW}{d\varepsilon} \right|_{\varepsilon=0} = -\lambda(t_1) x_\varepsilon(t_1, 0).$$

Two cases arise.

- $\hat{x}(t_1) > x_1$ (*the constraint is slack*). For small ε the terminal constraint continues to hold whichever sign $x_\varepsilon(t_1, 0)$ takes, so both signs are feasible; optimality requires $dW/d\varepsilon = 0$ for both, hence $\lambda(t_1) = 0$.
- $\hat{x}(t_1) = x_1$ (*the constraint binds*). Only variations with $x_\varepsilon(t_1, 0) \geq 0$ are feasible, so optimality requires $-\lambda(t_1) x_\varepsilon(t_1, 0) \leq 0$ for all such variations, that is $\lambda(t_1) \geq 0$; nothing forces it to vanish.

In both cases

$$\lambda(t_1) \geq 0 \quad \text{and} \quad \lambda(t_1)(\hat{x}(t_1) - x_1) = 0,$$

which is the complementary slackness condition of THEOREM 7.3.

Interpretation. $\lambda(t_1)$ is the shadow value of the state at the terminal date. If the planner ends with more of the state than required, the marginal unit must be worthless — otherwise it would have been converted into payoff earlier; if the terminal requirement binds, the marginal unit has positive value, exactly as the multiplier on a binding constraint in static optimization.

Exercise 7.11

The problem is

$$\max_{x,y} \int_0^{t_1} f(t, x(t), y(t)) dt \quad \text{s.t.} \quad \dot{x}(t) = G(t, x(t), y(t)), \quad x(0) = x_0,$$

with $x(t) \in X \subset \mathbb{R}^{K_x}$, $y(t) \in Y \subset \mathbb{R}^{K_y}$, and an interior continuous solution $(\hat{x}, \hat{y})$. The argument is the one-dimensional proof of THEOREM 7.1 written with inner products.

Variations. Take any piecewise continuous $\eta : [0, t_1] \rightarrow \mathbb{R}^{K_y}$ and set $y(t, \varepsilon) = \hat{y}(t) + \varepsilon \eta(t)$. Since $\hat{y}(t) \in \text{Int } Y$ and $[0, t_1]$ is compact, $y(t, \varepsilon) \in Y$ for $|\varepsilon|$ small. Let $x(t, \varepsilon)$ be the corresponding state path, which exists, is unique and is differentiable in ε because G is continuously differentiable; write $x_\varepsilon(t) \equiv \partial x(t, \varepsilon) / \partial \varepsilon|_{\varepsilon=0}$, which satisfies the linearized state equation

$$\dot{x}_\varepsilon(t) = D_x G(t, \hat{x}, \hat{y}) x_\varepsilon(t) + D_y G(t, \hat{x}, \hat{y}) \eta(t), \quad x_\varepsilon(0) = 0.$$

The derivative of the objective. For any continuously differentiable $\lambda : [0, t_1] \rightarrow \mathbb{R}^{K_x}$ the state equation lets us write, with H as in (7.28),

$$W(\varepsilon) = \int_0^{t_1} \left[H(t, x(t, \varepsilon), y(t, \varepsilon), \lambda) - \lambda(t) \cdot \dot{x}(t, \varepsilon) \right] dt,$$

and integrating the last term by parts,

$$W(\varepsilon) = \int_0^{t_1} \left[H + \dot{\lambda}(t) \cdot x(t, \varepsilon) \right] dt - \lambda(t_1) \cdot x(t_1, \varepsilon) + \lambda(0) \cdot x_0.$$

Differentiating at $\varepsilon = 0$,

$$\left. \frac{dW}{d\varepsilon} \right|_0 = \int_0^{t_1} \left[D_x H(t, \hat{x}, \hat{y}, \lambda) + \dot{\lambda}(t) \right] \cdot x_\varepsilon(t) dt + \int_0^{t_1} D_y H(t, \hat{x}, \hat{y}, \lambda) \cdot \eta(t) dt - \lambda(t_1) \cdot x_\varepsilon(t_1).$$

Choosing the costate. Let λ solve the linear terminal-value problem

$$\dot{\lambda}(t) = -D_x H(t, \hat{x}(t), \hat{y}(t), \lambda(t)), \quad \lambda(t_1) = 0,$$

which has a unique continuously differentiable solution on $[0, t_1]$ because the right-hand side is linear in λ with continuous coefficients. This is (7.30) and the terminal condition in (7.31), and it annihilates the first and third terms above, leaving

$$\left. \frac{dW}{d\varepsilon} \right|_0 = \int_0^{t_1} D_y H(t, \hat{x}, \hat{y}, \lambda) \cdot \eta(t) dt.$$

Conclusion. Since $(\hat{x}, \hat{y})$ is optimal and the variations are two-sided, this derivative must vanish for every admissible η . Taking η supported on a small interval around any t and equal to each coordinate direction in turn, and using continuity of $D_y H$ along the solution, gives

$$D_y H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0 \quad \text{for all } t \in [0, t_1],$$

which is (7.29). Finally $\dot{x} = D_\lambda H = G$ is the state equation with $x(0) = x_0$, completing (7.31). This proves THEOREM 7.7.

Exercise 7.12

Let $(\hat{x}, \hat{y})$ satisfy (7.29)–(7.31) and let X be convex. Write

$$M(t, x, \lambda) \equiv \max_{y \in Y} H(t, x, y, \lambda)$$

and suppose it is concave in x . Take any admissible pair (x, y) . As in Exercise 7.8, for every t ,

$$f(t, x, y) = H(t, x, y, \lambda) - \lambda(t) \cdot G(t, x, y) \leq M(t, x, \lambda) - \lambda(t) \cdot \dot{x}(t),$$

with equality along $(\hat{x}, \hat{y})$ since $\hat{y}$ maximizes H pointwise. Subtracting and using concavity of M in x together with the envelope relation $D_x M(t, \hat{x}, \lambda) = D_x H(t, \hat{x}, \hat{y}, \lambda) = -\dot{\lambda}(t)$,

$$\begin{aligned} W(x, y) - W(\hat{x}, \hat{y}) &\leq \int_0^{t_1} \left[D_x M(t, \hat{x}, \lambda) \cdot (x - \hat{x}) - \lambda \cdot (\dot{x} - \dot{\hat{x}}) \right] dt \\ &= \int_0^{t_1} \left[-\dot{\lambda} \cdot (x - \hat{x}) - \lambda \cdot (\dot{x} - \dot{\hat{x}}) \right] dt = - \left[\lambda(t) \cdot (x(t) - \hat{x}(t)) \right]_0^{t_1}. \end{aligned}$$

The lower limit vanishes because both paths start at x_0 , and the upper limit vanishes because $\lambda(t_1) = 0$ by (7.31). Hence $W(x, y) \leq W(\hat{x}, \hat{y})$ for every admissible pair: $(\hat{x}, \hat{y})$ is a global maximum.

If $M(t, \cdot, \lambda)$ is strictly concave, the first inequality is strict unless $x(t) = \hat{x}(t)$ almost everywhere, so the maximizer is unique (and then $y = \hat{y}$ wherever H has a unique maximizer).

Remark. Exercise 7.7 is what makes the hypothesis checkable: joint concavity of H in (x, y) — which follows from concavity of f , linearity of G in (x, y) , and $\lambda \geq 0$ — implies concavity of M in x . In the growth applications of the next chapters G is linear (capital accumulation) and f is concave (utility), so the conditions hold automatically.

Exercise 7.13

Let $(\hat{x}, \hat{y})$ be admissible and satisfy (7.39)–(7.41), let X be convex, let $M(t, x, \lambda)$ be concave in x for all t , and suppose

$$\lim_{t \rightarrow \infty} \lambda(t)(\hat{x}(t) - \tilde{x}(t)) \leq 0 \quad \text{for every admissible } \tilde{x}. \quad (*)$$

Take any admissible pair $(\tilde{x}, \tilde{y})$. Exactly as in Exercise 7.8, for each finite T ,

$$\int_0^T f(t, \tilde{x}, \tilde{y}) dt - \int_0^T f(t, \hat{x}, \hat{y}) dt \leq - \left[\lambda(t)(\tilde{x}(t) - \hat{x}(t)) \right]_0^T = \lambda(T)(\hat{x}(T) - \tilde{x}(T)),$$

using $\tilde{x}(0) = \hat{x}(0) = x_0$ for the lower limit. This is the “additional term” the hint refers to: in finite horizon it was eliminated by $\lambda(t_1) = 0$ or by a fixed endpoint, and here it survives.

Now let $T \rightarrow \infty$. The left-hand side converges to $W(\tilde{x}, \tilde{y}) - W(\hat{x}, \hat{y})$ whenever the two improper integrals exist (which admissibility guarantees), and the right-hand side is bounded above by zero by hypothesis (*). Hence

$$W(\tilde{x}, \tilde{y}) - W(\hat{x}, \hat{y}) \leq \limsup_{T \rightarrow \infty} \lambda(T)(\hat{x}(T) - \tilde{x}(T)) \leq 0,$$

so $(\hat{x}, \hat{y})$ achieves the global maximum of (7.32). If $M(t, \cdot, \lambda)$ is strictly concave, the inequality leading to the display is strict unless $\tilde{x} = \hat{x}$ almost everywhere, so the solution is unique.

Remark. Condition (*) is the honest price of infinite horizon, and the book rightly calls it hard to check: it involves *all* admissible comparison paths. Two special cases make it usable. If the state is nonnegative and $\lambda(t) \geq 0$ with $\lim_{t \rightarrow \infty} \lambda(t)\hat{x}(t) = 0$ — the transversality condition (7.69) — then $\lambda(T)(\hat{x}(T) - \tilde{x}(T)) \leq \lambda(T)\hat{x}(T) \rightarrow 0$ and (*) holds automatically. This is exactly how THEOREM 7.14 is used in the applications: check the transversality condition, and concavity does the rest.

Exercise 7.14

The discounted problem has $f(t, x, y) = e^{-\rho t} f(x, y)$ and $g(t, x, y) = g(x, y)$. Write the present-value Hamiltonian and its current-value counterpart,

$$H(t, x, y, \lambda) = e^{-\rho t} f(x, y) + \lambda(t)g(x, y), \quad \hat{H}(t, x, y, \mu) \equiv e^{\rho t} H(t, x, y, \lambda) = f(x, y) + \mu(t)g(x, y),$$

where the current-value costate is

$$\mu(t) \equiv e^{\rho t} \lambda(t) \quad \Longleftrightarrow \quad \lambda(t) = e^{-\rho t} \mu(t).$$

THEOREM 7.9 applies to the problem and gives, at points of continuity of $\hat{y}$, the necessary conditions (7.39)–(7.41). Translate each in turn.

(7.65). $H_y = 0$ reads $e^{-\rho t} f_y + \lambda g_y = 0$. Multiplying by $e^{\rho t} > 0$,

$$f_y(x, y) + \mu(t)g_y(x, y) = \hat{H}_y(t, \hat{x}, \hat{y}, \mu) = 0,$$

which is (7.65).

(7.66). Differentiating $\lambda = e^{-\rho t} \mu$,

$$\dot{\lambda}(t) = e^{-\rho t} [\dot{\mu}(t) - \rho \mu(t)],$$

while $H_x = e^{-\rho t} f_x + \lambda g_x = e^{-\rho t} [f_x + \mu g_x] = e^{-\rho t} \hat{H}_x$. Substituting into $\dot{\lambda} = -H_x$ and cancelling $e^{-\rho t} > 0$,

$$\dot{\mu}(t) - \rho \mu(t) = -\hat{H}_x(t, \hat{x}, \hat{y}, \mu) \quad \Longleftrightarrow \quad \rho \mu(t) - \dot{\mu}(t) = \hat{H}_x(t, \hat{x}, \hat{y}, \mu),$$

which is (7.66).

(7.67). The state equation is unchanged: $\dot{x} = H_\lambda = g(x, y) = \hat{H}_\mu$, with $x(0) = x_0$ and the limiting constraint $\lim_{t \rightarrow \infty} b(t)x(t) \geq x_1$ carried over from (7.41).

Since THEOREM 7.9 gives (7.39)–(7.41) as necessary conditions for an interior piecewise continuous solution, and the three translations above are equivalences, (7.65)–(7.67) are necessary as well.

Remark. The point of the current-value formulation is visible in (7.66): the term $\rho \mu$ appears because μ is measured in date- t rather than date-0 units, so its required rate of return has to cover the discount rate. Written as $\dot{\mu}/\mu = \rho - \hat{H}_x/\mu$, the condition is an asset-pricing equation — the shadow price of the state must appreciate at the discount rate less its “dividend”.

Exercise 7.15

(a) *Rewriting the objective.* For any continuously differentiable λ , the constraint $\dot{x} = g$ gives $f = H - \lambda\dot{x}$ with $H(t, x, y) \equiv f(t, x, y) + \lambda(t)g(t, x, y)$. Hence

$$W = \int_0^{t_1} f \, dt = \int_0^{t_1} [H - \lambda(t)\dot{x}(t)] \, dt,$$

and integrating the last term by parts, $\int_0^{t_1} \lambda\dot{x} \, dt = \lambda(t_1)x(t_1) - \lambda(0)x_0 - \int_0^{t_1} \dot{\lambda}x \, dt$, so

$$W(x(t), y(t)) = \int_0^{t_1} [H(t, x(t), y(t)) + \dot{\lambda}(t)x(t)] \, dt - \lambda(t_1)x(t_1) + \lambda(0)x_0,$$

as required.

(b) *Feasible variations.* Let $(\hat{x}, \hat{y}, \hat{t}_1)$ be optimal. Perturb the control on the original interval and extend the horizon:

$$y(t, \varepsilon) = \hat{y}(t) + \varepsilon\eta(t) \quad \text{on } [0, \hat{t}_1], \quad y(t, \varepsilon) = \hat{y}(\hat{t}_1) + \varepsilon\eta(t) \quad \text{on } [\hat{t}_1, \hat{t}_1 + \varepsilon\Delta t],$$

with $t_1 = \hat{t}_1 + \varepsilon\Delta t$. For $|\varepsilon|$ small this is feasible: $\hat{y}(t)$ is interior to Y and $\hat{x}(t)$ interior to X on a compact interval, so small perturbations stay inside, and the state path $x(t, \varepsilon)$ generated by $\dot{x} = g$ is well defined and differentiable in ε (the extension is over an interval of length $O(\varepsilon)$, on which the state moves by $O(\varepsilon)$).

(c) *The derivative.* Differentiating the expression in (a) at $\varepsilon = 0$, three groups of terms appear. The interior variation contributes $\int_0^{\hat{t}_1} [H_x + \dot{\lambda}]x_\varepsilon \, dt + \int_0^{\hat{t}_1} H_y\eta \, dt$. The moving upper limit contributes $H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1))\Delta t$ from the integral and $-\lambda(\hat{t}_1)\dot{x}(\hat{t}_1)\Delta t$ from $-\lambda(t_1)x(t_1)$; the endpoint term also contributes $-\lambda(\hat{t}_1)x_\varepsilon(\hat{t}_1)$. Since the second of these enters $-\lambda(t_1)x(t_1)$ with a minus sign and the total derivative of $x(t_1, \varepsilon)$ with respect to ε is $x_\varepsilon(\hat{t}_1) + \dot{x}(\hat{t}_1)\Delta t$, collecting terms gives exactly

$$\left. \frac{dW}{d\varepsilon} \right|_0 = \int_0^{\hat{t}_1} [H_x + \dot{\lambda}]x_\varepsilon \, dt + \int_0^{\hat{t}_1} H_y\eta \, dt + H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1))\Delta t - \lambda(\hat{t}_1)x_\varepsilon(\hat{t}_1) + \lambda(\hat{t}_1)\dot{x}(\hat{t}_1)\Delta t,$$

the expression in the exercise.

(d) *Why it vanishes.* $(\hat{x}, \hat{y}, \hat{t}_1)$ maximizes W over a set that contains the whole family $\{(x(\cdot, \varepsilon), y(\cdot, \varepsilon), \hat{t}_1 + \varepsilon\Delta t)\}$ for ε in a neighborhood of 0 — both signs are feasible, since the horizon may be lengthened or shortened and the control perturbed in either direction. A differentiable function of ε attaining an interior maximum at $\varepsilon = 0$ has zero derivative there.

The conditions of the Maximum Principle follow. Taking $\Delta t = 0$ and η arbitrary, and choosing λ to satisfy $\dot{\lambda} = -H_x$ with $\lambda(\hat{t}_1) = 0$, gives $H_y = 0$ as usual. Then, with $\lambda(\hat{t}_1) = 0$, the remaining terms leave $H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1))\Delta t = 0$ for every Δt , so

$$\boxed{H(\hat{t}_1, \hat{x}(\hat{t}_1), \hat{y}(\hat{t}_1), \lambda(\hat{t}_1)) = 0}$$

— the extra transversality condition associated with a free terminal date. Its meaning is an optimal stopping rule: continue as long as the flow value of continuing, measured by the Hamiltonian, is positive, and stop when it reaches zero.

(e) Letting $\hat{t}_1 \rightarrow \infty$. The condition becomes

$$\lim_{t \rightarrow \infty} H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0,$$

which is precisely the infinite-horizon transversality condition (7.56) of THEOREM 7.12. So the “vanishing Hamiltonian” that is usually presented as a separate infinite-horizon result is the same condition as the free-terminal-time condition of the finite-horizon problem: an infinite horizon is a horizon that is optimally never reached, and the value of stopping at the margin must be zero at every sufficiently distant date.

Exercise 7.16

The discounted problem has $f(t, x, y) = e^{-\rho t} f(x, y)$ and $g(t, x, y) = g(x, y)$: neither the payoff apart from the discount factor nor the constraint depends on calendar time.

Admissibility. The continuation satisfies $\dot{\hat{x}}(t) = g(\hat{x}(t), \hat{y}(t))$ for all $t \geq s$ and $\hat{x}(s)$ is its own initial condition, with $(\hat{x}(t), \hat{y}(t)) \in X \times Y$; so it is admissible for the problem started at s with initial state $\hat{x}(s)$.

Optimality. The objective of the date- s problem is

$$W_s(x, y) \equiv \int_s^\infty e^{-\rho(t-s)} f(x(t), y(t)) dt = e^{\rho s} \int_s^\infty e^{-\rho t} f(x(t), y(t)) dt,$$

so it ranks continuation plans exactly as the date-0 objective does: the two differ by the positive constant $e^{\rho s}$. Suppose the continuation were not optimal. Then there is an admissible $(\tilde{x}, \tilde{y})$ on $[s, \infty)$ with $\tilde{x}(s) = \hat{x}(s)$ and

$$\int_s^\infty e^{-\rho t} f(\tilde{x}, \tilde{y}) dt > \int_s^\infty e^{-\rho t} f(\hat{x}, \hat{y}) dt.$$

Splice the plans:

$$(\vec{x}(t), \vec{y}(t)) = \begin{cases} (\hat{x}(t), \hat{y}(t)), & t \in [0, s), \\ (\tilde{x}(t), \tilde{y}(t)), & t \geq s. \end{cases}$$

This is admissible from x_0 : it satisfies the state equation on each piece, and the two pieces agree at $t = s$ because $\tilde{x}(s) = \hat{x}(s)$, so $\vec{x}$ is continuous and satisfies $\dot{\vec{x}} = g(\vec{x}, \vec{y})$ throughout. Its value at date 0 is

$$\int_0^s e^{-\rho t} f(\hat{x}, \hat{y}) dt + \int_s^\infty e^{-\rho t} f(\tilde{x}, \tilde{y}) dt > \int_0^\infty e^{-\rho t} f(\hat{x}, \hat{y}) dt,$$

contradicting the optimality of $(\hat{x}, \hat{y})$ from date 0. Hence the continuation is optimal for the date- s problem.

Remark. Two features do the work, and both are needed: the objective is *exponentially* discounted, so that re-dating multiplies it by a constant and leaves the ranking of continuations unchanged; and g is autonomous, so that the constraint faced from s onward is the same correspondence, only with a different initial state. Drop the first and time consistency fails — this is exactly EXERCISE 7.22(b), where $\beta(t) = e^{-\rho t}$ turns out to be necessary as well as sufficient. The statement in the exercise writes the date- s initial condition as $x(s) = x_0$; what is meant, and what is proved above, is that the continuation is optimal for the problem started at the state the optimal path has reached, $x(s) = \hat{x}(s)$.

Exercise 7.20

Recall the problem: maximize $\int_0^\infty e^{-\rho t} u(y(t)) dt$ subject to $\dot{x} = -y$, $x(0) = 1$, $x(t) \geq 0$, with $u : [0, 1] \rightarrow \mathbb{R}$ strictly increasing, continuously differentiable and strictly concave. The Hamiltonian is $H(t, x, y, \lambda) = e^{-\rho t} u(y) - \lambda y$, and THEOREM 7.9 gives (7.58) and (7.59):

$$u'(\hat{y}(t)) = e^{\rho t} \lambda(t), \quad \dot{\lambda}(t) = 0 \implies \lambda(t) \equiv \lambda(0) > 0,$$

so that the Hotelling rule $\dot{\mu}/\mu = \rho$ holds for $\mu = e^{\rho t} \lambda$ and

$$\hat{y}(t) = (u')^{-1}[\lambda(0)e^{\rho t}].$$

Since u' is strictly decreasing and $\lambda(0)e^{\rho t} \rightarrow \infty$, extraction falls monotonically with $\hat{y}(t) \rightarrow 0$; the constant $\lambda(0)$ is fixed by exhaustion, $\int_0^\infty \hat{y}(t) dt = x(0)$.

Checking (7.56). The condition is $\lim_{t \rightarrow \infty} H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = 0$. Here

$$H(t, \hat{x}(t), \hat{y}(t), \lambda(t)) = e^{-\rho t} u(\hat{y}(t)) - \lambda(0) \hat{y}(t).$$

Both terms vanish in the limit:

- $\hat{y}(t) \rightarrow 0$ and u is continuous on $[0, 1]$, so $u(\hat{y}(t)) \rightarrow u(0)$, a finite number, and $e^{-\rho t} u(\hat{y}(t)) \rightarrow 0$;
- $\lambda(0) \hat{y}(t) \rightarrow 0$ because $\hat{y}(t) \rightarrow 0$ and $\lambda(0)$ is a constant.

Hence $\lim_{t \rightarrow \infty} H = 0$: the transversality condition (7.56) holds.

An explicit case. With $u(y) = \log y$ (unbounded below at 0, so strictly outside the stated domain, but the standard illustration), $\hat{y}(t) = e^{-\rho t} / \lambda(0)$, and exhaustion $\int_0^\infty \hat{y} dt = 1/(\rho \lambda(0)) = x(0)$ gives $\lambda(0) = 1/(\rho x(0))$ and $\hat{y}(t) = \rho x(0) e^{-\rho t}$: the resource is depleted at the constant proportional rate ρ . Then

$$H(t) = e^{-\rho t} \log(\rho x(0) e^{-\rho t}) - \frac{1}{\rho x(0)} \rho x(0) e^{-\rho t} = e^{-\rho t} [\log(\rho x(0)) - \rho t - 1] \longrightarrow 0,$$

since $te^{-\rho t} \rightarrow 0$.

Remark. Note which transversality condition is being verified. The *stronger* condition (7.69) requires $\lim_{t \rightarrow \infty} e^{-\rho t} \mu(t) \hat{x}(t)$ to vanish; since $e^{-\rho t} \mu(t) = \lambda(0)$ here, it reduces to $\lambda(0) \lim \hat{x}(t) = 0$, which holds precisely because the resource is fully exhausted, $\hat{x}(t) \rightarrow 0$. EXERCISE 7.18 modifies the problem so that a positive stock x_1 must be left behind. The stronger condition then fails, while the weaker one still holds — which is exactly why the book states (7.56) as the general result.

Exercise 7.22

(a) *Necessary conditions.* The Hamiltonian is

$$H(t, x, y, \lambda) = \beta(t) f(x, y) + \lambda(t) g(x, y),$$

and THEOREM 7.9 gives, along an interior solution,

$$H_y = 0 : \quad \beta(t) f_y(x, y) + \lambda(t) g_y(x, y) = 0, \quad \dot{\lambda} = -H_x : \quad \dot{\lambda}(t) = -[\beta(t) f_x + \lambda(t) g_x],$$

together with $\dot{x} = g(x, y)$, $x(0) = x_0$ and the transversality condition (7.56). Assume $g_y \neq 0$, so the first condition can be solved for the costate:

$$\lambda(t) = -\beta(t) \frac{f_y(x(t), y(t))}{g_y(x(t), y(t))} \equiv \beta(t) \Lambda(t), \quad (*)$$

where Λ depends on the path only, not on the discount function.

(b) *Time consistency forces exponential discounting.* Consider re-solving the problem at date $s > 0$ with the same structure — that is, with discount function $\beta(t - s)$ applied to returns $t - s$ away — starting from the state $\hat{x}(s)$ that the original plan has reached. Its necessary conditions are the same with $\beta(t)$ replaced by $\beta(t - s)$ and costate λ_s . Time consistency means that the original path $(\hat{x}, \hat{y})$ continues to solve this problem, so it must satisfy both sets of conditions. From (*) applied to each,

$$\lambda(t) = \beta(t)\Lambda(t), \quad \lambda_s(t) = \beta(t - s)\Lambda(t) \quad \implies \quad \lambda_s(t) = k(t)\lambda(t), \quad k(t) \equiv \frac{\beta(t - s)}{\beta(t)}.$$

Both costates satisfy their own costate equations along the *same* path:

$$\dot{\lambda} = -[\beta(t)f_x + \lambda g_x], \quad \dot{\lambda}_s = -[\beta(t - s)f_x + \lambda_s g_x] = -[k(t)\beta(t)f_x + k(t)\lambda g_x] = k(t)\dot{\lambda}(t).$$

But differentiating $\lambda_s = k\lambda$ directly gives $\dot{\lambda}_s = \dot{k}\lambda + k\dot{\lambda}$. Comparing, $\dot{k}(t)\lambda(t) = 0$, and since $\lambda(t) \neq 0$ along an interior solution, $\dot{k}(t) = 0$: the ratio $\beta(t - s)/\beta(t)$ is independent of t . Evaluating it at $t = s$ gives $k = \beta(0)/\beta(s)$, so for all $t \geq s \geq 0$,

$$\beta(t - s)\beta(s) = \beta(t)\beta(0).$$

Normalizing $\beta(0) = 1$, this is the Cauchy functional equation $\beta(t - s)\beta(s) = \beta(t)$; its only positive continuous solutions are

$$\beta(t) = e^{-\rho t} \quad \text{for some } \rho,$$

with $\rho \geq 0$ if the discount function is nonincreasing (and $\rho > 0$ needed for the objective to converge). Conversely, if $\beta(t) = e^{-\rho t}$, then $\beta(t - s) = e^{\rho s}\beta(t)$, the two systems differ by the positive constant $e^{\rho s}$, and the original path satisfies both — time consistency, which is Exercise 7.16.

(c) *Interpretation, and the contrast with LEMMA 7.1.* The result is the continuous-time counterpart of the discrete-time statement in EXERCISE 5.1: only exponential discounting makes the trade-off between two future dates independent of the date from which it is viewed. With any other β — hyperbolic discounting, say — the planner's relative valuation of dates t and t' changes as time passes, so at date s she would like to revise the plan made at date 0 even though nothing has been learned and no state has been surprised.

The contrast with LEMMA 7.1 is worth stating carefully, since the lemma is also called a principle of optimality and holds for general $f(t, x, y)$. LEMMA 7.1 says that the value function computed *from the date-0 perspective* is recursive: $V(t_0, x) = \int_{t_0}^{t_1} f dt + V(t_1, \hat{x}(t_1))$. That is a statement about one fixed objective, and it is true whatever the discount function, because $V(t_1, \cdot)$ is itself defined with the date-0 weights. Time consistency in the sense of part (b) is a different claim: that the plan optimal under the date-0 objective is still optimal under the *date-s* objective, which applies $\beta(t - s)$ rather than $\beta(t)$. Only exponential discounting makes the two objectives agree, up to scale, on the

ranking of continuations. In short, LEMMA 7.1 is about a consistent bookkeeping of one plan; part (b) is about whether a later self would keep it.

Exercise 7.27

The problem of Section 7.8 is

$$\max \int_0^\infty e^{-rt} [f(K) - I - \phi(I)] dt \quad \text{s.t.} \quad \dot{K} = I - \delta K, \quad K(0) > 0,$$

with f satisfying Assumptions 1 and 2 of Chapter 2 and ϕ strictly increasing, strictly convex, continuously differentiable, with $\phi(0) = \phi'(0) = 0$.

(a) ASSUMPTION 7.1 on a compact state space. Restrict $K \in [\varepsilon, \bar{k}]$ with $0 < \varepsilon < \bar{k} < \infty$; investment can then also be restricted to a compact interval, since $\dot{K} = I - \delta K$ and K is bounded, and since paying $I + \phi(I)$ for arbitrarily large I is never optimal. Now check the three parts:

- (1) *Monotonicity.* The payoff $f(K) - I - \phi(I)$ is increasing in K (as $f' > 0$) and decreasing in I (as $1 + \phi' > 0$), so it is weakly monotone in each argument; $g(K, I) = I - \delta K$ is increasing in I , decreasing in K , and independent of t .
- (2) $|g_I| \geq m$. Here $g_I \equiv 1$, so the condition holds with $m = 1$ — the control moves the state one for one.
- (3) $|f_I| \leq M$. Here $|f_I| = 1 + \phi'(I)$, which is bounded on the compact interval to which I has been restricted, so the condition holds with $M = 1 + \max \phi'$.

It is the third condition that fails on the unbounded state space, since ϕ' is unbounded; this is exactly why the compactification is needed before THEOREM 7.13 can be invoked.

(b) *From the modified problem to the original one.* Suppose the modified problem has a solution $(\hat{K}, \hat{I})$ with $\hat{K}(t) > \varepsilon$ for all t , so the artificial constraint never binds. Then $(\hat{K}, \hat{I})$ satisfies the necessary conditions (7.85) of the unconstrained problem, including the transversality condition $\lim_{t \rightarrow \infty} e^{-rt} q(t) \hat{K}(t) = 0$. Now appeal to sufficiency rather than necessity: the current-value Hamiltonian

$$\hat{H}(K, I, q) = f(K) - I - \phi(I) + q[I - \delta K]$$

is strictly concave in (K, I) — f is strictly concave, $-\phi$ is strictly concave, and the remaining terms are linear — so by Exercise 7.7 the maximized Hamiltonian is strictly concave in K , and THEOREM 7.14 applies on the *whole* feasible set $K \in \mathbb{R}_+$: any path satisfying the necessary conditions together with the transversality condition is the unique global maximum among all feasible paths, not merely among those confined to $[\varepsilon, \bar{k}]$. Hence $(\hat{K}, \hat{I})$ solves the original problem. (Note that sufficiency does not use ASSUMPTION 7.1 at all; the assumption was needed only to obtain the strong transversality condition as a *necessary* condition.)

(c) *Such an ε exists.* The steady state of the system (7.84), (7.88) satisfies

$$I^* = \delta K^*, \quad f'(K^*) = (r + \delta)(1 + \phi'(I^*)),$$

which has a unique positive solution because the left side is strictly decreasing in K from ∞ to 0 (Assumptions 1 and 2) while the right side is strictly increasing in K . The optimal path is the stable arm of a saddle point — the linearized system has one positive and one negative eigenvalue

— and $K(t)$ moves monotonically from $K(0)$ to K^* . Therefore

$$\inf_{t \geq 0} K(t) = \min\{K(0), K^*\} > 0,$$

and any $\varepsilon < \min\{K(0), K^*\}$ works: the modified problem's solution never touches the artificial lower bound, and part (b) applies.

With $f(K) = K^{1/3}$, $\phi(I) = I^2$, $\delta = r = 0.05$, the numbers are

$$K^* = 3.767391, \quad I^* = \delta K^* = 0.188370, \quad q^* = 1 + \phi'(I^*) = 1.376739,$$

and the eigenvalues of the linearized system are -0.108440 and $+0.158440$ — one of each sign, as the saddle-path structure requires. Figure 5 shows the phase diagram and the stable arm, which stays well away from $K = 0$.

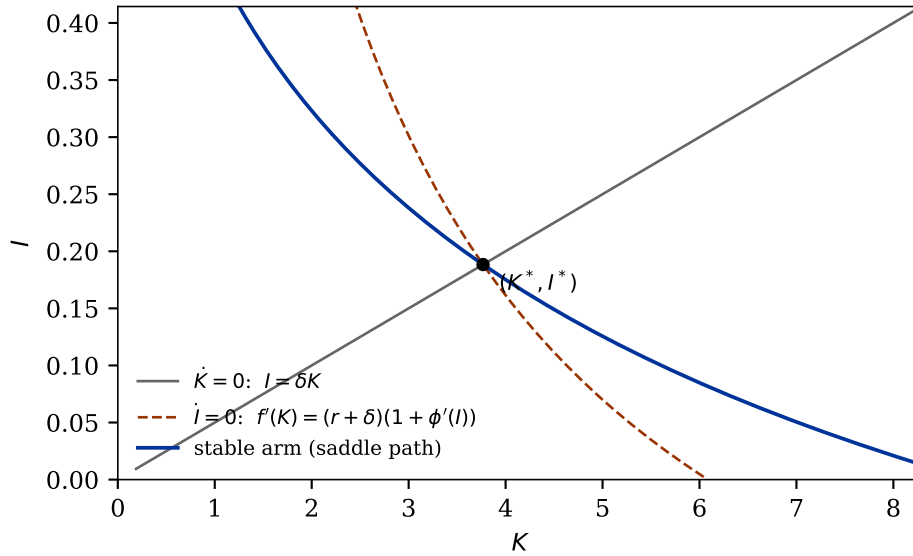


FIGURE 5. The q -theory phase diagram: the loci $\dot{K} = 0$ and $\dot{I} = 0$, their intersection at (K^*, I^*) , and the stable arm along which the firm approaches the steady state.

Remark. Adjustment costs lower the steady-state capital stock: with $\phi \equiv 0$ the steady state would solve $f'(K) = r + \delta$, giving $K = 6.085806$ in this parameterization, against 3.767391 with adjustment costs. The reason is visible in the steady-state condition — the required marginal product is scaled up by $q^* = 1 + \phi'(I^*) > 1$, the shadow price of installed capital, because replacing depreciated capital is itself costly.

Exercise 7.29

Suppose the candidate path has $I(t) = 0$ for all $t \geq t'$. Then from (7.84) its capital stock decays exponentially,

$$K(t) = K(t')e^{-\delta(t-t')} \longrightarrow 0 \quad \text{as } t \rightarrow \infty.$$

The deviation. Fix $T > t'$ and $\epsilon > 0$ small, and consider investing ϵ at every date $t \geq T$ instead of nothing. Two things make the accounting simple. First, the extra *cost* at each date is $\epsilon + \phi(\epsilon)$, and since $\phi(0) = \phi'(0) = 0$ we have $\phi(\epsilon) = o(\epsilon)$: to first order the adjustment cost of a small investment is zero, so the cost is ϵ per unit time. Second, the extra *capital* raises output at the margin by

$f'(K(t))$ per unit. To first order in ϵ the change in the present value of profits is therefore

$$\Delta = \epsilon \int_T^\infty e^{-rt} [f'(K(t)) - 1] dt + o(\epsilon).$$

The sign. By Assumption 2 of Chapter 2, $\lim_{K \rightarrow 0} f'(K) = \infty$. Since $K(t) \rightarrow 0$ along the candidate path, there is a $\tilde{K} > 0$ with $f'(K) > 1$ for all $K < \tilde{K}$, and a date T with $K(t) < \tilde{K}$ — hence $f'(K(t)) > 1$ — for all $t \geq T$. For that T the integrand is strictly positive throughout, so $\Delta > 0$ for ϵ small: the deviation strictly raises the present value of profits, and the candidate path is not optimal.

Quantitatively, with $f(K) = AK^\alpha$ the threshold is explicit: $f'(K) > 1$ if and only if $K < (A\alpha)^{1/(1-\alpha)}$, which equals 0.192450 for $A = 1$, $\alpha = 1/3$. Starting from the steady state $K^* = 3.767391$ and switching investment off, capital falls below that threshold after $\log(K^*/0.19245)/\delta \approx 59.5$ periods, and $f'(K(t)) \rightarrow \infty$ thereafter.

Conclusion. No optimal path sets investment to zero permanently after a finite date: gross investment is strictly positive in the long run, and indeed the optimal path converges to the steady state with $I^* = \delta K^* > 0$. The economics is the Inada condition doing its usual work — capital that has become scarce is arbitrarily productive at the margin, while the marginal cost of a small amount of investment is only its purchase price, adjustment costs being second order at $I = 0$.

Remark. The same argument shows why the constraint $I \geq 0$ studied in EXERCISE 7.28(f) — irreversible investment — binds only on the way *down*, never permanently: a firm may wish to disinvest faster than depreciation allows, but never to abstain from investment forever.

Chapter 8

The Neoclassical Growth Model

The twenty-four exercises of Chapter 8 left unsolved in the instructor's manual: Exercises 8.1, 8.3–8.6, 8.8–8.10, 8.12, 8.14, 8.16–8.18, 8.20–8.22, 8.24, 8.26, 8.28, 8.29, 8.32, 8.35, 8.36 and 8.39. The manual solves 8.2, 8.7, 8.11, 8.13, 8.15, 8.19, 8.23, 8.25, 8.27, 8.30, 8.31, 8.33, 8.34, 8.37 and 8.38. Notation follows the book: $a = A/L$ are per capita assets, k the capital-labor ratio (the effective capital-labor ratio K/AL from Section 8.7 on), $\tilde{c} = c/A$ normalized consumption, μ the current-value costate, and $\varepsilon_u(c) = -u''(c)c/u'(c)$. The numerical parts are computed by `code/ch8.py`.

Exercise 8.1

(a) *The household's problem.* The household has $L(t) = e^{nt}$ members, is fully altruistic and utilitarian, so at each date it values the sum of its members' instantaneous utilities, $\int_0^{L(t)} u(c_i(t)) di$, and discounts this flow at the subjective rate ρ . Its members share the total consumption $C(t)$ that the household's budget allows, and the household's assets accumulate at the market return. Hence

$$\max \int_0^\infty e^{-\rho t} \left[\int_0^{L(t)} u(c_i(t)) di \right] dt \quad \text{s.t.} \quad \int_0^{L(t)} c_i(t) di \leq C(t), \quad \dot{A}(t) = r(t)A(t) + W(t) - C(t),$$

with $W(t) = w(t)L(t)$ total labor income and $A(0)$ given.

(b) *Reduction.* Fix a date and a total $C(t)$. The inner problem is to maximize $\int_0^L u(c_i) di$ subject to $\int_0^L c_i di = C$. Since u is strictly concave, Jensen's inequality gives

$$\frac{1}{L} \int_0^L u(c_i) di \leq u\left(\frac{1}{L} \int_0^L c_i di\right) = u\left(\frac{C}{L}\right),$$

with equality if and only if c_i is the same for (almost) every member. So the cooperative household gives every member $c(t) \equiv C(t)/L(t)$, and the date- t flow of utility is

$$L(t)u(c(t)) = e^{nt}u(c(t)).$$

Discounting at ρ therefore gives the objective

$$\int_0^\infty e^{-\rho t} e^{nt} u(c(t)) dt = \int_0^\infty e^{-(\rho-n)t} u(c(t)) dt,$$

which is (8.3). For the constraint, divide $\dot{A} = rA + wL - cL$ by $L(t)$ and use $a = A/L$, so that $\dot{a} = \dot{A}/L - a\dot{L}/L$:

$$\dot{a}(t) = r(t)a(t) + w(t) - c(t) - na(t) = (r(t) - n)a(t) + w(t) - c(t),$$

which is (8.8).

Intuition. Two adjustments turn a growing household into a single infinitely lived consumer. The *effective discount rate* falls from ρ to $\rho - n$: the household discounts the future at ρ but there are more members to enjoy it, at the rate n , and with utilitarian altruism the two effects net out. The *effective return* falls from r to $r - n$: assets have to be spread over a population growing at n , so per

capita assets earn only $r - n$. Both adjustments make the population growth rate behave exactly like a shift in impatience and in the return — which is why Assumption 4' requires $\rho > n$, so that the transformed problem still discounts.

Exercise 8.3

Accumulation with zero consumption. With $c \equiv 0$ the law of motion (8.7) is the linear differential equation $\dot{A}(s) = r(s)A(s) + w(s)L(s)$. Its solution with $A(t) = 0$ is, by the integrating-factor formula,

$$A(T) = \int_t^T w(s)L(s) \exp\left(\int_s^T r(z) dz\right) ds,$$

so, discounting back to t ,

$$A(T) \exp\left(-\int_t^T r(z) dz\right) = \int_t^T w(s)L(s) \exp\left(-\int_t^s r(z) dz\right) ds \xrightarrow{T \rightarrow \infty} \bar{W}(t).$$

$\bar{W}(t)$ is thus the present value at t of the household's entire future labor income.

Why the limit. A household that never defaults must be able to repay its debt out of future resources. Consumption cannot be negative, so the largest present value it can ever devote to repayment is exactly $\bar{W}(t)$. Hence feasibility (in the absence of bankruptcy) requires

$$A(t) \geq -\bar{W}(t),$$

and any consumption-asset path violating this at some date is infeasible.

Per capita form (8.11). Divide by $L(t)$ and use $L(s) = L(t)e^{n(s-t)}$:

$$a(t) = \frac{A(t)}{L(t)} \geq -\int_t^\infty w(s) \frac{L(s)}{L(t)} e^{-\int_t^s r(z) dz} ds = -\int_t^\infty w(s) \exp\left(-\int_t^s (r(z) - n) dz\right) ds,$$

which is (8.11): population growth discounts the future wage stream at the *effective* rate $r - n$, exactly as it discounts utility at $\rho - n$.

Relation to (6.44). The discrete-time limit reads $a(t) \geq -\sum_{s \geq 0} [\prod_{j=0}^s (1 + r(t+j))^{-1}] w(t+s)$. Replacing the product of gross returns by the exponential of the integral of the instantaneous rate, and the sum by an integral, gives (8.11): the two are the same statement — borrowing is limited by the present value of the endowment stream — written in the two time conventions.

Exercise 8.4

Write $q(t) \equiv \exp(-\int_0^t (r(s) - n) ds)$ for the effective discount factor, and let

$$\tilde{a}(t) \equiv a(t)q(t)$$

be discounted per capita assets. From (8.8),

$$\dot{\tilde{a}}(t) = q(t)[\dot{a}(t) - (r(t) - n)a(t)] = q(t)[w(t) - c(t)] \leq q(t)w(t), \quad (*)$$

using $c \geq 0$.

Suppose (8.11) fails at some finite t' , that is

$$a(t') < -\int_{t'}^\infty w(s) \exp\left(-\int_{t'}^s (r(z) - n) dz\right) ds,$$

equivalently, multiplying by $q(t') > 0$,

$$\tilde{a}(t') < - \int_{t'}^{\infty} q(s)w(s) ds.$$

Integrating (*) from t' to T and letting $T \rightarrow \infty$,

$$\lim_{T \rightarrow \infty} \tilde{a}(T) \leq \tilde{a}(t') + \int_{t'}^{\infty} q(s)w(s) ds < 0.$$

So the discounted asset position converges to a strictly negative limit, which is precisely what (8.12) rules out: the household ends with debt whose present value does not vanish, so it could not have repaid it even with zero consumption from t' on. Hence (8.12) implies (8.11) at every date, and there is no loss of generality in imposing the single limiting constraint rather than a continuum of them.

Remark. The logic is the same one that makes a single terminal condition suffice in finite horizon: the budget set is defined by the flow constraint plus *one* solvency requirement, because the flow constraint already links all the dates. This is why the book can work with (8.12), and later with the no-Ponzi condition (8.14), rather than with the family (8.11).

Exercise 8.5

Equation (8.13) states, for every $T > 0$,

$$\int_0^T c(t)L(t)e^{\int_t^T r(s)ds} dt + A(T) = \int_0^T w(t)L(t)e^{\int_t^T r(s)ds} dt + A(0)e^{\int_0^T r(s)ds}.$$

Differentiate both sides with respect to T . For a term of the form $\Phi(T) = \int_0^T h(t)e^{\int_t^T r(s)ds} dt$, Leibniz's rule gives

$$\dot{\Phi}(T) = h(T) + r(T)\Phi(T),$$

since differentiating the exponent brings down $r(T)$ inside the integral. Applying this to both integrals and to the last term,

$$c(T)L(T) + r(T) \int_0^T c(t)L(t)e^{\int_t^T r} dt + \dot{A}(T) = w(T)L(T) + r(T) \int_0^T w(t)L(t)e^{\int_t^T r} dt + r(T)A(0)e^{\int_0^T r}.$$

Now use (8.13) itself to replace the two $r(T) \times (\cdot)$ terms: their difference is $r(T)[A(0)e^{\int_0^T r} + \int_0^T wLe^{\int_t^T r} - \int_0^T cLe^{\int_t^T r}] - r(T)A(0)e^{\int_0^T r} = r(T)A(T)$. Hence

$$\dot{A}(T) = r(T)A(T) + w(T)L(T) - c(T)L(T),$$

which is (8.7). Dividing by $L(T)$ and using $a = A/L$ with $\dot{L}/L = n$, as in Exercise 8.1, gives

$$\dot{a}(t) = (r(t) - n)a(t) + w(t) - c(t),$$

which is (8.8). The two formulations are therefore equivalent: the lifetime constraint carried forward to any date T is the integrated version of the flow constraint.

Exercise 8.6

The transversality condition (8.19) of the household problem is

$$\lim_{t \rightarrow \infty} \left[e^{-(\rho-n)t} \mu(t) a(t) \right] = 0.$$

From (8.24) — the integrated form of the costate equation (8.18) — the current-value costate is

$$\mu(t) = \mu(0) \exp\left(-\int_0^t (r(s) - \rho) ds\right), \quad \mu(0) = u'(c(0)) > 0.$$

Hence

$$e^{-(\rho-n)t} \mu(t) = \mu(0) \exp\left(-(\rho-n)t - \int_0^t r(s) ds + \rho t\right) = \mu(0) \exp\left(-\int_0^t (r(s) - n) ds\right),$$

so that (8.19) becomes

$$\mu(0) \lim_{t \rightarrow \infty} \left[a(t) \exp\left(-\int_0^t (r(s) - n) ds\right) \right] = 0.$$

Since $\mu(0) = u'(c(0)) > 0$, the limit itself must vanish:

$$\lim_{t \rightarrow \infty} \left[a(t) \exp\left(-\int_0^t (r(s) - n) ds\right) \right] = 0,$$

which is (8.16).

The role of (8.14) is to make this an equality rather than merely an upper bound. The no-Ponzi condition states that the same limit is ≥ 0 : the household may not end with debt of positive present value. Optimization then rules out the opposite slack: by the transversality condition the household does not end with *assets* of positive present value either, since those could have been consumed. Putting the two together gives (8.16), the statement that the lifetime budget constraint (8.15) holds with equality.

Remark. This is the continuous-time counterpart of the familiar finite-horizon simplification $A(T) \geq 0$ plus local nonsatiation implying $A(T) = 0$. Financial markets impose the inequality; the household's own optimization supplies the equality.

Exercise 8.8

(a) *One direction.* Suppose $\lim_{t \rightarrow \infty} a(t) \geq \hat{a} > -\infty$, and note that in any equilibrium $r(t) \rightarrow r^* > n$ (see (8.36)), so that $\int_0^t (r(s) - n) ds \rightarrow \infty$ and the effective discount factor $q(t) = \exp(-\int_0^t (r(s) - n) ds)$ tends to 0. Then for large t , $a(t) \geq \hat{a} - \varepsilon$, so

$$a(t)q(t) \geq (\hat{a} - \varepsilon)q(t) \rightarrow 0,$$

whence $\lim_{t \rightarrow \infty} a(t)q(t) \geq 0$, which is (8.14). A bounded debt is always repayable in present value.

The converse fails. Take $w(t) = e^{gt}w(0)$ and $r(t) = r$ constant, and consider the asset path

$$a(t) = -Me^{(r-n)t/2}, \quad M > 0.$$

Then $a(t)q(t) = -Me^{-(r-n)t/2} \rightarrow 0$, so (8.14) holds with equality, yet $a(t) \rightarrow -\infty$, so (8.12) cannot hold with any finite $\hat{a}$. Debt that grows, but more slowly than the interest factor, is ruled out by the natural debt limit only if that limit is finite — and the no-Ponzi condition permits it.

(b) *With sustained growth the natural debt limit is vacuous.* The relaxed limit (8.12) bounds assets below by

$$\hat{a} = -\lim_{t \rightarrow \infty} \bar{w}(t), \quad \bar{w}(t) \equiv \int_t^\infty w(s) \exp\left(-\int_t^s (r(z) - n) dz\right) ds.$$

Along the BGP of Section 8.7 wages grow at the rate g and the interest rate is constant at $r^* = \rho + \theta g$, so

$$\bar{w}(t) = \frac{w(t)}{r^* - n - g} = \frac{w(0)e^{gt}}{\rho - n - (1 - \theta)g},$$

which is finite at each date — exactly Assumption 4 — but grows without bound. Hence $\hat{a} = -\lim_{t \rightarrow \infty} \bar{w}(t) = -\infty$: the constraint set is unbounded below, so there is no lower bound on the state variable, and the compactness-type hypotheses behind THEOREMS 7.13 and 7.14 (in particular the requirement in Theorem 7.14 that $\lim e^{-(\rho-n)t} \mu(t)a(t) \geq 0$ for *all* feasible paths) cannot be verified from the natural debt limit. The no-Ponzi condition (8.14) supplies exactly that requirement directly, which is why the book adopts it.

Exercise 8.9

Let $r(t) \rightarrow r^* > n$ and $w(t) \rightarrow w < \infty$. Choose $\varepsilon > 0$ with $r^* - \varepsilon > n$ and a date T such that for all $s \geq T$,

$$r(s) \geq r^* - \varepsilon \quad \text{and} \quad w(s) \leq w + \varepsilon.$$

Then for every $t \geq T$,

$$\bar{w}(t) = \int_t^\infty w(s) \exp\left(-\int_t^s (r(z) - n) dz\right) ds \leq (w + \varepsilon) \int_t^\infty e^{-(r^* - \varepsilon - n)(s-t)} ds = \frac{w + \varepsilon}{r^* - \varepsilon - n} < \infty,$$

a bound independent of t . Hence

$$\hat{a} = -\lim_{t \rightarrow \infty} \bar{w}(t) \geq -\frac{w + \varepsilon}{r^* - \varepsilon - n} > -\infty.$$

The two hypotheses are exactly what is needed, and each is indispensable. If $r^* \leq n$ the geometric discounting at the effective rate $r - n$ fails and the integral diverges even for constant wages; if wages grow without bound — the case of technological progress in Exercise 8.8(b) — the numerator grows faster than the discounting shrinks. In the model without technological progress both hold at the steady state: $r^* = f'(k^*) - \delta = \rho > n$ by Assumption 4', and $w^* = f(k^*) - k^* f'(k^*) < \infty$.

Exercise 8.10

(a) *Same necessary conditions, same solution.* Consider maximizing (8.3) subject to (8.8) and $\lim_{t \rightarrow \infty} a(t) \geq \hat{a}$ with $\hat{a} > -\infty$. The current-value Hamiltonian is the same as in Section 8.2.2,

$$\hat{H}(t, a, c, \mu) = u(c) + \mu[w(t) + (r(t) - n)a - c],$$

and the state is now bounded below, so THEOREM 7.13 applies and yields the same candidate conditions (8.17)–(8.19): $u'(c) = \mu$, $\dot{\mu}/\mu = -(r - \rho)$ and $\lim e^{-(\rho-n)t} \mu(t)a(t) = 0$. Concavity of $\hat{H}$ in (a, c) then makes THEOREM 7.14 applicable, so these conditions are also sufficient.

It remains to check that the extra constraint does not bind. By Exercise 8.6, the transversality condition is equivalent to $\lim a(t)q(t) = 0$ with $q(t) = \exp(-\int_0^t (r - n))$. A path with $\lim a(t) = \hat{a}$ finite also has $\lim a(t)q(t) = 0$, so the constraint is satisfied with slack at the optimum: the household never wants to run its debt down to the natural limit, since doing so would require zero consumption forever, which is dominated whenever $u'(0) > 0$. Hence the two problems have the same solution, characterized by the Euler equation (8.22) and (8.16).

(b) *In equilibrium.* Market clearing gives $a(t) = k(t) \geq 0$, and the equilibrium path converges to the steady state with

$$r^* = f'(k^*) - \delta = \rho > n \quad \text{and} \quad w^* = f(k^*) - k^* f'(k^*) < \infty$$

by Assumption 4' and (8.36). These are precisely the hypotheses of Exercise 8.9, so $\hat{a} > -\infty$: in the model without technological progress the natural debt limit is a legitimate way to close the household problem, and it gives the same answer as the no-Ponzi condition. (With technological progress it is not, by Exercise 8.8(b).)

Exercise 8.12

The two loci in Figure 8.1 are

$$\dot{c} = 0 : \quad f'(k) = \rho + \delta \quad (\text{a vertical line at } k^*), \quad \dot{k} = 0 : \quad c = f(k) - (n + \delta)k.$$

The second is a hump-shaped curve whose maximum is at the golden rule capital-labor ratio, defined by

$$\frac{d}{dk} [f(k) - (n + \delta)k] = 0 \quad \Longleftrightarrow \quad f'(k_{\text{gold}}) = n + \delta.$$

The comparison. Assumption 4' states $\rho > n$, hence

$$f'(k^*) = \rho + \delta > n + \delta = f'(k_{\text{gold}}),$$

and since f is strictly concave, f' is strictly decreasing, so

$$k^* < k_{\text{gold}}.$$

The vertical line therefore always cuts the hump on its *upward-sloping* portion, strictly to the left of the peak. With the parameters of the accompanying program ($\alpha = 1/3$, $\delta = 0.05$, $\rho = 0.02$, $n = 0.01$, $g = 0.02$, $\theta = 1$, so that the relevant comparison is $f'(k^*) = \rho + \delta + \theta g = 0.09$ against $f'(k_{\text{gold}}) = n + g + \delta = 0.08$):

$$k^* = 7.1278 < k_{\text{gold}} = 8.5052.$$

Why they differ. The golden rule maximizes *steady-state* consumption, treating all dates alike. The household does not: it discounts future utility at $\rho > n$, so it is unwilling to give up consumption today in order to raise consumption in a distant steady state. The modified golden rule (8.35) equates the net marginal product of capital to the discount rate rather than to the replacement requirement, and since $\rho > n$ the household stops accumulating short of k_{gold} .

Two consequences are worth recording. First, the neoclassical economy is never dynamically inefficient: because $k^* < k_{\text{gold}}$, there is no way to raise consumption at all dates by cutting capital, in contrast to the Solow model with an arbitrarily high saving rate. Second, the gap between k^* and

k_{gold} is a pure measure of impatience — it closes as $\rho \downarrow n$ and widens as households become less patient.

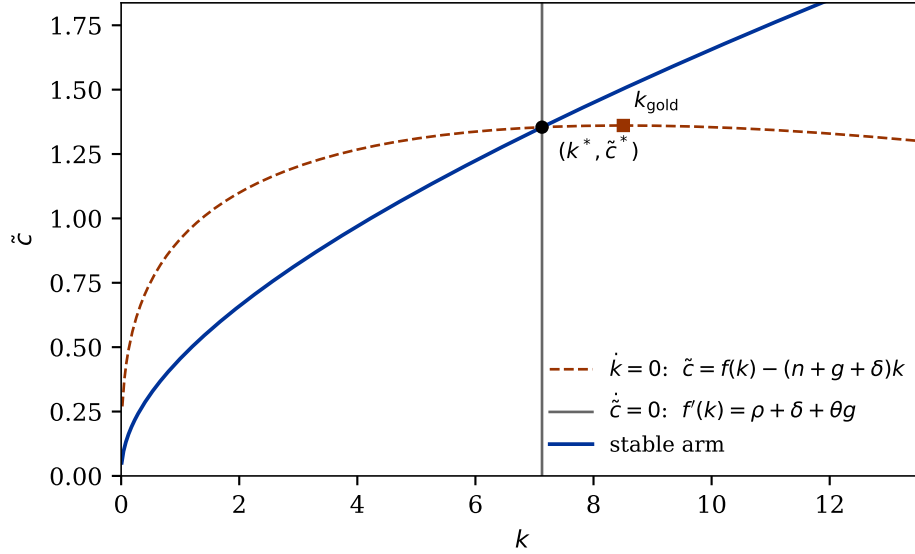


FIGURE 6. The neoclassical phase diagram: the vertical $\dot{c} = 0$ locus cuts the hump-shaped $\dot{k} = 0$ locus strictly to the left of k_{gold} , and the stable arm is the unique equilibrium path.

Exercise 8.14

(a) *What happens at and after T .* Output per capita is $f(k)$ with $f(0) = 0$ (Assumption 2), and capital cannot be negative, so once $k(T) = 0$ the economy has no resources: feasibility forces

$$c(t) = 0 \quad \text{and} \quad k(t) = 0 \quad \text{for all } t \geq T.$$

Along the approach, the interest rate is $r(t) = f'(k(t)) - \delta$, and the Inada condition $\lim_{k \rightarrow 0} f'(k) = \infty$ gives

$$\lim_{t \rightarrow T} r(t) = \infty.$$

Capital becomes infinitely scarce, so its rental rate explodes.

(b) *A profitable deviation.* Consider a household that, at some $t' < T$, reduces consumption by a small $\epsilon > 0$ for an instant and saves the proceeds until T . The utility cost, in date-0 units, is approximately

$$\text{cost} \simeq \epsilon e^{-(\rho-n)t'} u'(c(t')) < \infty,$$

which is finite because $c(t') > 0$ for $t' < T$. The saving accumulates at the market rate, so at T the household has $\epsilon \exp(\int_{t'}^T r(t) dt)$ extra units to consume, at a marginal utility of $u'(0) \geq u'(c(t'))$ (and $u'(0) = \infty$ if u satisfies an Inada condition). The utility gain is approximately

$$\text{gain} \simeq \epsilon \exp\left(-(\rho-n)T + \int_{t'}^T r(t) dt\right) u'(0).$$

Since $u'(0) \geq u'(c(t'))$, the deviation is profitable as soon as

$$\exp\left(-(\rho - n)T + \int_{t'}^T r(t) dt\right) > e^{-(\rho - n)t'} \iff \frac{1}{T - t'} \int_{t'}^T r(t) dt > \rho - n.$$

By part (a), $r(t) \rightarrow \infty$ as $t \rightarrow T$, so the average of r over $[t', T]$ diverges as $t' \uparrow T$, while $\rho - n$ is a fixed finite number. Hence for t' close enough to T the inequality holds strictly: every household strictly prefers to save rather than follow the candidate path.

Therefore the candidate is not consistent with household optimization and cannot be an equilibrium. Equilibrium paths never exhaust the capital stock in finite time — this is the content of the lower branch of Figure 8.1 being ruled out, and it is what leaves the stable arm as the unique equilibrium.

Exercise 8.16

The household now values only consumption *per member*, without weighting by the number of members: it is no longer utilitarian over its own members but treats the representative member's consumption as the object of interest. The budget constraint is unchanged, since it is an accounting identity: $\dot{a} = (r - n)a + w - c$.

Necessary conditions. The current-value Hamiltonian is

$$\hat{H}(t, a, c, \mu) = u(c) + \mu[w + (r - n)a - c],$$

discounted at ρ rather than $\rho - n$, so THEOREM 7.13 gives

$$u'(c) = \mu, \quad \rho\mu - \dot{\mu} = \mu(r - n) \iff \frac{\dot{\mu}}{\mu} = -(r - n - \rho),$$

and hence the Euler equation

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\varepsilon_u(c(t))} (r(t) - n - \rho).$$

The transversality condition becomes

$$\lim_{t \rightarrow \infty} \left[e^{-\rho t} \mu(t) a(t) \right] = 0,$$

with $e^{-\rho t}$ in place of $e^{-(\rho - n)t}$.

Steady state. Setting $\dot{c} = 0$ gives $r^* = \rho + n$, and with $r = f'(k) - \delta$,

$$f'(k^*) = \rho + n + \delta.$$

Comparing with (8.35): population growth now *does* affect the steady state, and

$$\frac{\partial k^*}{\partial n} = \frac{1}{f''(k^*)} < 0,$$

whereas in the baseline model $\partial k^* / \partial n = 0$ (PROPOSITION 8.3). Faster population growth lowers the steady-state capital-labor ratio and hence output per capita.

Why. In the baseline model two effects of population growth cancel exactly: assets must be spread over more people, which lowers the effective return to $r - n$, but the household also values the consumption of its new members, which lowers the effective discount rate to $\rho - n$. Here only the first effect is present — the household does not care that there are more members to enjoy consumption — so saving must earn $\rho + n$ rather than ρ to be worthwhile, and the capital stock is

correspondingly lower. Finally, discounting now requires only $\rho > 0$ rather than $\rho > n$: Assumption 4' can be weakened accordingly.

Exercise 8.17

With $f(k) = A\tilde{f}(k)$ the steady state (8.35) is

$$A\tilde{f}'(k^*) = \rho + \delta, \quad (*)$$

and steady-state consumption is $c^* = A\tilde{f}(k^*) - (n + \delta)k^*$ by (8.37).

The capital-labor ratio. Differentiating (*) implicitly, and using $\tilde{f}'' < 0$,

$$\frac{\partial k^*}{\partial A} = -\frac{\tilde{f}'(k^*)}{A\tilde{f}''(k^*)} > 0, \quad \frac{\partial k^*}{\partial \rho} = \frac{\partial k^*}{\partial \delta} = \frac{1}{A\tilde{f}''(k^*)} < 0, \quad \frac{\partial k^*}{\partial n} = 0,$$

the last because n does not appear in (*) at all: the effective return $r - n$ and the effective discount rate $\rho - n$ shift together, as Exercise 8.16 makes explicit.

Consumption. Differentiating c^* and using (*) to substitute $A\tilde{f}'(k^*) = \rho + \delta$,

$$\frac{\partial c^*}{\partial x} = [A\tilde{f}'(k^*) - (n + \delta)] \frac{\partial k^*}{\partial x} + \frac{\partial}{\partial x} \Big|_{k^* \text{ fixed}} = (\rho - n) \frac{\partial k^*}{\partial x} + \frac{\partial}{\partial x} \Big|_{k^* \text{ fixed}},$$

where the second term collects the direct effect. Since $\rho > n$ by Assumption 4', the coefficient $(\rho - n)$ is strictly positive, so the indirect effect always has the same sign as $\partial k^* / \partial x$. Hence

$$\begin{aligned} \frac{\partial c^*}{\partial A} &= \tilde{f}(k^*) + (\rho - n) \frac{\partial k^*}{\partial A} > 0, & \frac{\partial c^*}{\partial \rho} &= (\rho - n) \frac{\partial k^*}{\partial \rho} < 0, \\ \frac{\partial c^*}{\partial n} &= -k^* < 0, & \frac{\partial c^*}{\partial \delta} &= (\rho - n) \frac{\partial k^*}{\partial \delta} - k^* < 0. \end{aligned}$$

This is PROPOSITION 8.3.

Remark. The positivity of $\rho - n$ is doing real work here, and it is the same fact as Exercise 8.12: because $k^* < k_{\text{gold}}$, the economy is on the upward-sloping part of the steady-state consumption curve, so anything that raises k^* also raises c^* . In the Solow model, where the saving rate is a free parameter, this is not guaranteed — an economy above the golden rule raises consumption by saving *less* (Exercise 8.19 pursues this contrast).

Exercise 8.18

Without technological progress. The Euler equation (8.22) is

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\varepsilon_u(c(t))} (r(t) - \rho).$$

In a steady state consumption per capita is constant, $\dot{c} = 0$; since $\varepsilon_u(c)$ is finite and positive, the *only* way to have $\dot{c} = 0$ is

$$r^* = \rho \quad \Longleftrightarrow \quad f'(k^*) = \rho + \delta,$$

whatever the shape of u . The utility function drops out because it enters the Euler equation only through the factor $1/\varepsilon_u$ multiplying $r - \rho$, and multiplying zero by any finite number gives zero. What u does affect is the *speed* of the transition: ε_u governs how fast consumption responds to a given gap between r and ρ , and hence the eigenvalue of the linearized system.

With technological progress. Now per capita consumption grows at the rate g in the BGP, so the left-hand side of the Euler equation is no longer zero. With CRRA preferences (8.49), $\varepsilon_u \equiv \theta$, and a steady state with $\dot{c}/c = g$ requires

$$g = \frac{1}{\theta}(r^* - \rho) \quad \Longleftrightarrow \quad r^* = \rho + \theta g \quad \Longleftrightarrow \quad f'(k^*) = \rho + \delta + \theta g,$$

which is (8.53): the elasticity now appears. The economics is transparent. A growing consumption path means that marginal utility is falling over time, and the rate at which it falls is θg . The household is willing to hold capital only if its return compensates it both for impatience, ρ , and for the fact that the extra consumption arrives when it is already better off, θg . The less willing the household is to substitute consumption across time (the larger θ , the smaller the elasticity $1/\theta$), the larger the required return and the smaller the steady-state effective capital-labor ratio.

In one sentence: preferences matter for the steady state only when marginal utility is changing along it, and that happens exactly when there is growth.

Exercise 8.20

On the BGP consumption per capita grows at the constant rate g , so $c(t) = c(0)e^{gt}$, and with CRRA preferences (8.49) the objective is

$$U = \int_0^\infty e^{-(\rho-n)t} \frac{c(t)^{1-\theta} - 1}{1-\theta} dt = \frac{c(0)^{1-\theta}}{1-\theta} \int_0^\infty e^{-[(\rho-n)-(1-\theta)g]t} dt - \frac{1}{1-\theta} \int_0^\infty e^{-(\rho-n)t} dt.$$

The second integral converges if and only if $\rho > n$, which is Assumption 4' and is implied by Assumption 4 whenever $\theta \leq 1$; the first converges if and only if its exponent is negative:

$$(\rho - n) - (1 - \theta)g > 0 \quad \Longleftrightarrow \quad \rho - n > (1 - \theta)g,$$

which is exactly Assumption 4. When it holds,

$$U = \frac{c(0)^{1-\theta}}{(1-\theta)[\rho - n - (1-\theta)g]} - \frac{1}{(1-\theta)(\rho - n)} < \infty;$$

when it fails, the first integral diverges, and $U = +\infty$ for $\theta < 1$ (so no allocation can be optimal and the maximization problem is not well posed).

For the logarithmic case $\theta = 1$ the computation is direct: $u(c(t)) = \log c(0) + gt$, so

$$U = \int_0^\infty e^{-(\rho-n)t} [\log c(0) + gt] dt = \frac{\log c(0)}{\rho - n} + \frac{g}{(\rho - n)^2},$$

finite if and only if $\rho > n$, which is Assumption 4 with $\theta = 1$ since $(1 - \theta)g = 0$.

Remark. The condition has a clean reading: the effective discount rate $\rho - n$ must exceed the growth rate of *utility*, which is $(1 - \theta)g$ rather than g itself, because a household with a low intertemporal elasticity (θ large) values additional consumption less as it grows richer. For $\theta \geq 1$ Assumption 4 is implied by $\rho > n$, which is Exercise 8.29.

Exercise 8.21

The problem is to maximize (8.49) subject to (8.8) and the no-Ponzi condition (8.14). The current-value Hamiltonian is

$$\widehat{H}(t, a, c, \mu) = \frac{c^{1-\theta} - 1}{1-\theta} + \mu(t)[w(t) + (r(t) - n)a - c].$$

THEOREM 7.14 requires three things.

- (1) *A well-defined, finite objective.* By Exercise 8.20, Assumption 4 is exactly the condition under which the integral converges along the BGP (and hence along any path converging to it), so the value function is finite.
- (2) *Concavity.* $\widehat{H}$ is the sum of a strictly concave function of c and a function that is linear in (a, c) , hence concave in (a, c) and strictly concave in c . By Exercise 7.7 the maximized Hamiltonian is concave in the state, so the concavity hypothesis holds.
- (3) *The limiting condition.* THEOREM 7.14 needs $\lim_{t \rightarrow \infty} e^{-(\rho-n)t} \mu(t)a(t) \geq 0$ for every feasible $(a(t), c(t))$. Substituting $\mu(t) = \mu(0) \exp(-\int_0^t (r - \rho) ds)$ as in Exercise 8.6, this is

$$\lim_{t \rightarrow \infty} \left[a(t) \exp\left(-\int_0^t (r(s) - n) ds\right) \right] \geq 0,$$

which is precisely the no-Ponzi condition (8.14): no feasible path violates it, by construction of the constraint set.

Hence any path satisfying the necessary conditions (8.17)–(8.19) is a global maximum, and by strict concavity in c it is the unique one.

If Assumption 4 fails. Then $\rho - n \leq (1 - \theta)g$ and, by Exercise 8.20, utility along the BGP is infinite. Every feasible path that grows at rate g yields $+\infty$, so there is no meaningful ranking of allocations: the maximization problem has no solution in the usual sense, the value function is not finite, and THEOREM 7.14 — whose statement presumes a finite objective — cannot be applied. The transversality condition (8.52) also fails, since the exponent $\rho - n - (1 - \theta)g$ is no longer positive. Economically, the household is too patient relative to the growth rate of the economy: it would always postpone consumption, and no equilibrium with a well-defined intertemporal allocation exists.

Exercise 8.22

With labor-augmenting progress at rate g and CRRA preferences, the equilibrium is characterized by the planar system

$$\dot{k}(t) = f(k(t)) - \tilde{c}(t) - (n + g + \delta)k(t), \quad \frac{d\tilde{c}/dt}{\tilde{c}} = \frac{1}{\theta} [f'(k(t)) - \delta - \rho - \theta g], \quad (*)$$

with $k(0) > 0$ given and the transversality condition (8.52).

Step 1: a unique steady state with positive consumption. Setting the second equation to zero gives $f'(k^*) = \rho + \delta + \theta g$, which has a unique positive solution because f' is continuous, strictly decreasing and, by Assumption 2, ranges over $(0, \infty)$. Setting the first to zero at k^* gives $\tilde{c}^* =$

$f(k^*) - (n + g + \delta)k^*$, and this is strictly positive because

$$f'(k^*) = \rho + \delta + \theta g > n + g + \delta \quad \Longleftrightarrow \quad \rho - n > (1 - \theta)g,$$

which is Assumption 4. So $k^* < k_{\text{gold}}$ and the steady state lies on the upward-sloping part of the $\dot{k} = 0$ locus, where $f(k) - (n + g + \delta)k > 0$.

Step 2: the steady state is a saddle. The Jacobian of $(*)$ at $(k^*, \tilde{c}^*)$ is

$$J = \begin{pmatrix} f'(k^*) - (n + g + \delta) & -1 \\ \frac{\tilde{c}^* f''(k^*)}{\theta} & 0 \end{pmatrix}, \quad \text{tr } J = \rho - n - (1 - \theta)g > 0, \quad \det J = \frac{\tilde{c}^* f''(k^*)}{\theta} < 0,$$

using $f'' < 0$ and $\tilde{c}^* > 0$. A negative determinant means the two eigenvalues are real and of opposite sign,

$$\xi_1 = \frac{1}{2} \left[\text{tr } J - \sqrt{(\text{tr } J)^2 - 4 \det J} \right] < 0 < \xi_2,$$

so the steady state is a saddle point with a one-dimensional stable manifold. With the parameters of the program ($\alpha = 1/3$, $\delta = 0.05$, $\rho = 0.02$, $n = 0.01$, $g = 0.02$, $\theta = 1$) the eigenvalues are -0.1019 and $+0.1119$.

Step 3: the stable arm is the unique equilibrium. Fix $k(0) > 0$. There is exactly one value $\tilde{c}(0)$ on the stable manifold, and starting from it the path converges to $(k^*, \tilde{c}^*)$. Any other choice of $\tilde{c}(0)$ leads off the manifold, and the phase diagram leaves only two possibilities:

- $\tilde{c}(0)$ too high: the path crosses the $\dot{k} = 0$ locus and capital falls, reaching $k = 0$ in finite time. By Exercise 8.14 such a path cannot be an equilibrium — households would deviate, since the interest rate diverges as capital vanishes.
- $\tilde{c}(0)$ too low: consumption falls to zero while capital rises above k_{gold} toward the level $\bar{k}$ with $f'(\bar{k}) < \delta + g + n$. Then the exponent in the transversality condition (8.52) stays bounded away from $-\infty$, so $\lim k(t) \exp(-\int_0^t [f'(k) - g - \delta - n]) > 0$: the condition fails, and by THEOREM 7.14 the path is dominated. It is also plainly wasteful — the economy accumulates capital it never consumes.

Hence for each $k(0) > 0$ there is a unique equilibrium, the one starting on the stable arm.

Step 4: convergence is monotone. Along the stable arm the sign of $\dot{k}$ does not change: the arm lies below the $\dot{k} = 0$ locus for $k < k^*$ and above it for $k > k^*$, and it cannot cross either locus without leaving the manifold. Therefore $k(t)$ moves monotonically to k^* and, since $\tilde{c}$ is an increasing function of k along the arm (its slope is $\text{tr } J - \xi_1 > 0$, equal to 0.1119 in the numerical example, which the program confirms against a numerically traced arm, 0.1123), normalized consumption converges monotonically to $\tilde{c}^*$ as well. This establishes PROPOSITION 8.7.

Exercise 8.24

Set $\theta = 1$ in (8.55) and (8.56). With $x \equiv k^{\alpha-1}$ (so that $x = f(k)/k$ is the average product of capital) and $z \equiv \tilde{c}/k$, the system becomes

$$\frac{\dot{x}}{x} = -(1 - \alpha)[x - \delta - g - n - z], \quad \frac{\dot{z}}{z} = z - (1 - \alpha)x - (\rho - n). \quad (*)$$

The steady state in closed form. From the second equation $\dot{z} = 0$ gives $z^* = (1 - \alpha)x^* + \rho - n$, and from the first $x^* = \delta + g + n + z^*$. Solving the two,

$$x^* = \frac{\rho + \delta + g}{\alpha}, \quad z^* = x^* - (\delta + g + n) = \frac{\rho + \delta + g}{\alpha} - (\delta + g + n),$$

equivalently $k^* = [\alpha/(\rho + \delta + g)]^{1/(1-\alpha)}$ and $\tilde{c}^* = f(k^*) - (n + g + \delta)k^*$, which is (8.53)–(8.54) with $\theta = 1$.

The transition in closed form. The system (*) is a planar quadratic system and does not have an elementary general solution, but its behavior near the BGP does, and that is what the transitional dynamics require. Linearizing the original system in $(k, \tilde{c})$ at the steady state, as in Exercise 8.22 with $\theta = 1$,

$$\begin{aligned} \text{tr } J &= \rho - n, & \det J &= \tilde{c}^* f''(k^*) = -\frac{(1 - \alpha)(\rho + \delta + g)\tilde{c}^*}{k^*}, \\ \xi_{1,2} &= \frac{(\rho - n) \mp \sqrt{(\rho - n)^2 + 4(1 - \alpha)(\rho + \delta + g)\tilde{c}^*/k^*}}{2}, & \xi_1 &< 0 < \xi_2, \end{aligned}$$

and along the stable arm the solution is, in closed form,

$$\boxed{k(t) = k^* + (k(0) - k^*)e^{\xi_1 t}, \quad \tilde{c}(t) = \tilde{c}^* + (\rho - n - \xi_1)(k(0) - k^*)e^{\xi_1 t}}$$

the coefficient $\rho - n - \xi_1 = \text{tr } J - \xi_1 > 0$ being the slope of the stable arm, obtained from the eigenvector of J associated with ξ_1 . The speed of convergence is $|\xi_1|$, so the half-life of the transition is $\log 2/|\xi_1|$. With $\alpha = 1/3$, $\delta = 0.05$, $\rho = 0.02$, $n = 0.01$, $g = 0.02$:

$$k^* = 7.1278, \quad \tilde{c}^* = 1.3543, \quad \xi_1 = -0.1019, \quad \xi_2 = +0.1119, \quad \text{slope} = 0.1119, \quad \text{half-life} = 6.80.$$

The program checks the closed-form slope against a numerically traced stable arm (0.1123).

Remark. There is one parameter configuration in which (*) can be solved exactly rather than only near the BGP: when $\theta = \alpha$, the term $(\alpha - \theta)x$ in (8.56) vanishes and the z equation decouples,

$$\frac{\dot{z}}{z} = z - \frac{\rho}{\alpha} + n - \frac{(1 - \alpha)\delta}{\alpha},$$

a logistic equation with an explicit solution, after which $x(t)$ follows by quadrature. This is the closed-form case identified by Smith (2006); with logarithmic preferences ($\theta = 1$) the coupling is present and the linearized solution above is the appropriate closed form.

Exercise 8.26

The problem is

$$\max \sum_{t=0}^{\infty} \beta^t \log c(t) \quad \text{s.t.} \quad k(t+1) = k(t)^\alpha - c(t), \quad k(0) > 0,$$

full depreciation making the resource constraint this simple.

Euler equation and steady state. The Euler equation (6.50) is

$$\frac{1}{c(t)} = \beta \alpha k(t+1)^{\alpha-1} \frac{1}{c(t+1)}.$$

In a steady state $c(t) = c^*$ and $k(t) = k^*$, so $1 = \beta\alpha(k^*)^{\alpha-1}$ and

$$k^* = (\alpha\beta)^{\frac{1}{1-\alpha}}, \quad c^* = (k^*)^\alpha - k^* = (\alpha\beta)^{\frac{\alpha}{1-\alpha}}(1 - \alpha\beta).$$

This is the discrete-time counterpart of (8.35): the marginal product of capital, $\alpha(k^*)^{\alpha-1}$, equals the gross discount rate $1/\beta$.

The exact policy rule. Guess that a constant fraction of output is saved, $k(t+1) = sk(t)^\alpha$, so that $c(t) = (1-s)k(t)^\alpha$. Substituting into the Euler equation,

$$\frac{1}{(1-s)k(t)^\alpha} = \beta\alpha(sk(t)^\alpha)^{\alpha-1} \frac{1}{(1-s)(sk(t)^\alpha)^\alpha} \implies s = \alpha\beta,$$

after cancelling. So the exact optimal policy is

$$\boxed{c(t) = (1 - \alpha\beta)k(t)^\alpha, \quad k(t+1) = \alpha\beta k(t)^\alpha}$$

— the saving rate is constant at $\alpha\beta$, exactly as in the Solow model but here derived rather than assumed. (That the guess is the solution follows from uniqueness: the policy function of the concave dynamic programming problem is unique by COROLLARY 6.1, and this candidate satisfies the Euler equation and the transversality condition.)

The transition, exactly. Taking logarithms of the law of motion,

$$\log k(t+1) = \log(\alpha\beta) + \alpha \log k(t) \implies \log k(t) - \log k^* = \alpha^t [\log k(0) - \log k^*],$$

since $\log k^* = \log(\alpha\beta)/(1-\alpha)$ is the fixed point. The economy converges monotonically to k^* , and the log gap closes at the exact geometric rate α per period — a rare case in which the entire transition, not merely its linearization, is available in closed form. With $\alpha = 1/3$, $\beta = 0.96$ and $k(0) = 0.2k^*$ the iterated and closed-form paths agree to every printed digit ($k^* = 0.181019$, $k(1) = 0.105861$, $k(5) = 0.179824$).

Remark. The knife-edge assumption is full depreciation: with $\delta < 1$ the resource constraint becomes $k(t+1) = k(t)^\alpha + (1-\delta)k(t) - c(t)$, the constant-saving-rate guess fails, and no closed form is known. It is worth noting that here the neoclassical model reproduces the Solow model exactly, with the saving rate pinned down by preferences as $\alpha\beta$ rather than left free.

Exercise 8.28

(a) *Faster technological progress.* The steady state (8.53) is $f'(k^*) = \rho + \delta + \theta g$, so a rise from g to $g' > g$ raises the required marginal product and *lowers* the steady-state effective capital-labor ratio k^* ; it also lowers $\tilde{c}^* = f(k^*) - (n + g + \delta)k^*$, both because k^* falls (and the economy is to the left of the golden rule, where the curve is increasing) and because the replacement requirement $n + g + \delta$ rises. The growth rate of consumption per capita rises from g to g' .

On impact, k is predetermined and $\tilde{c}$ jumps onto the new stable arm. Since the economy starts at the old k^* , which exceeds the new k^* , it is to the *right* of the new steady state, and the arm is upward sloping there: normalized consumption therefore jumps above its new steady-state level, and k falls monotonically toward the new k^* .

Whether *actual* consumption $c(0) = A(0)\tilde{c}(0)$ rises or falls is genuinely ambiguous, and the ambiguity is the familiar one between income and substitution effects:

- faster growth makes the household richer in present-value terms, which pushes consumption up now (income effect);
- it also raises the equilibrium interest rate to $r^* = \rho + \theta g'$, making future consumption cheaper, which pushes consumption down now (substitution effect).

With CRRA preferences the two are balanced by θ : the substitution effect dominates when $\theta < 1$ (intertemporal elasticity $1/\theta > 1$), so consumption falls on impact and saving rises; the income effect dominates when $\theta > 1$, so consumption rises; and for $\theta = 1$ (log) the two exactly cancel in the sense that the consumption-wealth ratio is unchanged.

(b) *An announced future tax cut.* Assume, as in Section 8.8, that the tax proceeds are rebated lump-sum, so that there is no wealth effect and only the intertemporal price changes. With a capital tax the steady state (8.57) is

$$f'(k^*) = \frac{\delta + \rho + \theta g}{1 - \tau},$$

so a lower τ' lowers the required marginal product and *raises* k^* and $\tilde{c}^*$.

Between T and T' the old dynamics still apply — the tax is still τ — but the path must arrive exactly on the *new* stable arm at T' , since from then on the new dynamics govern and only the new arm converges. The new arm passes through a steady state with higher k^* , so at T' the economy must have accumulated capital. This requires $\dot{k} > 0$ between T and T' , that is, consumption below the $\dot{k} = 0$ locus. Hence at the moment of the announcement consumption *jumps down*: households start saving immediately in anticipation of the higher after-tax return, capital rises between T and T' , and after T' the economy travels up the new stable arm to the new steady state, with consumption rising throughout and ending above its original level.

The contrast between the two parts is instructive: in (a) the change is immediate and the sign of the impact effect depends on θ ; in (b) the change is anticipated, so the whole adjustment before T' is driven by the substitution effect alone, and its sign is unambiguous.

Exercise 8.29

With technological progress, the transversality condition (8.52) evaluated along the BGP reduces, as the book shows, to the requirement that

$$\rho - n - (1 - \theta)g > 0,$$

which is Assumption 4. If $\theta > 1$ then $(1 - \theta)g < 0$, so the requirement becomes

$$\rho - n > (1 - \theta)g, \quad \text{with a negative right-hand side,}$$

which is implied by Assumption 4', $\rho > n$ — the condition needed even in the model without growth. So for $\theta > 1$ Assumption 4 imposes nothing beyond ordinary discounting, and the transversality condition holds automatically.

The reason is visible in the utility function. For $\theta > 1$ the CRRA felicity $(c^{1-\theta} - 1)/(1 - \theta)$ is negative and *bounded above* by $1/(\theta - 1)$: as consumption grows without bound, flow utility rises toward a finite ceiling. Growth therefore cannot make the discounted integral diverge, and the only requirement is that the discount factor itself be summable, $\rho > n$. For $\theta < 1$ the felicity is

unbounded above and grows like $c^{1-\theta} = c(0)^{1-\theta} e^{(1-\theta)gt}$, so growth races against discounting and the extra condition $\rho - n > (1 - \theta)g$ is needed; see Exercise 8.20.

Remark. Empirically this is reassuring rather than restrictive: most estimates of the intertemporal elasticity of substitution $1/\theta$ lie below one, that is $\theta > 1$, so the canonical model's parameter restrictions are satisfied in the range of interest.

Exercise 8.32

Each country j is a closed neoclassical economy with the same technology and the same δ , n , g and θ , differing only in its discount rate ρ_j . Its BGP effective capital-labor ratio solves

$$f'(k_j^*) = \rho_j + \delta + \theta g,$$

so with the Cobb–Douglas form $f(k) = k^\alpha$,

$$k_j^* = \left[\frac{\alpha}{\rho_j + \delta + \theta g} \right]^{\frac{1}{1-\alpha}}, \quad y_j^* = (k_j^*)^\alpha = \left[\frac{\alpha}{\rho_j + \delta + \theta g} \right]^{\frac{\alpha}{1-\alpha}}.$$

All countries grow at the same rate g ; only the *level* of income per capita differs, and

$$\frac{y_j^*}{y_{j'}^*} = \left[\frac{\rho_{j'} + \delta + \theta g}{\rho_j + \delta + \theta g} \right]^{\frac{\alpha}{1-\alpha}}, \quad \frac{d \log y^*}{d \log (\rho + \delta + \theta g)} = -\frac{\alpha}{1-\alpha}.$$

With a capital share of about $\alpha = 1/3$ the elasticity is only $-1/2$, and it applies to $\rho + \delta + \theta g$, not to ρ alone — which is what makes the effect small.

The numbers. Take $\delta = 0.05$, $g = 0.02$, $\theta = 1$, and compare $\rho = 0.02$ with $\rho' = 0.022$ — a 10% difference in patience:

$$\frac{y^*(\rho')}{y^*(\rho)} = \left(\frac{0.090}{0.092} \right)^{1/2} = 0.9891,$$

a difference of only 1.1%. Even with no depreciation and no growth ($\delta = g = 0$), where the discount rate is the whole of the required return,

$$\frac{y^*(\rho')}{y^*(\rho)} = \left(\frac{0.020}{0.022} \right)^{1/2} = 0.9535,$$

a difference of 4.7%.

Interpretation. Preference heterogeneity of a plausible size cannot account for cross-country income differences: a tenfold income gap would require $(\rho_{j'} + \delta + \theta g)/(\rho_j + \delta + \theta g) = 10^2 = 100$, that is discount rates differing by orders of magnitude. The reason is the same diminishing-returns arithmetic that limits the Solow model's ability to explain income differences through saving rates: with a capital share of one third, the elasticity of steady-state income with respect to any determinant of the capital stock is only $\alpha/(1 - \alpha) = 1/2$. This is precisely why the book turns from proximate to fundamental causes, and later to models in which technology itself differs across countries.

Exercise 8.35

The two budget constraints. The government finances a constant flow of spending g per capita with lump-sum taxes $T(t)$ and debt $b(t)$:

$$\dot{b}(t) = r(t)b(t) + g - T(t), \quad \lim_{t \rightarrow \infty} \left[b(t) \exp\left(-\int_0^t (r(s) - n) ds\right) \right] = 0.$$

Integrating forward and using the no-Ponzi condition gives the government's intertemporal budget constraint,

$$\int_0^\infty T(t)q(t) dt = b(0) + \int_0^\infty g q(t) dt, \quad q(t) \equiv \exp\left(-\int_0^t (r(s) - n) ds\right). \quad (*)$$

The household holds assets $a(t) = k(t) + b(t)$ and faces

$$\dot{a}(t) = (r(t) - n)a(t) + w(t) - T(t) - c(t),$$

so, integrating forward with its own no-Ponzi condition,

$$\int_0^\infty c(t)q(t) dt = a(0) + \int_0^\infty [w(t) - T(t)]q(t) dt. \quad (**)$$

The proof. Substitute $(*)$ into $(**)$ and use $a(0) = k(0) + b(0)$:

$$\int_0^\infty c(t)q(t) dt = k(0) + b(0) + \int_0^\infty w(t)q(t) dt - \left[b(0) + \int_0^\infty g q(t) dt \right] = k(0) + \int_0^\infty [w(t) - g]q(t) dt.$$

The path of taxes has disappeared, and so has the initial debt: the household's lifetime budget constraint depends only on the initial *capital* stock and on the present value of wages net of government spending. The household's problem — maximize (8.3) subject to this constraint — is therefore unchanged by any reshuffling of $[T(t)]$ that satisfies $(*)$, so its consumption path is unchanged; and since $k(t) = a(t) - b(t)$ and the resource constraint $\dot{k} = f(k) - c - g - (n + \delta)k$ does not involve taxes either, the equilibrium path of the capital-labor ratio is unchanged as well. This is Ricardian equivalence.

Interpretation. Government debt is not net wealth. A tax cut today financed by borrowing must be paid for by higher taxes later whose present value is exactly equal; the infinitely lived household sees this, saves the entire tax cut, and leaves its consumption plan untouched. Only the *present value* of government spending matters, not how it is financed.

The result is a benchmark, not a description, and it is worth being explicit about what it assumes: taxes are lump-sum (distortionary taxes change relative prices, as in Exercise 8.39); the horizon is infinite with full altruism (in the overlapping generations models of Chapter 9 the future taxpayers are different people, and the equivalence fails); capital markets are perfect (binding borrowing constraints make the timing of taxes matter, as in Exercise 6.13); and there is no uncertainty about the incidence of future taxes.

Exercise 8.36

Capital jumps immediately. Firms are competitive and can rent as much capital as they like at the world rate, so the domestic marginal product must equal the world cost of capital at every instant:

$$f'(k(t)) - \delta = r^* \quad \implies \quad k(t) = k^o \equiv (f')^{-1}(r^* + \delta) \quad \text{for all } t > 0.$$

The capital-labor ratio is pinned down by the world interest rate alone, independently of preferences and of the country's own wealth, and it is reached *immediately*: if $k(0) < k^o$ the economy imports the difference on the first instant, borrowing $k^o - k(0)$ from abroad. There are no transitional dynamics in capital or in output — the entire adjustment happens at $t = 0$ — which is the sharpest contrast with the closed economy, where convergence takes decades.

Consumption. The Euler equation is $\dot{c}/c = (r^* - \rho)/\theta$, so consumption grows, is constant or falls according to whether $r^* \gtrless \rho$. A steady state with constant consumption therefore requires the knife-edge condition

$$r^* = \rho,$$

which is the standard assumption in small open economy models; otherwise the country accumulates foreign assets without bound ($r^* > \rho$) or runs its consumption down to zero ($r^* < \rho$), and no steady state exists. Assume $r^* = \rho$. Then $c(t) = c^*$ is constant and is determined by the country's intertemporal budget constraint: with net foreign debt $d(t) = k(t) - a(t)$ and the no-Ponzi condition,

$$c^* = (r^* - n) \left[a(0) + \int_0^\infty [w(t)] e^{-(r^* - n)t} dt \right],$$

that is, consumption equals the annuity value of national wealth — initial assets plus the present value of labor income.

How the debt is repaid. It is not repaid in the sense of being extinguished; it is *serviced* forever out of the higher output the imported capital makes possible. In the steady state,

$$c^* = f(k^o) - (n + \delta)k^o - (r^* - n)d,$$

so the country runs a permanent trade surplus of $(r^* - n)d$ — the interest bill net of the growth of the population over which it is spread — while its debt-to-labor ratio d stays constant. The no-Ponzi condition requires exactly this: the present value of the debt must go to zero, which permits debt growing at rate n but not faster.

Remark. The comparison with the closed economy is the point of the exercise. Openness removes the *capital accumulation* channel entirely: income per worker is $f(k^o)$ from the first instant, so cross-country differences in patience or initial capital show up in *consumption* and in net foreign asset positions, not in output. This is why the empirical failure of convergence is so hard to reconcile with open capital markets — the Lucas paradox of Chapter 1.

Exercise 8.39

The Solow model. Saving is a fixed fraction s of income, whatever the rate of return. A tax on capital income at the rate τ , rebated lump-sum to households, changes the *distribution* of income but not its total: output is still $f(k)$, and total saving is still $sf(k)$. The fundamental law of motion

is therefore unchanged,

$$\dot{k}(t) = sf(k(t)) - (n + \delta)k(t),$$

and so is the steady state $f(k^*)/k^* = (n + \delta)/s$. The tax is neutral. (If instead the revenue were thrown away or spent on something outside the model, the effect would come through the resource constraint, not through behavior; the point is that no *behavioral* margin responds.)

The neoclassical model. Here saving is chosen, and what the household compares is its discount rate with the *after-tax* return

$$r(t) = (1 - \tau)f'(k(t)) - \delta.$$

The Euler equation (8.50) then gives, in a BGP with consumption growing at g , $r^* = \rho + \theta g$, so that

$$f'(k^*) = \frac{\delta + \rho + \theta g}{1 - \tau},$$

which is (8.57). Since f' is strictly decreasing, a higher τ raises the required pre-tax marginal product and therefore *lowers* k^* , output per effective worker, and consumption. With $\alpha = 1/3$ the elasticity is again $\alpha/(1 - \alpha) = 1/2$, so a tax that raises the required return by 10% lowers steady-state income by about 5%.

Why the difference. In the Solow model the saving rate is a behavioral primitive that does not respond to prices, so a tax that changes the return to capital changes nothing that anyone decides. In the neoclassical model saving is the outcome of an intertemporal optimization, and the after-tax return *is* the relative price of future against present consumption. Taxing it drives a wedge between the marginal product of capital — what an extra unit of capital produces — and the rate at which households can convert consumption today into consumption tomorrow. Households equate their marginal rate of substitution to the after-tax return, firms equate the marginal product to the pre-tax return, and the wedge between the two shows up as a permanently lower capital stock.

This is a general lesson rather than a fact about capital taxes: whenever a model replaces a behavioral rule by optimization, policies that alter intertemporal prices acquire effects they did not have before. It is also why the neoclassical model, unlike the Solow model, can speak to the effects of policy and institutions on income levels — the theme the book takes up in Section 8.8 and in Part VIII.

Chapter 9

Growth with Overlapping Generations

The twenty-one exercises of Chapter 9 left unsolved in the instructor's manual: Exercises 9.2, 9.4, 9.5, 9.9–9.14, 9.18, 9.19, 9.22, 9.23, 9.25–9.31 and 9.33. The manual solves 9.1, 9.3, 9.6–9.8, 9.15–9.17, 9.20, 9.21, 9.24 and 9.32. Notation follows the book: $c_1(t)$ and $c_2(t+1)$ are the consumption of a generation born at t when young and old, $s(t)$ its saving, $k = K/L$ the capital-labor ratio, $R = 1 + r$ the gross return, ν the flow rate of death, n the arrival rate of new households, and $c(t | \tau)$ the consumption at t of a cohort born at τ . Capital depreciates fully in the two-period model ($\delta = 1$) unless stated otherwise. The numerical parts are computed by `code/ch9.py`.

Exercise 9.2

Recall Shell's economy of Section 9.1: households and commodities are both indexed by $i \in \mathbb{N}$, household i has preferences $u^i = c_i^i + c_{i+1}^i$, and the reallocated endowment vector $\tilde{\omega}^{i'}$ gives

- household $i < i'$: one unit of good i ; household i' : one unit each of goods i' and $i' + 1$;
household $i > i'$: one unit of good $i + 1$.

The allocation to be supported is $\tilde{x}^{i'}$: households $i < i'$ consume good i , household i' consumes one unit each of goods i' and $i' + 1$, and each household $i > i'$ consumes one unit of good $i + 1$.

Household optimization at $\tilde{p}$. Preferences are linear, so household i spends its entire income on whichever of the two goods it likes, i and $i + 1$, is cheaper, and is indifferent between them when their prices coincide.

- $i < i'$: both goods i and $i + 1$ satisfy $j \leq i'$, so $\tilde{p}_i = \tilde{p}_{i+1} = 1$. Income is $\tilde{p}_i = 1$, and the household is indifferent; consuming its own endowment, one unit of good i , is optimal.
- $i = i'$: $\tilde{p}_{i'} = 1$ and $\tilde{p}_{i'+1} = \rho^0 = 1$, so income is $\tilde{p}_{i'} + \tilde{p}_{i'+1} = 2$ and the two goods cost 1 each. Buying one unit of each is optimal.
- $i > i'$: the endowment is one unit of good $i + 1$, so income is $\tilde{p}_{i+1} = \rho^{i-i'}$, while the prices of the two goods the household cares about are $\tilde{p}_i = \rho^{i-i'-1}$ and $\tilde{p}_{i+1} = \rho^{i-i'}$. Since $\rho < 1$, good $i + 1$ is *strictly* cheaper, so the household spends everything on it and buys

$$\frac{\tilde{p}_{i+1}}{\tilde{p}_{i+1}} = 1 \text{ unit of good } i + 1,$$

which is exactly its assignment in $\tilde{x}^{i'}$, and now the optimum is unique.

Market clearing. $\tilde{x}^{i'}$ is the same allocation as in PROPOSITION 9.3, where each good j is consumed in total quantity one, matching the aggregate endowment. Hence $(\tilde{p}, \tilde{x}^{i'})$ is a competitive equilibrium given $\tilde{\omega}^{i'}$.

Remark. The geometric price vector does more than reproduce PROPOSITION 9.3 — it explains it. At $\tilde{p}$ the value of the aggregate endowment is

$$\sum_{j \in \mathbb{N}} \tilde{p}_j = \underbrace{i'}_{j \leq i'} + \sum_{j > i'} \rho^{j-i'-1} = i' + \frac{1}{1-\rho} < \infty,$$

whereas at the price vector $\bar{p} \equiv 1$ of PROPOSITION 9.1 it is $+\infty$. The finiteness of $\sum_h \sum_j p_j^* \omega_j^h$ is exactly the hypothesis of THEOREM 5.6, so at $\tilde{p}$ the extended First Welfare Theorem applies and *proves* that the equilibrium is Pareto optimal, while at $\bar{p}$ it does not apply and the equilibrium is in fact suboptimal. Exercise 9.11 shows that the same distinction — a finite or an infinite value of the aggregate endowment — separates the efficient from the inefficient equilibria of the overlapping generations model.

Exercise 9.4

The hint asks what happens to undepreciated capital, and that is the whole of the answer. At the end of date t the old own the capital stock $K(t)$, of which $(1-\delta)K(t)$ survives. Since the old consume everything and leave nothing behind, they must *sell* the undepreciated capital, and the only agents with any wealth to buy it are the young. Consequently the young's savings buy the whole capital stock of the next period: both the undepreciated part and the new investment.

Returns. A unit of capital rented out at $t+1$ earns the rental $f'(k(t+1))$ and is then worth $1-\delta$ units of the final good. The gross return to saving is therefore

$$R(t+1) = f'(k(t+1)) + 1 - \delta, \quad (*)$$

which replaces (9.3); the wage is still $w(t) = f(k(t)) - k(t)f'(k(t))$, as in (9.4), since the wage bill is unaffected by what happens to the capital afterwards.

Accumulation. Let $I(t+1)$ be new investment at $t+1$. The young at $t+1$ buy the surviving capital from the old and finance the new investment, so

$$s(t+1)L(t+1) = (1-\delta)K(t+1) + I(t+1) = K(t+2),$$

that is,

$$K(t+1) = L(t) s(w(t), R(t+1)),$$

which is (9.7) verbatim. Dividing by $L(t+1) = (1+n)L(t)$ gives the fundamental law of motion

$$k(t+1) = \frac{s(f(k(t)) - k(t)f'(k(t)), f'(k(t+1)) + 1 - \delta)}{1+n},$$

which is (9.8) with the return $(*)$ in place of $f'(k(t+1))$. Setting $\delta = 1$ recovers the text exactly.

Consistency with the resource constraint. At $t+1$ the economy has $F(K(t+1), L(t+1))$ of output and $(1-\delta)K(t+1)$ of surviving capital. The old consume $R(t+1)K(t+1)$, the young consume $[w(t+1) - s(t+1)]L(t+1)$, and $K(t+2) = s(t+1)L(t+1)$ is carried forward. Adding these and using Euler's theorem, $F = wL + f'(k)K$, gives

$$\begin{aligned} & [w(t+1) - s(t+1)]L(t+1) + [f'(k(t+1)) + 1 - \delta]K(t+1) + s(t+1)L(t+1) \\ & = F(K(t+1), L(t+1)) + (1-\delta)K(t+1), \end{aligned}$$

so the allocation is feasible.

The substantive point is that the two-period structure makes the young the owners of *all* the economy's wealth at every date, whatever the depreciation rate. Setting $\delta = 1$ is therefore a normalization of the return, not a restriction on the dynamics.

Exercise 9.5

Substituting the two budget constraints into the Euler equation (9.5), saving solves

$$\Phi(s; w, R) \equiv u'(w - s) - \beta R u'(Rs) = 0.$$

Since $u'' < 0$,

$$\Phi_s = -u''(w - s) - \beta R^2 u''(Rs) > 0,$$

so Φ is strictly increasing in s and the root is unique; the implicit function theorem applies.

Wages. $\Phi_w = u''(w - s) < 0$, hence

$$s_w = -\frac{\Phi_w}{\Phi_s} = \frac{-u''(c_1)}{-u''(c_1) - \beta R^2 u''(c_2)} \in (0, 1),$$

writing $c_1 = w - s$ and $c_2 = Rs$. Saving is strictly increasing in the wage, and by less than one for one: both goods are normal, so a richer household spreads the extra income over both periods of its life. This is the property the text uses to obtain $s_w \in (0, 1)$ in (9.12).

The interest rate. Differentiating,

$$\Phi_R = -\beta u'(Rs) - \beta R s u''(Rs) = -\beta u'(c_2) [1 - \varepsilon_u(c_2)], \quad \varepsilon_u(c) \equiv -\frac{c u''(c)}{u'(c)} > 0,$$

so that

$$s_R = -\frac{\Phi_R}{\Phi_s} = \frac{\beta u'(c_2) [1 - \varepsilon_u(c_2)]}{-u''(c_1) - \beta R^2 u''(c_2)}, \quad \text{sgn } s_R = \text{sgn} [1 - \varepsilon_u(c_2)].$$

Hence a sufficient (and, pointwise, necessary) condition for saving to be increasing in R is

$$\varepsilon_u(c) < 1 \quad \text{for all } c > 0 \quad \Longleftrightarrow \quad c \mapsto c u'(c) \text{ is strictly increasing,}$$

that is, the elasticity of intertemporal substitution $1/\varepsilon_u$ must exceed one. For the CRRA family (9.10), $\varepsilon_u \equiv \theta$, and this reproduces the text's $s_R > 0 \Leftrightarrow \theta < 1$, $s_R < 0 \Leftrightarrow \theta > 1$, $s_R = 0$ for log preferences.

Why. A higher R makes future consumption cheaper, which pushes saving up (substitution), but it also makes the household wealthier in present-value terms, which pushes consumption up in both periods and saving down (income). The elasticity ε_u measures how fast marginal utility falls, that is, how strong the income effect is; $\varepsilon_u = 1$ is the knife edge at which the two cancel.

Exercise 9.9

(a) *Equilibrium.* Preferences are $\log c_1(t) + \beta \log c_2(t+1)$, technology is $f(k) = k^\alpha$ with $\delta = 1$, and population grows at n . A competitive equilibrium is a sequence

$$\{K(t), c_1(t), c_2(t), R(t), w(t)\}_{t=0}^{\infty}$$

such that

- (1) factor prices are competitive: $R(t) = f'(k(t)) = \alpha k(t)^{\alpha-1}$ and $w(t) = f(k(t)) - k(t)f'(k(t)) = (1 - \alpha)k(t)^\alpha$, where $k(t) = K(t)/L(t)$;
 (2) each generation maximizes its utility subject to its two budget constraints,

$$c_1(t) + s(t) = w(t), \quad c_2(t+1) = R(t+1)s(t);$$

- (3) the capital market clears: $K(t+1) = L(t)s(t)$, with $K(0) > 0$ given.

A steady state is an equilibrium with $k(t) = k^*$ constant.

(b) *Characterization.* With log preferences the Euler equation gives $c_2(t+1) = \beta R(t+1)c_1(t)$, and substituting the budget constraints yields (9.19),

$$s(t) = \frac{\beta}{1+\beta} w(t) = \frac{\beta(1-\alpha)}{1+\beta} k(t)^\alpha,$$

a constant saving rate out of labor income, independent of R . Hence

$$k(t+1) = \frac{s(t)}{1+n} = \frac{\beta(1-\alpha)}{(1+n)(1+\beta)} k(t)^\alpha, \quad k^* = \left[\frac{\beta(1-\alpha)}{(1+n)(1+\beta)} \right]^{\frac{1}{1-\alpha}},$$

which is (9.20). The map is a concave power function through the origin, so convergence to k^* is monotone and global, exactly as in the Solow model. The steady-state gross return is

$$R^* = \alpha(k^*)^{\alpha-1} = \frac{\alpha(1+n)(1+\beta)}{\beta(1-\alpha)}.$$

Dynamic inefficiency means $r^* < n$, that is $R^* < 1+n$:

$$\frac{\alpha(1+\beta)}{\beta(1-\alpha)} < 1 \iff \alpha(1+\beta) < \beta(1-\alpha) \iff \alpha < \beta(1-2\alpha).$$

So, explicitly,

the economy overaccumulates $\iff \alpha < \frac{1}{2}$ and $\beta > \frac{\alpha}{1-2\alpha}$

Population growth drops out entirely: raising n lowers k^* and raises R^* in exactly the proportion in which it raises the golden-rule threshold $1+n$.

(c) *Is this plausible?* A period in this model is a working lifetime, say thirty years. The condition needs a low capital share and a very patient household:

α	0.20	0.25	0.30
$\beta > \alpha/(1-2\alpha)$	0.333	0.500	0.750
implied annual discount rate	< 3.73%	< 2.34%	< 0.96%

and at $\alpha = 1/3$ the condition becomes $\beta > 1$, that is a *negative* rate of time preference. With a conventional calibration — $\alpha = 1/3$, an annual discount rate of 2% ($\beta = 0.98^{30} = 0.545$) and annual population growth of 1% ($1+n = 1.01^{30}$, so $n = 0.348$) — the program gives

$$k^* = 0.0729 < k_{\text{gold}} = 0.1230, \quad R^* = 1.911 > 1+n = 1.348,$$

an annualized real interest rate of 2.18% against population growth of 1%: the economy is comfortably dynamically *efficient*. This matches what the data say — capital income exceeds investment

in essentially every developed economy — so dynamic inefficiency, while a genuine theoretical possibility, is not a description of any economy one would worry about.

(d) *Unfunded social security.* Let the government collect a constant d from each young individual and hand $(1+n)d$ to each old one. A generation now solves

$$\max \log(w(t) - s(t) - d) + \beta \log(R(t+1)s(t) + (1+n)d),$$

whose solution is

$$s(t) = \frac{\beta w(t) - [\beta + (1+n)/R(t+1)]d}{1+\beta}, \quad (1+n)k(t+1) = s(t). \quad (\dagger)$$

Saving falls, so the transfer reduces capital accumulation — which is precisely what is wanted when $k^* > k_{\text{gold}}$.

Two observations establish the Pareto improvement.

The steady-state comparison, analytically. Index steady states by k and use the Euler equation $c_2 = \beta f'(k)c_1$ together with the resource constraint $c_1 + c_2/(1+n) = C(k) \equiv f(k) - (1+n)k$. Solving,

$$c_1 = \frac{(1+n)C(k)}{(1+n) + \beta f'(k)}, \quad U(k) = (1+\beta) \log c_1 + \beta \log(\beta f'(k)) + \text{const},$$

and differentiating, after collecting the two terms in f'' ,

$$\frac{dU}{dk} = (1+\beta) \underbrace{\frac{f'(k) - (1+n)}{C(k)}}_{\text{resource effect}} + \underbrace{\frac{\beta f''(k)[(1+n) - f'(k)]}{f'(k)[(1+n) + \beta f'(k)]}}_{\text{price effect}}.$$

When $f'(k) < 1+n$ — exactly dynamic inefficiency — the first term is negative and, since $f'' < 0$ and $(1+n) - f'(k) > 0$, so is the second. Hence $dU/dk < 0$: *reducing* the steady-state capital-labor ratio strictly raises the lifetime utility of every generation living in the steady state. Both channels point the same way: less capital frees resources, and it raises the interest rate on which the old live.

The whole transition, numerically. Analysis of the steady states alone does not settle the fate of the generations passing through the transition, so the program iterates $(\dagger)$. Take $\alpha = 0.25$, $\beta = 0.9$, $n = 0$, which satisfies the condition in (b): $k^* = 0.2516 > k_{\text{gold}} = 0.1575$ and $R^* = 0.704 < 1$. Introducing at $t = 0$ a constant d equal to 2% of the steady-state wage gives

t	0	1	2	3	10	∞
$k(t)$	0.2516	0.2389	0.2358	0.2350	0.2347	0.2347
$U(t) - U^*$	+0.0488	+0.0329	+0.0288	+0.0277	+0.0274	+0.0274

Every generation from $t = 0$ on is strictly better off, the smallest gain over the entire transition being +0.0274; and the generation old at $t = 0$ receives $(1+n)d > 0$ out of nowhere. The allocation therefore Pareto dominates the laissez-faire equilibrium, which is PROPOSITION 9.8 made explicit.

Remark. The mechanism is a pyramid scheme that pays off. Unfunded social security transfers at the implicit rate n ; investing in capital pays r^* . When $r^* < n$ the pyramid is the better technology for moving resources across dates, so switching some resources into it makes everyone richer. When $r^* > n$ the same scheme is a bad investment, and Exercise 9.17 shows that it then necessarily hurts some future generation.

Exercise 9.10

Keep log preferences and $f(k) = k^\alpha$ with $\delta = 1$, and suppose an individual supplies one unit of labor when young and $z \geq 0$ units when old. ($z = 0$ is the baseline model of Section 9.3 and $z = 1$ the symmetric case.)

(a) *Equilibrium.* Total labor supply at t is

$$N(t) = L(t) + zL(t-1) = L(t) \frac{1+n+z}{1+n},$$

so it is natural to work with capital per unit of labor, $\hat{k} \equiv K/N$. Competitive factor prices are $R(t) = \alpha \hat{k}(t)^{\alpha-1}$ and $w(t) = (1-\alpha)\hat{k}(t)^\alpha$ per unit of labor. A generation born at t solves

$$\max \log c_1(t) + \beta \log c_2(t+1) \quad \text{s.t.} \quad c_1(t) + s(t) = w(t), \quad c_2(t+1) = R(t+1)s(t) + zw(t+1),$$

and the capital market clears with $K(t+1) = s(t)L(t)$: only the young hold assets, since the old consume their wage and their accumulated savings and leave nothing. A steady state is an equilibrium with $\hat{k}$ constant.

Note that the constraint $s(t) \geq 0$ is no longer automatic: an individual with substantial old-age earnings may wish to borrow against them. We proceed without imposing it and note below when it binds.

(b) *Characterization.* Lifetime wealth is $W(t) = w(t) + zw(t+1)/R(t+1)$, and log preferences give $c_1(t) = W(t)/(1+\beta)$, hence

$$s(t) = w(t) - \frac{W(t)}{1+\beta} = \frac{\beta w(t) - zw(t+1)/R(t+1)}{1+\beta}.$$

Because $w(t+1)/R(t+1) = (1-\alpha)\hat{k}(t+1)^\alpha / [\alpha \hat{k}(t+1)^{\alpha-1}] = (1-\alpha)\hat{k}(t+1)/\alpha$, this is linear in $\hat{k}(t+1)$. Using $K(t+1) = s(t)L(t)$ and $N(t+1) = L(t+1)(1+n+z)/(1+n) = L(t)(1+n+z)$,

$$(1+n+z)\hat{k}(t+1) = \frac{\beta(1-\alpha)\hat{k}(t)^\alpha - z(1-\alpha)\hat{k}(t+1)/\alpha}{1+\beta},$$

and collecting the terms in $\hat{k}(t+1)$,

$$\hat{k}(t+1) = A(z)\hat{k}(t)^\alpha, \quad A(z) = \frac{\alpha\beta(1-\alpha)}{\alpha(1+\beta)(1+n+z) + z(1-\alpha)}$$

Setting $z = 0$ gives $A(0) = \beta(1-\alpha)/[(1+\beta)(1+n)]$ and recovers (9.20).

The law of motion is again a concave power function through the origin, so there is a unique positive steady state

$$\hat{k}^* = A(z)^{\frac{1}{1-\alpha}},$$

and from any $\hat{k}(0) > 0$ the economy converges to it monotonically — the transitional dynamics are those of the Solow model, exactly as in PROPOSITION 9.5. Since $A(z)$ is strictly decreasing in z , the steady-state capital-labor ratio falls as the old earn more.

(c) *Overaccumulation.* The golden rule maximizes steady-state consumption per unit of labor, $f(\hat{k}) - (1+n)\hat{k}$, so $f'(\hat{k}_{\text{gold}}) = 1+n$ and $\hat{k}_{\text{gold}} = [\alpha/(1+n)]^{1/(1-\alpha)}$. Hence

$$\hat{k}^* > \hat{k}_{\text{gold}} \iff A(z) > \frac{\alpha}{1+n} \iff \beta(1-\alpha)(1+n) > \alpha(1+\beta)(1+n+z) + z(1-\alpha),$$

that is,

$$\text{overaccumulation} \iff z < \bar{z} \equiv \frac{(1+n)[\beta(1-2\alpha) - \alpha]}{1+\alpha\beta}$$

which requires $\beta(1-2\alpha) > \alpha$ — precisely the condition of Exercise 9.9 — and then caps z . So the answer is: yes, but only if old-age labor income is small. With $\alpha = 0.25$, $\beta = 0.9$, $n = 0$ the threshold is $\bar{z} = 0.163$, and the program gives

z	0	0.1	0.163	0.2	1.0
$\hat{k}^*/\hat{k}_{\text{gold}}$	1.598	1.177	1.000	0.918	0.292

The old supplying a sixth of a young worker's labor is already enough to eliminate overaccumulation; the symmetric case $z = 1$ is nowhere near it.

Why. In the baseline model the entire reason for saving is that the old have no income at all, so each generation must carry its whole retirement into the future as capital. Giving the old an income both reduces the amount that must be saved and — because the young can now capitalize it — lets them borrow against it. This is the same force that makes the continuous-time perpetual youth model of Section 9.8 always underaccumulate: there labor income is constant over the whole of life, which is the opposite extreme from the Samuelson–Diamond model. Exercise 9.30 completes the picture by showing that a *declining* path of labor income brings overaccumulation back.

Exercise 9.11

Write the OLG economy as an Arrow–Debreu economy: the commodities are dated consumption goods, and the households are the generations. Generation t is endowed with one unit of labor at date t and nothing else, and there are $L(t) = (1+n)^t L(0)$ of them. With the date-0 good as numeraire, the date- t price is

$$q(t) = \prod_{s=1}^t \frac{1}{R(s)} \quad \text{so that in a steady state} \quad q(t) = \frac{1}{(1+r^*)^t},$$

and the value of one unit of date- t labor is $w(t)q(t)$, which in a steady state is $w^*/(1+r^*)^t$ — the ratio in the hint. The total value of the economy's endowments is therefore

$$\sum_{h \in \mathcal{H}} \sum_{j=0}^{\infty} p_j^* \omega_j^h = \sum_{t=0}^{\infty} L(t) w(t) q(t) = L(0) w^* \sum_{t=0}^{\infty} \left(\frac{1+n}{1+r^*} \right)^t. \quad (*)$$

(a) $r^* < n$. Then $(1+n)/(1+r^*) > 1$, the terms of the geometric series in $(*)$ do not even tend to zero, and the sum is $+\infty$. The hypothesis of THEOREM 5.6 fails, so the theorem says nothing — and PROPOSITION 9.6 shows that the conclusion genuinely fails too: the equilibrium is Pareto suboptimal. The economics is transparent. Each date brings $(1+n)$ times as many endowment units as the last, while discounting shrinks their value only by $(1+r^*)$; when $r^* < n$ the economy's

aggregate endowment has infinite present value, and with infinite wealth there is room to give everyone more by rearranging *who* consumes *when*.

(b) $r^* > n$. Then $(1+n)/(1+r^*) < 1$ and

$$\sum_h \sum_j p_j^* \omega_j^h = \frac{L(0)w^*(1+r^*)}{r^* - n} < \infty.$$

(Outside the steady state the same conclusion holds as long as $r(t) \rightarrow r^* > n$: the tail of the series is dominated by a convergent geometric one, and the finitely many initial terms are finite.) Every hypothesis of THEOREM 5.6 is then met — preferences are continuous and locally nonsatiated, the allocation is a competitive equilibrium, and the value of the aggregate endowment is finite — so the theorem applies and the competitive equilibrium is Pareto efficient.

Remark. The comparison with Exercise 9.2 is exact. In Shell's economy, the equilibrium price vector $\bar{p} \equiv 1$ gives the aggregate endowment infinite value and the equilibrium is suboptimal; the geometric price vector $\tilde{p}$ gives it finite value and the equilibrium is optimal. In the OLG model the analogous role is played by the comparison of r^* with n . Note also what is *not* the source of the inefficiency: not market incompleteness (Shell's households can all trade with each other), and not externalities (there are none). It is only that the value of the aggregate endowment can be infinite, which is exactly what THEOREM 5.6 forbids.

Exercise 9.12

The theorem is silent. With $r^* = n$ the series (*) of Exercise 9.11 is $L(0)w^* \sum_{t \geq 0} 1 = +\infty$, so the hypothesis of THEOREM 5.6 fails. A sufficient condition failing proves nothing, so the theorem gives no information either way.

Pareto optimality when u and f are strictly concave. At $r^* = n$ the steady state is the golden rule: $f'(k^*) = 1 + n$. Its allocation is

$$s^* = (1+n)k^*, \quad c_1^* = w^* - s^*, \quad c_2^* = (1+n)s^*, \quad \text{so} \quad c_1^* + \frac{c_2^*}{1+n} = w^* = C^* \equiv f(k^*) - (1+n)k^*.$$

Suppose some feasible allocation $\{\hat{c}_1(t), \hat{c}_2(t), \hat{k}(t)\}$ with $\hat{k}(0) = k^*$ Pareto dominates it. Feasibility, per young person, is

$$f(\hat{k}(t)) = (1+n)\hat{k}(t+1) + \hat{c}_1(t) + \frac{\hat{c}_2(t)}{1+n}, \quad t \geq 0. \quad (\text{F})$$

Two *value losses* organize the argument. On the production side, strict concavity of f and $f'(k^*) = 1 + n$ give

$$\lambda(t) \equiv f(k^*) + (1+n)[\hat{k}(t) - k^*] - f(\hat{k}(t)) \geq 0, \quad \lambda(t) = 0 \iff \hat{k}(t) = k^*.$$

On the consumption side, note that at the golden rule the relative price of old-age consumption is $1/R^* = 1/(1+n)$, so the cost of generation t 's bundle is $\hat{m}(t) \equiv \hat{c}_1(t) + \hat{c}_2(t+1)/(1+n)$. Strict concavity of u makes the expenditure function strictly increasing in utility with a unique minimizer, so

$$\mu(t) \equiv \hat{m}(t) - w^* \geq 0, \quad \mu(t) = 0 \iff (\hat{c}_1(t), \hat{c}_2(t+1)) = (c_1^*, c_2^*),$$

for every generation that is at least as well off as in the steady state.

Now sum (F) over $t = 0, \dots, T$, rearranging the consumption terms by generation:

$$\sum_{t=0}^T \left[\widehat{c}_1(t) + \frac{\widehat{c}_2(t)}{1+n} \right] = \frac{\widehat{c}_2(0)}{1+n} + \sum_{t=0}^{T-1} \widehat{m}(t) + \widehat{c}_1(T),$$

while the production side gives

$$\sum_{t=0}^T \left[f(\widehat{k}(t)) - (1+n)\widehat{k}(t+1) \right] = (T+1)C^* + (1+n)[\widehat{k}(0) - \widehat{k}(T+1)] - \sum_{t=0}^T \lambda(t).$$

Equating the two, using $C^* = w^*$, $\widehat{k}(0) = k^*$, $\widehat{c}_2(0) \geq c_2^*$ and $c_2^*/(1+n) = (1+n)k^*$, and cancelling Tw^* from both sides,

$$\sum_{t=0}^T \lambda(t) + \sum_{t=0}^{T-1} \mu(t) + \frac{\widehat{c}_2(0) - c_2^*}{1+n} + \widehat{c}_1(T) + (1+n)\widehat{k}(T+1) \leq w^*. \quad (\dagger)$$

Every term is nonnegative and the bound is uniform in T , so both series converge; hence $\lambda(t) \rightarrow 0$ and $\mu(t) \rightarrow 0$. Since λ vanishes only at k^* and is continuous with $\lambda(k) \rightarrow \infty$ as $k \rightarrow \infty$, this forces $\widehat{k}(t) \rightarrow k^*$; and since the minimum-cost bundle at utility U^* is unique, $\mu(t) \rightarrow 0$ forces $(\widehat{c}_1(t), \widehat{c}_2(t+1)) \rightarrow (c_1^*, c_2^*)$, so $\widehat{c}_1(T) \rightarrow c_1^*$. Letting $T \rightarrow \infty$ in $(\dagger)$ and using $w^* = c_1^* + (1+n)k^*$,

$$\sum_{t=0}^{\infty} \lambda(t) + \sum_{t=0}^{\infty} \mu(t) + \frac{\widehat{c}_2(0) - c_2^*}{1+n} \leq 0.$$

A sum of nonnegative terms bounded above by zero is zero term by term: $\lambda \equiv 0$, $\mu \equiv 0$ and $\widehat{c}_2(0) = c_2^*$. So $\widehat{k}(t) = k^*$ for all t and every generation has exactly the steady-state bundle — the supposed improvement is the steady state itself, and no one is strictly better off. The golden-rule equilibrium is Pareto optimal. ■

An example with Pareto suboptimality. Strict concavity is what makes the argument work, and dropping it breaks the conclusion. Let u be linear, $u(c) = c$, with $\beta = 1/(1+n)$, and let f be strictly concave with $f'(k^*) = 1+n$. (With $\beta(1+n) = 1$ each generation is exactly indifferent along its budget line, so savings are indeterminate and k^* is one of a continuum of equilibria.) Lifetime utility is then

$$U_t = c_1(t) + \beta c_2(t+1) = c_1(t) + \frac{c_2(t+1)}{1+n},$$

which is precisely the *cost* of the bundle at the golden-rule prices. Now take any $\varepsilon \in (0, c_1^*]$ and consider

$$\widehat{k}(t) = k^*, \quad \widehat{c}_1(t) = c_1^* - \varepsilon, \quad \widehat{c}_2(t) = c_2^* + (1+n)\varepsilon \quad \text{for all } t \geq 0.$$

This is feasible: at every date $\widehat{c}_1(t) + \widehat{c}_2(t)/(1+n) = c_1^* + c_2^*/(1+n) = C^*$. Each generation $t \geq 0$ has

$$\widehat{U}_t = (c_1^* - \varepsilon) + \frac{c_2^* + (1+n)\varepsilon}{1+n} = c_1^* + \frac{c_2^*}{1+n} = U^*,$$

so no one is worse off; and the generation already old at $t = 0$ consumes $c_2^* + (1+n)\varepsilon > c_2^*$ and is strictly better off. The equilibrium is therefore Pareto *suboptimal*.

Remark. The example is Samuelson's transfer scheme in its purest form: a permanent gift from each young generation to the old, at the implicit rate of return n . When $r^* = n$ the scheme is exactly break-even for

everyone born afterwards, and pure profit for the generation alive when it starts. With strictly concave u the same perturbation moves each generation *off* its optimum along a budget line of unchanged value and so makes it strictly worse off — which is exactly the $\mu(t) > 0$ term in (‡), and exactly why strict concavity restores optimality. The knife edge is genuinely a knife edge.

Exercise 9.13

Under a fully funded system the government collects $d(t)$ from each young individual, invests it in capital, and returns $R(t+1)d(t)$ in old age. The generation's problem is

$$\max u(c_1(t)) + \beta u(c_2(t+1)) \quad \text{s.t.} \quad c_1(t) + s(t) + d(t) \leq w(t), \quad c_2(t+1) \leq R(t+1)(s(t) + d(t)).$$

Part 2: no constraint on s . Change variables to *total* saving $\hat{s}(t) \equiv s(t) + d(t)$. The two constraints become

$$c_1(t) + \hat{s}(t) \leq w(t), \quad c_2(t+1) \leq R(t+1)\hat{s}(t),$$

which are the constraints of the problem *without* social security, in the variable $\hat{s}$. Since $d(t)$ has disappeared from the problem entirely, the optimal $\hat{s}(t)$ is the same function $s(w(t), R(t+1))$ of (9.6), and the individual simply offsets the contribution one for one:

$$s(t) = s(w(t), R(t+1)) - d(t).$$

Total investment is $\hat{s}(t)L(t) = (1+n)k(t+1)L(t)$, exactly (9.7). Hence the sequences $\{k(t)\}$, $\{w(t)\}$, $\{R(t)\}$ and all consumption levels are identical with and without the program: the set of competitive equilibria is the same, and the program changes only the *label* attached to a part of each household's portfolio.

Part 1: $s(t) \geq 0$ imposed. If, given the feasible sequence $\{d(t)\}$, the optimal saving sequence satisfies $s(t) > 0$ for all t , then the constraint is slack at the optimum, so the solution coincides with the unconstrained one, and part 2 applies verbatim. Conversely, take any competitive equilibrium without social security; there $\hat{s}(t) = s(w(t), R(t+1)) > 0$, and provided $d(t) < s(w(t), R(t+1))$ — which is what the hypothesis $s(t) > 0$ says — the same allocation is a competitive equilibrium with social security. The two sets coincide. ■

The result is the exact analogue of Ricardian equivalence (Exercise 8.35), and for the same reason: a fully funded system makes no transfer between generations. Each generation receives back, at the market rate, precisely what it paid in, so its lifetime budget constraint $c_1 + c_2/R = w$ is untouched. Only the composition of saving — private versus public — changes, and nothing in the model depends on that label. Exercise 9.14 shows that this is not merely a neutrality result: the program cannot help even when it is forced to bind.

Exercise 9.14

If the constraint never binds — $d(t) < s(w(t), R(t+1))$ at every date — then by PROPOSITION 9.7 the allocation is exactly the laissez-faire one, so there is nothing to improve. Suppose then that the constraint binds somewhere, and let T be the first date at which it does, so that $s(T) = 0$ and total saving is $d(T)$.

Nothing changes before T . For $t < T$ the constraint is slack, so by PROPOSITION 9.7 the capital-labor ratios $k(t)$, $t \leq T$, and hence the factor prices $w(t)$, $R(t)$ for $t \leq T$, coincide with the laissez-faire ones. In particular generation $T - 1$, whose consumption depends on $w(T - 1)$ and $R(T)$, is exactly as well off as before, and so are all earlier generations and the initial old. The initial old, incidentally, could only gain from a transfer, and a fully funded system makes none.

Generation T is strictly worse off. Its wage $w(T)$ is unchanged, since $k(T)$ is. Total saving is $(1 + n)k(T + 1) = d(T)$. Write $g(k) \equiv s(w(T), f'(k)) - (1 + n)k$, so that the laissez-faire $k_{\text{LF}}(T + 1)$ is the root of g , and the binding of the constraint says $g(d(T)/(1 + n)) < 0$. Under the stability condition maintained throughout the chapter (under which g crosses zero once, from above — it holds automatically whenever $s_R \geq 0$, and in particular in PROPOSITION 9.4), this gives

$$k(T + 1) = \frac{d(T)}{1 + n} > k_{\text{LF}}(T + 1) \quad \implies \quad R(T + 1) = f'(k(T + 1)) < R_{\text{LF}}(T + 1).$$

Generation T therefore faces the budget set

$$\left\{ (c_1, c_2) \geq 0 : c_1 + \frac{c_2}{R(T + 1)} \leq w(T) \right\} \subsetneq \left\{ (c_1, c_2) \geq 0 : c_1 + \frac{c_2}{R_{\text{LF}}(T + 1)} \leq w(T) \right\},$$

a *strict* subset — the same intercept $w(T)$ on the c_1 axis, a strictly lower one on the c_2 axis. And within that smaller set it is not even free to choose: the constraint $s(T) \geq 0$ forces it to the point $c_1(T) = w(T) - d(T)$, which is not its optimum. Both effects push the same way, so generation T is strictly worse off than in the laissez-faire equilibrium.

Hence no fully funded program yields a Pareto improvement. ■

The general lesson. A fully funded system is, by construction, an intergenerational transfer of zero present value; it cannot relax anybody's lifetime budget constraint, and the only thing it can do is restrict choice. Exploiting dynamic inefficiency requires transferring resources *from* later generations *to* earlier ones at a rate above the market return, and only an unfunded system does that (Exercise 9.9(d) and PROPOSITION 9.8). Forcing additional saving in an economy that already saves too much is, if anything, exactly the wrong intervention.

Exercise 9.18

An individual dies at the end of each period with probability ν , independently across periods. The probability of living exactly ℓ periods is therefore

$$\Pr[\text{life} = \ell] = (1 - \nu)^{\ell-1} \nu, \quad \ell = 1, 2, \dots,$$

a geometric distribution: surviving the first $\ell - 1$ periods and then dying. These probabilities sum to $\nu \sum_{\ell \geq 1} (1 - \nu)^{\ell-1} = \nu/[1 - (1 - \nu)] = 1$, as they must. Expected life is

$$\mathbb{E}[\text{life}] = \nu + 2(1 - \nu)\nu + 3(1 - \nu)^2\nu + \dots = \nu \sum_{\ell=1}^{\infty} \ell q^{\ell-1}, \quad q \equiv 1 - \nu \in (0, 1).$$

To evaluate the sum, differentiate the geometric series $\sum_{\ell \geq 0} q^\ell = 1/(1-q)$ term by term (legitimate inside the radius of convergence):

$$\sum_{\ell=1}^{\infty} \ell q^{\ell-1} = \frac{d}{dq} \left[\frac{1}{1-q} \right] = \frac{1}{(1-q)^2} = \frac{1}{\nu^2}.$$

Hence

$$E[\text{life}] = \frac{\nu}{\nu^2} = \frac{1}{\nu} < \infty,$$

which is (9.31).

Perpetual youth. What makes the name apt is that the geometric distribution is memoryless: conditional on having survived to any date, the remaining life is again geometric with the same parameter, so the expected further life is $1/\nu$ at every age. Age carries no information about longevity, which is why in Section 9.7 the annuity payment (9.33) depends on assets alone and not on when the individual was born, and why in Section 9.8 the Euler equation (9.43) is the same for every cohort. The continuous-time counterpart is the exponential distribution with rate ν , whose mean is likewise $1/\nu$ (Exercise 9.25(b)).

Exercise 9.19

Individual choice. The function $c^\eta b^{1-\eta}$ is a strictly increasing transform of $\eta \log c + (1-\eta) \log b$, so it represents the same preferences as the logarithmic specification with β replaced by $(1-\eta)/\eta$. Maximizing subject to $c^i(t) + b^i(t) \leq y^i(t)$ gives the constant bequest share

$$b^i(t) = (1-\eta)y^i(t) = (1-\eta)[w(t) + R(t)b^i(t-1)], \quad c^i(t) = \eta y^i(t)$$

which is (9.27) with $\beta/(1+\beta)$ replaced by $1-\eta$.

Equilibrium. A competitive equilibrium is, as in DEFINITION 9.2, a sequence of consumption and bequest levels $\{[c^i(t)]_{i \in [0,1]}, [b^i(t)]_{i \in [0,1]}\}_{t \geq 0}$ solving the above, a sequence of capital-labor ratios given by $k(t+1) = \int_0^1 b^i(t) di$, and factor prices $w(t) = f(k(t)) - k(t)f'(k(t))$, $R(t) = f'(k(t))$.

Aggregate dynamics. Integrating the bequest rule across the unit measure of individuals and using Euler's theorem, $w(t) + R(t)k(t) = f(k(t))$,

$$k(t+1) = \int_0^1 b^i(t) di = (1-\eta)[w(t) + R(t)k(t)] = (1-\eta)f(k(t)),$$

the analogue of (9.28). As there, the aggregate law of motion is independent of the distribution of bequests, and it is a strictly concave function of $k(t)$ through the origin. By Assumptions 1 and 2 there is a unique positive steady state

$$k^* = (1-\eta)f(k^*),$$

to which the economy converges monotonically from any $k(0) > 0$ — Solow dynamics again.

The distribution of wealth. At the steady-state factor prices $w^* = f(k^*) - k^*f'(k^*)$ and $R^* = f'(k^*)$, individual bequests follow the affine recursion $b^i(t) = (1-\eta)[w^* + R^*b^i(t-1)]$. Its slope satisfies

$$(1-\eta)R^* = (1-\eta)f'(k^*) < (1-\eta)\frac{f(k^*)}{k^*} = 1$$

by strict concavity of f and the definition of k^* , so the recursion is a contraction and every individual's bequest converges to the same level,

$$b^* = \frac{(1-\eta)w^*}{1-(1-\eta)R^*},$$

the analogue of (9.30). Deviations from b^* shrink by the factor $(1-\eta)R^*$ per generation, so the cross-sectional variance of wealth falls at the rate $[(1-\eta)R^*]^2$. With $f(k) = Ak^\alpha$ one computes $R^* = \alpha/(1-\eta)$, so the contraction factor is exactly α : wealth inequality dies out at the rate of the capital share, independently of η and A . The economy therefore reproduces PROPOSITION 9.9 in full.

Remark. Nothing substantive changes, and the reason is worth stating plainly: the individual's problem is static, so only the *ordinal* content of preferences matters, and $c^\eta b^{1-\eta}$ is ordinally identical to the logarithmic case. Exercise 9.20, which allows $u_1(c) + u_2(b)$ with general u_1 and u_2 , is where the additional structure is genuinely needed: there the bequest share varies with income, aggregate dynamics can depend on the distribution, and stability and convergence to equality have to be assumed rather than derived.

Exercise 9.22

Population is constant and capital fully depreciates, so steady-state consumption is $f(k) - k$ and the golden rule is $f'(k_{\text{gold}}) = 1$. Since f' is strictly decreasing,

$$k^* > k_{\text{gold}} \iff f'(k^*) < 1.$$

From (9.29), $k^* = [\beta/(1+\beta)]f(k^*)$, that is $f(k^*)/k^* = (1+\beta)/\beta$. Writing $\phi(k) \equiv kf'(k)/f(k)$ for the elasticity of output with respect to capital,

$$f'(k^*) = \phi(k^*) \frac{f(k^*)}{k^*} = \phi(k^*) \frac{1+\beta}{\beta},$$

so the condition becomes

$$k^* > k_{\text{gold}} \iff \phi(k^*) < \frac{\beta}{1+\beta}$$

the elasticity of output with respect to capital must fall short of the warm glow bequest rate.

Cobb–Douglas. With $f(k) = Ak^\alpha$, $\phi \equiv \alpha$ and $k^* = [\beta A/(1+\beta)]^{1/(1-\alpha)}$, so

$$f'(k^*) = \frac{\alpha(1+\beta)}{\beta} \quad \text{and} \quad k^* > k_{\text{gold}} \iff \alpha(1+\beta) < \beta \iff \alpha < \frac{\beta}{1+\beta},$$

independently of A . With $\alpha = 1/3$ this needs only $\beta > 1/2$; taking $\beta = 0.6$, the program gives $k^* = 0.2296 > k_{\text{gold}} = 0.1925$ and $f'(k^*) = 0.889 < 1$. So yes, the warm glow economy overaccumulates for entirely ordinary parameters.

Comparison with the baseline model. Exercise 9.9 showed that the Samuelson–Diamond model overaccumulates only if $\alpha(1+2\beta) < \beta$. Side by side,

β	0.4	0.6	1.0
warm glow: $\alpha <$	0.286	0.375	0.500
baseline: $\alpha <$	0.222	0.273	0.333

The warm glow condition is always the weaker one. The reason is visible in the two accumulation equations: here $k(t+1) = [\beta/(1+\beta)]f(k(t))$, a constant fraction of *total* income, whereas in the baseline model $k(t+1) = \beta w(t)/[(1+n)(1+\beta)]$, a constant fraction of *labor* income alone. Bequests are made out of capital income too, so the economy saves a strictly larger share of output and reaches a higher capital-labor ratio.

Remark. This matters for how one reads the dynamic inefficiency literature. The baseline OLG model makes overaccumulation look like a parameter curiosity, needing implausible patience (Exercise 9.9(c)); the warm glow model, which is no less reasonable a description of bequests, produces it routinely. Whether an economy overaccumulates is therefore a question about the *structure* of saving — out of what income, for what motive — rather than about preferences alone.

Exercise 9.23

Population evolves by (9.34): a fraction ν of the $L(t)$ individuals alive at t dies, and $nL(t)$ new households arrive, so

$$L(t+1) = (1-\nu)L(t) + nL(t) = (1+n-\nu)L(t), \quad L(0) = 1 \quad \implies \quad L(t) = (1+n-\nu)^t.$$

Cohort sizes. The individuals who are of age s at date t were born at date $t-s+1$, when the population was $L(t-s)$, so at birth they numbered

$$nL(t-s) = n(1+n-\nu)^{t-s}.$$

Deaths are independent across individuals and dates with probability ν , so a fraction $(1-\nu)$ of the cohort survives each period. By date t the cohort has lived through $s-1$ such periods, so its size is

$$n(1+n-\nu)^{t-s}(1-\nu)^{s-1},$$

which is the claim. Setting $s = 1, 2, 3$ gives $n(1+n-\nu)^{t-1}$, $n(1+n-\nu)^{t-2}(1-\nu)$ and $n(1+n-\nu)^{t-3}(1-\nu)^2$, the three cases listed in the text.

Consistency. The cohorts born after date 0 have ages $s = 1, \dots, t$, and the survivors of the initial population have age t and number $(1-\nu)^t L(0) = (1-\nu)^t$. Writing $r \equiv (1-\nu)/(1+n-\nu)$, so that $1-r = n/(1+n-\nu)$, the total is

$$\sum_{s=1}^t n(1+n-\nu)^{t-s}(1-\nu)^{s-1} + (1-\nu)^t = n(1+n-\nu)^{t-1} \frac{1-r^t}{1-r} + (1-\nu)^t,$$

and since $n(1+n-\nu)^{t-1}/(1-r) = (1+n-\nu)^t$, this equals

$$(1+n-\nu)^t [1-r^t] + (1-\nu)^t = (1+n-\nu)^t - (1-\nu)^t + (1-\nu)^t = (1+n-\nu)^t = L(t),$$

as it must.

What the formula says. The age distribution at date t is geometric: the share of s -year-olds relative to $(s+1)$ -year-olds is $(1+n-\nu)/(1-\nu) > 1$, a constant. The population therefore has a stationary age structure from the first date onward, which is precisely the tractability the perpetual youth assumption buys: one never has to keep track of who is how old, because the shape of the age pyramid never changes. Exercise 9.25 derives the continuous-time counterpart, where the age density is exactly $ne^{-n \cdot \text{age}}$.

Exercise 9.25

(a) *Cohort sizes.* Aggregate population obeys (9.38), $\dot{L}(t) = (n - \nu)L(t)$, which with $L(0) = 1$ integrates to

$$L(\tau) = \exp((n - \nu)\tau).$$

New households arrive at the exponential rate n , so the *flow* of births at τ is

$$L(\tau | \tau) = nL(\tau) = n \exp((n - \nu)\tau),$$

which is the initial size (strictly, the density) of the cohort born at τ .

Remark. The exercise as stated omits the factor n ; (9.39) carries it, and it must, since the cohort born in an instant is a flow. The quantity $\exp((n - \nu)\tau)$ in the exercise is the population $L(\tau)$, of which the newborn cohort is the fraction n per unit time.

(b) *Survival.* Let $S(x)$ be the probability of being alive x units of time after birth. The Poisson death rate ν means $S(x + dx) = S(x)(1 - \nu dx) + o(dx)$, so $\dot{S} = -\nu S$ with $S(0) = 1$, whence

$$S(x) = e^{-\nu x}, \quad \text{and for a cohort born at } \tau, \quad S(t - \tau) = \exp(-\nu(t - \tau)).$$

(Equivalently, length of life is exponential with rate ν , mean $1/\nu$, memoryless — the continuous-time form of Exercise 9.18.)

(c) *Equation (9.39).* The cohort born at τ has size $n \exp((n - \nu)\tau)$ at birth, and a fraction $\exp(-\nu(t - \tau))$ of it is still alive at t . Multiplying,

$$L(t | \tau) = n \exp(-\nu(t - \tau) + (n - \nu)\tau),$$

which is (9.39). It integrates correctly: since the economy starts at $t = 0$ with $L(0) = 1$ newborns,

$$\int_0^t L(t | \tau) d\tau + e^{-\nu t} = n e^{-\nu t} \int_0^t e^{n\tau} d\tau + e^{-\nu t} = e^{-\nu t} (e^{nt} - 1) + e^{-\nu t} = e^{(n-\nu)t} = L(t).$$

A useful by-product: the density of individuals of age $a = t - \tau$ in the population is $L(t | t - a)/L(t) = n e^{-na}$, an exponential age distribution with mean age $1/n$ — the continuous-time counterpart of the geometric age pyramid of Exercise 9.23.

(d) *An arbitrary initial age distribution.* Formula (9.39) describes the age structure generated by letting the birth-and-death process run from the infinite past (or, as here, from a date at which everybody is newborn). Suppose instead that at $t = 0$ the economy contains a mass $M(0)$ of individuals with an arbitrary age distribution. At any later t the population splits into two groups:

- the survivors of the date-0 population, of mass $M(0)e^{-\nu t}$, whose ages are the initial ages shifted up by t and whose distribution is therefore the initial one shifted, not the exponential one;
- the cohorts born in $(0, t]$, which do follow (9.39).

Since $M(0)e^{-\nu t} > 0$ for every finite t , the first group never disappears, so the age distribution never exactly equals the stationary one at any finite date. It only converges: total population still

satisfies $\dot{L} = (n - \nu)L$, so $L(t) = M(0)e^{(n-\nu)t}$ and the share of the initial group is

$$\frac{M(0)e^{-\nu t}}{M(0)e^{(n-\nu)t}} = e^{-nt} \xrightarrow[t \rightarrow \infty]{} 0,$$

at the exponential rate n — the rate at which new households arrive and wash the initial composition out. The stationary age structure of (9.39) is thus a long-run property of the demographic process, not an identity, which is why the text is careful to start the economy with $L(0) = 1$ newborns.

Exercise 9.26

Step 1: the lifetime budget constraint. The flow constraint (9.40) is $\dot{a}(t | \tau) = (r(t) + \nu)a(t | \tau) - c(t | \tau) + w(t)$. Multiplying by the integrating factor $\exp(-\int_t^x (r(z) + \nu) dz)$ and integrating from t to T ,

$$a(T | \tau) \exp\left(-\int_t^T (r(z) + \nu) dz\right) - a(t | \tau) = \int_t^T [w(x) - c(x | \tau)] \exp\left(-\int_t^x (r(z) + \nu) dz\right) dx.$$

The transversality condition (9.44) sends the first term to zero as $T \rightarrow \infty$, so

$$a(t | \tau) + \underbrace{\int_t^\infty w(x) \exp\left(-\int_t^x (r(z) + \nu) dz\right) dx}_{= \omega(t)} = \int_t^\infty c(x | \tau) \exp\left(-\int_t^x (r(z) + \nu) dz\right) dx. \quad (*)$$

The left side is total wealth: assets plus human wealth. Human wealth is discounted at $r + \nu$, not r , because the individual loses the earnings stream when he dies, at the rate ν ; and it does not depend on τ , since every survivor faces the same future wages.

Step 2: substitute the Euler equation. Integrating (9.43), $\dot{c}(x | \tau)/c(x | \tau) = r(x) - \rho$, gives

$$c(x | \tau) = c(t | \tau) \exp\left(\int_t^x (r(z) - \rho) dz\right).$$

Putting this into the right side of (*), the interest rates cancel exactly:

$$\int_t^\infty c(t | \tau) \exp\left(\int_t^x (r(z) - \rho) dz - \int_t^x (r(z) + \nu) dz\right) dx = c(t | \tau) \int_t^\infty e^{-(\rho+\nu)(x-t)} dx = \frac{c(t | \tau)}{\rho + \nu}.$$

Therefore

$$\boxed{c(t | \tau) = (\rho + \nu)[a(t | \tau) + \omega(t)]}$$

which is (9.45). ■

Two features of this consumption function drive everything in Section 9.8. It is *linear* in wealth — a consequence of logarithmic preferences, which make the propensity to consume independent of the interest rate path — so individual consumption functions can simply be added up to give the aggregate (9.46), even though cohorts differ in their assets. And the propensity to consume out of wealth is the *effective* discount rate $\rho + \nu$ rather than ρ : the shorter the expected life, the faster wealth is run down. The two facts together produce the extra term $-(\rho + \nu)nk/c$ in the aggregate Euler equation (9.48): the newborn arriving at rate n hold no assets, only human wealth, so their consumption is below average and drags aggregate consumption growth down.

Exercise 9.27

Setup. Let $A(t) = A(0)e^{gt}$ multiply labor, and let $k \equiv K/(AL)$ be the effective capital-labor ratio and $\tilde{c} \equiv c/A$ normalized average consumption per capita. Factor prices are $R(t) = f'(k(t))$, $r(t) = f'(k(t)) - \delta$ and $w(t) = A(t)[f(k(t)) - k(t)f'(k(t))]$. Effective labor $A(t)L(t)$ grows at $n - \nu + g$, so (9.42) becomes

$$\dot{k}(t) = f(k(t)) - (n - \nu + g + \delta)k(t) - \tilde{c}(t). \quad (\text{K})$$

The aggregate Euler equation. Nothing in the derivation of the text changes until the last step: individual consumption is still $c(t | \tau) = (\rho + \nu)[a(t | \tau) + \omega(t)]$ by Exercise 9.26, average assets still satisfy $\dot{a} = (r - (n - \nu))a + w - c$ and human wealth still satisfies $(r + \nu)\omega = \dot{\omega} + w$, so

$$\dot{c}(t) = (r(t) - \rho)c(t) - (\rho + \nu)na(t), \quad a(t) = A(t)k(t).$$

Writing $c = A\tilde{c}$, so that $\dot{c} = A(\dot{\tilde{c}} + g\tilde{c})$, and dividing by $A\tilde{c}$,

$$\frac{\dot{\tilde{c}}(t)}{\tilde{c}(t)} = f'(k(t)) - \delta - \rho - g - (\rho + \nu)n \frac{k(t)}{\tilde{c}(t)}, \quad (\text{C})$$

which is (9.48) with the extra $-g$: growth raises the required return exactly as in the neoclassical model with logarithmic preferences (Exercise 8.18).

Uniqueness of the steady state. Setting (C) to zero and substituting into (K),

$$\frac{\tilde{c}^*}{k^*} = \frac{(\rho + \nu)n}{f'(k^*) - \delta - \rho - g}, \quad \Psi(k) \equiv \frac{f(k)}{k} - (n - \nu + g + \delta) - \frac{(\rho + \nu)n}{f'(k) - \delta - \rho - g} = 0. \quad (\text{S})$$

Consider Ψ on $(0, k_{\text{mgr}})$, where $f'(k_{\text{mgr}}) = \rho + \delta + g$ so that the denominator is positive. There $f(k)/k$ is strictly decreasing, and $f'(k) - \delta - \rho - g$ is strictly decreasing and positive, so the subtracted fraction is strictly increasing: Ψ is strictly decreasing. By Assumption 2, $f(k)/k \rightarrow \infty$ and $f'(k) \rightarrow \infty$ as $k \rightarrow 0$, so $\Psi \rightarrow +\infty$; and as $k \uparrow k_{\text{mgr}}$ the denominator tends to 0^+ and $\Psi \rightarrow -\infty$. Hence there is exactly one root k^* , and it satisfies

$$k^* < k_{\text{mgr}} :$$

underaccumulation relative to the modified golden rule persists, exactly as in PROPOSITION 9.10.

Saddle-path stability. The Jacobian of (K)–(C) at $(k^*, \tilde{c}^*)$ is

$$J = \begin{pmatrix} f'(k^*) - (n - \nu + g + \delta) & -1 \\ \tilde{c}^* f''(k^*) - (\rho + \nu)n & f'(k^*) - \delta - \rho - g \end{pmatrix}.$$

Using (S) to substitute $(\rho + \nu)n = [f(k^*)/k^* - (n - \nu + g + \delta)][f'(k^*) - \delta - \rho - g]$, the determinant collapses to

$$\det J = \left[f'(k^*) - \frac{f(k^*)}{k^*} \right] [f'(k^*) - \delta - \rho - g] + \tilde{c}^* f''(k^*) < 0,$$

since $f'(k^*) < f(k^*)/k^*$ by strict concavity with $f(0) = 0$, the second bracket is positive because $k^* < k_{\text{mgr}}$, and $f'' < 0$. A negative determinant means two real eigenvalues of opposite sign: the steady state is a saddle with a one-dimensional stable manifold. Globally, the $\dot{k} = 0$ locus is concave through the origin and the $\dot{\tilde{c}} = 0$ locus $\tilde{c} = (\rho + \nu)nk/[f'(k) - \delta - \rho - g]$ is convex through the origin and diverges at k_{mgr} , so they cross exactly once; above the stable arm feasibility fails, below

it consumption goes to zero and (9.44) is violated. Convergence is monotone along the unique upward-sloping stable arm.

A higher g . Differentiating (S),

$$\frac{\partial \Psi}{\partial g} = -1 - \frac{(\rho + \nu)n}{[f'(k) - \delta - \rho - g]^2} < 0, \quad \frac{\partial \Psi}{\partial k} < 0 \quad \implies \quad \frac{dk^*}{dg} = -\frac{\partial \Psi / \partial g}{\partial \Psi / \partial k} < 0.$$

So the effective capital-labor ratio *falls*, and with it output per effective worker; since $k^* < k_{\text{mgr}} < k_{\text{gold}}$, the $\dot{k} = 0$ locus is increasing at k^* , so $\tilde{c}^* = f(k^*) - (n - \nu + g + \delta)k^*$ falls too — both because k^* falls and because the replacement requirement rises. Meanwhile consumption and output *per capita* now grow at g instead of being constant. Faster technological progress thus lowers the level of the normalized variables and raises the growth rate of the actual ones, precisely as in Exercise 8.28(a) with $\theta = 1$: the higher required return $\rho + \delta + g$ is what does the work, and with logarithmic preferences the income and substitution effects on the impact consumption jump exactly cancel.

Exercise 9.28

This is the case $g = 0$ of Exercise 9.27. The system is

$$\dot{k} = f(k) - (n - \nu + \delta)k - c, \quad \dot{c} = c[f'(k) - \delta - \rho] - (\rho + \nu)nk,$$

the second being (9.48) cleared of the denominator. Its Jacobian at the steady state is

$$J = \begin{pmatrix} f'(k^*) - (n - \nu + \delta) & -1 \\ c^* f''(k^*) - (\rho + \nu)n & f'(k^*) - \delta - \rho \end{pmatrix},$$

with

$$\text{tr } J = 2f'(k^*) - (n - \nu + \delta) - \delta - \rho, \quad \det J = [f'(k^*) - (n - \nu + \delta)][f'(k^*) - \delta - \rho] + c^* f''(k^*) - (\rho + \nu)n.$$

Steady state (9.50) states $(\rho + \nu)n = [f(k^*)/k^* - (n - \nu + \delta)][f'(k^*) - \delta - \rho]$; substituting, the terms in $(n - \nu + \delta)$ cancel and

$$\det J = \left[f'(k^*) - \frac{f(k^*)}{k^*} \right] [f'(k^*) - \delta - \rho] + c^* f''(k^*) < 0$$

because $f'(k^*) < f(k^*)/k^*$ (strict concavity with $f(0) = 0$), $f'(k^*) > \delta + \rho$ (the text's $k^* < k_{\text{mgr}}$), $c^* > 0$ and $f'' < 0$. Since $\det J$ is the product of the eigenvalues and the discriminant $(\text{tr } J)^2 - 4\det J > 0$, the eigenvalues are real, nonzero and of opposite sign. The steady state is a saddle: exactly one initial value of $c(0)$ puts the economy on the stable manifold, which is the unique equilibrium path, and every other choice violates either feasibility or the transversality condition (9.44).

With $f(k) = k^{1/3}$, $\delta = 0.06$, $\rho = 0.02$, $n = \nu = 0.02$ the program gives

$$(k^*, c^*) = (7.8829, 1.5172), \quad \text{tr } J = +0.0283, \quad \det J = -0.0115,$$

with eigenvalues -0.0940 and $+0.1223$, and the closed-form determinant above reproduces -0.0115 exactly. For reference $k_{\text{mgr}} = 8.5052$ and $k_{\text{gold}} = 13.0946$, so $k^* < k_{\text{mgr}} < k_{\text{gold}}$, as PROPOSITION 9.10 requires.

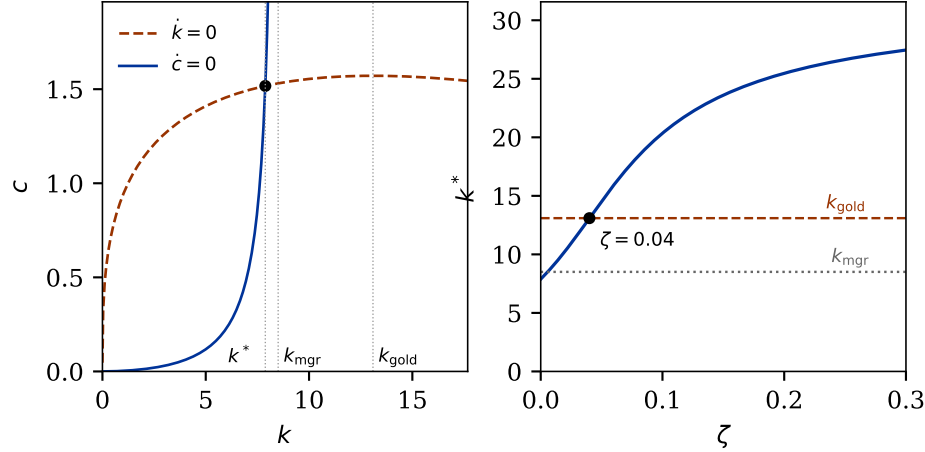


FIGURE 7. Left: the continuous-time perpetual youth model. The convex $\dot{c} = 0$ locus cuts the concave $\dot{k} = 0$ locus once, at a k^* below both k_{mgr} and k_{gold} : there is always underaccumulation. Right: with labor income declining at the rate ζ (Exercise 9.30), k^* rises with ζ and passes k_{gold} at $\zeta = 0.04$.

Exercise 9.29

Write the steady-state condition (9.50) as

$$\Phi(k; n, \nu) \equiv \frac{f(k)}{k} - (n - \nu + \delta) - \frac{(\rho + \nu)n}{f'(k) - \delta - \rho} = 0,$$

which, as in Exercise 9.27, is strictly decreasing in k on $(0, k_{\text{mgr}})$: $\partial\Phi/\partial k < 0$. Hence $\text{sgn}(dk^*/dx) = \text{sgn}(\partial\Phi/\partial x)$ for $x \in \{n, \nu\}$.

The arrival rate of new households.

$$\frac{\partial\Phi}{\partial n} = -1 - \frac{\rho + \nu}{f'(k^*) - \delta - \rho} < 0 \quad \implies \quad \frac{dk^*}{dn} < 0.$$

Both terms push the same way: a larger n raises the effective replacement requirement $n - \nu + \delta$, and it increases the flow of newborns who arrive with no assets and therefore consume less than average, which is the dilution term $(\rho + \nu)nk/c$.

The death rate. Here the two forces conflict — a larger ν raises the effective discount rate $\rho + \nu$ but lowers net population growth $n - \nu$ — and

$$\frac{\partial\Phi}{\partial \nu} = 1 - \frac{n}{f'(k^*) - \delta - \rho}.$$

To sign it, use the steady-state relations. From (9.49), $(\rho + \nu)n/[f'(k^*) - \delta - \rho] = c^*/k^*$, so $n/[f'(k^*) - \delta - \rho] = c^*/[k^*(\rho + \nu)]$; and from (9.46) with $a^* = k^*$, $c^* = (\rho + \nu)(k^* + \omega^*)$, where $\omega^* > 0$ is steady-state human wealth per capita. Therefore

$$\boxed{\frac{\partial\Phi}{\partial \nu} = 1 - \frac{k^* + \omega^*}{k^*} = -\frac{\omega^*}{k^*} < 0 \quad \implies \quad \frac{dk^*}{d\nu} < 0.}$$

The discounting effect dominates, and by exactly the ratio of human to physical wealth: shorter expected lives make households consume a larger fraction of their wealth, and the larger their human wealth relative to their assets, the more this matters.

Consumption. Differentiating $c^* = f(k^*) - (n - \nu + \delta)k^*$ and using $f'(k^*) > n - \nu + \delta$ (which holds because $k^* < k_{\text{gold}}$),

$$\frac{dc^*}{dn} = [f'(k^*) - (n - \nu + \delta)] \frac{dk^*}{dn} - k^* < 0,$$

unambiguously negative: both the fall in k^* and the higher replacement requirement reduce consumption. For ν ,

$$\frac{dc^*}{d\nu} = [f'(k^*) - (n - \nu + \delta)] \frac{dk^*}{d\nu} + k^*,$$

a negative indirect effect against a positive direct one (a higher death rate means slower population growth and hence less capital widening). With the parameters of Exercise 9.28 the program gives

$$\frac{dk^*}{dn} = -30.27, \quad \frac{dc^*}{dn} = -8.61, \quad \frac{dk^*}{d\nu} = -10.86, \quad \frac{dc^*}{d\nu} = +7.62,$$

and $\omega^* = c^*/(\rho + \nu) - k^* = 30.05$, so $-\omega^*/k^* = -3.81 < 0$ as claimed. The direct term dominates in $dc^*/d\nu$ by a wide margin: $f'(k^*) - (n - \nu + \delta) = 0.024$ is small because k^* is not far below k_{gold} in consumption terms, while $k^* = 7.88$ is large.

Remark. It is worth being clear about what ν is not doing. In the neoclassical growth model of Chapter 8 the analogous parameter would simply add to the discount rate and nothing else; here it enters three times — through the effective discount rate $\rho + \nu$, through population growth $n - \nu$, and through the annuity return $r + \nu$ — and the three effects have to be disentangled before the sign is clear. That n and ν both lower k^* while moving c^* in opposite directions is a reminder that finite lives and the arrival of new households are distinct forces, a theme the book pursues in Exercises 9.30 and 9.32.

Exercise 9.30

(a) *Human wealth and the aggregate Euler equation.* Each individual is born with one unit of effective labor, which decays at the rate $\zeta > 0$, so an individual of cohort τ earns $e^{-\zeta(t-\tau)}w(t)$ at t , where w is the wage per efficiency unit. Repeating Exercise 9.26 with this earnings stream, the present value of future earnings is

$$\omega(t | \tau) = \int_t^\infty \exp\left(-\int_t^s (r(z) + \nu) dz\right) e^{-\zeta(s-\tau)} w(s) ds,$$

which is (9.51), and logarithmic preferences again give $c(t | \tau) = (\rho + \nu)[a(t | \tau) + \omega(t | \tau)]$. Factoring out the cohort's current efficiency, $\omega(t | \tau) = e^{-\zeta(t-\tau)}\tilde{\omega}(t)$ with $\tilde{\omega}(t) \equiv \int_t^\infty \exp(-\int_t^s (r + \nu + \zeta))w(s) ds$, so that differentiating,

$$\dot{\tilde{\omega}}(t) = (r(t) + \nu + \zeta)\tilde{\omega}(t) - w(t).$$

The effective discount rate applied to labor income is now $r + \nu + \zeta$: earnings are lost both when the individual dies, at rate ν , and as his productivity decays, at rate ζ . Averaging over cohorts (the age density is ne^{-na} by Exercise 9.25, so average efficiency and average human wealth carry the same constant factor $n/(n + \zeta)$, which cancels below), average human wealth per capita obeys $\dot{\omega} = (r + \nu + \zeta)\omega - \bar{w}$, where $\bar{w}$ is average labor income per capita, while average assets obey

$\dot{a} = (r - (n - \nu))a + \bar{w} - c$. Hence

$$\dot{c} = (\rho + \nu)(\dot{a} + \dot{w}) = (\rho + \nu)\left[(r + \nu + \zeta)(a + \omega) - (n + \zeta)a - c\right] = (r + \zeta - \rho)c - (\rho + \nu)(n + \zeta)a,$$

using $c = (\rho + \nu)(a + \omega)$ in the last step. Dividing by c and setting $a = k$, $r = f'(k) - \delta$,

$$\frac{\dot{c}(t)}{c(t)} = f'(k(t)) - \delta - \rho + \zeta - (\rho + \nu)(n + \zeta)\frac{k(t)}{c(t)},$$

which is (9.52); (9.42) is unchanged. Setting $\zeta = 0$ recovers (9.48).

(b) *Overaccumulation*. The steady state now solves

$$\Phi(k; \zeta) \equiv \frac{f(k)}{k} - (n - \nu + \delta) - \frac{(\rho + \nu)(n + \zeta)}{f'(k) - \delta - \rho + \zeta} = 0, \quad c^* = f(k^*) - (n - \nu + \delta)k^*,$$

and $\partial\Phi/\partial k < 0$ as before. Differentiating in ζ ,

$$\frac{\partial\Phi}{\partial\zeta} = -\frac{(\rho + \nu)[f'(k) - \delta - \rho - n]}{[f'(k) - \delta - \rho + \zeta]^2} > 0 \quad \text{whenever } f'(k) < n + \delta + \rho,$$

so k^* is increasing in ζ over the relevant range. As $\zeta \rightarrow \infty$ the fraction tends to $\rho + \nu$, so the steady-state condition becomes

$$\frac{f(k)}{k} = n - \nu + \delta + \rho + \nu = n + \delta + \rho,$$

which defines a limiting k_∞ . Since k_{gold} solves $f'(k_{\text{gold}}) = n - \nu + \delta$,

$$k_\infty > k_{\text{gold}} \iff f'(k_\infty) < n - \nu + \delta,$$

and with $f(k) = Ak^\alpha$ one has $f'(k_\infty) = \alpha f(k_\infty)/k_\infty = \alpha(n + \delta + \rho)$, so the condition is the transparent

$$\boxed{\alpha(n + \delta + \rho) < n - \nu + \delta}$$

For $\alpha = 1/3$, $\delta = 0.06$, $\rho = 0.02$, $n = \nu = 0.02$ the left side is 0.0333 and the right side 0.06: the condition holds, and the program traces

ζ	0	0.02	0.05	0.10	0.25	$\rightarrow \infty$
k^*	7.88	10.27	14.54	20.35	26.65	31.62
k^*/k_{gold}	0.60	0.78	1.11	1.55	2.04	2.42

with $k^* = k_{\text{gold}}$ exactly at $\zeta = 0.04$ and overaccumulation above it (see the right panel of Figure 7). Note how little decay is needed: $\zeta = 0.04$ means effective labor halves about every seventeen years, which is not an extreme life-cycle earnings profile.

Intuition. With a flat earnings path an individual can finance old-age consumption out of current wages and has no need to accumulate; that is why the model of Section 9.8 always underaccumulates. A declining path forces saving for the future, and the steeper the decline the more must be saved. As $\zeta \rightarrow \infty$ all labor income is concentrated at birth, which is exactly the Samuelson–Diamond configuration — income when young, none when old — and overaccumulation duly reappears. Exercise 9.10 makes the same point from the other direction, by letting the old earn. The book completes the argument with Exercise 9.32: overaccumulation requires $n > 0$ and survives even with $\nu = 0$, so it is the arrival of new individuals rather than finite lives that is essential.

Exercise 9.31

The government. Let $b = B/L$ be debt per capita, $g = G/L$ spending per capita and $T(t)$ the lump-sum tax levied on each individual alive at t . The government's flow constraint and no-Ponzi condition are

$$\dot{b}(t) = (r(t) - (n - \nu))b(t) + g - T(t), \quad \lim_{t \rightarrow \infty} b(t) \exp\left(-\int_0^t (r(s) - (n - \nu)) ds\right) = 0,$$

which integrate to the intertemporal budget constraint

$$\int_0^\infty T(t) q(t) dt = b(0) + \int_0^\infty g q(t) dt, \quad q(t) \equiv \exp\left(-\int_0^t (r(s) - (n - \nu)) ds\right). \quad (G)$$

Any tax path satisfying (G) is feasible.

The household. An individual of cohort τ now faces $\dot{a}(t | \tau) = (r(t) + \nu)a(t | \tau) - c(t | \tau) + w(t) - T(t)$, and Exercise 9.26 gives

$$c(t | \tau) = (\rho + \nu)[a(t | \tau) + \omega(t)], \quad \omega(t) = \int_t^\infty \exp\left(-\int_t^s (r(z) + \nu) dz\right) [w(s) - T(s)] ds.$$

Here is the wedge. The household discounts future taxes at $r + \nu$, because it may be dead when they fall due; the government discounts them at $r - (n - \nu)$, because it can levy them on people not yet born. The two rates differ by

$$(r + \nu) - (r - (n - \nu)) = n,$$

the arrival rate of new households — and by nothing else.

Failure of equivalence. Consider a tax cut $dT < 0$ at date 0 financed by a path of higher taxes $\{dT(s)\}_{s>0}$ satisfying (G), so that $\int_0^\infty q(s) dT(s) ds = -dT > 0$. The household's human wealth changes by

$$d\omega(0) = -dT - \int_0^\infty \exp\left(-\int_0^s (r + \nu)\right) dT(s) ds > -dT - \int_0^\infty q(s) dT(s) ds = 0,$$

the strict inequality because $\exp(-\int_0^s (r + \nu)) < q(s)$ for every $s > 0$ when $n > 0$. Households value the deferred tax bill at strictly less than the government's cost of deferring it, so they perceive themselves as richer and raise consumption immediately.

The same point appears in the aggregate dynamics. Redoing the derivation of (9.48) with taxes and debt, average assets are $a = k + b$ and

$$\dot{a} = (r - (n - \nu))a + w - T - c, \quad \dot{\omega} = (r + \nu)\omega - (w - T),$$

so that $\dot{c} = (\rho + \nu)(\dot{a} + \dot{\omega}) = (r - \rho)c - (\rho + \nu)na$ and

$$\boxed{\dot{k} = f(k) - (n - \nu + \delta)k - c - g, \quad \frac{\dot{c}}{c} = f'(k) - \delta - \rho - (\rho + \nu)n \frac{k(t) + b(t)}{c(t)}}.$$

Government debt enters the equilibrium conditions *on its own*, not merely through the present value of spending. Two tax paths with the same present value generate different debt paths $\{b(t)\}$, hence different $\{k(t), c(t)\}$. Ricardian equivalence fails. ■

Interpretation, and the contrast with Chapter 8. In the neoclassical growth model (Exercise 8.35) the household that receives the tax cut is the same infinitely lived entity that will pay the higher taxes, and it discounts both at the same rate; debt is therefore not net wealth and the timing of taxes is irrelevant. Here part of the future tax bill falls on individuals who are not yet alive, and the current generation does not internalize their burden. Government debt is net wealth to the extent of that externality, a deficit today raises current consumption, crowds out capital and lowers the capital-labor ratio.

The magnitude of the wedge is exactly n , which delivers a sharp corollary: if $n = 0$ — nobody new ever arrives — the household's discount rate $r + \nu$ and the government's $r + \nu$ coincide, and Ricardian equivalence holds *even with* $\nu > 0$, that is, even with finite lives. What breaks equivalence is not mortality but the arrival of new households, the same conclusion the book reaches about overaccumulation in Exercise 9.32.

Exercise 9.33

The technology is $Y(t) = AK(t)^{1-\alpha}L(t)^\alpha$ with $L \equiv 1$ and full depreciation, so $Y(t) = AK(t)^{1-\alpha}$, the capital share is $1 - \alpha$, and competitive factor prices are

$$R(t) = (1 - \alpha)AK(t)^{-\alpha}, \quad w(t) = \alpha AK(t)^{1-\alpha}.$$

Capital income is taxed at τ and the proceeds, $\tau R(t)K(t)$, are consumed by the government, so the private resource constraint is $c(t) + K(t+1) = [1 - \tau(1 - \alpha)]AK(t)^{1-\alpha}$.

Economy 1: infinitely lived households. The household maximizes $\sum_{t \geq 0} \beta^t \log c(t)$ taking w and the after-tax return $(1 - \tau)R$ as given, so the Euler equation is

$$\frac{1}{c(t)} = \beta(1 - \tau)R(t+1) \frac{1}{c(t+1)} = \beta(1 - \tau)(1 - \alpha)AK(t+1)^{-\alpha} \frac{1}{c(t+1)}.$$

Guess a constant saving rate out of after-tax income; substituting $c(t) = (1 - \sigma)[1 - \tau(1 - \alpha)]AK(t)^{1-\alpha}$ into the Euler equation and cancelling gives

$$K(t+1) = \beta(1 - \tau)(1 - \alpha)AK(t)^{1-\alpha}, \quad K^* = [\beta(1 - \tau)(1 - \alpha)A]^{-\frac{1}{\alpha}}, \quad Y^* = A(K^*)^{1-\alpha}$$

(the guess is the solution because the policy function of this strictly concave dynamic programming problem is unique, by COROLLARY 6.1). Hence

$$\frac{d \log Y^*}{d \log(1 - \tau)} = \frac{1 - \alpha}{\alpha} > 0 :$$

taxation strictly reduces the steady-state capital stock and output.

Economy 2: overlapping generations. A generation born at t works when young, saves $s(t)$ and consumes the after-tax proceeds when old:

$$\max \log c_1(t) + \beta \log c_2(t+1) \quad \text{s.t.} \quad c_1(t) = w(t) - s(t), \quad c_2(t+1) = (1 - \tau)R(t+1)s(t).$$

With log preferences the Euler equation gives $c_2(t+1) = \beta(1 - \tau)R(t+1)c_1(t)$ and hence

$$s(t) = \frac{\beta}{1 + \beta} w(t) = \frac{\beta \alpha A}{1 + \beta} K(t)^{1-\alpha},$$

in which τ does not appear: the after-tax return cancels, because with log preferences income and substitution effects offset exactly. Since $K(t+1) = s(t)$,

$$K(t+1) = \frac{\beta\alpha A}{1+\beta} K(t)^{1-\alpha}, \quad K^* = \left[\frac{\beta\alpha A}{1+\beta} \right]^{\frac{1}{\alpha}}, \quad \frac{dY^*}{d\tau} = 0.$$

Taxation is completely neutral for output. With $\alpha = 2/3$ (a capital share of $1/3$), $\beta = 0.98^{30}$ and $A = 1$, the program gives

τ	0	0.10	0.25	0.50
Y^* (economy 1)	0.4264	0.4045	0.3693	0.3015
Y^* (economy 2)	0.4851	0.4851	0.4851	0.4851

a tax of 50% costing the first economy 29.3% of output and the second nothing.

Interpretation. In the OLG economy the tax is levied on the return received by the *old*, whose saving decision is already behind them, and it is financed out of a saving rate that — with log preferences — does not respond to the return. It is effectively a lump-sum levy on a generation that has no margin left to distort. In the infinite-horizon economy the same household earns the wage, owns the capital and is on the intertemporal margin at *every* date; the after-tax return is the price at which it converts consumption today into consumption tomorrow, so taxing it drives a permanent wedge between the marginal product of capital and the intertemporal rate of substitution, and the capital stock falls accordingly.

Remark. Two caveats sharpen the lesson. First, the OLG neutrality is a knife edge: with CRRA preferences of curvature θ , saving is $s(t) = w(t)/\{1 + \beta^{-1/\theta}[(1-\tau)R(t+1)]^{-(1-\theta)/\theta}\}$, which does depend on τ unless $\theta = 1$. What is robust is not the exact zero but the fact that the effect runs through a single, weak channel rather than through a repeated intertemporal margin. Second, even economy 1 generates only modest differences: the elasticity $(1-\alpha)/\alpha$ equals $1/2$ at a capital share of one third, so accounting for a tenfold income gap would require $(1-\tau)$ to differ by a factor of 100. Together with Exercise 8.32, which reaches the same conclusion for discount rates, this is decisive evidence that differences in capital income taxation cannot explain the cross-country income distribution — and part of the case for the book's turn, in Part VIII, toward technology and institutions.

Chapter 10

Human Capital and Economic Growth

The fourteen exercises of Chapter 10 left unsolved in the instructor's manual: Exercises 10.1, 10.3–10.5, 10.8–10.13, 10.15–10.17 and 10.19. The manual solves 10.2, 10.6, 10.7, 10.14, 10.18 and 10.20. Notation follows the book: $s(t)$ is the fraction of time spent in schooling, h human capital, $\eta(S)$ the human capital produced by S years of full-time schooling, ϕ the Ben-Porath human capital production function, $f(k, h)$ the per capita production function of Section 10.4, $\kappa = K/H$ the effective capital-labor ratio of Section 10.5, and λ the worker's share of output in Section 10.6. The numerical parts are computed by `code/ch10.py`.

Exercise 10.1

The environment. Time is discrete, $t = 0, 1, \dots, T$ with $T \leq \infty$. The individual discounts at the factor $\beta \in (0, 1)$ and faces a constant probability of death $\nu \in [0, 1]$ per period, so his objective is

$$\sum_{t=0}^T [\beta(1 - \nu)]^t u(c(t)), \quad (\text{D1})$$

with u strictly increasing. Human capital evolves according to

$$h(t+1) = G(t, h(t), s(t)), \quad h(0) \text{ given}, \quad (\text{D2})$$

where $s(t) \in [0, 1]$ is the fraction of time devoted to schooling, subject to the further restriction

$$s(t) \in S(t) \subset [0, 1]. \quad (\text{D3})$$

The individual faces an exogenous path of wages per unit of human capital $[w(t)]_{t=0}^T$, an exogenous path of raw labor $[\omega(t)]_{t=0}^T$ and a constant gross interest rate $1 + r > 1$ at which he may borrow and lend freely. His lifetime budget constraint is

$$\sum_{t=0}^T \frac{c(t)}{(1+r)^t} \leq \sum_{t=0}^T \frac{w(t)[1-s(t)][h(t) + \omega(t)]}{(1+r)^t} \equiv Y([s, h]). \quad (\text{D4})$$

(As in Exercises 8.5 and 9.5, (D4) is the integrated form of the flow constraint $a(t+1) = (1+r)a(t) + w(t)[1-s(t)][h(t) + \omega(t)] - c(t)$ together with $a(0) = 0$ and the terminal condition $\lim_{t \rightarrow \infty} a(t)/(1+r)^t \geq 0$; free borrowing and lending at the single rate r is what collapses the $T+1$ flow constraints into the one constraint (D4).)

Separation Theorem (discrete time). Suppose u is strictly increasing and $r > 0$.

Then $[\hat{c}(t), \hat{s}(t), \hat{h}(t)]_{t=0}^T$ solves the maximization of (D1) subject to (D2)–(D4) if and only if

- (1) $[\hat{s}(t), \hat{h}(t)]_{t=0}^T$ maximizes the present value of earnings $Y([s, h])$ subject to (D2) and (D3); and
- (2) $[\hat{c}(t)]_{t=0}^T$ maximizes (D1) subject to $\sum_t (1+r)^{-t} c(t) \leq \hat{Y} \equiv Y([\hat{s}, \hat{h}])$.

Proof of “only if”. Suppose $[\hat{c}, \hat{s}, \hat{h}]$ solves the full problem but $[\hat{s}, \hat{h}]$ does not maximize Y . Then there is a feasible $[\tilde{s}, \tilde{h}]$ with $\tilde{Y} \equiv Y([\tilde{s}, \tilde{h}]) > \hat{Y}$. Feasibility of $[\hat{c}, \hat{s}, \hat{h}]$ gives $\sum_t (1+r)^{-t} \hat{c}(t) \leq \hat{Y}$. For $\varepsilon > 0$ set $c(t) = \hat{c}(t) + \varepsilon$ for all t , together with $s = \tilde{s}$ and $h = \tilde{h}$. Then

$$\sum_{t=0}^T \frac{c(t)}{(1+r)^t} = \sum_{t=0}^T \frac{\hat{c}(t)}{(1+r)^t} + \varepsilon A(T), \quad A(T) \equiv \sum_{t=0}^T \frac{1}{(1+r)^t} = \frac{1 - (1+r)^{-(T+1)}}{1 - (1+r)^{-1}} \leq \frac{1+r}{r} < \infty,$$

the annuity factor — the discrete-time counterpart of $[1 - e^{-rT}]/r$, and finite even for $T = \infty$ precisely because $r > 0$. Hence $\sum_t (1+r)^{-t} c(t) \leq \hat{Y} + \varepsilon A(T)$, and choosing $\varepsilon < (\tilde{Y} - \hat{Y})/A(T)$ makes this at most $\tilde{Y}$. So $[c, \tilde{s}, \tilde{h}]$ is feasible; and since u is strictly increasing and $c(t) > \hat{c}(t)$ at every date, it is strictly preferred to $[\hat{c}, \hat{s}, \hat{h}]$, contradicting optimality. Therefore $[\hat{s}, \hat{h}]$ maximizes Y , and given that, $[\hat{c}]$ must maximize (D1) on the budget set it generates.

Proof of “if”. Suppose $[\hat{s}, \hat{h}]$ maximizes Y , with value $\hat{Y}$, and $[\hat{c}]$ maximizes (D1) subject to $\sum_t (1+r)^{-t} c(t) \leq \hat{Y}$. Take any feasible $[c, s, h]$ for the full problem. Its schooling plan yields $Y([s, h]) \leq \hat{Y}$, so the budget constraint (D4) gives $\sum_t (1+r)^{-t} c(t) \leq Y([s, h]) \leq \hat{Y}$; hence $[c]$ lies in the budget set over which $[\hat{c}]$ was optimal, and so $[c]$ is weakly worse than $[\hat{c}]$. Therefore $[\hat{c}, \hat{s}, \hat{h}]$ solves the full problem. ■

What the proof uses. Only three things: (i) u is strictly increasing, so a uniformly larger consumption path is strictly better; (ii) there is a *single* lifetime budget constraint, which requires borrowing and lending at a common rate with no quantity limits; and (iii) schooling enters the problem *only* through that constraint — there is no leisure, so s does not appear in u . Exercise 10.2 shows that the theorem fails when (ii) is dropped (a borrowing constraint makes the timing of earnings matter, not just their present value) or when (iii) is dropped (with $u(c, 1-l)$ the marginal rate of substitution between schooling and consumption is no longer independent of the consumption plan).

Remark. This is Fisher separation for human capital: with a complete market for intertemporal trade, the individual’s “firm” — his stock of human capital — should be run to maximize its present value, and preferences determine only how that value is spread across dates. The discrete-time proof is if anything cleaner than the continuous-time one, since the annuity factor $A(T)$ is an explicit finite sum.

Exercise 10.3

By THEOREM 10.1 the optimal S maximizes the present value of earnings, (10.8):

$$V(S) = \int_S^\infty e^{-(r+\nu)t} w(t) h(t) dt.$$

Two substitutions do the work. Wages grow at g_w by (10.7), so $w(t) = w(0)e^{g_w t}$. Human capital equals $\eta(S)$ at graduation and grows at g_h thereafter by (10.6), so for $t \geq S$,

$$h(t) = \eta(S)e^{g_h(t-S)}.$$

Hence the integrand is

$$e^{-(r+\nu)t} w(0)e^{g_w t} \eta(S)e^{g_h(t-S)} = w(0)\eta(S)e^{-g_h S} e^{-(r+\nu-g_w-g_h)t},$$

and since $g_w + g_h < r + \nu$ the integral converges:

$$V(S) = w(0)\eta(S)e^{-g_h S} \frac{e^{-(r+\nu-g_w-g_h)S}}{r+\nu-g_h-g_w} = \frac{\eta(S) w(0) \exp(-(r+\nu-g_w)S)}{r+\nu-g_h-g_w},$$

the two exponents in S combining because $-g_h S - (r + \nu - g_w - g_h)S = -(r + \nu - g_w)S$. This is (10.9). With $r = 0.10$, $\nu = g_w = 0.02$, $g_h = 0$ and $S = 8$ the program evaluates the integral numerically as 15.158858689 against the closed form 15.158858689.

Reading the formula. The numerator is the human capital the individual owns on graduation day, valued at the current wage and discounted for the delay S at the rate $r + \nu - g_w$; the denominator turns the flow into a stock, an annuity factor at the effective rate $r + \nu - g_h - g_w$. The first-order condition is then (10.10),

$$\frac{\eta'(S^*)}{\eta(S^*)} = r + \nu - g_w,$$

which equates the *proportional* gain in earnings from an extra year of schooling to the cost of postponing the whole earnings stream by a year.

Remark. The text justifies the uniqueness of S^* by saying that “since $\eta(S)$ is concave, the objective function in (10.9) is strictly concave.” That is not quite the right argument: for $\Psi(S) = \eta(S)e^{-aS}$ one has $\Psi'' = e^{-aS}[\eta'' - 2a\eta' + a^2\eta]$, whose sign is not determined by $\eta'' < 0$. What is true, and is all that is needed, is that Ψ is strictly *log*-concave: $\log \Psi(S) = \log \eta(S) - aS$ and $\log \eta$ is strictly concave whenever η is positive, increasing and strictly concave. Equivalently η'/η is strictly decreasing, so (10.10) has at most one solution, and the objective is strictly quasiconcave, which gives a unique maximizer. The same observation is what makes the comparative static of Exercise 10.4 unambiguous.

Exercise 10.4

Higher r , less schooling. By (10.10) the individual's schooling solves

$$\Lambda'(S^*) = r + \nu - g_w, \quad \Lambda(S) \equiv \log \eta(S),$$

and $\Lambda' = \eta'/\eta$ is strictly decreasing, since

$$\frac{d}{dS} \frac{\eta'}{\eta} = \frac{\eta''\eta - (\eta')^2}{\eta^2} < 0.$$

Implicit differentiation therefore gives

$$\frac{dS^*}{dr} = \frac{1}{[\eta''\eta - (\eta')^2]/\eta^2} < 0.$$

Impatient individuals weigh the foregone earnings of an extra year more heavily against the proportional wage gain it delivers, and stop schooling sooner. (The same computation shows $dS^*/d\nu < 0$ and $dS^*/dg_w > 0$.)

What the wage regression recovers. Cross-sectionally, log earnings of an individual with schooling S_j and experience e_j are, from the derivation of (10.12),

$$\log W_j = \text{constant} + \Lambda(S_j) + g_h e_j.$$

The relation between log wages and schooling is thus the *human capital production function* $\Lambda = \log \eta$, whatever generates the variation in S . Estimating the linear equation (10.12) by least squares

(controlling for experience) delivers

$$\hat{\gamma}_s = \frac{\text{Cov}(\Lambda(S_j), S_j)}{\text{Var}(S_j)},$$

which, since Λ is strictly concave, is a weighted average of the individual marginal returns

$$\Lambda'(S_j) = r_j + \nu - g_w.$$

Three conclusions follow.

(i) *The regression is consistent, but for a population average.* Because r_j affects wages *only* through schooling — it does not enter the wage equation — there is no omitted-variable bias of the usual “ability bias” kind, and $\hat{\gamma}_s$ estimates the average marginal return in the population. Had the heterogeneity been in η (ability) rather than in r , S_j would be correlated with the error and OLS would be inconsistent. The source of the variation, not its size, is what determines whether the regression means anything.

(ii) *But what it estimates is a preference parameter, not a technological one.* The marginal return is $r_j + \nu - g_w$, the individual’s own discount rate adjusted for mortality and wage growth. A cross section in which schooling varies only because patience varies traces out the *marginal return schedule* Λ' , so the returns to schooling coefficient reveals how impatient people are, not how productive schooling is. The empirical range $\gamma_s \in [0.06, 0.10]$ then says that effective discount rates lie in roughly that range, which is exactly the reading the text gives when it calibrates $r \approx 0.10$, $\nu \approx 0.02$ and $g_w \approx 0.02$ to obtain $\gamma_s \approx 0.10$.

(iii) *A single number conceals very heterogeneous returns.* Take $\eta(S) = \exp(aS - \frac{b}{2}S^2)$, so that $\Lambda'(S) = a - bS$ and $S^*(r) = [a - (r + \nu - g_w)]/b$ is linear in r . With $a = 0.20$, $b = 0.012$, $\nu = g_w = 0.02$ and r spread uniformly over $[0.04, 0.16]$, the program gives

r	0.04	0.07	0.10	0.13	0.16
S^* (years)	13.33	10.83	8.33	5.83	3.33
marginal return	0.040	0.070	0.100	0.130	0.160

and an OLS Mincer coefficient of $\hat{\gamma}_s = 0.100$ — the population mean. But the marginal return ranges from 4% for the most-schooled to 16% for the least-schooled. A policy that induces the impatient to stay one more year earns 16%; one that extends the schooling of the already highly educated earns 4%. Neither is $\hat{\gamma}_s$. Figure 8 shows the gap: the true relation $\Lambda(S)$ is concave, and the fitted line crosses it twice.

Remark. This is the theoretical core of the returns-to-schooling literature that follows Card (1999): when marginal returns are heterogeneous, an instrument that shifts the *cost* of schooling recovers a local average of the returns for the individuals it moves, not a common parameter — and it can exceed the OLS estimate precisely because the individuals whose schooling responds to cost shifters are the ones with the highest marginal returns. The model here is the simplest environment in which that statement is exactly true.

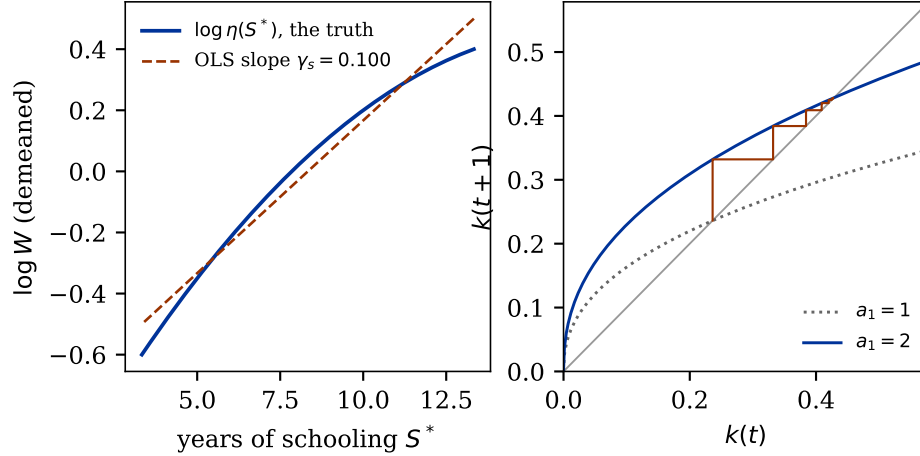


FIGURE 8. Left (Exercise 10.4): when schooling varies only because discount rates vary, the true relation between log wages and schooling is the concave function $\log \eta(S)$, and the Mincer coefficient is the average of the individual marginal returns. Right (Exercise 10.19): in the closed economy of Section 10.6, a rise in one group's ability shifts the accumulation map upward, and the economy climbs to a higher k^* — raising the *other* group's human capital from the next period on.

Exercise 10.5

The problem is

$$\max_{[s(t)]} \int_0^\infty e^{-(r+\nu)t} (1-s(t))h(t) dt \quad \text{s.t.} \quad \dot{h}(t) = \phi(s(t)h(t)) - \delta_h h(t), \quad s(t) \in [0, 1], \quad h(0) > 0.$$

Step 1: the state is confined to a compact interval. Setting $s = 1$ gives the fastest possible accumulation, $\dot{h} \leq \phi(h) - \delta_h h$. Since ϕ is strictly concave, increasing and eventually flat, $\phi(h) - \delta_h h$ is negative for large h ; let $\bar{h}$ solve $\phi(\bar{h}) = \delta_h \bar{h}$. Then

$$h(t) \leq h_{\max} \equiv \max\{h(0), \bar{h}\} \quad \text{for all } t.$$

For a lower bound, note that the optimal path converges monotonically to the steady state $h^* = \phi(x^*)/\delta_h > 0$ of (10.17)–(10.18), so $\inf_t h(t) = \min\{h(0), h^*\} > 0$; pick ϵ strictly below that number. This is the argument of Exercise 7.27(c) applied here, and it is what the hint means by imitating Section 7.7. The state therefore lives in the compact interval $[\epsilon, h_{\max}] \subset (0, \infty)$, and the objective is bounded:

$$\left| \int_0^\infty e^{-(r+\nu)t} (1-s)h dt \right| \leq \frac{h_{\max}}{r+\nu} < \infty.$$

Step 2: THEOREM 7.13 gives the necessary conditions. The control set $[0, 1]$ is compact, $G(h, s) = \phi(sh) - \delta_h h$ is continuously differentiable, the state is confined to a compact interval, and the objective converges. Theorem 7.13 therefore applies to the current-value Hamiltonian

$$\mathcal{H}(h, s, \mu) = (1-s)h + \mu[\phi(sh) - \delta_h h]$$

and yields (10.14): $\mathcal{H}_s = 0$, $\mathcal{H}_h = (r+\nu)\mu - \dot{\mu}$ and $\lim_{t \rightarrow \infty} e^{-(r+\nu)t} \mu(t)h(t) = 0$. The first is an equality rather than an inequality because the optimum is interior: $\mathcal{H}_s = -h + \mu h \phi'(sh)$ is $+\infty$ at

$s = 0$ (since $\phi'(0) = \infty$) and negative at $s = 1$ (since ϕ' has fallen to zero by then), so no corner can be optimal.

Step 3: THEOREM 7.14 gives sufficiency — after a change of variable. Theorem 7.14 requires the maximized Hamiltonian to be concave in the state. As written, $\mathcal{H}$ is *not* jointly concave in (h, s) : the term $(1 - s)h$ is bilinear, with Hessian $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ and eigenvalues ± 1 , and $\phi(sh)$ is not jointly concave either. Take instead the control

$$x(t) \equiv s(t)h(t) \in [0, h(t)],$$

which is the transformation the text introduces immediately after (10.14). The problem becomes

$$\max \int_0^\infty e^{-(r+\nu)t} [h(t) - x(t)] dt \quad \text{s.t.} \quad \dot{h} = \phi(x) - \delta_h h, \quad 0 \leq x(t) \leq h(t),$$

whose Hamiltonian

$$\mathcal{H}(h, x, \mu) = h - x + \mu[\phi(x) - \delta_h h]$$

is linear in h and, since $\mu = 1/\phi'(x) > 0$ by (10.15), strictly concave in x : jointly concave in (h, x) . The constraint set $\{(h, x) : 0 \leq x \leq h\}$ is convex, and the change of variable is a bijection for $h > 0$, so the two problems have the same solutions.

Finally, Theorem 7.14 needs $\lim_{t \rightarrow \infty} e^{-(r+\nu)t} \mu(t)h(t) \geq 0$ along every feasible path, which is immediate here because $h \geq 0$ and $\mu > 0$. Hence the conditions (10.14) are sufficient as well as necessary, and strict concavity of ϕ makes the maximizing x — and therefore the whole optimal path — unique.

Remark. The change of variable is usually presented as algebraic convenience, a way of reducing (10.14) to the two equations (10.15) and (10.16). This exercise shows that it is doing something more substantial: it is what convexifies the problem. Whenever a control enters multiplicatively with the state, it is worth asking whether the product is the natural control.

Exercise 10.8

Write $z = (k, h, i_k, i_h)$ and

$$\begin{aligned} \widehat{H}(z; \mu_k, \mu_h) &= \underbrace{u(f(k, h) - i_h - i_k)}_{= (u \circ g)(z)} + \underbrace{\mu_h(i_h - \delta_h h) + \mu_k(i_k - \delta_k k)}_{\text{linear in } z}, \quad g(z) \equiv f(k, h) - i_k - i_h. \end{aligned}$$

The second group of terms is linear in z for any values of the costates, hence concave, so it suffices to show that $u \circ g$ is concave.

g is concave. f is jointly strictly concave in (k, h) by Assumptions 1' and 2', and $-i_k - i_h$ is linear; a concave function plus a linear one is concave.

A concave increasing function of a concave function is concave. Let z_0, z_1 and $\lambda \in [0, 1]$, and write $z_\lambda = \lambda z_1 + (1 - \lambda)z_0$. Then

$$g(z_\lambda) \geq \lambda g(z_1) + (1 - \lambda)g(z_0)$$

by concavity of g ; applying u , which is strictly increasing,

$$u(g(z_\lambda)) \geq u(\lambda g(z_1) + (1 - \lambda)g(z_0)) \geq \lambda u(g(z_1)) + (1 - \lambda)u(g(z_0)),$$

the second inequality by concavity of u . Hence $u \circ g$ is concave, and so is $\hat{H}$. ■

Both hypotheses are needed: if u were decreasing the first inequality would reverse, and if u were convex the second would. Assumption 3 supplies exactly $u' > 0$ and $u'' < 0$.

Concavity of $\hat{H}$ in the states *and* the controls jointly implies concavity of the maximized Hamiltonian in the states (Exercise 7.7), which is what THEOREM 7.14 requires; with the transversality conditions, the conditions listed after (10.24) therefore characterize the optimum.

Remark. $\hat{H}$ is concave but not *strictly* so, and the failure is informative. Along any direction that changes i_k and i_h in opposite directions by the same amount, g is constant and $u \circ g$ is unchanged; only the linear terms move. The composition of investment is therefore not pinned down by concavity, which is precisely why the first-order conditions deliver $\mu_k = \mu_h$ — the two investments are perfect substitutes in the resource constraint — and why, as the text observes after PROPOSITION 10.1, the economy can jump instantaneously to the balanced ratio with $i_k(0)$ very negative and $i_h(0)$ very large.

Exercise 10.9

Write (10.25) as $\Delta(k, h) = 0$ with

$$\Delta(k, h) \equiv f_k(k, h) - f_h(k, h) - (\delta_k - \delta_h).$$

Uniqueness. For fixed $k > 0$,

$$\frac{\partial \Delta}{\partial h} = f_{kh}(k, h) - f_{hh}(k, h) > 0,$$

since $f_{kh} > 0$ by the maintained complementarity assumption and $f_{hh} < 0$ by strict concavity. So $\Delta(k, \cdot)$ is strictly increasing and has at most one zero.

Existence. By Assumption 2', $f_h(k, h) \rightarrow \infty$ as $h \rightarrow 0$ while $f_k(k, h)$ stays finite, so $\Delta(k, h) \rightarrow -\infty$; and as $h \rightarrow \infty$, $f_h(k, h) \rightarrow 0$ while $f_k(k, h)$ increases without bound (since $f_{kh} > 0$; for $f = k^\alpha h^\beta$, $f_k = \alpha k^{\alpha-1} h^\beta \rightarrow \infty$), so $\Delta(k, h) \rightarrow +\infty$. By continuity there is exactly one root, which defines $\xi(k)$.

Monotonicity and smoothness. f is twice continuously differentiable and $\Delta_h \neq 0$, so the implicit function theorem (THEOREM A.25) makes ξ continuously differentiable, with

$$\xi'(k) = -\frac{\Delta_k}{\Delta_h} = \frac{f_{kh}(k, h) - f_{kk}(k, h)}{f_{kh}(k, h) - f_{hh}(k, h)} \bigg|_{h=\xi(k)} > 0$$

because $f_{kk} < 0$, $f_{hh} < 0$ and $f_{kh} > 0$ make both numerator and denominator strictly positive.

An illustration. With $f(k, h) = k^\alpha h^\beta$, $\alpha = \beta = 1/3$, $\delta_k = 0.08$ and $\delta_h = 0.04$, the program gives

k	0.5	1.0	2.0	5.0
$\xi(k)$	0.5508	1.1302	2.3353	6.1819
$\xi'(k)$	1.1377	1.1772	1.2294	1.3267

matching a numerical derivative to six digits. In the special case $\delta_k = \delta_h$ the locus is exactly linear, $h = (\beta/\alpha)k$; a higher depreciation rate on physical capital tilts the economy toward human capital, and here $h^*/k^* = 1.667$ rather than 1.

What the result says. Physical and human capital are always in balance, at a ratio determined entirely by technology and depreciation, because the two are produced from the same good at the same price and can be converted into each other freely. The economy therefore has only one effective state variable — which is the content of PROPOSITION 10.1 and the reason imbalances require irreversibility (Exercise 10.14) or labor market frictions (Section 10.6).

Exercise 10.10

(a) *Consumption.* The first two conditions after (10.24) give $\mu_k(t) = \mu_h(t) = u'(c(t)) \equiv \mu(t)$. The third then reads

$$f_k(k, h)u'(c) - \mu\delta_k = \rho\mu - \dot{\mu} \quad \implies \quad \frac{\dot{\mu}}{\mu} = \rho + \delta_k - f_k(k, h).$$

Since $\mu = u'(c)$,

$$\frac{\dot{\mu}}{\mu} = \frac{u''(c)\dot{c}}{u'(c)} = -\varepsilon_u(c)\frac{\dot{c}}{c}, \quad \varepsilon_u(c) = -\frac{u''(c)c}{u'(c)},$$

so that, substituting $h = \xi(k)$ from Exercise 10.9,

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\varepsilon_u(c(t))} \left[f_k(k(t), \xi(k(t))) - \delta_k - \rho \right].$$

Capital. Along the optimal path $h(t) = \xi(k(t))$, so $\dot{h} = \xi'(k)\dot{k}$. From (10.21) and (10.22),

$$i_k = \dot{k} + \delta_k k, \quad i_h = \dot{h} + \delta_h h = \xi'(k)\dot{k} + \delta_h \xi(k),$$

and substituting into the resource constraint (10.23), which binds,

$$c + \dot{k} + \delta_k k + \xi'(k)\dot{k} + \delta_h \xi(k) = f(k, \xi(k)).$$

Collecting the terms in $\dot{k}$,

$$\dot{k}(t) = \frac{1}{1 + \xi'(k(t))} \left[f(k(t), \xi(k(t))) - \delta_h \xi(k(t)) - \delta_k k(t) - c(t) \right],$$

which is the second equation of PROPOSITION 10.1. The factor $1/[1 + \xi'(k)]$ is the price of raising k by one unit: doing so requires $\xi'(k)$ additional units of human capital to keep the economy on the locus.

Transversality. Integrating $\dot{\mu}/\mu = \rho + \delta_k - f_k$,

$$e^{-\rho t} \mu(t) = \mu(0) \exp \left(- \int_0^t [f_k(k(s), \xi(k(s))) - \delta_k] ds \right),$$

so $\lim_{t \rightarrow \infty} e^{-\rho t} \mu_k(t)k(t) = 0$ is equivalent to

$$\lim_{t \rightarrow \infty} k(t) \exp \left(- \int_0^t [f_k(k(s), \xi(k(s))) - \delta_k] ds \right) = 0,$$

and the second transversality condition, in h , is then implied by $h = \xi(k)$ and $\mu_h = \mu_k$. The system is a planar one in (k, c) with exactly the structure of the one-sector neoclassical model of Chapter 8, so it has a unique steady state at $f_k(k^*, \xi(k^*)) = \delta_k + \rho$, saddle-path stable, with monotone convergence. ■

Remark. The transversality condition as printed in PROPOSITION 10.1 omits δ_k from the exponent. The derivation above shows that the correct exponent carries the depreciation rate, $-\int_0^t [f_k - \delta_k] ds$, that is $-\int_0^t r(s) ds$ with $r = f_k - \delta_k$ the net interest rate, exactly as in (8.32) and PROPOSITION 8.7. Since $f_k - \delta_k \rightarrow \rho > 0$ along the equilibrium path, both versions are satisfied by the optimal path and nothing substantive turns on it.

(b) *The other form.* Repeat the argument with the *fourth* condition after (10.24) in place of the third:

$$f_h(k, h)u'(c) - \mu\delta_h = \rho\mu - \dot{\mu} \quad \implies \quad \frac{\dot{\mu}}{\mu} = \rho + \delta_h - f_h(k, h),$$

so that

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\varepsilon_u(c(t))} \left[f_h(k(t), \xi(k(t))) - \delta_h - \rho \right].$$

The two expressions are identical because subtracting the two costate equations gives

$$f_k(k, h) - \delta_k = f_h(k, h) - \delta_h,$$

which is exactly (10.25). The program confirms it: with $f = k^{1/3}h^{1/3}$, $\delta_k = 0.08$, $\delta_h = 0.04$, the columns $f_k - \delta_k$ and $f_h - \delta_h$ agree to six digits at every k on the locus, and the steady state has

$$k^* = 61.728, \quad h^* = 102.881, \quad f_k(k^*, h^*) - \delta_k = f_h(k^*, h^*) - \delta_h = \rho = 0.02.$$

Why. Both capitals are assets, both are produced from the consumption good one for one, and both can be accumulated or decumulated freely. No-arbitrage therefore forces their net rates of return to be equal, and that common value *is* the interest rate $r(t)$ the household faces. The Euler equation compares r with ρ ; it is indifferent about which of the two assets one uses to measure r .

Exercise 10.11

Modify the resource constraint to $c + (1 + \tau)(i_k + i_h) \leq f(k, h)$, as in Section 8.10, and repeat the optimization. The current-value Hamiltonian is

$$\hat{H} = u(f(k, h) - (1 + \tau)(i_k + i_h)) + \mu_k(i_k - \delta_k k) + \mu_h(i_h - \delta_h h),$$

and the first-order conditions in the two investments give

$$\mu_k = \mu_h = (1 + \tau)u'(c).$$

The costate equation in k is $f_k u'(c) - \mu_k \delta_k = \rho \mu_k - \dot{\mu}_k$, so

$$\frac{\dot{\mu}_k}{\mu_k} = \rho + \delta_k - \frac{f_k(k, h)}{1 + \tau},$$

the distortion entering because a unit of forgone consumption now buys only $1/(1 + \tau)$ units of capital. In a steady state $\dot{c} = 0$, hence $\dot{\mu}_k = 0$, and likewise for h :

$$f_k(k^*, h^*) = (1 + \tau)(\rho + \delta_k), \quad f_h(k^*, h^*) = (1 + \tau)(\rho + \delta_h). \quad (*)$$

Cobb–Douglas. With $Y = K^\alpha H^\beta L^{1-\alpha-\beta}$ and $L = 1$, so $y = k^\alpha h^\beta$, the two conditions in $(*)$ read $\alpha y/k = A_k$ and $\beta y/h = A_h$, where $A_k \equiv (1 + \tau)(\rho + \delta_k)$ and $A_h \equiv (1 + \tau)(\rho + \delta_h)$. Hence $k = \alpha y/A_k$

and $h = \beta y / A_h$, and substituting back into $y = k^\alpha h^\beta$,

$$y = \left(\frac{\alpha y}{A_k}\right)^\alpha \left(\frac{\beta y}{A_h}\right)^\beta = y^{\alpha+\beta} \left(\frac{\alpha}{A_k}\right)^\alpha \left(\frac{\beta}{A_h}\right)^\beta \implies y = \left[\left(\frac{\alpha}{A_k}\right)^\alpha \left(\frac{\beta}{A_h}\right)^\beta\right]^{\frac{1}{1-\alpha-\beta}}.$$

Both A_k and A_h are proportional to $1 + \tau$, so $y \propto (1 + \tau)^{-(\alpha+\beta)/(1-\alpha-\beta)}$ and

$$\frac{Y(\tau)}{Y(\tau')} = \left(\frac{1 + \tau'}{1 + \tau}\right)^{\frac{\alpha+\beta}{1-\alpha-\beta}}$$

which is (10.26). ■

How much this adds. The elasticity is $(\alpha + \beta)/(1 - \alpha - \beta)$ where the one-capital model of Chapter 8 gave $\alpha/(1 - \alpha)$:

(α, β)	(1/3, 0)	(1/3, 1/6)	(1/3, 1/3)
exponent	0.50	1.00	2.00
eightfold distortion	2.83	8.00	64.00

Adding a human capital share equal to the physical capital share quadruples the exponent, and the income ratio a given distortion can generate rises from under threefold to sixty-fourfold.

Remark. The mechanism deserves scepticism, and the text supplies it. The large number comes entirely from an extremely elastic human capital response: the model has human capital adjusting to after-tax returns exactly as physical capital does, so a tax on investment compounds across both margins. Whether schooling decisions really respond to corporate taxes or to corruption in that way is doubtful, and the exercise's arithmetic is the same arithmetic that makes the Mankiw–Romer–Weil accounting attribute so much to human capital — which Chapter 3 found the evidence does not support. What the calculation establishes is a logical possibility, not a measurement.

Exercise 10.12

(a) *The consumer.* The representative household owns the two capital stocks and rents them to firms at $R_k(t)$ and $R_h(t)$, and supplies its unit of raw labor at the wage $w(t)$. It solves

$$\max \int_0^\infty e^{-\rho t} u(c(t)) dt \quad \text{s.t.} \quad c(t) + i_k(t) + i_h(t) \leq R_k(t)k(t) + R_h(t)h(t) + w(t),$$

$$\dot{k}(t) = i_k(t) - \delta_k k(t), \quad \dot{h}(t) = i_h(t) - \delta_h h(t), \quad k(0), h(0) > 0 \text{ given,}$$

together with the no-Ponzi conditions $\lim_{t \rightarrow \infty} k(t)e^{-\int_0^t (R_k - \delta_k)} \geq 0$ and the analogue in h . Prices are taken as given.

(b) *Equilibrium.* A competitive equilibrium is a path

$$[c(t), i_k(t), i_h(t), k(t), h(t), R_k(t), R_h(t), w(t)]_{t=0}^\infty$$

such that

- (1) the household's path solves the problem in (a) given prices;
- (2) firms maximize: with $F(K, H, L)$ constant returns to scale and $L = 1$,

$$R_k(t) = f_k(k(t), h(t)), \quad R_h(t) = f_h(k(t), h(t)), \quad w(t) = f(k(t), h(t)) - k(t)f_k - h(t)f_h;$$

- (3) markets clear: the factor markets by (2), and the goods market by $c(t) + i_k(t) + i_h(t) = f(k(t), h(t))$, which follows from (2) and Euler's theorem.

(c) *Characterization.* The household's current-value Hamiltonian is

$$\hat{H} = u(R_k k + R_h h + w - i_k - i_h) + \mu_k(i_k - \delta_k k) + \mu_h(i_h - \delta_h h),$$

with R_k, R_h, w exogenous to the household. The conditions in the two investments give $\mu_k = \mu_h = u'(c)$ again, and the two costate equations give

$$\frac{\dot{\mu}_k}{\mu_k} = \rho + \delta_k - R_k(t), \quad \frac{\dot{\mu}_h}{\mu_h} = \rho + \delta_h - R_h(t).$$

Equality of the costates forces the *no-arbitrage condition*

$$R_k(t) - \delta_k = R_h(t) - \delta_h \equiv r(t),$$

and substituting the firms' conditions from (2) turns this into $f_k(k, h) - f_h(k, h) = \delta_k - \delta_h$, which is (10.25); so $h(t) = \xi(k(t))$ in equilibrium too. The Euler equation is

$$\frac{\dot{c}}{c} = \frac{1}{\varepsilon_u(c)} [r(t) - \rho] = \frac{1}{\varepsilon_u(c)} [f_k(k, \xi(k)) - \delta_k - \rho],$$

the resource constraint reproduces the second equation of PROPOSITION 10.1, and the transversality conditions are the ones derived in Exercise 10.10.

Every equilibrium condition is therefore identical to an optimality condition of the planner's problem, and conversely; since both problems have a unique solution given $k(0)$, the competitive equilibrium and the optimal growth path coincide. ■

Why they coincide, and what is doing the work. There are no externalities, technology has constant returns, markets are complete and the household is a price taker, so the First and Second Welfare Theorems of Chapter 5 apply. What is worth noticing is the interpretation of (10.25): in the planner's problem it is a first-order condition, an instruction to keep the two capitals in balance; in the market it is a *no-arbitrage condition*, forced by the fact that both assets are produced from the same good at the same price and can be converted at will. The balance the planner chooses is the balance arbitrage imposes.

Exercise 10.13

Let $Y(t) = F(K(t), H(t), A(t)L)$ with $A(t) = A(0)e^{gt}$, $L = 1$ and F constant returns to scale in its three arguments; preferences are CRRA with curvature θ .

(a) *Equilibrium* is defined exactly as in Exercise 10.12, with the firm's conditions

$$R_k(t) = F_K, \quad R_h(t) = F_H, \quad w(t) = F_L \cdot A(t),$$

where w is now the wage per unit of raw labor and grows over time.

(b) *Transformed variables.* Because progress is labor augmenting and F has constant returns, the natural normalization is by effective labor:

$$\hat{k} \equiv \frac{K}{AL}, \quad \hat{h} \equiv \frac{H}{AL}, \quad \hat{c} \equiv \frac{c}{A}, \quad f(\hat{k}, \hat{h}) \equiv F(\hat{k}, \hat{h}, 1),$$

and these are the variables constant in a steady state; c , k , h and y all grow at g . The accumulation equations become

$$\dot{\hat{k}} = \hat{v}_k - (\delta_k + g)\hat{k}, \quad \dot{\hat{h}} = \hat{v}_h - (\delta_h + g)\hat{h},$$

effective labor growing at g adding to the effective depreciation rate of both stocks.

(c) *Steady state and dynamics.* The two investment conditions again give $\mu_k = \mu_h$, and the two costate equations again give

$$f_k(\hat{k}, \hat{h}) - \delta_k = f_h(\hat{k}, \hat{h}) - \delta_h,$$

which is (10.25) unchanged: the balance locus $\hat{h} = \xi(\hat{k})$ is the *same function* as without technological progress, since g enters both stocks symmetrically. The Euler equation with CRRA preferences is

$$\frac{\hat{c}}{\bar{c}} = \frac{1}{\theta} [f_k(\hat{k}, \xi(\hat{k})) - \delta_k - \rho - \theta g],$$

so the steady state is

$$\boxed{f_k(\hat{k}^*, \xi(\hat{k}^*)) = \rho + \delta_k + \theta g, \quad f_h(\hat{k}^*, \hat{h}^*) = \rho + \delta_h + \theta g, \quad \hat{h}^* = \xi(\hat{k}^*),}$$

the modified golden rule of Chapter 8 applied to each capital stock, and

$$\dot{\hat{k}} = \frac{1}{1 + \xi'(\hat{k})} [f(\hat{k}, \xi(\hat{k})) - (\delta_h + g)\xi(\hat{k}) - (\delta_k + g)\hat{k} - \hat{c}].$$

This is a planar system in $(\hat{k}, \hat{c})$ with exactly the structure of PROPOSITION 8.7: a unique steady state, saddle-path stable, with monotone convergence along the stable arm. A higher g raises the required marginal product $\rho + \delta_k + \theta g$ and therefore *lowers* $\hat{k}^*$ and $\hat{h}^*$; the technology-adjusted levels fall while the growth rates rise. Assumption 4 — $\rho > (1 - \theta)g$ with $n = 0$ — is again needed for finite utility.

(d) *Why human capital accumulates faster.* Two observations combine.

First, along the balanced growth path $\hat{h} = H/(AL)$ is constant, so the *stock* of human capital grows at exactly the rate g . A higher g therefore means permanently faster human capital accumulation, not just faster output growth.

Second, the mechanism is complementarity. Human capital is complementary with effective labor, $F_{HA} > 0$; as A rises, the marginal product of a given stock of human capital rises with it, and by (10.25) the return to human capital must remain tied to the return to physical capital. So the return to human capital investment rises one for one with the rate of technological progress, and the household responds by accumulating H at the rate g . Since $\hat{h} = \xi(\hat{k})$ is a fixed relation, human capital tracks physical capital exactly: anything that raises the growth rate of K raises that of H by the same amount. With $f = \hat{k}^\alpha \hat{h}^\beta$ and $\delta_k = \delta_h$ the ratio is literally constant, $\hat{h}^*/\hat{k}^* = \beta/\alpha$, regardless of g .

Remark. This is the model's answer to a recurring empirical question — whether skill-biased technological change should raise or lower the incentive to invest in skills. Here the answer is unambiguous, because progress is factor-neutral in the relevant sense and the two capitals are tied by (10.25). Models in which technology and skills are chosen jointly, and in which the *direction* of technological change responds to relative supplies, are the subject of Chapter 15, and there the answer is considerably less simple.

Exercise 10.15

By constant returns to scale and Euler's theorem,

$$F_K(K, H)K + F_H(K, H)H = F(K, H) \quad \text{for all } K, H > 0.$$

Differentiate with respect to K :

$$F_{KK}K + F_K + F_{HK}H = F_K \quad \implies \quad F_{KK}K + F_{KH}H = 0,$$

that is

$$F_{KH} = -\frac{K}{H}F_{KK} \geq 0$$

since F is concave in K (Assumption 1), so $F_{KK} \leq 0$, and $K, H > 0$. If F is *strictly* concave in K the inequality is strict. ■

The symmetric computation, differentiating Euler's identity with respect to H , gives $F_{KH} = -\frac{H}{K}F_{HH}$, and multiplying the two,

$$F_{KH}^2 = F_{KK}F_{HH} \quad \implies \quad \det \begin{pmatrix} F_{KK} & F_{KH} \\ F_{KH} & F_{HH} \end{pmatrix} = 0.$$

The Hessian of a constant-returns production function is always singular, with null direction (K, H) — the direction in which the function is linear.

Verification. For the CES $F = (K^\rho + H^\rho)^{1/\rho}$ with $\rho = -1/2$ (elasticity of substitution 2/3), the program computes F_{KK} and F_{KH} by finite differences:

(K, H)	(1, 1)	(1, 3)	(4, 1)
F_{KK}	-0.093750	-0.139899	-0.009260
F_{KH}	+0.093750	+0.046634	+0.037037
$-(K/H)F_{KK}$	+0.093750	+0.046633	+0.037039

What this means. With exactly two factors and constant returns, capital and human capital *cannot* be substitutes in the sense $F_{KH} < 0$; complementarity is not an assumption but a theorem. The intuition is that constant returns tie the two marginal products together: doubling both factors doubles output and leaves both marginal products unchanged, so if raising K alone lowers F_K , raising H must raise it. Note that this argument is special to two factors — with three or more, a pair of factors can be substitutes while constant returns still hold, which is why the production function $F(K, H, L)$ of Section 10.4 needs $f_{kh} > 0$ as a separate assumption.

Exercise 10.16

As in Exercise 9.4, the accounting question is who holds the undepreciated capital. Each adult of generation t owns $b_i(t-1)$, rents it to firms, and at the end of the period receives the rental $f'(\kappa(t))b_i(t-1)$ together with the $(1-\delta)b_i(t-1)$ that has not depreciated. His gross return is therefore

$$R(t) = f'(\kappa(t)) + 1 - \delta,$$

replacing (10.32), while the wage per unit of human capital is unchanged,

$$w(t) = f(\kappa(t)) - \kappa(t)f'(\kappa(t)).$$

What does not change. The individual's problem is still to maximize $m_i(t) - \gamma(h_i(t)/a)$ by Exercise 10.17, and $m_i(t) = w(t)h_i(t) + R(t)b_i(t-1)$ is linear in h_i with coefficient $w(t)$, which does not involve δ . So the human capital condition (10.35) and its solution (10.36),

$$aw(t) = \gamma' \left(\frac{h_i(t)}{a} \right), \quad h_i(t) = h(t) \equiv a \gamma'^{-1} \left(a [f(\kappa(t)) - \kappa(t)f'(\kappa(t))] \right),$$

are word for word the same. Every individual still chooses the same human capital, and it still depends on nothing but the effective capital-labor ratio: the two substantive conclusions of Section 10.5 survive untouched.

What changes, and how little. Bequests are still $b_i(t) = (1 - \eta)m_i(t)$, so

$$K(t+1) = \int_0^1 b_i(t) di = (1 - \eta)[w(t)H(t) + R(t)K(t)] = (1 - \eta)[Y(t) + (1 - \delta)K(t)],$$

using Euler's theorem $w(t)H(t) + f'(\kappa(t))K(t) = Y(t)$. Dividing by $H(t+1) = h(t+1)$ and using $Y(t) = f(\kappa(t))h(t)$,

$$\kappa(t+1) = \frac{(1 - \eta)[f(\kappa(t)) + (1 - \delta)\kappa(t)]h(t)}{h(t+1)},$$

which is (10.37) with f replaced by

$$\tilde{f}(\kappa) \equiv f(\kappa) + (1 - \delta)\kappa$$

in the accumulation term only (the factor prices still involve f itself). The steady state is

$$\kappa^* = (1 - \eta)\tilde{f}(\kappa^*) = (1 - \eta)[f(\kappa^*) + (1 - \delta)\kappa^*],$$

in place of (10.38). This equation has a unique positive root for exactly the same reason: $\tilde{f}$ is strictly concave with $\tilde{f}(0) = 0$ and $\tilde{f}'(0) = \infty$, and $(1 - \eta)\tilde{f}'(\kappa) \rightarrow (1 - \eta)(1 - \delta) < 1$, so $(1 - \eta)\tilde{f}(\kappa) - \kappa$ is positive near zero and eventually negative. PROPOSITION 10.2 therefore holds verbatim with $\tilde{f}$ in place of f , and the conclusion drawn from it — that capital-skill complementarity pins down a target ratio κ^* and so rules out large imbalances — is unaffected.

Remark. The exercise says the dynamics “remain unchanged”; strictly they do not — the steady-state condition acquires the term $(1 - \delta)\kappa^*$, and κ^* is larger for $\delta < 1$ than for $\delta = 1$, since more capital survives each period. What is unchanged is everything the section uses: the human capital decision, its independence of the wealth distribution, the uniqueness of κ^* , and the balance result. Setting $\delta = 1$ is a normalization that shortens (10.38), not a substantive restriction.

Exercise 10.17

Preferences (10.28) are

$$\eta^{-\eta}(1 - \eta)^{-(1-\eta)}c_i(t)^\eta b_i(t)^{1-\eta} - \gamma \left(\frac{h_i(t)}{a} \right),$$

and the budget constraint (10.29) is $c_i(t) + b_i(t) \leq m_i(t)$.

(10.33). The effort term does not involve c_i or b_i , so the consumption-bequest split solves $\max c^\eta b^{1-\eta}$ subject to $c + b = m$ (the constraint binds since the objective is strictly increasing). Taking logs and differentiating,

$$\frac{\eta}{c} = \frac{1-\eta}{b} \implies b = \frac{1-\eta}{\eta} c, \quad \text{so} \quad c \left(1 + \frac{1-\eta}{\eta}\right) = \frac{c}{\eta} = m,$$

giving

$$c_i(t) = \eta m_i(t), \quad b_i(t) = (1-\eta) m_i(t),$$

which is (10.33): constant shares, as always with Cobb–Douglas preferences.

(10.34). Substituting back,

$$\eta^{-\eta} (1-\eta)^{-(1-\eta)} (\eta m)^\eta ((1-\eta)m)^{1-\eta} = \eta^{-\eta} (1-\eta)^{-(1-\eta)} \eta^\eta (1-\eta)^{1-\eta} m^{\eta+(1-\eta)} = m,$$

every constant cancelling against the normalizing factor. Hence the indirect utility function is

$$m_i(t) - \gamma \left(\frac{h_i(t)}{a} \right)$$

which is (10.34). ■

The point of the normalization. The factor $\eta^{-\eta} (1-\eta)^{-(1-\eta)}$ is chosen so that indirect utility is exactly *linear* in income, with a coefficient of one. That makes the human capital decision a pure income-maximization problem: from (10.34) the individual chooses h_i to maximize $w(t)h_i - \gamma(h_i/a)$, which involves neither his inherited wealth $b_i(t-1)$ nor any preference parameter other than γ . Consequently

- every individual chooses the *same* human capital, (10.36), however unequal their bequests — there are no income effects in schooling;
- aggregate dynamics do not depend on the distribution of wealth, which is what makes (10.37) a scalar difference equation;
- the human capital choice “separates from” the consumption-bequest choice.

The last point is THEOREM 10.1 in a two-period discrete-time setting: here it is delivered not by perfect capital markets but by quasi-linear preferences, which serve the same purpose of making the marginal value of income independent of how much of it there is. Exercise 10.20 in Section 9.6 — general $u_1(c) + u_2(b)$ — is where dropping this structure starts to matter.

Exercise 10.19

Closing the economy. Suppose there are two groups, of measures π and $1-\pi$, with abilities a_1 and a_2 . Everything in Section 10.6 is unchanged except that the interest rate no longer comes from world markets: capital per firm is whatever last period’s bequests supply,

$$\widehat{k}(t) = \int_0^1 b_i(t-1) di = K(t),$$

and $R(t)$ adjusts to clear the capital market, $R(t) = (1-\lambda) \int_0^1 F_k(\widehat{k}(t), h_i(t)) di$. The essential change is that $\widehat{k}(t)$ is *predetermined*: firms invest before they know which worker they will meet, and the aggregate they can invest was fixed a period earlier.

Given $\widehat{k}(t)$, each worker chooses human capital from

$$\lambda a_i F_h(\widehat{k}(t), h_i) = \gamma' \left(\frac{h_i}{a_i} \right), \quad (\text{W})$$

which defines $\widehat{h}_i(\widehat{k}(t); a_i)$. Differentiating (W) in $\widehat{k}$ and in a_i ,

$$\frac{\partial \widehat{h}_i}{\partial \widehat{k}} = \frac{\lambda a_i F_{hk}}{\gamma''/a_i - \lambda a_i F_{hh}} > 0, \quad \frac{\partial \widehat{h}_i}{\partial a_i} = \frac{\lambda F_h + \gamma'' h_i / a_i^2}{\gamma''/a_i - \lambda a_i F_{hh}} > 0,$$

both denominators positive because γ is strictly convex and F concave in h , and both numerators positive because $F_{hk} \geq 0$ by Exercise 10.15. With Cobb–Douglas bequests as in Section 10.5, total income equals total output and

$$\widehat{k}(t+1) = (1-\eta) \Phi(\widehat{k}(t)), \quad \Phi(\widehat{k}) \equiv \int_0^1 F(\widehat{k}, \widehat{h}_i(\widehat{k})) di = Y(t). \quad (\text{K})$$

The externality. Raise a_1 . At the impact date $t = 0$:

- group 1's human capital rises, by $\partial \widehat{h}_1 / \partial a_1 > 0$;
- current output $Y(0) = \Phi(\widehat{k}(0))$ rises, since F is increasing in h ;
- group 2's human capital is *unchanged*, because $\widehat{h}_2$ depends on a_2 and on $\widehat{k}(0)$, and $\widehat{k}(0)$ is predetermined.

From $t = 1$ on, by (K), the higher output raises bequests and therefore $\widehat{k}(1)$; the higher capital per firm raises $\widehat{h}_i$ for *every* i , including group 2, which raises output again, and so on. The long-run effect is

$$\frac{d\widehat{k}^*}{da_1} = \frac{(1-\eta) \partial \Phi / \partial a_1}{1 - (1-\eta) \Phi'(\widehat{k}^*)} > 0, \quad \Phi'(\widehat{k}) = \int_0^1 \left[F_k + F_h \frac{\partial \widehat{h}_i}{\partial \widehat{k}} \right] di > \int_0^1 F_k di,$$

with the multiplier $1/[1 - (1-\eta) \Phi'(\widehat{k}^*)] > 1$ under the stability condition $(1-\eta) \Phi'(\widehat{k}^*) < 1$. Group 2's steady-state human capital rises by $(\partial \widehat{h}_2 / \partial \widehat{k}) d\widehat{k}^* / da_1 > 0$.

An explicit example. Take $F(k, h) = k^{1/3} h^{2/3}$, $\gamma(e) = e^3/3$, $\lambda = 1/2$, $\eta = 0.4$, $\pi = 1/2$, $a_1 = a_2 = 1$. Then (W) solves in closed form, $\widehat{h}_i = [\frac{2}{3} \lambda a_i^3 \widehat{k}^{1/3}]^{3/7} \propto a_i^{9/7} \widehat{k}^{1/7}$, and (K) becomes $\widehat{k}(t+1) = (1-\eta) C \widehat{k}(t)^{3/7}$, a contraction with a unique stable steady state. Raising a_1 by 10% gives

t	0	1	2	3	5	∞
$\widehat{k}(t)$	0.23616	0.24621	0.25065	0.25257	0.25376	0.25403
$\widehat{h}_2(t)$ gain	0.000%	0.597%	0.854%	0.965%	1.032%	1.047%

The closed forms are $d \log \widehat{k}^* / d \log a_1 = 0.750$ and $d \log \widehat{h}_2^* / d \log a_1 = 0.107$, which the numerical experiment reproduces (0.765 and 0.109 for a discrete 10% change). Group 2 ends up with over 1% more human capital, without any change in its own ability, preferences or constraints. The right panel of Figure 8 shows the accumulation map (K) shifting up and the economy climbing to the new steady state.

Why this is different from the open economy. In the open economy of PROPOSITION 10.3, $\widehat{k}$ is pinned down by $(1-\lambda) \int F_k(\widehat{k}, \widehat{h}_i(\widehat{k})) di = R^*$, so a rise in a_1 does raise $\widehat{k}$ and hence $\widehat{h}_2$ — but instantaneously, and financed from abroad. Here the adjustment must be financed out of the

economy's own saving, so it takes time, arrives only from the next period, and compounds: higher output this period buys more capital next period, which raises everyone's human capital, which raises output again. That is the sense in which the externality is *dynamic*.

Remark. The externality exists at all only because of the labor market friction. With competitive labor markets (Sections 10.4 and 10.5) each worker is paid his marginal product and internalizes the full return to his human capital, so raising one group's ability changes the others' decisions only through factor prices in the usual, second-order way. Here random matching plus a fixed sharing rule λ means the return each worker captures depends on the capital *other* people's investments have called forth; complementarity then turns a private improvement into a public one. This is the mechanism Chapter 21 develops into a full theory of human capital externalities and virtuous cycles.

Chapter 11

First-Generation Models of Endogenous Growth

The thirteen exercises of Chapter 11 left unsolved in the instructor's manual: Exercises 11.1–11.3, 11.5–11.7, 11.9–11.13, 11.19 and 11.20. The manual solves 11.4, 11.8, 11.14–11.18 and 11.21. Notation follows the book: A is the productivity of the linear technology, g^* the equilibrium growth rate, θ the inverse of the intertemporal elasticity of substitution, κ the share of capital in the investment sector of Section 11.3, and $\tilde{f}(L) \equiv F(1, BL)$ the normalized production function of the Romer model of Section 11.4. The numerical parts are computed by `code/ch11.py`.

Exercise 11.1

Write $\lambda \equiv A - \delta - n$ and $g \equiv \theta^{-1}(A - \delta - \rho)$, so that (11.13) is the first-order linear nonautonomous equation

$$\dot{k}(t) = \lambda k(t) - c(0)e^{gt}.$$

Multiply by the integrating factor $e^{-\lambda t}$:

$$\frac{d}{dt} [k(t)e^{-\lambda t}] = -c(0)e^{(g-\lambda)t},$$

and integrate. Provided $g \neq \lambda$,

$$k(t)e^{-\lambda t} = \kappa - \frac{c(0)}{g - \lambda} e^{(g-\lambda)t} \quad \implies \quad k(t) = \kappa e^{\lambda t} + \frac{c(0)}{\lambda - g} e^{gt},$$

where κ is a constant of integration. Finally,

$$\lambda - g = (A - \delta - n) - \frac{A - \delta - \rho}{\theta} = \frac{(A - \delta)(\theta - 1) + \rho}{\theta} - n = (A - \delta)(\theta - 1)\theta^{-1} + \rho\theta^{-1} - n,$$

so that

$$k(t) = \kappa e^{(A-\delta-n)t} + [(A - \delta)(\theta - 1)\theta^{-1} + \rho\theta^{-1} - n]^{-1} [c(0) \exp(\theta^{-1}(A - \delta - \rho)t)],$$

which is (11.14). ■

The sign of $\lambda - g$. The second inequality in (11.12) states $\rho + \delta > (1 - \theta)(A - \delta) + \theta n + \delta$, that is $\rho > (1 - \theta)(A - \delta) + \theta n$; dividing by $\theta > 0$ gives exactly $\lambda - g > 0$. (This is also what makes the household's utility finite: along the equilibrium path consumption grows at g , and $\int_0^\infty e^{-(\rho-n)t} c(t)^{1-\theta} dt$ converges if and only if $\rho - n > (1 - \theta)g$, which rearranges to the same inequality.)

Why the constant vanishes. Substituting (11.14) into the transversality condition (11.10), and noting that $k(t)e^{-\lambda t} = \kappa + \frac{c(0)}{\lambda - g} e^{-(\lambda - g)t}$,

$$\lim_{t \rightarrow \infty} k(t)e^{-\lambda t} = \kappa,$$

because $\lambda - g > 0$ kills the second term. The transversality condition therefore forces $\kappa = 0$, leaving

$$k(t) = \frac{c(0)}{\lambda - g} e^{gt} = k(0) e^{gt}, \quad c(0) = (\lambda - g)k(0),$$

which are (11.15) and (11.16). Capital, consumption and output all grow at g from the very first instant: there are no transitional dynamics.

Remark. The knife-edge case $g = \lambda$ is excluded by (11.12), but it is worth knowing what it looks like: the particular solution is then $-c(0)te^{\lambda t}$ and $k(t) = [\kappa - c(0)t]e^{\lambda t}$, which is eventually negative for any κ . The condition $\lambda > g$ is thus not a technicality — it is what makes a positive capital stock consistent with consumption growing at the Euler rate forever.

Exercise 11.2

Two proofs; the direct one is more informative.

The optimal growth problem. The planner maximizes (11.1) subject to the resource constraint $\dot{k} = (A - \delta - n)k - c$ and $k(0) > 0$. The current-value Hamiltonian is

$$\hat{H}(k, c, \mu) = \frac{c^{1-\theta} - 1}{1-\theta} + \mu[(A - \delta - n)k - c],$$

so the conditions of THEOREM 7.13 are

$$c(t)^{-\theta} = \mu(t), \quad \mu(t)(A - \delta - n) = (\rho - n)\mu(t) - \dot{\mu}(t), \quad \lim_{t \rightarrow \infty} e^{-(\rho-n)t} \mu(t)k(t) = 0.$$

The second gives $\dot{\mu}/\mu = \rho - (A - \delta)$, and since $\mu = c^{-\theta}$ implies $\dot{\mu}/\mu = -\theta\dot{c}/c$,

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\theta}(A - \delta - \rho),$$

which is (11.9). For the transversality condition, substitute $\mu(t) = c(0)^{-\theta} e^{-(A-\delta-\rho)t}$:

$$e^{-(\rho-n)t} \mu(t)k(t) = c(0)^{-\theta} k(t) e^{-(A-\delta-n)t},$$

so the condition is exactly (11.10). Every condition of the planner's problem therefore coincides with an equilibrium condition.

Sufficiency and uniqueness. $\hat{H}$ is the sum of a strictly concave function of c and a function linear in (k, c) , hence concave in (k, c) and strictly concave in c ; and $\lim_{t \rightarrow \infty} e^{-(\rho-n)t} \mu(t)k(t) \geq 0$ for every feasible path because $\mu > 0$ and $k \geq 0$. THEOREM 7.14 therefore applies: the conditions above are sufficient, and by strict concavity in c the optimum is unique. Since the competitive equilibrium of PROPOSITION 11.1 satisfies them, it *is* the optimum, and in particular it is Pareto optimal. ■

The Welfare Theorem route. Alternatively, the economy has a representative household, all markets are competitive and there are no externalities, so THEOREM 5.6 applies provided the value of the aggregate endowment at equilibrium prices is finite. Here the household's only endowment is its initial capital $k(0)$ — labor earns nothing, since $Y = AK$ does not use labor — so that value is $k(0) < \infty$, trivially. Hence the equilibrium is Pareto optimal.

Remark. It is worth being explicit about why sustained endogenous growth and Pareto optimality coexist here, when they will not in Section 11.4. Growth in the *AK* model comes from the *absence of diminishing returns*, not from any externality: each firm's private marginal product of capital, A , is also its social marginal

product. In the Romer model the two differ, and PROPOSITION 11.6 shows the equilibrium then grows too slowly. Endogenous growth is not, in itself, a source of inefficiency.

Exercise 11.3

(a) *Equilibrium.* With $Y(t) = AK(t) + BL(t)$, both factors have constant marginal products, so competitive pricing gives

$$R(t) = A, \quad r(t) = r = A - \delta, \quad w(t) = w = B \quad \text{for all } t.$$

A competitive equilibrium is a path $[c(t), k(t), w(t), R(t)]_{t=0}^{\infty}$ such that the representative household maximizes $\int_0^{\infty} e^{-(\rho-n)t} \frac{c^{1-\theta}-1}{1-\theta} dt$ subject to $\dot{a} = (r-n)a + w - c$ and the no-Ponzi condition, given $k(0)$; factor prices are as above; and $a(t) = k(t)$.

(b) *Characterization.* The current-value Hamiltonian is

$$\hat{H}(a, c, \mu) = \frac{c^{1-\theta} - 1}{1-\theta} + \mu[(r-n)a + w - c],$$

giving $c^{-\theta} = \mu$, $\dot{\mu}/\mu = \rho - r$ and the transversality condition. Hence

$$\frac{\dot{c}(t)}{c(t)} = \frac{1}{\theta}(A - \delta - \rho) \equiv g,$$

a constant, exactly as in the baseline AK model. The capital accumulation equation is now

$$\dot{k}(t) = \lambda k(t) + B - c(0)e^{gt}, \quad \lambda \equiv A - \delta - n,$$

which differs from (11.13) by the constant B — labor income. Solving as in Exercise 11.1,

$$k(t) = \kappa e^{\lambda t} - \frac{B}{\lambda} + \frac{c(0)}{\lambda - g} e^{gt},$$

and the transversality condition again forces $\kappa = 0$, so that

$$\boxed{k(t) + \frac{B}{\lambda} = \left[k(0) + \frac{B}{\lambda}\right] e^{gt}, \quad c(0) = (\lambda - g) \left[k(0) + \frac{B}{\lambda}\right].}$$

The object that grows at the constant rate g is not capital but *total wealth*: physical capital plus B/λ , the present value of the constant labor income stream discounted at the effective rate $\lambda = r - n$.

Transitional dynamics. Since $k(t)$ itself is total wealth minus a constant,

$$\frac{\dot{k}(t)}{k(t)} = g \left[1 + \frac{B}{\lambda k(t)} \right] > g,$$

which is strictly decreasing in k and tends to g only as $t \rightarrow \infty$. Capital therefore grows *faster* than consumption at every finite date, at a rate that falls monotonically toward g : the economy has nontrivial transitional dynamics even though consumption growth is constant. Output $y = Ak + B$ behaves the same way,

$$\frac{\dot{y}}{y} = \frac{A\dot{k}}{Ak + B} = g \frac{A(k + B/\lambda)}{Ak + B} > g \quad (\text{because } A/\lambda > 1),$$

again falling monotonically to g . With $A = 0.15$, $B = 0.30$, $\delta = 0.05$, $\rho = 0.02$, $n = 0.01$, $\theta = 3$ and $k(0) = 1$ the program gives $g = 0.02667$ and

t	0	25	50	100	200
$\dot{k}/k$	0.1156	0.0441	0.0334	0.0282	0.0268
$\dot{y}/y$	0.0385	0.0317	0.0290	0.0272	0.0267

(c) *The labor share.* Labor income is BL and output is $AK + BL$, so

$$\frac{w(t)L(t)}{Y(t)} = \frac{B}{Ak(t) + B},$$

which is strictly decreasing in k and, since $k(t) \rightarrow \infty$, falls monotonically to zero. In the calibration above it goes from 0.667 at $t = 0$ to 0.132 at $t = 50$ and 0.0022 at $t = 200$ (Figure 9, right panel). The capital share rises to one. This is the baseline AK model's counterfactual prediction arriving more slowly rather than being avoided: adding raw labor delays the outcome without changing it, because labor is a factor in fixed supply whose marginal product does not grow.

(d) *An unanticipated rise in B .* The growth rate g depends only on A , δ , ρ and θ , so it is unchanged — permanently. What changes is the level. Human wealth B/λ jumps up, so consumption jumps up immediately by $(\lambda - g)\Delta B/\lambda$; capital is predetermined and does not jump, but its subsequent growth rate $g[1 + B/(\lambda k)]$ is higher at every k , so k and y rise over time relative to the old path. The labor share jumps up and then resumes its decline to zero. So the effect is a level effect asymptotically, arriving through a temporary increase in the growth rate — the opposite pattern from a change in A , which changes the asymptotic growth rate itself.

(e) *Pareto optimality.* The planner's resource constraint is $\dot{k} = (A - \delta - n)k + B - c$ and its Hamiltonian is concave in (k, c) , strictly so in c ; repeating Exercise 11.2 word for word gives the same Euler equation, the same transversality condition and hence the same allocation. THEOREM 7.14 makes the conditions sufficient, so the equilibrium is the unique optimum. Equivalently, THEOREM 5.6 applies because the value of the aggregate endowment at equilibrium prices, $k(0) + B/\lambda$, is finite — which is exactly the condition $\lambda > g > 0$ established in Exercise 11.1.

Exercise 11.5

The growth rate. With the budget constraint (11.18), $\dot{a} = ((1 - \tau)r - n)a + w - c$, the current-value Hamiltonian is

$$\widehat{H} = \frac{c^{1-\theta} - 1}{1 - \theta} + \mu[(1 - \tau)r - n)a + w - c],$$

so $c^{-\theta} = \mu$ and

$$\frac{\dot{\mu}}{\mu} = (\rho - n) - ((1 - \tau)r - n) = \rho - (1 - \tau)r.$$

Using $\dot{\mu}/\mu = -\theta\dot{c}/c$ and $r = A - \delta$,

$$g = \frac{\dot{c}}{c} = \frac{(1 - \tau)(A - \delta) - \rho}{\theta},$$

which is (11.19). The tax reduces growth one for one with the after-tax return, and $dg/d\tau = -(A - \delta)/\theta < 0$.

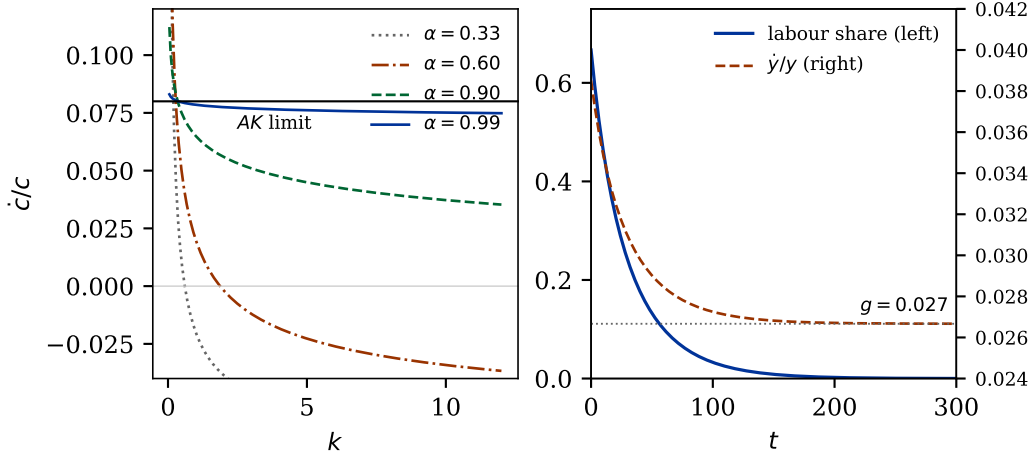


FIGURE 9. Left (Exercise 11.6): the growth rate of consumption in the neoclassical model with $y = Ak^\alpha$ and log preferences. As $\alpha \rightarrow 1$ the curve flattens onto the AK line $A - \delta - \rho$ and the steady state disappears to infinity. Right (Exercise 11.3): in the $AK + BL$ economy the labor share falls to zero while output growth declines monotonically to g — nontrivial transitional dynamics in a model with a constant consumption growth rate.

The saving rate. Assume, as in Section 8.10, that the revenue is rebated lump-sum, so the *aggregate* resource constraint is still $\dot{k} = (A - \delta - n)k - c$. Solving it as in Exercise 11.1 with g now given by (11.19), the transversality condition again eliminates the homogeneous solution and k grows at g . The saving rate is therefore

$$s = \frac{\dot{K} + \delta K}{Y} = \frac{\dot{k}/k + n + \delta}{A} = \frac{1}{A} \left[\frac{(1 - \tau)(A - \delta) - \rho}{\theta} + n + \delta \right].$$

Multiplying numerator and denominator by θ and collecting the terms in δ ,

$$\theta A s = (1 - \tau)A - (1 - \tau)\delta - \rho + \theta n + \theta\delta = (1 - \tau)A - \rho + \theta n - (1 - \tau - \theta)\delta,$$

that is

$$s = \frac{(1 - \tau)A - \rho + \theta n - (1 - \tau - \theta)\delta}{\theta A}$$

which is (11.20); setting $\tau = 0$ recovers (11.17). Differentiating,

$$\frac{\partial s}{\partial \tau} = \frac{-A + \delta}{\theta A} = -\frac{A - \delta}{\theta A} < 0 \quad \text{provided } A > \delta,$$

as the text states.

What is new relative to Chapter 8. There the saving rate was also endogenous, but a capital tax changed only the *level* of the steady state. Here the saving rate is constant *and* the growth rate depends on it, so a permanent difference in τ becomes a permanent difference in growth — and hence an unbounded difference in income. Exercise 11.7 puts numbers on this.

Exercise 11.6

The equilibrium. With $\theta = 1$ the equilibrium is the planar system

$$\frac{\dot{c}(t)}{c(t)} = \alpha A k(t)^{\alpha-1} - \delta - \rho, \quad \dot{k}(t) = A k(t)^\alpha - c(t) - (n + \delta)k(t),$$

with $k(0) > 0$ given and the transversality condition of Chapter 8. There is a unique steady state,

$$\alpha A (k^*)^{\alpha-1} = \rho + \delta \quad \Longleftrightarrow \quad k^* = \left[\frac{\alpha A}{\rho + \delta} \right]^{\frac{1}{1-\alpha}}, \quad c^* = A (k^*)^\alpha - (n + \delta)k^*,$$

saddle-path stable with monotone convergence along the stable arm (PROPOSITION 8.7 with $\theta = 1$, $g = 0$).

The limit $\alpha \rightarrow 1$. Three things happen at once.

- (1) *The Euler equation becomes the AK Euler equation.* For any fixed k ,

$$\alpha A k^{\alpha-1} - \delta - \rho \rightarrow A - \delta - \rho,$$

which is (11.9) with $\theta = 1$: the growth rate becomes independent of k .

- (2) *The steady state runs off to infinity.* Provided $A > \rho + \delta$, the base $\alpha A / (\rho + \delta)$ exceeds one for α close to one while the exponent $1/(1-\alpha)$ diverges, so $k^* \rightarrow \infty$: there is no finite point at which accumulation stops.

- (3) *Convergence becomes infinitely slow.* The stable eigenvalue of the linearized system is $\xi_1 = \frac{1}{2} [\text{tr } J - \sqrt{(\text{tr } J)^2 - 4 \det J}]$ with $\text{tr } J = \rho - n$ and, using $\alpha A (k^*)^{\alpha-1} = \rho + \delta$ to eliminate k^* ,

$$\det J = c^* f''(k^*) = (\alpha - 1)(\rho + \delta) \left[\frac{\rho + \delta}{\alpha} - (n + \delta) \right] \rightarrow 0,$$

so $\xi_1 \rightarrow 0$ and the half-life $\log 2 / |\xi_1| \rightarrow \infty$.

With $A = 0.15$, $\delta = 0.05$, $\rho = 0.02$, $n = 0$ the program gives

α	1/3	0.6	0.9	0.99	0.9999
$\log_{10} k^*$	-0.22	0.27	2.85	32.66	3309.50
ξ_1	-0.0770	-0.0343	-0.0072	-0.00070	-0.0000070
half-life	9.0	20.2	96.8	990	99,021
$\dot{c}/c$ at $k = 1$	-0.020	0.020	0.065	0.0785	0.07999

against the AK growth rate $A - \delta - \rho = 0.08$. The left panel of Figure 9 shows the curve $\dot{c}/c$ flattening onto the AK line.

Interpretation. The AK model is not a different model but the boundary case of the neoclassical one. What stops growth in the neoclassical model is diminishing returns — the failure of f' to stay bounded away from zero — and $\alpha \rightarrow 1$ is precisely the limit in which that force disappears. Three familiar statements are the same statement: “the Inada condition $\lim_{k \rightarrow \infty} f'(k) = 0$ fails,” “the steady state is at infinity,” and “there are no transitional dynamics.”

This also explains why the convergence evidence of Chapter 3 bears directly on endogenous growth. Convergence rates of about 2% a year correspond to a capital share well below one; a capital share

close to one would predict essentially no convergence at all. The debate over whether α should be measured as the narrow physical capital share ($\approx 1/3$) or a broad share including human capital ($\approx 2/3$ or more) is, in this light, a debate about how close the world is to the AK boundary.

Exercise 11.7

The arithmetic. By (11.19),

$$g(\tau) = \frac{(1 - \tau)(A - \delta) - \rho}{\theta} = \frac{(1 - \tau)(0.10) - 0.02}{3},$$

so

$$g(0.2) = \frac{0.08 - 0.02}{3} = 0.0200, \quad g(0.4) = \frac{0.06 - 0.02}{3} = 0.0133,$$

a growth gap of 0.00667 a year. Starting from the same income in 1900, by 2000

$$\frac{Y(0.2, 100)}{Y(0.4, 100)} = e^{0.00667 \times 100} = e^{2/3} = 1.948,$$

so the low-tax country is almost exactly twice as rich. And the gap keeps widening without bound:

horizon (years)	50	100	200	500
income ratio	1.396	1.948	3.794	28.03

The saving rates, from (11.20), are 0.467 and 0.422: a difference of four and a half percentage points, sustained forever.

Comparison with the neoclassical model. The same tax difference in the model of Chapter 8 produces only a permanent *level* difference, $[(1 - \tau)/(1 - \tau')]^{\alpha/(1-\alpha)}$, which is 1.155 at $\alpha = 1/3$ and 1.333 at $\alpha = 1/2$. The AK model converts a bounded 15% level effect into an unbounded growth effect.

Is it plausible? The mechanism is coherent but the implications are hard to accept, for four reasons.

First, the world income distribution would have to diverge without limit. Tax rates, expropriation risk and other determinants of the after-tax return differ across countries by far more than twenty percentage points, and they have done so for decades. The model then predicts that relative incomes should be spreading apart at an accelerating rate. The evidence in Chapter 1 is that the postwar world income distribution has been roughly stable, with substantial mobility but no systematic divergence of the whole distribution.

Second, the capital share must tend to one, since $Y = AK$ leaves nothing for labor. Factor shares have been roughly constant, which is the first of the Kaldor facts. Exercise 11.3 shows that adding raw labor only postpones this problem; Section 11.2 solves it, but by making human capital accumulate linearly, which relocates rather than removes the difficulty.

Third, the sensitivity is extreme. Here $dg/d\tau = -(A - \delta)/\theta = -0.033$, so each *percentage point* of capital tax costs a third of a tenth of a percentage point of growth forever — about 3.4% of income per century. No estimate of the effect of taxation on growth is anywhere near this large, and cross-country growth regressions find policy variables explaining level differences far better than they explain persistent growth differences.

Fourth, and most fundamentally, the result is an artifact of exact linearity. The reason small policy differences compound forever is that the marginal product of capital never falls. Any diminishing

returns at all — $\alpha < 1$ in Exercise 11.6 — restores a steady state and converts the growth effect into a level effect, however slowly. A model in which the answer flips discontinuously at $\alpha = 1$ is not a good guide to a world in which α is measured with error.

Remark. The honest conclusion is the one the text draws: the *AK* class shows that endogenous growth and policy-sensitive growth rates are *possible* in a neoclassical environment, which was worth establishing. It does not show that they are what generates the observed cross-country income distribution. That is why Part IV turns to models in which technology is chosen purposefully and in which the linearity that sustains growth is derived from the economics of innovation rather than assumed.

Exercise 11.9

The effective capital-labor ratio is constant. The household's conditions (11.25) are $\mu_a = \mu_h$, $w(t) - \delta_h = r(t)$ and $\dot{c}/c = \theta^{-1}(r(t) - \rho)$. The first says that the shadow values of a unit of physical and a unit of human capital must coincide, since both are bought with the same good and neither is subject to a nonnegativity constraint; the second is the no-arbitrage condition it implies, equating the net returns. Substituting the competitive factor prices (11.24), $R = f'(k)$ and $w = f(k) - kf'(k)$ with $k = K/H$, the no-arbitrage condition reads

$$f'(k(t)) - \delta_k = f(k(t)) - k(t)f'(k(t)) - \delta_h. \quad (*)$$

The left side is strictly decreasing in k (derivative $f'' < 0$) and the right side strictly increasing (derivative $-kf'' > 0$), so $(*)$ has at most one solution. It has at least one: as $k \rightarrow 0$, Assumption 2 gives $f'(k) \rightarrow \infty$ while $f(k) - kf'(k) \rightarrow 0$, so the left side exceeds the right; as $k \rightarrow \infty$, $f'(k) \rightarrow 0$ while $f(k) - kf'(k) \rightarrow \infty$, so the inequality reverses. Hence there is a unique k^* solving (11.26), and

$$k(t) = k^* \quad \text{for all } t \geq 0,$$

including $t = 0$: the economy jumps immediately to the balanced ratio by reallocating its initial stocks, which is feasible because investment in either capital may be negative.

Growth. With k constant, $r(t) = r^* = f'(k^*) - \delta_k$ is constant, so the Euler equation gives the constant growth rate

$$g^* = \frac{f'(k^*) - \delta_k - \rho}{\theta}.$$

Since $k = K/H$ is constant, K and H grow at a common rate; output $Y = F(K, H) = Hf(k^*)$ grows at that same rate; and the aggregate resource constraint

$$\dot{K} + \dot{H} = F(K, H) - C - \delta_k K - \delta_h H$$

then reads, after dividing by H and using $K = k^*H$,

$$g(1 + k^*) = f(k^*) - \frac{C}{H} - (\delta_k k^* + \delta_h),$$

so C/H is constant if and only if g is, and the Euler equation pins $g = g^*$.

No transitional dynamics and uniqueness. Given g^* , the accumulation equation for H is linear and nonautonomous, exactly as in Exercise 11.1; its homogeneous solution is eliminated by the transversality condition, so H (and therefore K , Y and C) grows at g^* from the first instant. The

current-value Hamiltonian is concave in states and controls and strictly concave in c (the argument of Exercise 10.8), so THEOREM 7.14 makes the conditions sufficient and the path unique.

The factor shares. The share of capital in national income is

$$\frac{R(t)K(t)}{Y(t)} = \frac{f'(k^*)k^*}{f(k^*)} \in (0, 1),$$

constant over time and strictly less than one by strict concavity of f with $f(0) = 0$. This is the model's advantage over the baseline AK economy.

The conditions. $f'(k^*) > \rho + \delta_k$ gives $g^* > 0$, and $\rho + \delta_k > (1 - \theta)(f'(k^*) - \delta_k) + \delta_k$, that is $\rho > (1 - \theta)(f'(k^*) - \delta_k)$, is equivalent to $\rho > (1 - \theta)g^*$ — the finiteness of utility and the transversality condition, exactly as in Exercise 11.1. ■

Remark. The mechanism is worth naming. The economy behaves like an AK model because, once k is pinned at k^* , the pair (K, H) accumulates linearly: $\dot{K} + \dot{H}$ is proportional to $K + H$. Diminishing returns to each factor separately are perfectly consistent with constant returns to the reproducible factors taken together, and it is the latter that sustains growth. This is the observation the text makes at the start of Section 11.3: what matters is that the *accumulation* technology is linear, not that the production function is AK .

Exercise 11.10

The planner's problem. With $n = 0$ and $L = 1$, the planner maximizes (11.1) subject to

$$\dot{K} = i_K - \delta_k K, \quad \dot{H} = i_H - \delta_h H, \quad c + i_K + i_H \leq F(K, H),$$

given $K(0), H(0) > 0$. The current-value Hamiltonian, after substituting the binding resource constraint, is

$$\hat{H} = \frac{c^{1-\theta} - 1}{1 - \theta} + \mu_K(i_K - \delta_k K) + \mu_H(i_H - \delta_h H), \quad c = F(K, H) - i_K - i_H.$$

The conditions in the two investments give

$$\mu_K = \mu_H = c^{-\theta} \equiv \mu,$$

and the two costate equations give

$$\frac{\dot{\mu}_K}{\mu_K} = \rho + \delta_k - F_K(K, H), \quad \frac{\dot{\mu}_H}{\mu_H} = \rho + \delta_h - F_H(K, H).$$

Equality of the costates therefore forces

$$F_K(K, H) - \delta_k = F_H(K, H) - \delta_h.$$

Since F is homogeneous of degree one, $F_K(K, H) = f'(k)$ and $F_H(K, H) = f(k) - kf'(k)$ with $k = K/H$, so this is exactly (11.26) and hence $k(t) = k^*$ for all t , by the uniqueness established in Exercise 11.9. The Euler equation is

$$\frac{\dot{c}}{c} = \frac{1}{\theta} [F_K(K, H) - \delta_k - \rho] = \frac{1}{\theta} [r^* - \rho],$$

and the two transversality conditions are those of the household's problem.

Coincidence. Every condition of the planner's problem is an equilibrium condition of Section 11.2 and conversely: (11.24) turns F_K into R and F_H into w , so the planner's no-arbitrage condition is the market's, and the planner's Euler equation is the household's. $\widehat{H}$ is concave in (K, H, i_K, i_H) by the argument of Exercise 10.8 — a concave increasing u composed with the concave $F(K, H) - i_K - i_H$, plus linear terms — so THEOREM 7.14 applies and both problems have the same unique solution. The competitive equilibrium is therefore the optimal growth path, and in particular Pareto optimal. ■

The Welfare Theorem route. Alternatively, the economy has a representative household, competitive markets and no externalities, and the value of its endowment at equilibrium prices — $a(0)$ plus the value of $h(0)$, both finite — is finite, so THEOREM 5.6 applies.

Remark. The comparison with Exercise 10.12 is exact and worth making. There the same no-arbitrage condition $F_K - \delta_k = F_H - \delta_h$ defined a *locus* $h = \xi(k)$ along which the economy travelled to a steady state; here, because F is homogeneous of degree one in the two *reproducible* factors, the same condition pins down a single *ratio* k^* and the economy grows along it forever. The difference between a neoclassical model and an endogenous growth model, in this family, is entirely whether the third factor — raw labor in Chapter 10 — is present.

Exercise 11.11

The result. In Section 11.1 the equilibrium growth rate of consumption, capital and output per capita is

$$g^* = \frac{A - \delta - \rho}{\theta},$$

in which n does not appear. In Section 11.2 it is

$$g^* = \frac{f'(k^*) - \delta_k - \rho}{\theta}, \quad f'(k^*) - \delta_k = f(k^*) - k^* f'(k^*) - \delta_h,$$

and again n appears nowhere: $k^* = K/H$ is a ratio of two accumulated stocks, so reinstating population growth in that section changes nothing.

Why. The growth rate of consumption per capita comes from the Euler equation, which compares one number, the net return r , with another, the discount rate ρ . In these models r is determined by technology alone — $A - \delta$, or $f'(k^*) - \delta_k$ — because returns to the reproducible factors do not diminish. Population growth affects neither. More precisely, n enters the household's problem twice and the two effects cancel exactly: it lowers the effective return on assets from r to $r - n$ (assets must be spread over more people) and it lowers the effective discount rate from ρ to $\rho - n$ (there are more people to enjoy consumption), so the difference $r - \rho$ that drives the Euler equation is untouched. This is the same cancellation as in Exercise 8.1.

Population growth is not entirely without effect. From (11.17),

$$s = \frac{A - \rho + \theta n + (\theta - 1)\delta}{\theta A},$$

which is strictly increasing in n : a faster-growing population requires more capital widening, so the economy saves more. And aggregate output grows at $g^* + n$ rather than g^* . What is invariant is the growth rate of the *per capita* variables.

Is it plausible? The prediction sits at a knife edge between two forces that the models of this chapter both switch off.

Against, from above: scale effects. In the Romer model of Section 11.4 the growth rate is $\theta^{-1}(\tilde{f}(L) - L\tilde{f}'(L) - \delta - \rho)$, which is strictly increasing in L , so a larger population grows faster — and if population were growing the growth rate would rise without bound (Exercise 11.18). Any model in which ideas are nonrival has this feature: more people means more people generating ideas that everyone can use. The very long-run evidence discussed in Chapter 4 — accelerating growth over millennia as population grew — favors some scale effect. Chapter 13 shows how to keep the idea without the counterfactual acceleration.

Against, from below: diminishing returns to a fixed factor. In a Malthusian economy with land, faster population growth lowers income per capita. The AK models have no fixed factor at all — indeed Section 11.1 has no factor but capital — so this channel is absent too.

The models of this chapter therefore predict exact neutrality precisely because they contain neither of the two mechanisms that would break it. Empirically the cross-country correlation between population growth and growth in income per capita is weakly negative rather than zero, though, as Chapter 4 discusses, the direction of causation is very hard to establish. Exact neutrality is best read as a benchmark, not a prediction.

Exercise 11.12

Replace (11.27) by a general constant-returns technology in the consumption sector, $C(t) = G(K_C(t), L_C(t))$, keeping $I(t) = AK_I(t)$. Write $g(k_C) \equiv G(k_C, 1)$ for the intensive form, with $k_C \equiv K_C/L_C$.

Along any growth path, $k_C \rightarrow \infty$. Labor is used only in the consumption sector and is in fixed supply, so $L_C = L$ is constant. On a BGP the share κ of capital in the investment sector is constant, so $K_C = (1 - \kappa)K$ grows at the rate $g_K > 0$ of the capital stock. Hence

$$k_C(t) = \frac{(1 - \kappa)K(t)}{L} \rightarrow \infty.$$

The factor shares. Labor income arises only in the consumption sector, so with $p_C = 1$ the wage bill is

$$w(t)L = G(K_C, L) - K_C G_K(K_C, L) = [1 - \sigma(k_C)]C(t), \quad \sigma(k) \equiv \frac{k g'(k)}{g(k)},$$

σ being the elasticity of the consumption technology with respect to capital. GDP at current prices is $p_C C + p_I I$, so the labor share of GDP is

$$\frac{w(t)L}{C(t) + p_I(t)I(t)} = [1 - \sigma(k_C(t))] \cdot \frac{C(t)}{C(t) + p_I(t)I(t)}.$$

On a BGP the two components of GDP must grow at the same rate — otherwise their ratio diverges and the composition of output is not constant — so the second factor is constant. The labor share of GDP is therefore constant if and only if $\sigma(k_C(t))$ is constant.

Only Cobb–Douglas has constant σ . Suppose $\sigma(k) = \alpha$ for all k in an interval. Then

$$\frac{d \log g(k)}{d \log k} = \alpha \quad \implies \quad \log g(k) = \alpha \log k + \text{const} \quad \implies \quad g(k) = Bk^\alpha,$$

that is $G(K_C, L_C) = BK_C^\alpha L_C^{1-\alpha}$. Since k_C ranges over an unbounded interval along the BGP, constancy of σ over that whole range forces Cobb–Douglas globally. Conversely, for any non-Cobb–Douglas constant-returns G the share drifts. For the CES family $g(k) = B[\gamma k^{(\sigma_E-1)/\sigma_E} + 1 - \gamma]^{\sigma_E/(\sigma_E-1)}$ with elasticity of substitution σ_E ,

$$\sigma(k) \longrightarrow \begin{cases} 1 & \text{if } \sigma_E > 1, \\ 0 & \text{if } \sigma_E < 1, \end{cases} \quad \text{as } k \rightarrow \infty,$$

so the capital share in the consumption sector tends to one or to zero. ■

Why this matters. The two-sector model was introduced precisely to deliver sustained growth *with* constant factor shares, avoiding the baseline AK model’s capital share of one. This exercise shows the achievement is fragile: it rests on Cobb–Douglas, which is exactly the functional form that makes factor shares independent of factor proportions. Since the model’s whole point is that the capital-labor ratio in the consumption sector rises without bound, any departure from unit elasticity of substitution lets the shares drift — and if $\sigma_E > 1$ the model reproduces the very defect it was built to avoid.

Exercise 11.13

From (11.33) and (11.34),

$$g_K^* = \frac{A - \delta - \rho}{1 - \alpha(1 - \theta)}, \quad g_C^* = \alpha g_K^* = \frac{\alpha(A - \delta - \rho)}{1 - \alpha(1 - \theta)}.$$

Differentiating, and writing $D \equiv 1 - \alpha(1 - \theta) > 0$,

$$\boxed{\frac{\partial g_K^*}{\partial \alpha} = \frac{(A - \delta - \rho)(1 - \theta)}{D^2}, \quad \frac{\partial g_C^*}{\partial \alpha} = \frac{A - \delta - \rho}{D^2} > 0.}$$

So consumption, output and wages always grow faster when α is larger, but the capital stock does so only when $\theta < 1$; for $\theta > 1$ capital grows *more slowly*, and for $\theta = 1$ its growth rate is independent of α altogether. With $A = 0.15$, $\delta = 0.05$, $\rho = 0.02$ the program gives

	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 0.7$
$\theta = 0.5 \quad (g_K^*, g_C^*)$	(0.0941, 0.0282)	(0.1067, 0.0533)	(0.1231, 0.0862)
$\theta = 1$	(0.0800, 0.0240)	(0.0800, 0.0400)	(0.0800, 0.0560)
$\theta = 2$	(0.0615, 0.0185)	(0.0533, 0.0267)	(0.0471, 0.0329)
$\theta = 3$	(0.0500, 0.0150)	(0.0400, 0.0200)	(0.0333, 0.0233)

Why consumption growth rises. Two channels, both positive. The *technological* channel is direct: α is the elasticity of consumption output with respect to capital, and $g_C = \alpha g_K$, so a given rate of capital accumulation buys more consumption growth. The *price* channel works through the return. By (11.30) the relative price of investment goods falls at the rate $(1 - \alpha)g_K$, so the consumption-denominated return is

$$r_C = A - \delta - (1 - \alpha)g_K.$$

A higher α means the consumption sector uses capital more intensively, so the same capital deepening depresses the price of investment goods by less, and r_C is higher. A higher return raises consumption growth through the Euler equation.

Why capital growth can fall. The higher return raises g_C , but $g_K = g_C/\alpha$ and α has risen too. Which effect wins is exactly the usual income-versus-substitution question: when $\theta > 1$ the income effect of a higher return dominates, the household does not increase saving proportionately, and g_K falls even though g_C rises. Intuitively, when capital is more productive in making consumption goods, the economy needs *less* capital accumulation to deliver any given consumption growth, and a sufficiently income-driven household takes part of the gain as a lower rate of accumulation.

Remark. The exercise asks “Why does it increase the rate of capital accumulation?” — which is correct only for $\theta < 1$. The unambiguous statement is about consumption (and hence output and wages), whose growth rate rises with α at every θ . Since most estimates put $\theta > 1$, the empirically relevant case is the one in which α and g_K^* move in *opposite* directions. The rate of capital deepening, $g_K^* - g_C^* = (1 - \alpha)g_K^*$, falls with α for every θ : a more capital-intensive consumption sector means less divergence between the capital stock and output, which is the sense in which raising α moves the model back toward the Kaldor facts.

Exercise 11.19

The wedge. By (11.39) and PROPOSITION 11.6,

$$g_C^* = \frac{1}{\theta}(\tilde{f}(L) - L\tilde{f}'(L) - \delta - \rho), \quad g_C^S = \frac{1}{\theta}(\tilde{f}(L) - \delta - \rho),$$

so the entire gap is the term

$$L\tilde{f}'(L) = \frac{w(t)L}{K(t)},$$

using $w(t) = K(t)\tilde{f}'(L)$ from (11.37). Firms pay capital its *private* marginal product $\tilde{f}(L) - L\tilde{f}'(L)$, but a unit of capital also raises $A = BK$ and so raises every firm’s output; the external part of its social marginal product is exactly the wage bill per unit of capital. Any policy that closes the gap must raise the household’s net return from $\tilde{f}(L) - L\tilde{f}'(L) - \delta$ to $\tilde{f}(L) - \delta$.

Policy 1: a Pigouvian subsidy to capital. Pay a flow subsidy of ς units of output per unit of capital, financed by lump-sum taxes, and set

$$\varsigma^* = L\tilde{f}'(L).$$

The household’s net return becomes $\tilde{f}(L) - L\tilde{f}'(L) + \varsigma^* - \delta = \tilde{f}(L) - \delta$, so its Euler equation delivers g_C^S and the equilibrium coincides with the optimum. As a proportional subsidy to capital income the rate is

$$\varsigma^* = \frac{L\tilde{f}'(L)}{\tilde{f}(L) - L\tilde{f}'(L)} = \frac{\text{labor share}}{\text{capital share}},$$

which *exceeds* one whenever the labor share exceeds the capital share. With $F = K^\alpha(AL)^{1-\alpha}$, $A = BK$ and $\alpha = 1/3$, the program gives $\varsigma^* = 2$: the subsidy must be 200% of capital income.

The financing has a striking property. The total cost of the subsidy is $\varsigma^*K(t) = L\tilde{f}'(L)K(t) = w(t)L$ — exactly the wage bill, at every date. Since labor is supplied inelastically, a tax on labor income is lump-sum here, so

a 100% tax on labor income, handed to capital, balances the budget instant by instant and implements the optimum.

Calibrating $\tilde{f}(L) = 0.30$ (a capital-output ratio of 3.3) with $\alpha = 1/3$, $\delta = 0.05$, $\rho = 0.02$ and $\theta = 2$, the program gives $R = 0.10$, an equilibrium growth rate of 1.5% and an optimal growth rate of 11.5%.

Policy 2: a consumption tax falling over time. Let consumers face the price $q(t)$ per unit of consumption, so that the budget constraint is $\dot{a} = ra + w - q(t)c$. The current-value Hamiltonian gives $c^{-\theta} = \mu q$, hence

$$\frac{\dot{c}}{c} = \frac{1}{\theta} \left(r - \rho - \frac{\dot{q}}{q} \right).$$

Setting

$$\frac{\dot{q}(t)}{q(t)} = -L\tilde{f}'(L)$$

— a consumption tax whose rate declines, equivalently a consumption subsidy growing at the rate $L\tilde{f}'(L)$, financed by lump-sum taxes — again yields $g_C = g_C^S$. This policy leaves the return to capital untouched and works purely on the intertemporal price of consumption.

Why these are genuinely different. Policy 1 corrects the externality where it arises, at the margin at which capital is rewarded; it would remain correct if the household's preferences changed. Policy 2 corrects the household's intertemporal margin instead, and its required rate happens to coincide only because the two margins are linked one for one in this model. A third variant — an investment subsidy at a *constant* rate, lowering the purchase price of capital goods once and for all — does *not* work: a constant relative price leaves the rate of return unchanged, so it produces a level effect and no growth effect at all.

Remark. The size of the required intervention is the real content here. The optimal subsidy is not a small correction but a transfer of the entire wage bill to capital, and it triples the growth rate in the calibration above. That is a measure of how much work the externality (11.36) is doing: assuming $A = BK$ makes the social return to capital the whole output-capital ratio rather than the capital share of it. It is a reason to read the Romer model of this section as a tractable illustration rather than a quantitative theory — and a reason why Part IV models the returns to knowledge through the profits of innovators, where the wedge is disciplined by the market structure rather than assumed.

Exercise 11.20

(a) *Equilibrium.* Firms take $A(t)$ as given, so

$$R(t) = F_K(K(t), A(t)L), \quad w(t) = A(t)F_L(K(t), A(t)L),$$

where F_L denotes the derivative with respect to the second argument. Imposing $A(t) = K(t)$ and writing $\tilde{f}(L) \equiv F(1, L)$, homogeneity of degree one gives $F(K, KL) = K\tilde{f}(L)$, while F_K is homogeneous of degree zero, so

$$R(t) = F_K(1, L) = \tilde{f}(L) - L\tilde{f}'(L) \equiv R, \quad w(t) = K(t)\tilde{f}'(L),$$

both exactly as in continuous time. R is a constant, independent of K . A competitive equilibrium is a path $[C(t), K(t), R(t), w(t)]_{t=0}^{\infty}$ such that the household maximizes $\sum_t \beta^t \frac{C(t)^{1-\theta}-1}{1-\theta}$ subject to

$$C(t) + K(t+1) = R(t)K(t) + w(t)L$$

taking (R, w) as given; firms maximize as above; and $A(t) = K(t)$. (Full depreciation makes R the gross return.) Market clearing gives the resource constraint

$$C(t) + K(t+1) = RK(t) + K(t)L\tilde{f}'(L) = \tilde{f}(L)K(t) = Y(t).$$

(b) *The BGP.* The household's Euler equation is

$$C(t)^{-\theta} = \beta R C(t+1)^{-\theta} \quad \implies \quad \boxed{\frac{C(t+1)}{C(t)} = (\beta R)^{1/\theta} \equiv 1 + g}$$

a constant, because R does not depend on K . Positive growth requires

$$\beta R > 1,$$

and finite utility (equivalently the transversality condition $\lim_{t \rightarrow \infty} \beta^t C(t)^{-\theta} K(t+1) = 0$, which reduces to $\beta(1+g)^{1-\theta} < 1$) requires

$$\beta R^{1-\theta} < 1.$$

Together,

$$\frac{1}{R} < \beta < R^{\theta-1},$$

an interval that is nonempty if and only if $R^{\theta} > 1$, that is $R > 1$: the private gross return must exceed one, which with full depreciation is the requirement that capital more than replace itself. Positive consumption also needs $\tilde{f}(L) > 1 + g$, which holds automatically for $\theta \geq 1$ since then $(\beta R)^{1/\theta} \leq \beta R < R < \tilde{f}(L)$. With $F = K^{\alpha}(AL)^{1-\alpha}$, $\alpha = 1/3$ and $\tilde{f}(L) = 4$, so that $R = \alpha \tilde{f}(L) = 4/3$, the program gives

θ	0.5	1	2	3
admissible β	(0.750, 0.866)	(0.750, 1.000)	(0.750, 1.333)	(0.750, 1.778)
$1 + g$ at $\beta = 0.8$	1.1378	1.0667	1.0328	1.0217
planner's $1 + g$	10.24	3.200	1.7889	1.4736

The planner internalizes $A = K$ and so faces the return $\tilde{f}(L)$ rather than R , giving $(\beta \tilde{f}(L))^{1/\theta} > 1 + g$ — when her problem is well posed, which for $\theta < 1$ it need not be (at $\theta = 0.5$ and $\beta = 0.8$, $\beta \tilde{f}(L)^{1-\theta} = 1.6 > 1$ and her objective is infinite).

(c) *No transitional dynamics.* The Euler equation holds at *every* date, and its right side is a constant, so consumption grows at $1 + g$ from $t = 0$ whatever $K(0)$ is. Substituting $C(t) = C(0)(1 + g)^t$ into the resource constraint gives the linear difference equation

$$K(t+1) = \tilde{f}(L)K(t) - C(0)(1 + g)^t,$$

whose general solution is

$$K(t) = \kappa \tilde{f}(L)^t + \frac{C(0)}{\tilde{f}(L) - (1 + g)}(1 + g)^t.$$

Now use the transversality condition. Since $\beta^t C(t)^{-\theta} = C(0)^{-\theta} [\beta(1+g)^{-\theta}]^t = C(0)^{-\theta} R^{-t}$, the condition $\lim \beta^t C(t)^{-\theta} K(t+1) = 0$ becomes

$$\lim_{t \rightarrow \infty} R^{-t} K(t+1) = 0.$$

The homogeneous part contributes $\tilde{f}(L) [\tilde{f}(L)/R]^t \rightarrow \infty$, because $\tilde{f}(L) > R$; so $\kappa = 0$. The particular part contributes $(1+g) [(1+g)/R]^t \rightarrow 0$ precisely when $1+g < R$, which is the condition $\beta < R^{\theta-1}$ already imposed. Hence

$$K(t) = K(0)(1+g)^t, \quad C(0) = [\tilde{f}(L) - (1+g)]K(0),$$

and the economy is on the balanced growth path from the first period. ■

Remark. The exercise's part (c) refers to “the BGP characterized in part c”, evidently a slip for part (b). The substantive point is the discrete-time counterpart of Exercise 11.1: a constant private return makes the Euler equation autonomous, so consumption growth is immediately constant, and the transversality condition then eliminates the explosive homogeneous solution for capital. The linearity that produces endogenous growth is also what produces the absence of transitional dynamics; the two are the same property seen from different sides.

Chapter 12

Modeling Technological Change

The eight exercises of Chapter 12 left unsolved in the instructor's manual: Exercises 12.1, 12.3, 12.4, 12.6–12.8, 12.10 and 12.12. The manual solves 12.2, 12.5, 12.9, 12.11, 12.13 and 12.14. Notation follows the book: ψ is the pre-innovation marginal cost, $\lambda > 1$ the size of a process innovation, μ its cost, $D(p)$ demand, $\varepsilon_D(p) = -pD'(p)/D(p)$ the elasticity of demand, p^M the post-innovation monopoly price, $\hat{p}^M$ the pre-innovation one, and ε the elasticity of substitution between Dixit–Stiglitz varieties. It is convenient throughout to write

$$\Pi^M(c) \equiv \max_p D(p)(p - c)$$

for the gross monopoly profit at marginal cost c , with maximizer $p^M(c)$. The numerical parts are computed by `code/ch12.py`.

Exercise 12.1

After a successful innovation the firm's marginal cost is $\lambda^{-1}\psi$, and as an unconstrained monopolist it solves

$$\max_p \pi(p) = D(p)(p - \lambda^{-1}\psi).$$

The first-order condition is

$$D'(p)(p - \lambda^{-1}\psi) + D(p) = 0 \quad \implies \quad p - \lambda^{-1}\psi = -\frac{D(p)}{D'(p)}.$$

Writing the elasticity of demand as $\varepsilon_D(p) \equiv -pD'(p)/D(p) > 0$, we have $-D(p)/D'(p) = p/\varepsilon_D(p)$, so

$$p\left(1 - \frac{1}{\varepsilon_D(p)}\right) = \lambda^{-1}\psi \quad \implies \quad \boxed{p^M = \frac{\lambda^{-1}\psi}{1 - \varepsilon_D(p^M)^{-1}}}$$

which is (12.2). ■

Three remarks complete the derivation. First, a positive price requires $\varepsilon_D(p^M) > 1$: a monopolist never operates on the inelastic part of its demand curve, because raising the price there would raise revenue and cut cost simultaneously. Second, the first-order condition is sufficient when marginal revenue $p(1 - \varepsilon_D(p)^{-1})$ is increasing in p , which holds for all the demand curves used below. Third, the same computation with $\lambda = 1$ gives the pre-innovation monopoly price $\hat{p}^M = p^M(\psi)$ of (12.6).

The drastic threshold. The innovation is drastic when firm 1 can charge p^M without being undercut by rivals still producing at cost ψ , that is when $p^M \leq \psi$. Substituting (12.2),

$$\frac{\lambda^{-1}\psi}{1 - \varepsilon_D(p^M)^{-1}} \leq \psi \quad \iff \quad \lambda \geq \lambda^* \equiv \frac{1}{1 - \varepsilon_D(p^M)^{-1}},$$

as the text states. The threshold is exactly the monopoly markup factor: the innovation must cut cost by at least as much as the monopolist would like to mark up.

Remark. With the constant-elasticity demand $D(p) = Ap^{-\varepsilon}$ the formula collapses to $p^M = \frac{\varepsilon}{\varepsilon-1}\lambda^{-1}\psi$, a fixed proportional markup, and $\lambda^* = \varepsilon/(\varepsilon - 1)$. This is the same expression as the Dixit–Stiglitz price (12.15)

of Section 12.4, and for the same reason: (12.10) makes each variety's demand isoelastic with elasticity ε . The two halves of the chapter share one formula.

Exercise 12.3

Deriving (12.5). Before the innovation the industry is competitive at price ψ , so producer surplus is zero. Measure everything as a change from that benchmark.

Under monopoly pricing (the drastic case, $p^M \leq \psi$), firm 1 sells $D(p^M)$ at the markup $p^M - \lambda^{-1}\psi$, so producer surplus rises by $D(p^M)(p^M - \lambda^{-1}\psi)$; the price falls from ψ to p^M , so consumer surplus rises by $\int_{p^M}^{\psi} D(p) dp$; and the innovation costs μ . Hence

$$\tilde{S}_1^I = D(p^M)(p^M - \lambda^{-1}\psi) + \int_{p^M}^{\psi} D(p) dp - \mu.$$

Under limit pricing (the non-drastic case), the price stays at ψ , so consumers gain nothing and the whole gain is the cost saving on the units still sold:

$$S_1^I = D(\psi)(\psi - \lambda^{-1}\psi) - \mu = D(\psi)\lambda^{-1}(\lambda - 1)\psi - \mu,$$

which is (12.5). Note that $S_1^I = \pi_1^I$ exactly: with limit pricing the private and social values of the innovation coincide, because the innovator captures the entire gain.

Proving PROPOSITION 12.2. It is cleanest to treat the two regimes as one function of the size of the innovation. For each $\lambda > 1$ let

$$p_1(\lambda) \equiv \min\{p^M(\lambda^{-1}\psi), \psi\}$$

be the equilibrium price, and define the gross (pre- μ) private value and equilibrium surplus

$$\Pi(\lambda) = D(p_1)(p_1 - \lambda^{-1}\psi), \quad \Sigma(\lambda) = \Pi(\lambda) + \int_{p_1}^{\psi} D(p) dp, \quad S(\lambda) = \int_{\lambda^{-1}\psi}^{\psi} D(p) dp,$$

so that $\Pi = \pi_1^I + \mu$ and $\Sigma = S_1^I + \mu$ for $\lambda < \lambda^*$, while $\Pi = \hat{\pi}_1^I + \mu$ and $\Sigma = \tilde{S}_1^I + \mu$ for $\lambda \geq \lambda^*$, and $S = S^I + \mu$ throughout.

Step 1: $\Pi(\lambda) < \Sigma(\lambda) < S(\lambda)$ at every λ . The first inequality is immediate, since $\Sigma - \Pi = \int_{p_1}^{\psi} D \geq 0$, with equality only under limit pricing. For the second, split the integral at p_1 :

$$S(\lambda) - \Sigma(\lambda) = \int_{\lambda^{-1}\psi}^{p_1} D(p) dp - D(p_1)(p_1 - \lambda^{-1}\psi) > 0,$$

because D is strictly decreasing, so $D(p) > D(p_1)$ for every $p < p_1$ and the integral strictly exceeds the rectangle. This is the deadweight loss of pricing above marginal cost. Combining, $\Pi < \Sigma < S$, which gives both $\hat{\pi}_1^I < S^I$ and $\tilde{S}_1^I < S^I$, and (for $\lambda < \lambda^*$, where $\Pi = \Sigma$) $\pi_1^I = S_1^I < S^I$.

Step 2: both are strictly increasing in λ . For $\lambda < \lambda^*$, $\Pi(\lambda) = \Sigma(\lambda) = D(\psi)\psi(1 - \lambda^{-1})$, plainly increasing. For $\lambda \geq \lambda^*$, $\Pi(\lambda) = \Pi^M(\lambda^{-1}\psi)$ and the envelope theorem gives

$$\frac{d\Pi}{d\lambda} = \frac{D(p^M)\psi}{\lambda^2} > 0, \quad \frac{d\Sigma}{d\lambda} = \frac{D(p^M)\psi}{\lambda^2} - D(p^M)\frac{dp^M}{d\lambda} > 0,$$

the second term positive because a larger λ lowers marginal cost and hence p^M . Both functions are continuous at λ^* (there $p^M = \psi$), so they are strictly increasing throughout. Since $\hat{\pi}_1^I$ and $\tilde{S}_1^I$ refer

to values of λ above λ^* while π_1^I and S_1^I refer to values below it,

$$\pi_1^I < \hat{\pi}_1^I \quad \text{and} \quad S_1^I < \tilde{S}_1^I,$$

which completes PROPOSITION 12.2. ■

With $D(p) = p^{-3}$ and $\psi = 1$, so that $\lambda^* = 1.5$, the program gives

λ	1.1	1.3	1.5	1.8	2.5
private	0.0909	0.2308	0.3333	0.4800	0.9259
equilibrium surplus	0.0909	0.2308	0.3333	0.7000	1.8148
full social value	0.1050	0.3450	0.6250	1.1200	2.6250

— the three columns coinciding in the limit-pricing region and fanning out once monopoly pricing takes over (Figure 10).

Reading the proposition. The first chain is the *appropriability effect*: even with a perfectly enforced patent, the innovator captures only part of the surplus its innovation creates, because it cannot price-discriminate and so leaves consumer surplus on the table. The second is the familiar deadweight loss: conditional on innovating, monopoly pricing does not realize the whole gain. Patents therefore buy innovation at the price of static inefficiency, and they still buy too little of it.

Exercise 12.4

Subtracting the two definitions,

$$\tilde{S}_1^I - \hat{\pi}_1^I = \left[D(p^M)(p^M - \lambda^{-1}\psi) + \int_{p^M}^{\psi} D(p) dp - \mu \right] - \left[D(p^M)(p^M - \lambda^{-1}\psi) - \mu \right] = \int_{p^M}^{\psi} D(p) dp.$$

In the regime where $\hat{\pi}_1^I$ is the relevant profit the innovation is drastic, so $p^M \leq \psi$ and the integral is over an interval of nonnegative length with a nonnegative integrand. Hence

$$\tilde{S}_1^I \geq \hat{\pi}_1^I > 0. \quad \blacksquare$$

The non-drastic branch is even simpler: there $S_1^I = \pi_1^I$ identically, so the implication holds with equality. With $\lambda = 1.8$ and $D(p) = p^{-3}$ the program reports a gap of 0.2200, the consumer surplus gain.

Why this matters. The difference between the social and private values is exactly the consumer surplus created, and in this environment that difference is a *gain*, never a loss. So in the competitive industry of Section 12.3.3 there can be no *excessive* innovation: any innovation a patent-holding firm is willing to pay for is one society should want. All the inefficiency runs one way, toward too little innovation. Exercise 12.8 asks whether this survives when the innovator displaces an incumbent monopolist rather than a competitive fringe, and the answer turns out to be more delicate than the text suggests.

Exercise 12.6

The comparison is between two firms facing the same innovation opportunity but different market positions. A firm in the competitive industry of Section 12.3.3 earns nothing before innovating; the monopolist of Section 12.3.4 already earns $\Pi^M(\psi) > 0$ and has no rivals, so after innovating it

prices freely at $p^M(\lambda^{-1}\psi)$ in either regime. Their gross values of the innovation are

$$\begin{aligned} \text{competitive firm: } V_C(\lambda) &= \begin{cases} D(\psi)(\psi - \lambda^{-1}\psi) & \lambda < \lambda^*, \\ \Pi^M(\lambda^{-1}\psi) & \lambda \geq \lambda^*, \end{cases} \\ \text{monopolist: } V_I(\lambda) &= \Pi^M(\lambda^{-1}\psi) - \Pi^M(\psi). \end{aligned}$$

The drastic branch. Here

$$V_C - V_I = \Pi^M(\psi) > 0,$$

so $\Delta\hat{\pi}_1^I = V_I - \mu < V_C - \mu = \hat{\pi}_1^I$. The gap is *exactly* the profit the monopolist destroys by innovating — Arrow's replacement effect in its purest form.

The limit-pricing branch. Now $V_C = D(\psi)(\psi - \lambda^{-1}\psi)$, which is *smaller* than $\Pi^M(\lambda^{-1}\psi)$, so the comparison is no longer immediate. Use the envelope theorem on Π^M : since $\partial[D(p)(p - c)]/\partial c = -D(p)$,

$$\frac{d\Pi^M(c)}{dc} = -D(p^M(c)) \quad \implies \quad V_I = \Pi^M(\lambda^{-1}\psi) - \Pi^M(\psi) = \int_{\lambda^{-1}\psi}^{\psi} D(p^M(c)) dc.$$

Meanwhile

$$V_C = D(\psi)(\psi - \lambda^{-1}\psi) = \int_{\lambda^{-1}\psi}^{\psi} D(\psi) dc.$$

Compare the integrands. In the non-drastic case $p^M(\lambda^{-1}\psi) > \psi$, and $p^M(\cdot)$ is increasing, so $p^M(c) \geq p^M(\lambda^{-1}\psi) > \psi$ for every $c \in [\lambda^{-1}\psi, \psi]$; since D is strictly decreasing,

$$D(p^M(c)) < D(\psi) \quad \text{for all such } c,$$

and therefore $V_I < V_C$, that is

$$\boxed{\Delta\hat{\pi}_1^I < \pi_1^I} \quad \blacksquare$$

Combining the two branches with PROPOSITION 12.2's $\pi_1^I < \hat{\pi}_1^I$ gives the full chain of PROPOSITION 12.3. With $D(p) = p^{-3}$, $\psi = 1$ and $\Pi^M(\psi) = 0.1481$, the program gives

λ	1.1	1.3	1.8	4.0
competitive	0.0909	0.2308	0.4800	2.3704
incumbent	0.0311	0.1022	0.3319	2.2222
gap	0.0598	0.1285	0.1481	0.1481

the gap settling at exactly $\Pi^M(\psi)$ once the innovation becomes drastic.

Two readings of the same integral. The envelope formula says that the value of a cost reduction is the quantity sold, integrated over the cost range — the innovator saves D per unit of cost reduction. The competitive firm computes that quantity at the *competitive* price ψ ; the monopolist computes it at its own, higher, monopoly price, where it sells less. The monopolist therefore values the same cost reduction less, both because it sells fewer units and because part of its gain merely replaces profits it was already making. Hence COROLLARY 12.1: entrants innovate more than incumbents, which is the germ of the Schumpeterian growth models of Chapter 14.

Exercise 12.7

The threshold. Write $\Pi_E \equiv V_C(\lambda)$ for the entrant's gross value — it earns nothing today and becomes the seller if it innovates — and $V_I = \Pi^M(\lambda^{-1}\psi) - \Pi^M(\psi)$ for the incumbent's, as in Exercise 12.6. Net values are $\Pi_E - \chi\mu$ and $V_I - \mu$, so the entrant has the stronger incentive when

$$\Pi_E - \chi\mu > V_I - \mu \quad \Longleftrightarrow \quad (\chi - 1)\mu < \Pi_E - V_I,$$

that is

$$\chi < \bar{\chi} \equiv 1 + \frac{\Pi_E - V_I}{\mu}$$

Exercise 12.6 established $\Pi_E - V_I > 0$ in both pricing regimes, so $\bar{\chi} > 1$ always: a *slightly* handicapped entrant still out-invests the incumbent, and the size of the handicap it can absorb is the replacement effect measured in units of the research cost. In the drastic case $\Pi_E - V_I = \Pi^M(\psi)$ exactly, so $\bar{\chi} = 1 + \Pi^M(\psi)/\mu$.

The elasticity of demand. Take $D(p) = p^{-\varepsilon}$ with $\psi = 1$, for which

$$p^M(c) = \frac{\varepsilon}{\varepsilon - 1}c, \quad \Pi^M(c) = \frac{c^{1-\varepsilon}}{\varepsilon - 1} \left(\frac{\varepsilon - 1}{\varepsilon} \right)^\varepsilon \quad \Longrightarrow \quad \frac{\Pi^M(\lambda^{-1})}{\Pi^M(1)} = \lambda^{\varepsilon-1}.$$

In the drastic region the ratio of the two gross incentives is therefore

$$\frac{\Pi_E}{V_I} = \frac{\Pi^M(\lambda^{-1})}{\Pi^M(\lambda^{-1}) - \Pi^M(1)} = \frac{\lambda^{\varepsilon-1}}{\lambda^{\varepsilon-1} - 1},$$

which is strictly *decreasing* in ε , and so is $\bar{\chi} - 1 = \Pi^M(1)/\mu$, since $\Pi^M(1) = \frac{1}{\varepsilon-1} \left(\frac{\varepsilon-1}{\varepsilon} \right)^\varepsilon$ falls with ε . With $\lambda = 1.8$ and $\mu = 0.05$ the program gives

ε	1.5	2	3	5	10
$\bar{\chi}$	7.26	5.89	3.96	2.64	1.77
Π_E/V_I	3.38	2.22	1.45	1.11	1.005

Intuition. The incumbent's handicap is the rent it destroys, $\Pi^M(\psi)$, and that rent is a markup times a quantity. More elastic demand means a smaller markup: the incumbent has less to lose, so the replacement effect weakens and the entrant's relative advantage shrinks toward nothing as $\varepsilon \rightarrow \infty$ (where the incumbent is effectively a competitive firm and the two coincide). Read in reverse: the replacement effect is strongest precisely where market power is strongest, so it is in the most concentrated, least elastic industries that one should expect innovation to come from outside.

Remark. This gives some content to a familiar empirical puzzle. Incumbents do a great deal of R&D, which the replacement effect alone would not predict. The model points to the answer: incumbents innovate when their cost advantage in research is large (χ well above $\bar{\chi}$), and $\bar{\chi}$ is small exactly where demand is elastic and markups thin. Both incumbent-led incremental innovation and entrant-led radical innovation are consistent with the same mechanism at different parameter values.

Exercise 12.8

(a) *First, the environment of Section 12.3.4 will not produce it.* Suppose an entrant makes a process innovation and displaces a monopolist whose marginal cost is ψ . Measure surplus against

the *pre-innovation* outcome, in which the incumbent charges $\hat{p}^M = p^M(\psi)$. After entry the price is $p_1 = \min\{p^M(\lambda^{-1}\psi), \psi\} \leq \psi < \hat{p}^M$, so

$$\Delta S = \underbrace{\Pi^M(\lambda^{-1}\psi) - \Pi^M(\psi)}_{\text{producer surplus}} + \underbrace{\int_{p_1}^{\hat{p}^M} D(p) dp}_{\text{consumer surplus}} - \mu, \quad \text{entrant's value} = \Pi_E - \mu,$$

and in the drastic case $\Pi_E = \Pi^M(\lambda^{-1}\psi)$, so

$$\Delta S - (\Pi_E - \mu) = \int_{p_1}^{\hat{p}^M} D(p) dp - \Pi^M(\psi).$$

But $\Pi^M(\psi) = D(\hat{p}^M)(\hat{p}^M - \psi)$, and since D is decreasing and $p_1 \leq \psi$,

$$\int_{p_1}^{\hat{p}^M} D(p) dp \geq \int_{\psi}^{\hat{p}^M} D(p) dp > D(\hat{p}^M)(\hat{p}^M - \psi) = \Pi^M(\psi),$$

so $\Delta S > \Pi_E - \mu$: the social value *exceeds* the private value, and excessive innovation is impossible. The same computation with the limit-pricing Π_E gives the same conclusion. The program confirms it at every λ for both isoelastic and linear demand.

The reason is structural. A process innovation under Bertrand competition must *undercut* the incumbent to be worth anything, so it necessarily drags the price from $\hat{p}^M$ down to at most ψ , and the consumer surplus that releases always outweighs the rent it destroys.

An environment in which it does occur. What the business-stealing intuition requires is an innovation that captures the incumbent's rents *without* competing them into consumer surplus. Suppose the innovation is a new product that replaces the incumbent's rather than undercutting it: the entrant's good has the same marginal cost ψ but is $\lambda > 1$ times more appealing, in the sense that its demand curve is $\lambda D(p)$. The entrant becomes the monopolist and, since scaling demand does not change the optimal price, charges the same $\hat{p}^M$. Then

$$\text{social gain} = (\lambda - 1)[\Pi^M(\psi) + CS(\hat{p}^M)] - \mu, \quad \text{private gain} = \lambda \Pi^M(\psi) - \mu,$$

where $CS(p) = \int_p^\infty D(q) dq$. The private gain exceeds the social gain when

$$\lambda \Pi^M(\psi) > (\lambda - 1)[\Pi^M(\psi) + CS(\hat{p}^M)] \iff (\lambda - 1)CS(\hat{p}^M) < \Pi^M(\psi),$$

that is, whenever

$$\lambda < 1 + \frac{\Pi^M(\hat{p}^M)}{CS(\hat{p}^M)}$$

For $D(p) = p^{-\varepsilon}$ the ratio Π^M/CS equals $\psi/\hat{p}^M = (\varepsilon - 1)/\varepsilon$, so the condition is $\lambda < 2 - 1/\varepsilon$. Taking $\varepsilon = 3$, $\psi = 1$ and $\lambda = 1.2$, the program gives $\Pi^M(\psi) = 0.14815$ and $CS(\hat{p}^M) = 0.22222$, hence

$$\text{social gain} = 0.074074 - \mu, \quad \text{private gain} = 0.177778 - \mu.$$

For any $\mu \in (0.0741, 0.1778)$ — at $\mu = 0.12$, a private value of +0.0578 against a social value of −0.0459 — the entrant innovates and total surplus *falls*. This proves PROPOSITION 12.4. ■

Remark. The text asserts PROPOSITION 12.4 with $\tilde{S}_1^I$ as defined in (12.5), but that quantity measures surplus against a *competitive* benchmark priced at ψ , and Exercise 12.4 shows it can never fall short of $\hat{\pi}_1^I$.

Measured correctly against the incumbent monopolist's outcome, the first calculation above shows that a process innovation cannot be excessive either. The proposition is nonetheless true, and the intuition the text gives for it is exactly right — “the social planner values the profits made by the monopolist, the entrant only values the profits it will make” — but it needs an environment in which the entrant takes those profits without passing them to consumers. The quality-ladder models of Chapter 14, where each innovator limit-prices at its predecessor's price, are of precisely that kind.

(b) *What makes it more likely.* The condition $(\lambda - 1)CS < \Pi^M$ names the forces directly.

- *Small innovations.* The social gain is proportional to $\lambda - 1$ while the business stolen is proportional to λ ; as $\lambda \downarrow 1$ the genuine improvement vanishes but the rents transferred do not. Excessive innovation is a phenomenon of incremental, rent-seeking improvements, not of radical ones.
- *Large incumbent rents relative to consumer surplus.* The threshold is $1 + \Pi^M/CS$, so anything that raises profit relative to consumer surplus at the monopoly price widens the region. With isoelastic demand $\Pi^M/CS = (\varepsilon - 1)/\varepsilon$, so, perhaps surprisingly, *more* elastic demand makes excessive innovation more likely: a flat demand curve leaves little consumer surplus above the monopoly price relative to the rectangle of profit.
- *Innovations that displace rather than undercut.* This is the decisive structural condition, and it is what fails in part (a). New varieties, quality improvements sold at the old price, and business-model innovations all transfer rents without a price war; pure cost reductions under Bertrand competition do not.
- *Duplication of research effort.* Not modeled here, but the most important force in practice: if n firms each spend μ racing for the same prize, society pays $n\mu$ for one innovation while each firm compares its own μ with its chance of winning the whole rent. Chapter 14 returns to this.

Against all of these stands the appropriability effect of PROPOSITION 12.2, which pushes the other way. The honest conclusion — the one the text draws — is that the sign of the wedge between private and social innovation incentives is not determined by theory, and must be settled empirically, innovation by innovation.

Exercise 12.10

The consumption index is

$$C = \left[\sum_{i=1}^N c_i^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

and (12.10) states $(c_i/C)^{-1/\varepsilon} = p_i/P$, that is

$$c_i = C \left(\frac{p_i}{P} \right)^{-\varepsilon}. \quad (*)$$

(This is the definition of P : the first-order condition between varieties, $(c_i/c_{i'})^{-1/\varepsilon} = p_i/p_{i'}$, says that $p_i(c_i/C)^{1/\varepsilon}$ is the same for every i , and P is that common value. It uses $\partial C/\partial c_i = (C/c_i)^{1/\varepsilon}$, which follows from differentiating the index.)

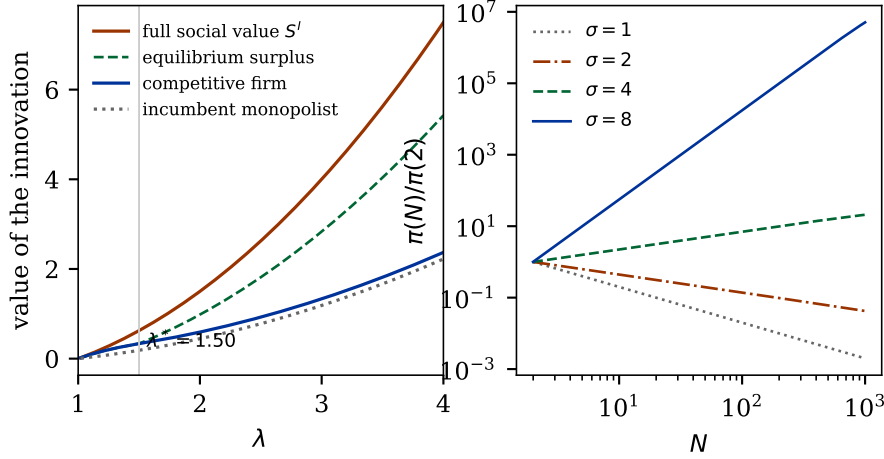


FIGURE 10. Left (Exercises 12.3, 12.4 and 12.6): private and social values of a process innovation with isoelastic demand. Below λ^* limit pricing makes the competitive firm's private value equal to the equilibrium surplus; above it the curves fan out, and the incumbent monopolist's value falls short of the competitive firm's by exactly $\Pi^M(\psi)$. Right (Exercise 12.12): Dixit–Stiglitz profits against the number of varieties, for a differentiated sector with a small expenditure share. Profits fall with N when $\sigma = 1$ and rise once σ clears the threshold.

Substituting (*) into the definition of C ,

$$C = \left[\sum_{i=1}^N \left(C(p_i/P)^{-\varepsilon} \right)^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}} = C \left[\sum_{i=1}^N \left(\frac{p_i}{P} \right)^{1-\varepsilon} \right]^{\frac{\varepsilon}{\varepsilon-1}},$$

so the bracket must equal one:

$$\sum_{i=1}^N \left(\frac{p_i}{P} \right)^{1-\varepsilon} = 1 \quad \Longleftrightarrow \quad P^{\varepsilon-1} \sum_{i=1}^N p_i^{1-\varepsilon} = 1 \quad \Longleftrightarrow \quad \boxed{P = \left[\sum_{i=1}^N p_i^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}}$$

which is (12.11). ■

P deserves its name. Using (*) again, total spending on the differentiated varieties is

$$\sum_{i=1}^N p_i c_i = C P^{\varepsilon} \sum_{i=1}^N p_i^{1-\varepsilon} = C P^{\varepsilon} P^{-(\varepsilon-1)} = PC,$$

so the budget constraint (12.9) collapses to $PC + y \leq m$, which is (12.12): P is exactly the price at which the composite C may be treated as a single good. This is what makes the two-stage budgeting of Section 12.4 legitimate, and it is why every model in the rest of the book can write demand for a variety as $(p_i/P)^{-\varepsilon}C$ and be done with it.

Love of variety in one line. If all prices are equal at p , then $P = N^{1/(1-\varepsilon)}p = pN^{-1/(\varepsilon-1)}$, which is strictly decreasing in N since $\varepsilon > 1$. Adding varieties lowers the true cost of a unit of the composite even though no individual price has moved — the price-index counterpart of $U(\bar{C}/N, \dots, \bar{C}/N, y) = u(N^{1/(\varepsilon-1)}\bar{C}, y)$. The whole of the aggregate demand externality of Exercise 12.12, and much of the growth-through-variety literature of Chapter 13, rests on this single exponent.

Exercise 12.12

Setting up. With the symmetric price (12.15) the ideal price index is $P = N^{-1/(\varepsilon-1)} \frac{\varepsilon}{\varepsilon-1} \psi$ by (12.16), and the text's expression for profits is

$$\pi = \frac{1}{\varepsilon N} [m - g(P, m)] = \frac{PC}{\varepsilon N},$$

using $PC = m - g(P, m)$ from (12.13). Two forces act when N rises: the term $1/N$ divides a given amount of spending over more firms, while P falls, which raises spending on the composite if households substitute toward it.

The condition. Differentiate, using $\partial P / \partial N = -P / [(\varepsilon - 1)N]$:

$$\frac{\partial \pi}{\partial N} = \frac{1}{\varepsilon} \left[\frac{-g_P(P, m) \partial P / \partial N}{N} - \frac{m - g(P, m)}{N^2} \right] = \frac{1}{\varepsilon N^2} \left[\frac{P g_P(P, m)}{\varepsilon - 1} - PC \right],$$

so

$$\boxed{\frac{\partial \pi}{\partial N} > 0 \iff g_P(P, m) > (\varepsilon - 1) C}$$

Spending on the outside good must respond strongly enough to the price index of the differentiated bundle. Equivalently, writing $E \equiv PC$ for expenditure on the composite and $\eta_E \equiv d \log E / d \log P = -g_P / C$,

$$\frac{d \log \pi}{d \log N} = -1 - \frac{\eta_E}{\varepsilon - 1} > 0 \iff \eta_E < -(\varepsilon - 1) \iff \left| \frac{d \log C}{d \log P} \right| > \varepsilon.$$

The demand for the composite must be more elastic than the substitution among the varieties within it.

In terms of u . Let u be CES with elasticity of substitution σ between C and y , and let $s_C = PC/m$ be the expenditure share of the differentiated sector. Then $\eta_E = -(\sigma - 1)(1 - s_C)$, and the condition becomes

$$\sigma > 1 + \frac{\varepsilon - 1}{1 - s_C}.$$

Three cases are worth naming. If $\sigma = 1$ (Cobb–Douglas u) expenditure on the composite is fixed, $\eta_E = 0$, and $d \log \pi / d \log N = -1$: profits always fall, whatever ε or s_C . If $\sigma < 1$ they fall faster still. Profits rise only when C and y are strong substitutes, and the requirement tightens as the differentiated sector grows — a sector that already absorbs all of spending has nowhere to draw further demand from. With $\varepsilon = 3$, $N = 10$ and $m = 1$ the program gives

σ	1	2	4	8
$d \log \pi / d \log N$ (s_C small)	−1.00	−0.50	+0.50	+2.50
$d \log \pi / d \log N$ ($s_C \approx 0.5$ –1)	−1.00	−0.84	−0.86	−0.98

The aggregate demand externality. When the condition holds, each firm's entry makes the *other* firms more profitable, through a purely pecuniary channel: more varieties lower the ideal price index, which draws spending into the sector. In a competitive, complete-market economy such pecuniary externalities would net out and the First Welfare Theorem would apply; here they do not, because

prices already exceed marginal cost. That is what makes the effect a genuine externality with first-order welfare consequences, and it is the mechanism behind the multiple equilibria and poverty traps of Chapter 21 — and, in a different guise, behind the scale effects of Chapter 13.